

SANGGALA CORRIDOR PROJECT



**SANGGALA
CORRIDOR
PROJECT**



Project title	Sanggala Corridor Project
Project ID	5458
Crediting period	16 March 2023 to 15 March 2073
Project lifetime	16 March 2023 to 15 March 2073 (50 years)
(CCB) GHG accounting period	16 March 2023 to 15 March 2073 (50 years)
Original date of issue	03 March 2025 for pipeline listing
Most recent date of issue	03 March 2025
Version	1.0
VCS Standard version	VCS Version 4.7
CCB Standards version	CCB Version 3.1
Project location	Indonesia, West Kalimantan
Project proponent(s)	PT CarbonX Bumi Harmoni Dessi Yuliana

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Validation/verification body	AENOR, S.A.U
History of CCB status	This is the first version for seeking validation under both the Climate Community, and Biodiversity Standard, as well as the Verified Carbon Standard.
Gold Level criteria	The project is seeking the Triple Gold Level criteria for Climate Change Adaptation Benefits (GL1), Exceptional Benefits for Community (GL2) and Biodiversity (GL3). GL1: The project will implement activities that increase the resilience of local communities and biodiversity to climate change, including activities that support livelihood diversification, habitat conservation, and restoration. GL2: A total of 10 villages adjacent to the project area are included as the target beneficiaries of this project. Over 50% of the villagers in the project zone are below the national poverty line. The project will provide tangible benefits to local communities living around the project area through the creation of program to improve their livelihoods, health, education, and well-being. GL3: The project is home to 59 threatened species, including <i>Presbytis chrysomelas</i>, <i>Manis Javanica</i>, and <i>Hylobates muelleri abbotti</i>. The project activities will contribute to the conservation of these species and their habitat and restore degraded habitats to improve connectivity between

	each forest patches, enhance habitat quality, and reduce threats to biodiversity.
Expected verification schedule	N/A
Prepared by	EcoSecurities

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1 SUMMARY OF PROJECT BENEFITS

The Sanggala Corridor Project is a forest conservation and restoration project in Sanggau and Landak Regencies, West Kalimantan Province, Indonesia. The project area is located within a state-designated production forest, where the forest management company, PT Citra Mulia Inti (PT CMI), holds the concession ownership permit, which cover an area of 20,645 hectares. While the projects area itself encompasses 16,285.68 hectares.

Prior to the project initiation, PT CMI had an approved plan to convert the entire project area into an Industrial Timber Plantation. However, the Sanggala Corridor project aims to protect and restore the natural forest ecosystem within this area, thereby contributing to long-term conservation goals and supporting sustainable development in the region.

The Sanggala Corridor Project is being managed by the Indonesian company PT CarbonX Bumi Harmoni (CarbonX). Currently, CarbonX is the majority shareholder in PT CMI and is overseeing the transition from a forest conversion and Industrial Timber Plantation program to sustainable forest management practice that includes forest conservation, restoration, and community development program.

The project is expected to deliver a range of climate, community, and biodiversity benefits by preserving the remaining forest area and restoring the non-forest area through restoration and agroforestry activities. In terms of climate benefits, it is estimated that the Sanggala Corridor Project will result in net emission reduction and removals of 8,875,703 tCO₂e over the project's 50-year lifespan, with an average annual emission reduction and removal of 177,514 tCO₂e. These outcomes will be achieved through activities aligned with the VM0007 REDD+ Methodology framework (Avoided Planned Deforestation) and VM0047 Afforestation, Reforestation, and Revegetation. Additionally, the project will deliver climate change adaptation benefits by increasing the resilience of local communities and biodiversity to climate change.

The project meets the CCB criteria for exceptional community benefits. There are 10 villages within the project zone, with over 50% of the villagers living below the national poverty line of IDR 2,592,657 per month per household¹. The project activities will improve access to education and health services, create employment opportunities, and offer alternative livelihoods to the communities. These activities are expected to positively impact the communities' livelihoods and overall well-being.

Additionally, the Sanggala Corridor Project ecosystem is home to 59 globally threatened species, including *Presbytis chrysomelas*, *Manis Javanica*, and *Hylobates muelleri abbotti*. The project will contribute to the conservation of these species and their habitat by improving habitat quality and reducing threats to biodiversity.

¹ In 2023, the national poverty line was set at a per capita expenditure of IDR 550,458 per month per capita or 2,592,657 per household (4.71 members per household). [Indonesia Poverty Profile in March 2023 - BPS-Statistics Indonesia](#).

1.1 Unique Project Benefits

Outcome or impact estimated by the end of project lifetime	Section Reference
1) The project is expected to maintain the High Conservation Value identified in the project zone. By retaining the High Conservation Value area, the project is expected to increase the community's well-being and biodiversity conservation.	2.1.17, 4.1.3, 4.2.4
2) The project is expected to increase community resilience to climate change by diversifying agricultural activity through agroforestry.	2.1.17, 5.1.2, 5.2.4,
3) The project is expected to maintain and enhance the population density of three trigger species, <i>Presbytis chrysomelas</i> , <i>Manis javanica</i> , and <i>Hylobates muelleri abbotti</i> .	2.1.17, 5.5.2

1.2 Standardized Benefit Metrics

Category	Metric	Estimated by the end of project lifetime	Section reference
GHG emission reductions or carbon dioxide removals	Net estimated removals in the project area, measured against the without-project scenario	3,596,079 tCO ₂ e	3.2.5
	Net estimated reductions in the project area, measured against the without-project scenario	5,279,624 tCO ₂ e	3.2.5
Forest ² cover	For REDD ³ projects: Estimated number of hectares of reduced forest loss in the project area, measured against the without-project scenario	9,309.46 hectares	3.1.3
	For ARR ⁴ projects: Estimated number of hectares with increased forest cover in the project area, measured against the without-project scenario	6,976.22 hectares	3.1.3
Improved land management	Number of hectares of existing production forest land where IFM ⁵ practices are expected to occur as a result of project activities, measured against the without-project scenario	Not Applicable	-
	Number of hectares of non-forest land where improved land management practices are expected to occur as a result of	Not applicable	-

² Land with woody vegetation that meets an internationally accepted definition (e.g., UNFCCC, FAO, or IPCC) of what constitutes a forest, which includes threshold parameters, such as minimum forest area, tree height and level of crown cover, and may include mature, secondary, degraded and wetland forests (VCS Program Definitions)

³ Reduced emissions from deforestation and forest degradation (REDD) - Activities that reduce GHG emissions by slowing or stopping conversion of forests to non-forest land and/or reduce the degradation of forest land where forest biomass is lost (VCS Program Definitions)

⁴ Afforestation, reforestation and revegetation (ARR) - Activities that increase carbon stocks in woody biomass (and in some cases soils) by establishing, increasing and/or restoring vegetative cover through the planting, sowing and/or human-assisted natural regeneration of woody vegetation (VCS Program Definitions)

⁵ Improved forest management (IFM) - Activities that change forest management practices and increase carbon stock on forest lands managed for wood products such as saw timber, pulpwood, and fuelwood (VCS Program Definitions)

Category	Metric	Estimated by the end of project lifetime	Section reference
	project activities, measured against the without-project scenario		
Training	Total number of community members who are expected to have improved skills and/or knowledge resulting from training provided as part of project activities	2,400 people	2.1.17, 4.2.1
	Number of female community members who are expected to have improved skills and/or knowledge resulting from training as part of project activities	1,152 people	2.1.17, 4.2.1
Employment	Total number of people expected to be employed in project activities ⁶ , expressed as number of full-time employees ⁷	1,236 people	2.1.17, 4.2.1
	Number of women expected to be employed as a result of project activities, expressed as number of full-time employees	589 women	2.1.17, 4.2.1
Livelihoods	Total number of people expected to have improved livelihoods ⁸ or income generated as a result of project activities	2,400 people	2.1.17, 4.2.1
	Number of women expected to have improved livelihoods or	1,152 people	2.1.17, 4.2.1

⁶ Employed in project activities means people directly working on project activities in return for compensation (financial or otherwise), including employees, contracted workers, sub-contracted workers and community members that are paid to carry out project-related work.

⁷ Full time equivalency is calculated as the total number of hours worked (by full-time, part-time, temporary and/or seasonal staff) divided by the average number of hours worked in full-time jobs within the country, region or economic territory (adapted from the UN System of National Accounts (1993) paragraphs 17.14[15.102];[17.28])

⁸ Livelihoods are the capabilities, assets (including material and social resources) and activities required for a means of living (Krantz, Lasse, 2001. The Sustainable Livelihood Approach to Poverty Reduction. SIDA). Livelihood benefits may include benefits reported in the Employment metrics of this table.

Category	Metric	Estimated by the end of project lifetime	Section reference
	income generated as a result of project activities		
Health	Total number of people for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	11,750 people	2.1.17, 4.2.1
	Number of women for whom health services are expected to improve as a result of project activities, measured against the without-project scenario	5,591 people	2.1.17, 4.2.1
Education	Total number of people for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	5,969 people	2.1.17, 4.2.1
	Number of women and girls for whom access to, or quality of, education is expected to improve as result of project activities, measured against the without-project scenario	2,865 women	2.1.17, 4.2.1
Water	Total number of people who are expected to experience increased water quality and/or improved access to drinking water as a result of project activities, measured against the without-project scenario	5,865 people	-
	Number of women who are expected to experience increased water quality and/or improved access to drinking water as a result of project	2,729 people	-

Category	Metric	Estimated by the end of project lifetime	Section reference
	activities, measured against the without-project scenario		
Well-being	Total number of community members whose well-being ⁹ is expected to improve as a result of project activities	11,750 people	2.1.17, 4.2.1
	Number of women whose well-being is expected to improve as a result of project activities	5,591 people	2.1.17
Biodiversity conservation	Expected change in the number of hectares managed significantly better by the project for biodiversity conservation ¹⁰ , measured against the without-project scenario	16,285 hectares	3.1.3, 5.4.1
	Expected number of globally Critically Endangered or Endangered species ¹¹ benefiting from reduced threats as a result of project activities ¹² , measured against the without-project scenario	19 species	5.5

⁹ Well-being is people's experience of the quality of their lives. Well-being benefits may include benefits reported in other metrics of this table (e.g. Training, Employment, Livelihoods, Health, Education and Water), and may also include other benefits such as strengthened legal rights to resources, increased food security, conservation of access to areas of cultural significance, etc.

¹⁰ Managed for biodiversity conservation in this context means areas where specific management measures are being implemented as a part of project activities with an objective of enhancing biodiversity conservation, e.g. enhancing the status of endangered species

¹¹ Per IUCN's Red List of Threatened Species

¹² In the absence of direct population or occupancy measures, measurement of reduced threats may be used as evidence of benefit

2 PROJECT DETAILS

2.1 Project Goals, Design and Long-Term Viability

2.1.1 Summary Description of the Project (VCS, 3.2, 3.6, 3.10, 3.11, 3.13, 3.14; CCB, G1.2)

The Sanggala Corridor Project is a forest conservation and restoration initiative located in Sanggau and Landak Regencies of West Kalimantan Province, Indonesia. Covering an area of 16,285.68 hectares, the project is situated entirely within a state-designated production forest where the forest management company, PT Citra Mulia Inti (PT CMI) holds the concession ownership permit. PT CMI is legally authorized to convert the existing standing forest to an Industrial Timber Plantation under its Forest Utilization Business License, issued on 2015 by the Minister of Environment and Forestry Decree SK 2/1/IUPHHK-HTI/PDMN/2015 (and amended by Ministry of Environment and Forestry Decree SK.1039/Menlhkh/Setjen/HPL.01/11/2021).

Prior to the project's initiation, PT CMI had an approved plan to clear the forest and establish an Industrial Timber Plantation, in the current forested and non-forested area, with *Anthocephalus cadamba* as the main industrial tree crop and *Havea brasiliensis* (Para rubber) as the livelihood crop¹³.

Early 2023 marked a turning point for the operational of the PT CMI concession area, as PT Carbonx Bumi Harmoni (CarbonX) acquired the majority shareholding of the company and initiated The Sanggala Corridor Project. CarbonX is also overseeing the transition from the original forest conversion and Industrial Timber Plantation program initiated by previous management to forest protection, restoration, and community development program.

The project objective is to reduce emissions and increase carbon sequestration, conserve biodiversity and protect the forest ecosystem, and improve the community livelihood and wellbeing.

The project zone includes 10 villages that play a critical role in the success of the Sanggala Corridor Project. To support community development and enhance resilience, the project will implement a comprehensive, community-based agroforestry program. This program along with other community development initiatives, will be co-designed with the local communities to ensure their relevance and sustainability.

In addition to agroforestry, the project will introduce alternative livelihoods programs such as ecotourism, agricultural commodities production, and direct employment opportunities within the project itself. These initiatives aim to diversify income sources and improve the community livelihoods and well-being. With regards to the well-being of the community, the project will improve access on healthcare and education by enhancing of the already available program designed by the government and providing reasonable equivalent interventions.

¹³ Livelihood crops are crops cultivated to improve community welfare. These can include primary crops that produce timber or perennial trees that produce non-timber forest products, or a combination of both. In PT CMI's management plan, or RKU, PT CMI have committed to supporting the community by partnering with them in the establishment and management of Para Rubber through a partnership scheme.

Situated in the Borneo Lowland Rainforest ecoregion, the Sanggala Corridor Project is adjacent to a protected forest and the Gunung Nyiut Nature Reserve. The project area is home to several threatened species including *Presbytis chrysomelas* (Cross-marked Langur), *Manis javanica* (Sunda Pangolin), and *Hylobates muelleri abbotti* (Western Grey Gibbon). Forest conservation related activities such as forest and fire patrolling, restoration and biodiversity monitoring will be conducted to conserve the biodiversity and the forest ecosystem.

The project is expected to reduce emissions by avoiding planned deforestation and implementing an afforestation, reforestation, and revegetation project in degraded forest areas and community-identified farmlands. The project will generate net Green House Gas (GHG) emission reduction and removals of 8,875,703 tCO₂e over a 50-year crediting period, with an average annual reduction of 177,514 tCO₂e.

It is important to note that the Sanggala Corridor Project area is not located in jurisdictional REDD+ program. This independent project seeks to make a significant contribution to emissions reduction efforts and sustainable land use management, while fostering environmental stewardship and socio-economic development within the region.

2.1.2 Audit History (VCS, 4.1)

Audit type	Period	Program	Validation/verification body name	Number of years
Validation		Not a jurisdictional REDD+ program	AENOR, S.A.U	

2.1.3 Sectoral Scope and Project Type (VCS, 3.2)

Sectoral scope	14: Agriculture, forestry, and other land use
AFOLU project category ¹⁴	Reduced Emission from Deforestation and Degradation (REDD) Afforestation, Reforestation, and Revegetation (ARR)
Project activity type	Avoiding Planned Deforestation (APD) Afforestation, Reforestation, and Revegetation (ARR)

2.1.4 Project Eligibility (VCS, 3.1, 3.6, 3.8, 3.18, 4.1; CCB Program Rules, 4.2.4, 4.6.4)

This project utilizes eligible methodologies under the VCS Program, namely VM0007 (REDD project activity) and VM0047 (ARR project activity). The project applies the latest versions of these applicable methodologies and complies fully with these methodologies, including the application of all referenced tools and modules. The project meets all eligibility criteria under both VM0047 and VM0007 methodologies:

¹⁴ See Appendix 1 of the VCS Standard

- The ARR project activity will increase carbon sequestration and/or reduce GHG emissions by establishing, increasing, or restoring vegetative cover (forest or non-forest) through the planting, sowing, or human-assisted natural regeneration of woody vegetation. This condition meets the eligibility criteria set in the VCS Standard 4.7 Appendix 1.
- The project area designated for ARR category occurs within an area that has been classified as non-forest land for the past ten years.
- The project uses an area-based approach for the ARR category, which requires continuous tree and/or shrub cover on any contiguous area exceeding one hectare. In this project, all the polygons that will be subject to restoration under the area-based approach are greater than one hectare.
- ARR project activities do not involve mechanical vegetation removal to offsite locations or burning any pre-existing dead wood for site preparation. All materials produced during site preparation will remain within the project boundary.
- The ARR project is not located on organic soils or in wetlands. The absence of organic soils is indicated by no histosol soil class identified in the project area. The project area only contains ultisol, oxisol, and inceptisol (see Map 3).
- ARR project activity will not take place in tidal wetland.
- ARR project activity will include direct (manual planting) and indirect activity (facilitating natural regeneration).
- As per VCS Standard 4.7 Appendix 1., the APD project area meets an internationally accepted definition of forest of Indonesia country threshold (a collection of trees with height greater than 5 meters and having more than 30% canopy cover) and qualifies as forest for a minimum of 10 years before the project start date (See Section 3.1.3). The APD project activity involves avoiding planned deforestation, which permanently halting or reducing deforestation on forest land that is legally authorized and legally documented for conversion.

2.1.5 Transfer Project Eligibility (VCS, 3.23, Appendix 2)

This item does not apply to the project since it is not a transfer project or CPA-seeking registration.

2.1.6 Project Design (VCS, 3.6)

Indicate if the project has been designed as:

- ☐ Single location or installation
- ☒ Multiple locations or project activity instances (but not a grouped project)
- ☐ Grouped project

2.1.7 Project Proponent (VCS, 3.7; CCB, G1.1)

Organization name	PT CarbonX Bumi Harmoni
Contact person	Dessi Yuliana
Title	President Director
Address	Treasury Tower 33 rd Floor, District 8, SCBD Lot.28 Jl. Jendral Sudirman Kav. 52-53 Jakarta 12190 – Indonesia
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Organization name	PT Citra Mulia Inti
Contact person	Dessi Yuliana
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Telephone	+62 50200 839 ext. 209
Email	dessi@carbonxco.com

2.1.8 Other Entities Involved in the Project

Organization name	Ecosecurities Group Limited (EGL)
Role in the project	Carbon technical services
Contact person	Peter Stephen
Title	Asia Pacific Regional Director
Address	Rue de la Faïencerie 2, 1227 Carouge, Switzerland
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Email	peter.stephen@ecosecurities.com

2.1.9 Project Ownership (VCS, 3.2, 3.7, 3.10; CCB, G5.8)

PT CarbonX Bumi Harmoni is the legitimate owner of the project and has responsibility and rights over the project. This is based on PT CarbonX Bumi Harmoni 99% shareholding of PT CMI, who holds the forest concession right based on the Minister of Environment and Forestry Decree SK 2/1/IUPHHK-HTI/PDMN/2015 (and amended by Ministry of Environment and Forestry Decree SK.1039/Menlhkh/Setjen/HPL.01/11/2021). The proof of ownership can be seen in Annex 1.

2.1.10 Project Start Date (VCS, 3.8)

Project start date	16 March 2023
Justification	The project start date corresponds with CarbonX's shares acquisition of PT CMI and the immediate decision to stop logging in the PT CMI concession.

2.1.11 Benefits Assessment and Project Crediting Period (VCS, 3.9; CCB, G1.9)

Crediting period	16 March 2023 to 15 March 2073 (50 years) which meets the VCS standard of a minimum 40-year project.
Start date of first or fixed crediting period	16 March 2023 to 15 March 2073
CCB benefits assessment period	The benefit assessment period is from 16 March 2023 to 15 March 2073, where climate change adaptive capacity and resilience, biodiversity and community well-being resulting from the project activities will be monitored.

2.1.12 Differences in Assessment/Project Crediting Periods (CCB, G1.9)

There are no differences between the assessment period for GHG emissions accounting, climate adaptive capacity and resilience, community and biodiversity.

2.1.13 Project Scale and Estimated Reductions or Removals (VCS, 3.10)

☒ < 300,000 tCO_{2e}/year (project)

☐ ≥ 300,000 tCO_{2e}/year (large project)

Calendar year of crediting period	Estimated reductions or removals (tCO _{2e})
16 March 2023 to 15 March 2024	935,019
16 March 2024 to 15 March 2025	1,166,586

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
16 March 2025 to 15 March 2026	974,094
16 March 2026 to 15 March 2027	880,138
16 March 2027 to 15 March 2028	561,811
16 March 2028 to 15 March 2029	169,753
16 March 2029 to 15 March 2030	164,588
16 March 2030 to 15 March 2031	176,478
16 March 2031 to 15 March 2032	188,844
16 March 2032 to 15 March 2033	201,576
16 March 2033 to 15 March 2034	186,873
16 March 2034 to 15 March 2035	160,229
16 March 2035 to 15 March 2036	138,755
16 March 2036 to 15 March 2037	119,892
16 March 2037 to 15 March 2038	110,083
16 March 2038 to 15 March 2039	111,289
16 March 2039 to 15 March 2040	112,404
16 March 2040 to 15 March 2041	113,032
16 March 2041 to 15 March 2042	113,242
16 March 2042 to 15 March 2043	113,093
16 March 2043 to 15 March 2044	105,741
16 March 2044 to 15 March 2045	105,057
16 March 2045 to 15 March 2046	104,156
16 March 2046 to 15 March 2047	103,068
16 March 2047 to 15 March 2048	101,819
16 March 2048 to 15 March 2049	100,432
16 March 2049 to 15 March 2050	98,928
16 March 2050 to 15 March 2051	97,326
16 March 2051 to 15 March 2052	95,641
16 March 2052 to 15 March 2053	93,887
16 March 2053 to 15 March 2054	80,534
16 March 2054 to 15 March 2055	78,912

Calendar year of crediting period	Estimated reductions or removals (tCO ₂ e)
16 March 2055 to 15 March 2056	77,261
16 March 2056 to 15 March 2057	75,588
16 March 2057 to 15 March 2058	73,900
16 March 2058 to 15 March 2059	72,203
16 March 2059 to 15 March 2060	70,502
16 March 2060 to 15 March 2061	68,803
16 March 2061 to 15 March 2062	67,109
16 March 2062 to 15 March 2063	65,425
16 March 2063 to 15 March 2064	49,775
16 March 2064 to 15 March 2065	48,483
16 March 2065 to 15 March 2066	47,205
16 March 2066 to 15 March 2067	45,943
16 March 2067 to 15 March 2068	44,699
16 March 2068 to 15 March 2069	43,474
16 March 2069 to 15 March 2070	42,268
16 March 2070 to 15 March 2071	41,084
16 March 2071 to 15 March 2072	39,921
16 March 2072 to 15 March 2073	38,780
Total estimated ERRs during the first or fixed crediting period	8,875,703
Total number of years	50
Average annual ERRs	177,514

2.1.14 Physical Parameters (CCB, G1.3)

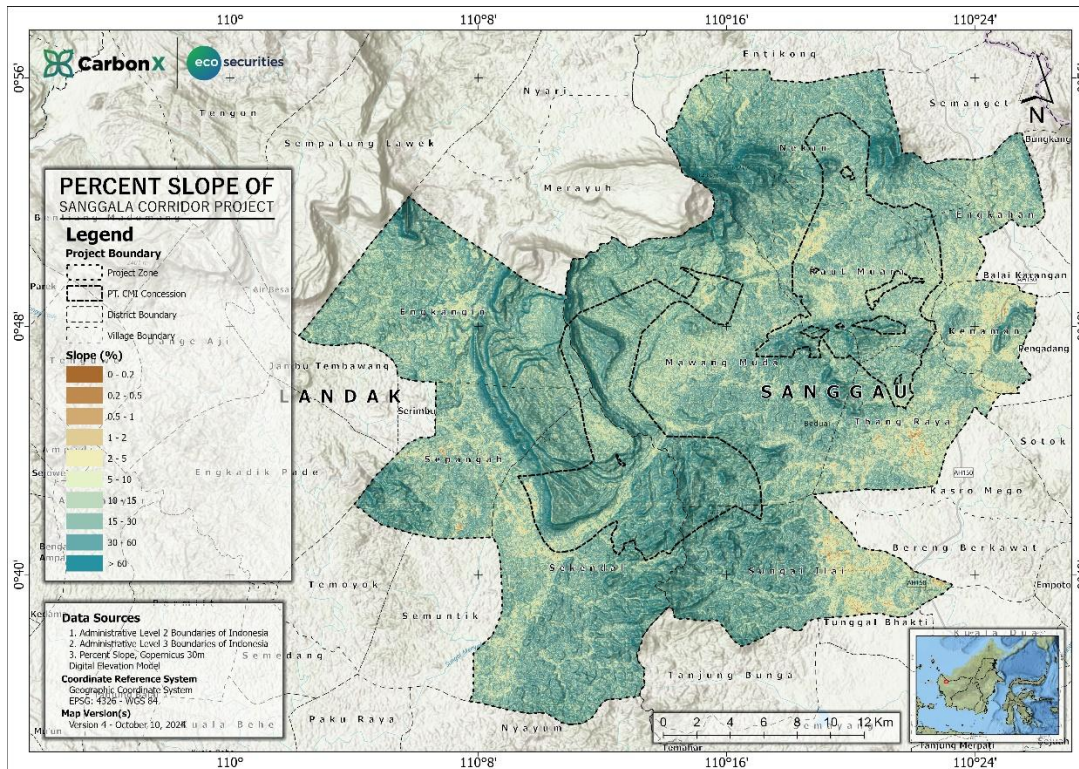
2.1.14.1 Slopes

The Sanggala Corridor Project encompasses a range of slopes within its project zone (see Map 1). Following are the description of the Sanggala Corridor Project's project zone slopes that following slope gradient class from FAO¹⁵. Flat areas, with slopes between 0 – 0.2%, make up a minimal 0.13% of the total area, while level areas (0.2% -0.5%) represent 0.21%. Then, the nearly level areas, with slopes between 0.5% - 1%, cover 0.45% of the project zone areas.

¹⁵ FAO, 2006: Guideline for Soil Description. Fourth Edition. Rome, Italy

Very gently sloping land (1%-2%) accounts for 1.25% of the total project zone area, while gently sloping areas, with slopes between 2%-5%, cover 5.53 of the landscape. Sloping areas (5%-10% make 11.83% of the project zones.

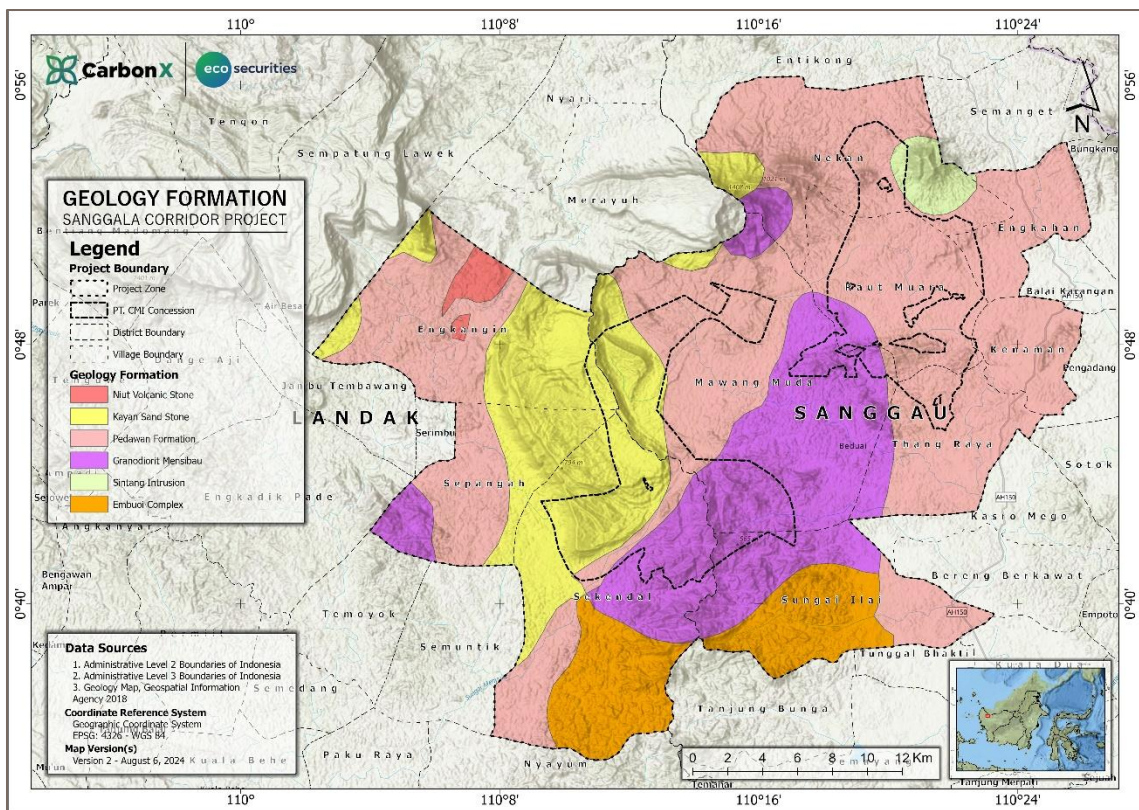
The majority of the project area consists of steeper slopes, strongly sloping areas (10%-15%) account for 13.25% of the total area, while moderately steep terrain (15%-30%) covers the largest portion at 33.42%. Steep terrain, with slope between 30-60%, makes up 28.79% of the project zones area. Lastly, the very steep areas, with slopes over 60%, account for 5.13%. This distribution of slopes reflects the varied topography of the project zone.



Map 1. The slope map of the project zone

2.1.14.2 Geological Feature

The geological formation of the project zone predominantly consists of sedimentary rock, with the Pedawan Formation covering 51% of the area and Granodiorit Mensibau accounting for 20% of the project area (see Map 2). The remainder of the project zone consists of Kayan Sandstone (17%) and the Embuoi Complex (9%), along with smaller areas of intrusive and extrusive rock, such as the Sintang Intrusion (2%) and Niut Volcanic Stone (1%).



Map 2. The geological formation of the project zone

2.1.14.3 Soil

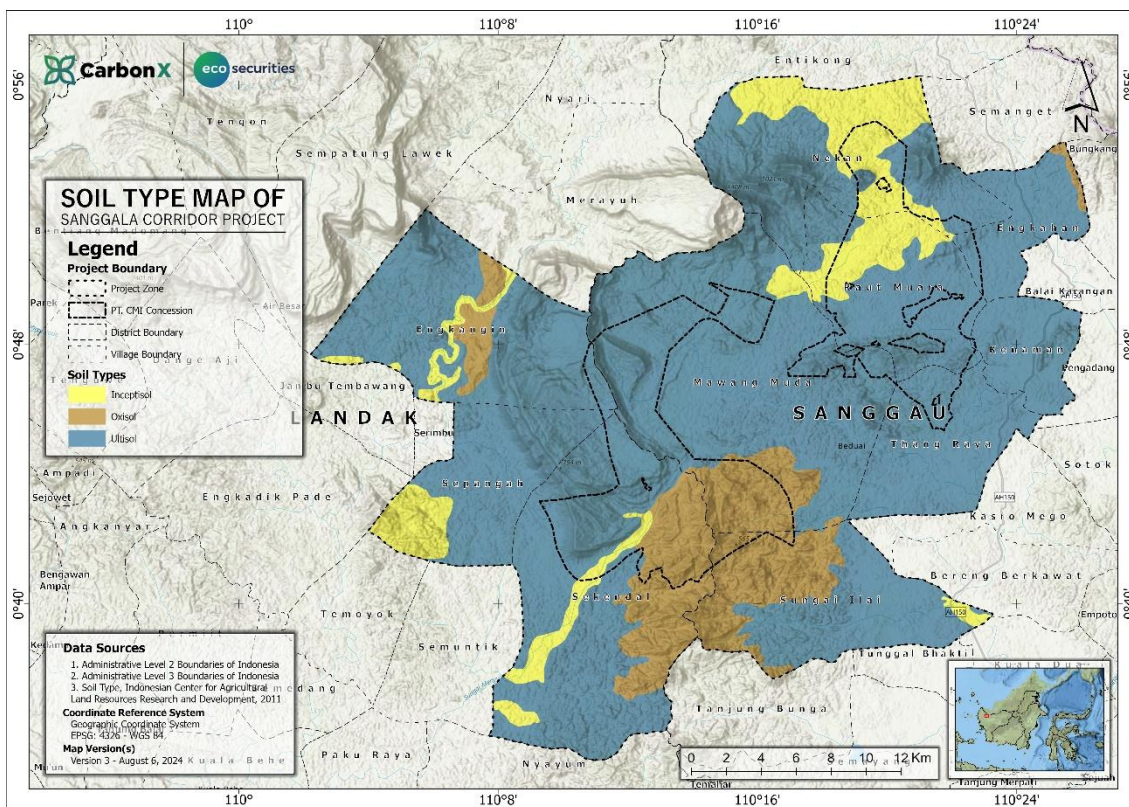
According to the data obtained from The Indonesian Center for Agricultural Land Resources Research and Development, the soil types in the project zone consists of ultisol (77.3%), oxisol (12%) and inceptisol soils (17%) (see Map 3).

Ultisol soils are commonly known as red clay soils, which are mineral soils lacking calcium carbonate, less than 10% weatherable minerals in the top layer of the soil and has less than 35% throughout the soil profile¹⁶. Utisols are typically associated with natural tropical forests with understory vegetation of reeds, fens, and shrubs.

Oxisol soils are commonly associated with tropical rain forests. They are characterized by containing no more than 10% weatherable mineral at all depths and having low cation exchange capacity¹⁵. Oxisol soils are typically red or yellowish owing to their high concentration of iron, aluminum oxides and hydroxides.

Inceptisol soils are young soils with no accumulation of clays, iron oxides, aluminum oxides or organic matter¹⁵. These soils are generally characterized with a high fertility capacity and therefore tend to support agricultural activity.

¹⁶ Food and Agriculture Organization of the United Nations, World Reference Base for Soil Resources: A Framework for International Classification, Correlation, and Communication, Rome: Food and Agriculture Organization of the United Nations, 2006

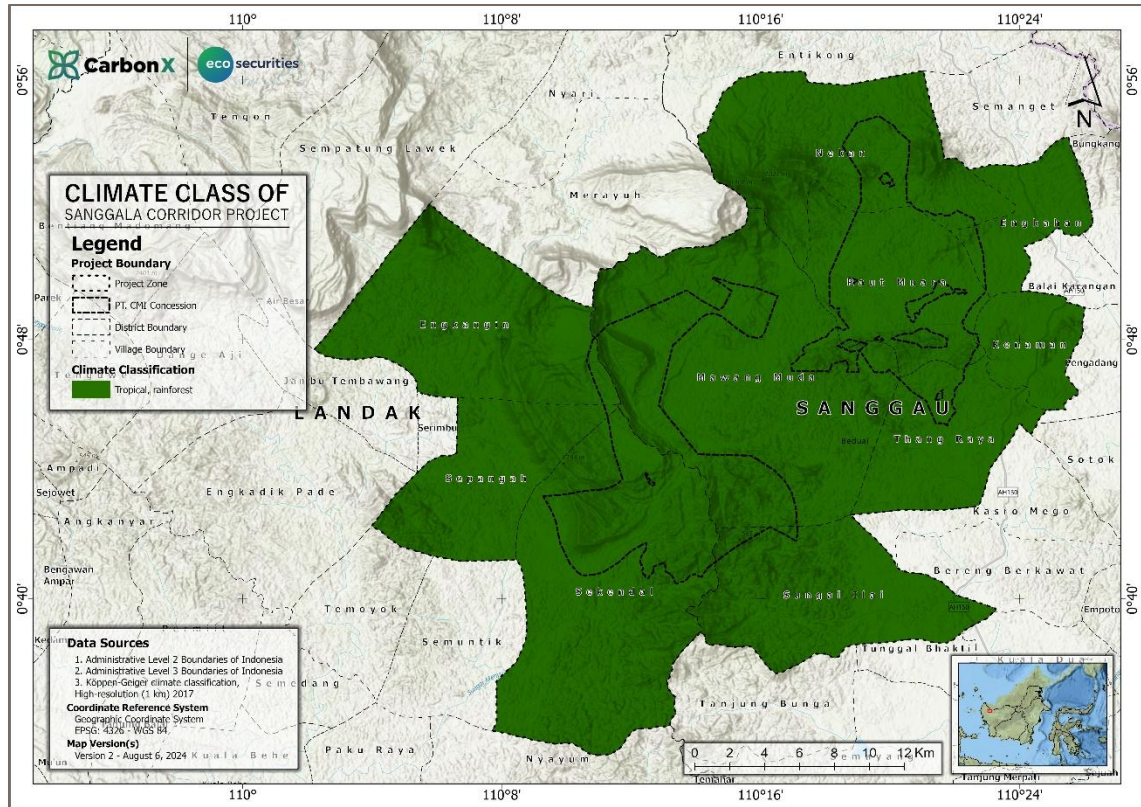


Map 3. The soil typology of the project zone

2.1.14.4 Climate

The project zone has a tropical rainforest climate (Af) as stated under the Köppen Climate Classification¹⁷ (see Map 4) and is characterized by warm and humid conditions through the whole year. The air temperature is relatively stable throughout the year, with an average of 28.8 °C.

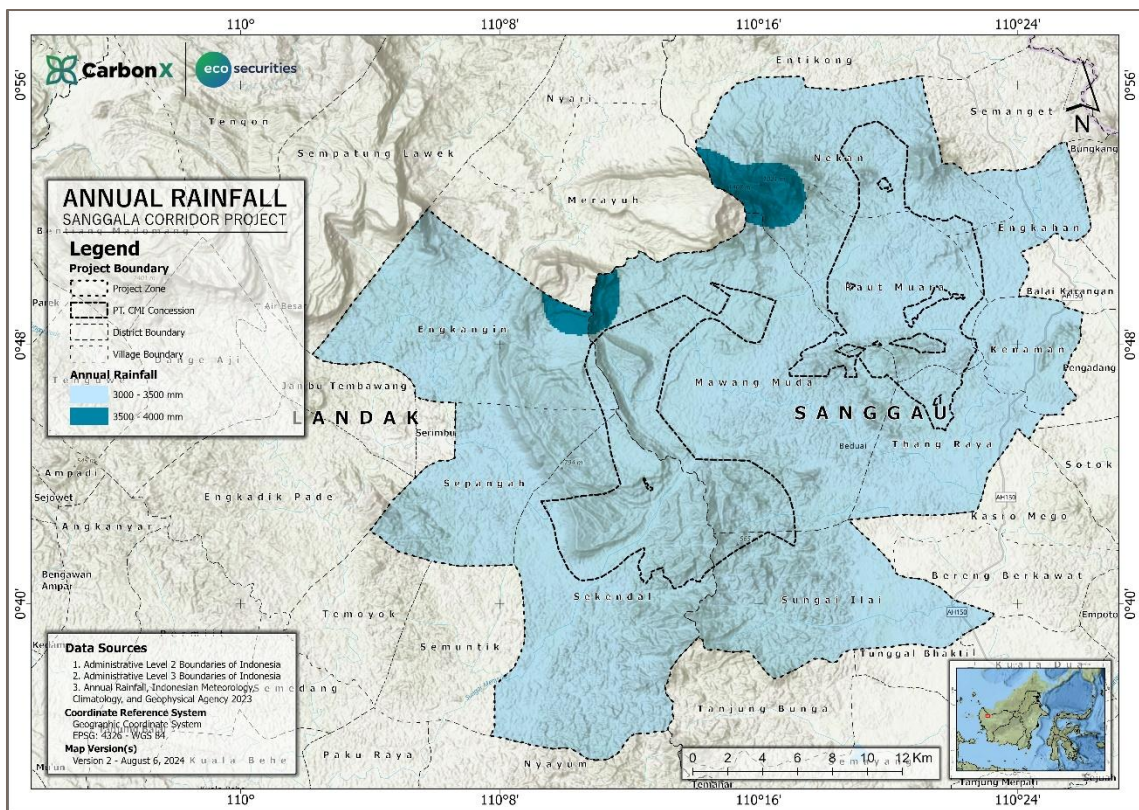
¹⁷ Beck, H.E., N.E. Zimmermann, T.R. McVicar, N. Vergopolan, A. Berg, E.F. Wood. Present and future Köppen-Geiger climate classification maps at 1-km resolution, Nature Scientific Data, 2018.



Map 4. The Köppen-Geiger climate classification

The project zone experiences relatively high average annual rainfall, ranging from 3,000 to 3,500 mm per year (see Map 5). This abundant rainfall likely plays a significant role in sustaining the extensive tropical forest areas within the region¹⁸.

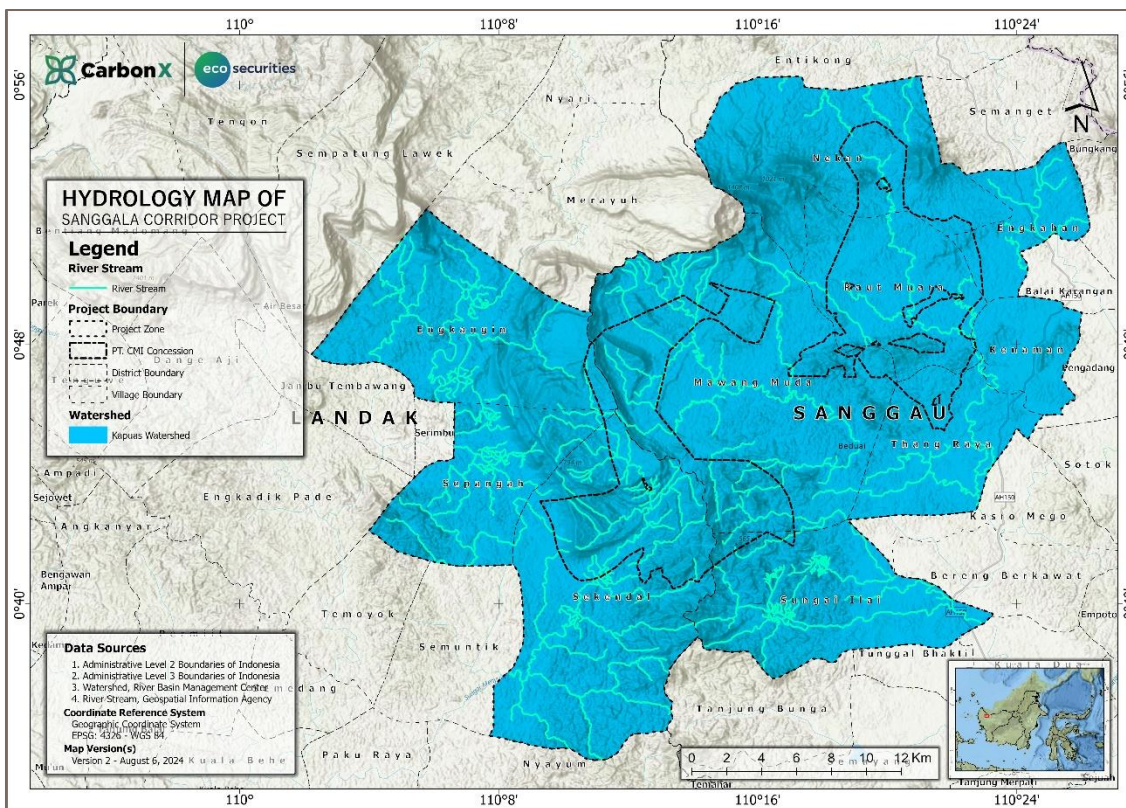
¹⁸ Kabupaten Landak Dalam Angka 2023



Map 5. The average annual rainfall of the project zone

2.1.14.5 Hydrology

The dominant hydrological system in West Kalimantan is the Kapuas River, the longest river in Indonesia stretching 1,143 km and reaching up to 700 m wide at its delta. The Kapuas River basin covers more than 67% of West Kalimantan with numerous rivers, including the Landak River, feeding into the Kapuas River. The project zone is situated within the Landak River basin (see Map 6) and the Kapuas watershed.

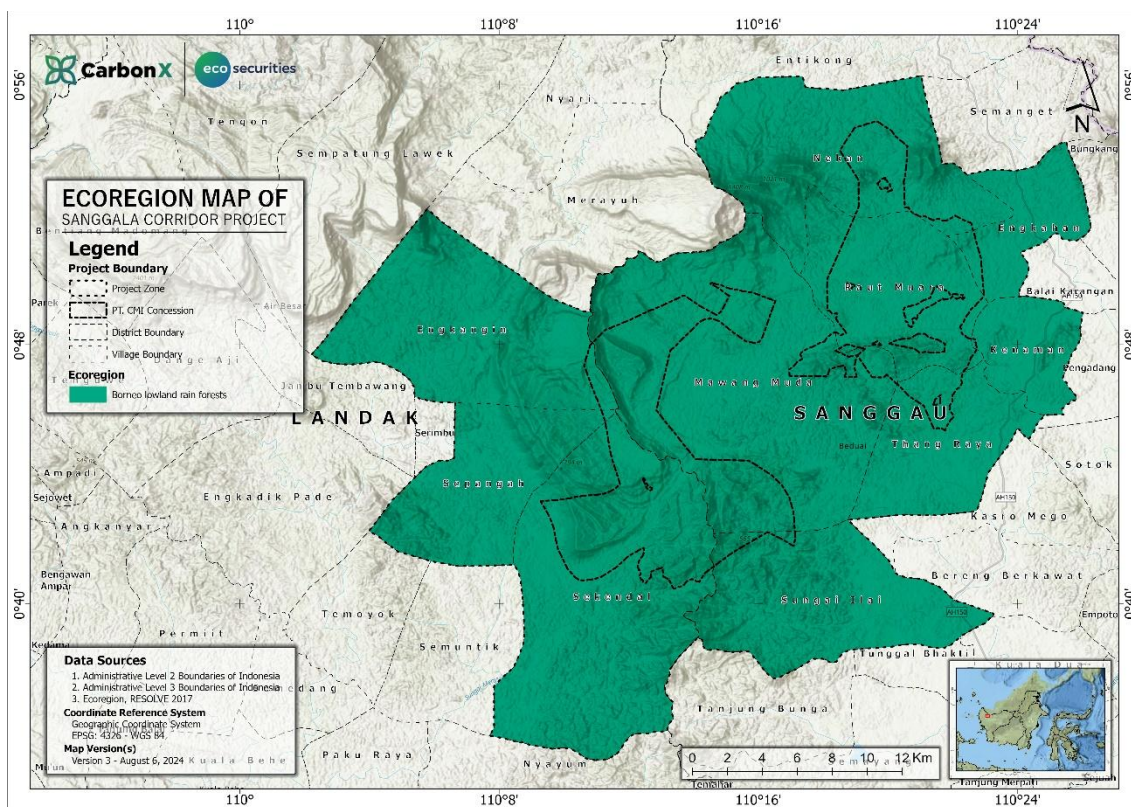


Map 6. The hydrological system of the project zone

2.1.14.6 Ecoregion

According to the data from RESOLVE 2018 ecoregions, the project zone is situated within the Borneo Lowland Rainforest ecoregion (see Map 7), which is also the dominant ecoregion of the island of Borneo. This ecoregion spans an area of 428,000 square kilometers, covering 57% of Borneo Island and is part of the tropical and subtropical moist broadleaf forest biome. The ecoregion also supports around 15,000 plant species, 380 bird species, and several mammal species but is under significant threat from logging, hunting, and conversion to agricultural and other commercial land uses.¹⁹

¹⁹ Southeastern Asia: Indonesia and Malaysia | Ecoregions | WWF (worldwildlife.org)

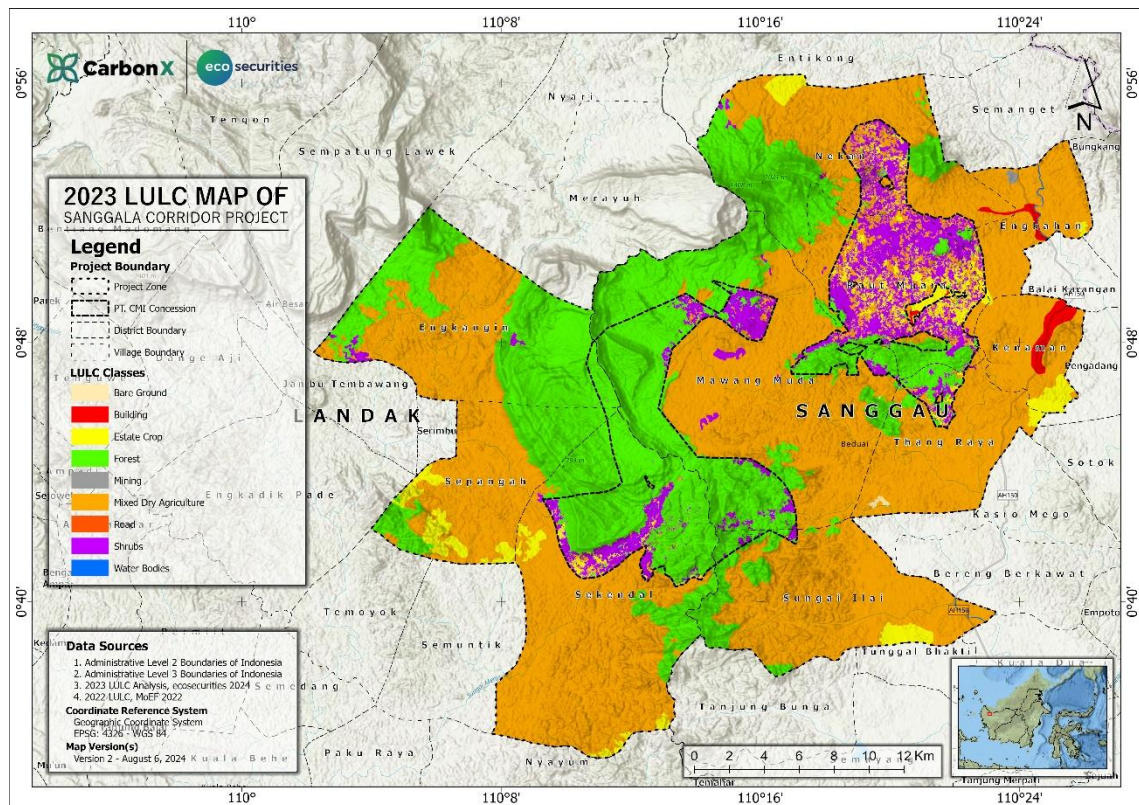


2.1.14.7 Land cover and vegetation type

The concession area is predominantly covered by forest (54.16 %), while the remaining area consists of non-forest land, mainly shrubs (29.05%) and mixed dryland agriculture (10.03%) (see Table 1). The plant species that dominate the forest ecosystem in the project area include Meranti (*Shorea sp*), Medang (*Litsea sp*), Jambu-Jambuan (*Syzygium sp*), and Keruing (*Dipterocarpus sp*). These species are generally found throughout the island of Borneo.

Table 1. Area of land use and land cover in the concession

Land Cover	Area (Ha)	Percentage
Built-up	38.38	0.19%
Estate Crop	1,342.55	6.52%
Forest	11,159.70	54.16%
Mixed Dry Agriculture	2065.84	10.03%
Shrubs	5,986.46	29.05%
Water Bodies	12.56	0.06%
Total	20,605.53	100%



Map 8. The 2023 land cover map

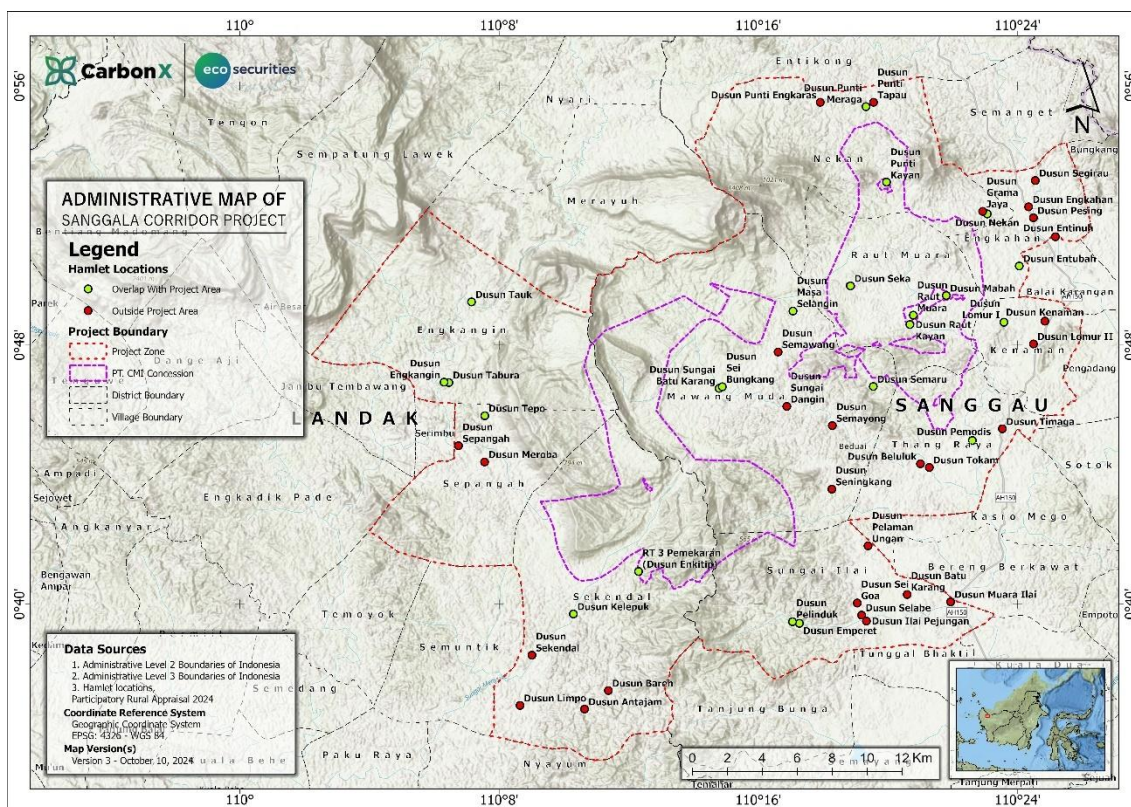
2.1.15 Social Parameters (VCS, 3.18; CCB, G1.3)

2.1.15.1 Main Settlement

The project area is located within the sub-districts of Air Besar, Beduai, Entikong, and Sekayam of Landak and Sanggau Regencies, West Kalimantan Province, Indonesia. There are 10 villages overlapping with the project area: Engkangin, Sekendal, Sepangah, Mawang Muda, Sungai Ilai, Thang Raya, Nekan, Engkahan, Kenaman, and Raut Muara villages. These 10 villages are further subdivided into 50 sub-villages (hamlets), 20 which have boundaries that overlap with the project area. Detailed information regarding the villages can be found in Table 2 and the village boundaries are shown on Map 9.

Table 2. The list of main settlements within the project zone

Regency	Sub-district	Village	Hamlets that overlap with the Project Area	Hamlets that is outside the Project Area
Landak	Air Besar	Engkangin	Engkangin Tabura Tauk	-
Landak	Air Besar	Sekendal	Kelepuk (RT Engkitip)	Limpo Sekendal Bareh Antajam
Landak	Air Besar	Sepangah	Tepo	Sepangah Meroba
Sanggau	Beduai	Mawang Muda	Sungai Bungkarang, Sungai Batu Karang (RT Tunguh)	Semayong Semaru Seningkang Sei Dangin Sei Mawang
Sanggau	Beduai	Sungai Ilai	Pelinduk Emperet	Muara Ilai Batu Karang Sei Goa Pelaman Ungan Selabe Ilai Pejungan
Sanggau	Beduai	Thang Raya	Pemodis	Beluluk Tokam Muara Kayan Timaga
Sanggau	Entikong	Nekan	Nekan, Punti Kayan Punti Meraga	Grama Jaya Punti Tapau Punti Engkaras
Sanggau	Sekayam	Engkahan	Entubah	Segirau Engkahan Pesing Entinuh
Sanggau	Sekayam	Kenaman	Lomur I (RT 07)	Kenaman Lomur II
Sanggau	Sekayam	Raut Muara	Raut Muara, Raut Kayan, Seka, Masa Selangin, Mabah	



2.1.15.2 Land Use and Economic Activities

Agriculture is the primary land use and economic activity for the local communities in the 10 villages. Traditionally, these communities practice shifting cultivation based on subsistence farming patterns, with knowledge and techniques passed down through generations and deeply embedded in local culture. Key subsistence crops include local rice varieties (paddy rice), CGPRT crops (coarse grains, pulses, roots, and tubers such as corn, ground nuts, sweet cassava, soybean, and *palawija*), pepper, and vegetables. These crops are supplemented by foraging for non-timber forest products and hunting in nearby forests. Figure 1 Illustrated the community-owned paddy fields within the project area.



Figure 1. Paddy field owned by the community in the project area

More permanent cash crops in the area include oil palm and rubber plantations. Figure 2 below illustrates the primary agricultural commodities cultivated by the communities throughout the project zone.

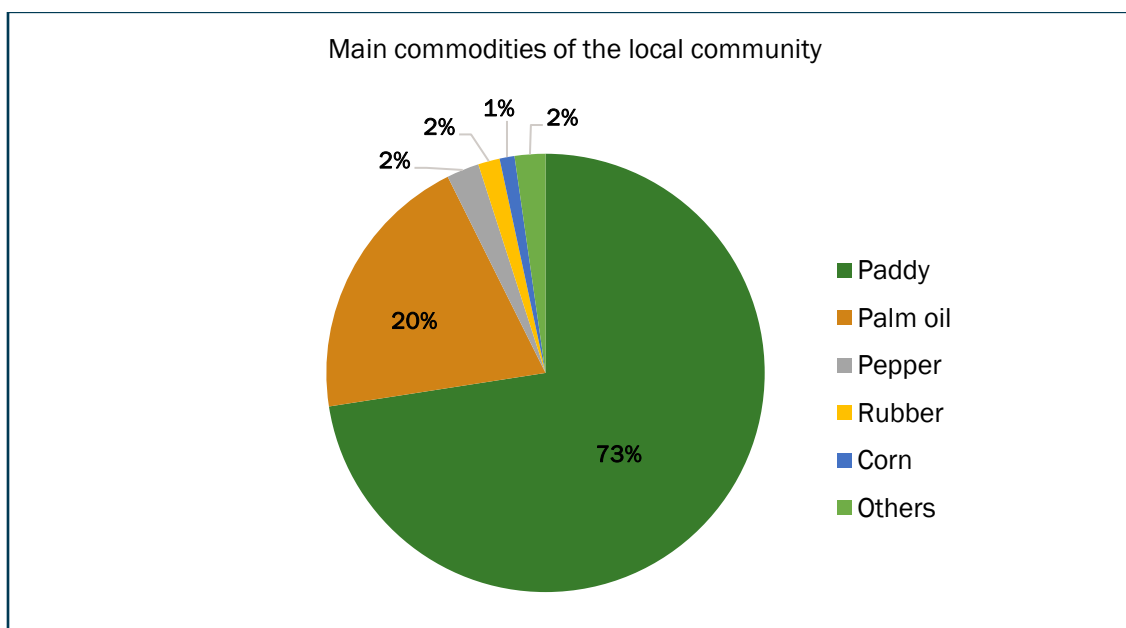


Figure 2. The main commodities of the local community within the project zone

In addition to agriculture, the local population engages in small-scale home industries, producing products such as *tempoyak* (a condiment made from fermented Durian) and *tuak* (a local alcoholic drink), which are sold in local village markets and nearby towns. The villages of Engkangin, Sepangah, Mawang Muda, Sungai Illai, and Kenaman have also established a village-owned enterprise (*Badan Usaha Milik Desa*, or BUMDES) that aims to support the creation, development and promotion of village-based businesses.

2.1.15.3 Relevant Historical Conditions

The Indonesian Government designated the project area as a production forest in the 1980s, making it legal for companies to apply for rights to manage the area. The first company having a

right to manage the area was PT Yayasan Maju Kerja²⁰, Yayasan Maju Kerja works around 1980s to 1990s, and then it is estimated that the license held by the organisation was expired, making the area up for grabbed for other entity to apply for license to operate. which then in 2015 the concession was owned by PT CMI.

In 2015, the previous owner of PT CMI completed the application process for the rights to manage the forest area in the concession area, intending to develop industrial forest plantation in the area. Later in 2022, PT Carbonx Bumi Harmoni completed the acquisition process of PT CMI by owning 99% of PT CMI shares and later transformed the focus of the company into conservation and restoration.

Historically, the area has been inhabited by Dayak communities, with six Dayak sub-tribes now living in the project zone. These communities are believed to have originated from the mountains of what is now Brunei Darussalam and later migrated to the Sekayam District, particularly to an area called by locals as Tampun Juah in the Segumon region of Sanggau regency. Over time, small settlements formed across the area. According to local legend passed down through generations, a war between broke out between the Dayak communities and supernatural beings or “ghosts”, forcing many families to leave the area and relocate within the project zone. Since then, these communities have permanently settled in the area, forging economic and cultural ties to the land and its natural resources.

2.1.15.4 Social and Cultural Conditions

Information on the social and cultural conditions of the project zone is provided below. The information was compiled from the community socio-economic baseline as well as HCV study that was conducted during the project development.

2.1.15.4.1 Population

Based on the community assessment conducted by the project team, the total population in the 10 villages is 25,028 people. Sekendal village has the largest population with 3,927 residents, while Engkangin village has the smallest with only 1,236 residents. In all villages, the male population exceeds the female population with an average sex ratio of 111 male residents for every 100 females across the 10 villages. The detailed population data for each village can be seen in Table 3.

Table 3. The population data for the ten villages within the project zone

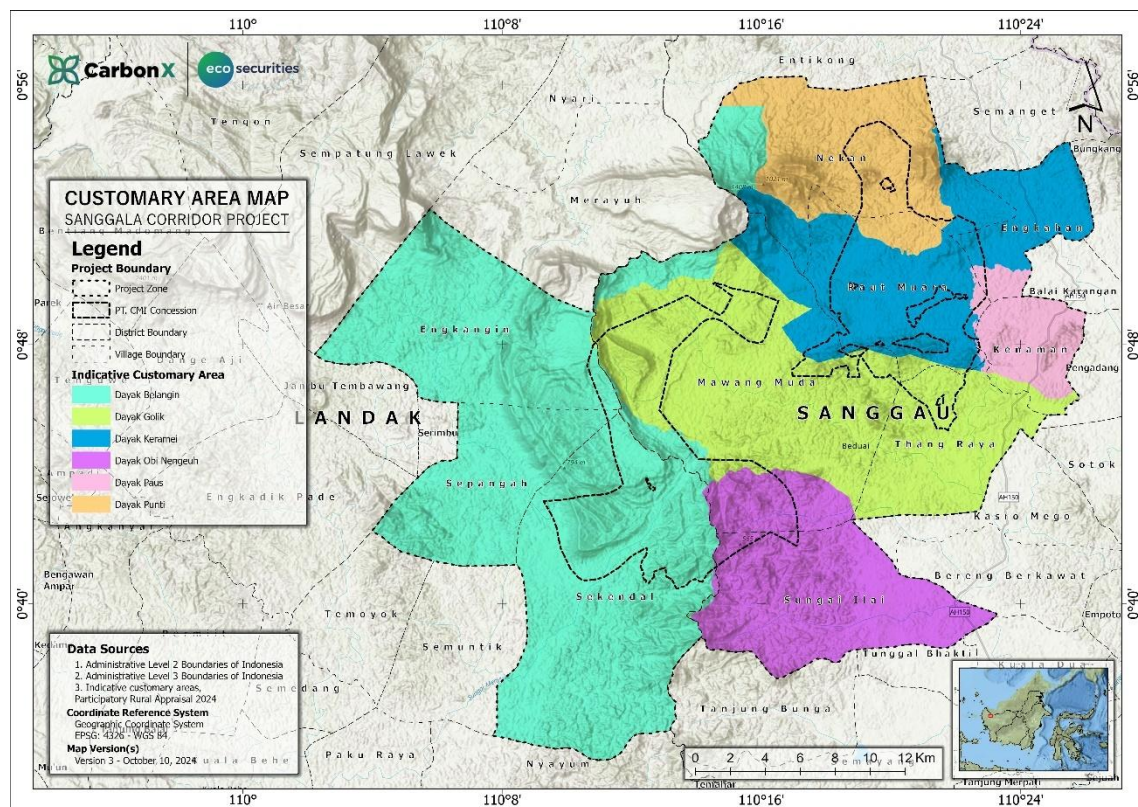
Village	Male	Female	Population	Sex Ratio	Number of Families
Engkangin	695	541	1,236	128	355
Sepangah	864	638	1,502	135	506
Sekendal	2,082	1,845	3,927	113	1,098
Mawang Muda	1,167	1,110	2,277	105	576
Sungai Ilai	1,797	1,584	3,381	113	922
Thang Raya	1,011	924	1,935	109	543
Kenaman	1,316	1,250	2,566	105	694

²⁰ PT CMI previous management plan

Village	Male	Female	Population	Sex Ratio	Number of Families
Nekan	1,279	1,218	2,497	105	774
Engkahan	1,358	1,293	2,651	105	811
Raut Muara	1,580	1,476	3,056	107	791
Total	13,149	11,879	25,028	111	7,070

2.1.15.4.2 Ethnicity

The whole project area and the broader project zone are customary areas of local Dayak communities, the indigenous people of Kalimantan. Six sub-Dayak tribes live within the project zone, namely: Dayak Belangin, Dayak Obi Nengeuh, Dayak Golik, Dayak Keremai, Dayak Paus, and Dayak Puntii (see Map 10 for indicative map of customary areas within the project zone). The exact dates when these Dayak communities settled in the area are unknown, but village elders clearly state that it was long before Indonesia gained independence in 1945.



Map 10. The indicative maps of customary areas within the project zone

2.1.15.4.3 Income

The average monthly household income across the ten villages is IDR 1,222,500 (~ USD 75). The Nekan village has the lowest average monthly household income at IDR 925,000 (~ USD 56), as the community primarily grows rice and pepper, which are less profitable compared to oil palm crops. In contrast, the Sungai Ilai village has the highest average monthly household income at IDR 1,700,000 (~ USD 104) since most of the villagers cultivate oil palms. Despite this, all village

household incomes fall well below Indonesia's national poverty line of IDR 2,592,657 per month per household²¹.

To gain additional income, the communities living in the project zone turn their attention to the production of palm fruit which would then be sold to palm oil production company located in the proximity of project area, namely PT. Borneo Ketapang Permai (PT BKP). It is a palm oil company owned by First Resources Limited (<https://www.first-resources.com/>), based in Sungai Ilai village. To support the development of oil palm plantations, PT BKP has established an out-grower scheme to help the local community cultivate their own oil palm plantations. The profit-sharing agreement under this arrangement is 70:30 (70% for the company and 30% for the community).

2.1.15.4.4 Education

Elementary school facilities are available in all ten villages, while junior high schools are only located in Sepangah, Sekendal, Mawang Muda, Sungai Ilai, and Kenaman Villages. High school institutions are only in Sepangah and Kenaman Villages. A detailed list of educational facilities situated in the ten villages within the project area is provided in Table 4.

Table 4. The educational facilities situated in the ten villages within the project zone

Village	Primary School	Junior High School	Senior High School
Engkangin	SDN 14 Dsn Tauk, SDN 26 Dsn Engkangin, SDN 10 Dsn Engkangin	-	-
Sepangah	SDN 8 Dsn Sepangah, SDN Dsn Tepo, SDN Dsn Meroba	SMPN 1 Air Besar (Dusun Sepangah)	SMAN 1 Air Besar (Dusun Sepangah)
Sekendal	SDN 2 Dsn Sekendal, SDN 16 Dsn Kelepuk, SDN Dsn Bareh, SDN Dsn Entajam, SDN Dsn Limpo	SMPN Satu Atap Sekendal, SMPN 5 Air Besar, SMPN 5 Sekendal	-
Mawang Muda	SDN 9 Dsn Sei Dangin, SDN 15 Dsn Seningkang, SDN 03 Dsn Semayong,	SMPN 3 Satu Atap, SMPN 03 Dsn Sungai Dangin	-

²¹ In 2023, the national poverty line was set at a per capita expenditure of IDR 550,458 per month per capita or 2,592,657 per household (4.71 members per household). [Indonesia Poverty Profile in March 2023 - BPS-Statistics Indonesia](#).

Village	Primary School	Junior High School	Senior High School
	SDN 07 Dsn Sungai Bungkang		
Sungai Ilai	SDN 12 Dsn Pelaman Ungan, SDN 05 Dsn Muara Ilai, SDN 10 Dsn Ilai Pejungan, SDN 16 Dsn Sei Goa, SDN 11 Dsn Emperet	SMPN 02 Beduai	-
Thang Raya	SDN 06 Pemodis	-	-
Kenaman	SDN 05 Dsn Lomur	SMPN 1 Sekayam, SMPN 3 Kenaman	SMAN 1 Sekayam, SMAN 2 Sekayam
Nekan	SDN 3 Dsn Punti Kayan, SDN 8 Dsn Nekan		
Engkahan	SDN 18 Dsn Entubah, SDN 13 Dsn Pesing		
Raut Muara	SDN 06 Dsn Raut Muara, SD Mini (Dsn Seka), SDN 24 Dsn Masa Selangai	SMPN Satu Atap, SMPN 06 Dsn Raut Muara	

The location and accessibility of educational facilities across the project zone significantly affects educational opportunities for the various communities. Communities located farther from village centers and educational facilities tend to have lower levels of education compared to those who are closer. Additionally, many villagers lack the financial resources to send their children to school and often prioritize household and farming activities that provide immediate returns over long-term investment in their children's education. As a result, only 44% of the community completes elementary education, with illiteracy being common across the project zone. Table 5 outlines the highest education levels obtained by the community in the project zone.

Table 5. Percentage of the highest educational level obtained for each village.

Highest Educational Level				
Village	Elementary School	Junior High School	Senior High School	University
Engkangin	34%	24%	34%	8%
Sepangah	30%	27%	32%	10%

Highest Educational Level				
Village	Elementary School	Junior High School	Senior High School	University
Sekendal	43%	19%	30%	8%
Mawang Muda	48%	22%	25%	4%
Sungai Ilai	40%	19%	36%	6%
Thang Raya	50%	18%	26%	6%
Kenaman	54%	16%	23%	7%
Engkahan	44%	24%	23%	9%
Nekan	53%	20%	25%	3%
Raut Muara	49%	21%	19%	6%
Average	44%	21%	27%	7%

2.1.15.4.5 Health

Several health care services can be accessed by the community across the project zone. These include:

- Integrated health care centers (*Posyandu*) at the sub-village (or hamlet) level offering services such as maternal and child health care, monitoring of children's growth and development, immunization, family planning, basic medical care, and community outreach services.
- Village level, health care facilities (*Puskesmas*), village maternity huts (*Polindes*) and auxiliary health care centers (*Pustu*). These centers provide the following services:
 - Puskesmas (Pusat Kesehatan Desa)*:** These are primary healthcare services in rural Indonesian communities, offering diagnosis and treatment for common illnesses, maternal and child health care services (such as antenatal and postnatal care), immunizations, and family planning. They also focus on nutrition and growth monitoring of children, health education and promotion, disease prevention and control, environmental health services such as sanitation, and emergency and referral services to higher healthcare facilities.
 - Polindes (Pondok Bersalin Desa)*:** These are village maternity posts in Indonesia, primarily focused on maternal and child health. Services provided at *Polindes* include antenatal care, birthing assistance, and postnatal care for both mother and child. *Polindes* also administer immunizations to children and pregnant women, provide family planning services, and conduct health education sessions on nutrition, breastfeeding, and hygiene. Additionally, *Polindes* play a critical role in monitoring child growth and development.
 - Pustu (Puskesmas Pembantu)*:** These are auxiliary community health care centers in Indonesia that support the primary community health centers (*Puskesmas*), by providing basic medical consultations and treatments, maternal and child health services (such as antenatal and postnatal care), immunizations, and family planning. *Pustu* also engage in health education and promotion, focusing on hygiene, nutrition, and disease prevention.

They perform routine health screenings, manage minor illnesses, and refer patients to higher-level facilities when necessary.

- At the district level, community health centers (Puskesmas, or Pusat Kesehatan Masyarakat) provide comprehensive healthcare services to the community, including medical consultations, treatment, maternal and child health services, immunizations, and family planning. Puskesmas play a key role in disease prevention and health education, promoting healthy lifestyles and preventative healthcare through community outreach programs. Additionally, these centers also address environmental health issues such as sanitation and clean water supply, offer emergency care, and manage satellite clinics like Pustu and Polindes to extend their reach into rural and remote areas. Only seven of the ten villages located in the project zone have Puskesmas: Engkangin, Sepangah, Sungai Illai, Thang Raya, Kenaman, Nekan, and Engkahan villages.
- A hospital is located in Kenaman village.

Table 6 below details the health facilities available in each village.

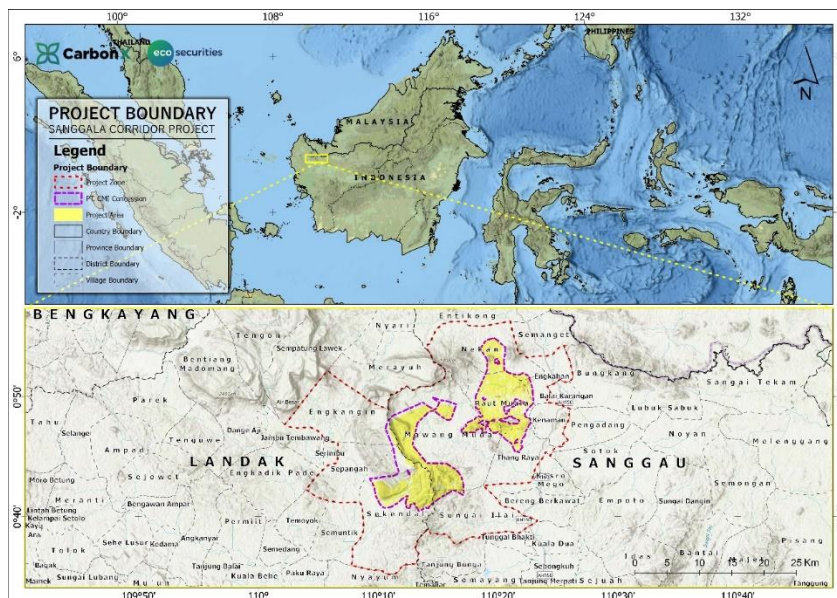
Table 6. List of health facilities available in each village.

Village	Health Facilities Level		
	Hamlet	Village	District
Engkangin	Posyandu	Puskesmas/Polindes/Pustu	Puskesmas
Sepangah	Posyandu	Puskesmas/Polindes/Pustu	Puskesmas
Sekendal	Posyandu	Puskesmas/Polindes/Pustu	-
Mawang Muda	Posyandu	Puskesmas/Polindes/Pustu	-
Sungai Illai	Posyandu	Puskesmas/Polindes/Pustu	Puskesmas
Thang Raya	Posyandu	Puskesmas/Polindes/Pustu	Puskesmas
Kenaman	Posyandu	Puskesmas/Polindes/Pustu	Puskesmas and Hospital
Nekan	Posyandu	Puskesmas/Polindes/Pustu	Puskesmas
Engkahan	Posyandu	Puskesmas/Polindes/Pustu	Puskesmas
Raut Muara	Posyandu	Puskesmas/Polindes/Pustu	-

Even though the community has access to various medical services and medical units, several obstacles prevent the community from accessing these services. The main reason is the lack of funds and high cost for medical treatment. Another significant barrier is the lack of transportation and associated cost to access these health services. Although government health insurance (BPJS) is available, it is only accessible in Sungai Illai, Thang Raya, Kenaman, Engkahan, and Raut Muara villages. As such, the community often relies on traditional medicine, typically sourced from neighboring forest areas.

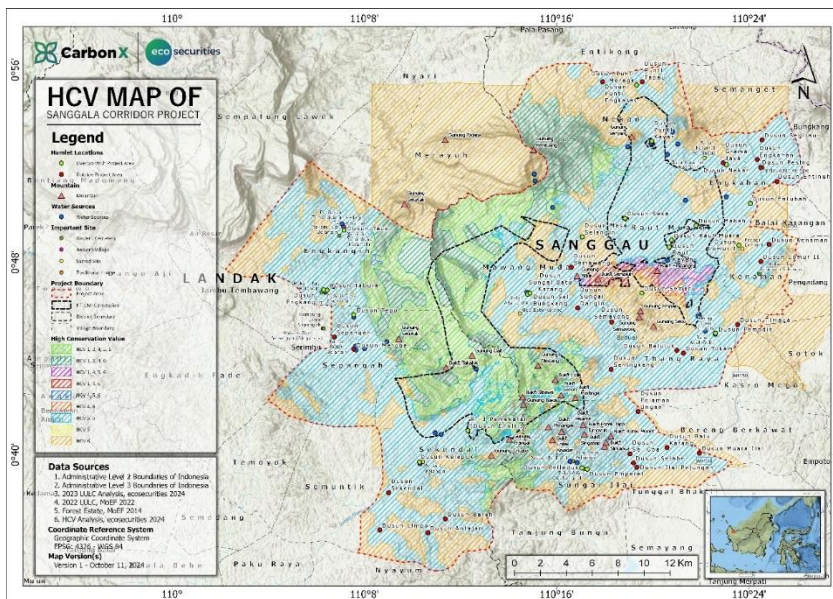
2.1.16 Project Zone Map and Project Location (VCS, 3.11, 3.18; CCB, G1.4-7, G1.13, CM1.2, B1.2)

The project zone is defined as the area encompassing the project area(s) in which project activities that directly affect land and associated resources, including activities such as those related to provision of alternate livelihoods and community development, are implemented. In this context, the project area covers 16,285 hectares within the concession area of PT CMI, located in Sanggau and Landak Regencies, West Kalimantan, Indonesia. Section 2.1. provides an overview of PT CMI's ownership. This is where project activities that generate net climate benefits will take place. Subsequently, the boundaries of the project zone were determined based on village administrative borders which overlapping with PT CMI concession area, to include all villages that have some economic, cultural or social, and jurisdictional connections to the project area. The map detailing the project area and project zone is shown in Map 11.



Map 11. The project area and project zone

The HCV area, including location where communities living in the project zone can be seen in Section 4.1.3 and 5.1.2.

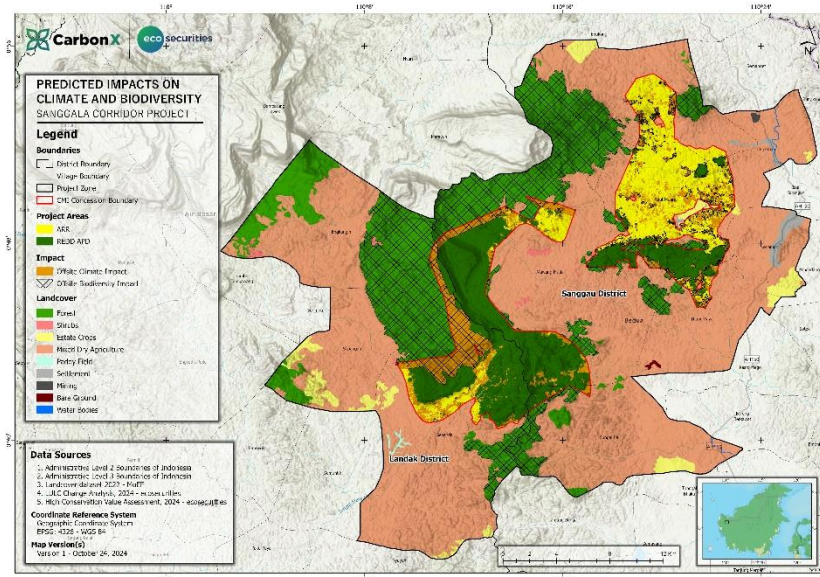


Map 12. HCV area in the project zone

The project strives to deliver positive climate impacts throughout the broader project zone; however, conservatively, it anticipates that these benefits will at least occur within the PT CMI concession area. These impacts result from forest protection and restoration efforts, including microclimate cooling through activities like restoration and agroforestry, reduced flood risks due to increased tree cover that improves water infiltration and reduces runoff during heavy rainfall, and enhanced water retention that mitigates drought conditions by improving soil moisture retention and maintaining water availability during dry periods and reducing the severity of droughts.

Similar to the anticipated climate impacts beyond project area, the project aims for biodiversity benefits to extend across the entire project zone. However, it conservatively expects that biodiversity impacts will primarily occur in the forested and non-forested areas adjacent to the project area, both within and outside the concession area. These impacts are driven by forest protection and restoration activities that foster the development of biodiversity corridors, enhancing habitat connectivity. Additionally, community-focused initiatives such as education, training, and livelihood improvement are predicted to reduce threats to biodiversity in these forest areas.

Map 13 illustrates the areas where climate and biodiversity impacts are anticipated beyond the project area. In conclusion, the project expects that the positive impacts generated by project activities could bring positive benefit to communities and other stakeholders (Section 2.3.1). These benefits include enhanced ecosystem services resulting from key activities that not limited to restoration of degraded areas, the creation of alternative livelihoods, improved effort on wildlife conservation, and monitoring of illegal activities.



Map 13. Predicted offsite climate and biodiversity impact

2.1.17 Project Activities and Theory of Change (VCS, 3.6; CCB, G1.8)

The forests in the project area are under direct threat from legally sanctioned plans to convert the current standing forest to industrial plantations. The project's goal is to halt these plans, thereby reducing emissions from planned deforestation while also removing greenhouse gases from the atmosphere through afforestation, reforestation, and revegetation activities. Additionally, the project will provide tangible benefits to the community through agroforestry, agricultural development, employment opportunities, biodiversity protection, and improved access to education and healthcare facilities across the project zone. These efforts are critical to the project's success, as limited community economic development opportunities may indirectly threaten the project's integrity through continued agriculture expansion and possible illegal logging.

In the baseline scenario, the establishment of an industrial timber plantation in the project area may offer some limited livelihood opportunities for local communities through the hiring of labor. However, it will not address the more significant issues faced by the locals, such as limited access to agricultural markets, education, healthcare services, and the loss of culturally and economically vital forest resources. Therefore, the project aims to conserve and restore the forest ecosystem, protect biodiversity, as well as improve the well-being of the local communities by enhancing livelihoods and increasing access to education and healthcare.

The project's primary goal objective is to produce GHGs emission reductions/removals, conserve biodiversity and strengthen community resilience and livelihoods through improved natural resource management and the development of a new localized "green economy". This goal will be achieved through six integrated and connected outcomes, as outlined in Figure 3. These outcomes are expected to address the root causes of deforestation, and reduce biodiversity loss, improve local community well-being, decrease GHG emissions and increase GHG removals,

through the APD and ARR activities. For the complete project activities and indicator, please see Annex 2.

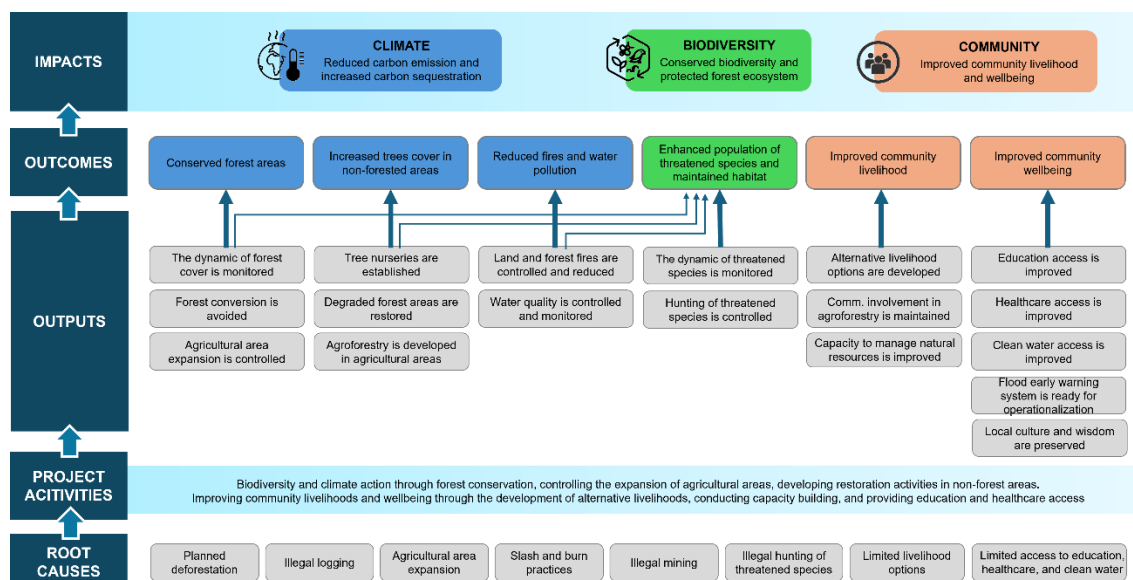


Figure 3. The theory of change

Outcome 1. Conserved forest areas

The project will terminate all forest conversion plans and restrict agricultural expansion, thereby permanently protecting the remaining forest areas within the project zone. Three key outputs are integral to achieving this outcome. These include:

1) The dynamic of forest cover is monitored

The project team will develop an annual land-use and land-cover (LULC) map and conduct analyses to monitor forest cover dynamics. In parallel, a community-based forest monitoring program will be initiated using participatory rural appraisal tools and techniques to track forest degradation, thereby complementing the remote sensing work required for the LULC analysis.

2) Forest conversion is avoided

Under the revised August 2023 *Rencana Kerja Usaha* (RKU)²² or forest concession management plan (Annex 3), the project team will shift plans from the forest conversion for industrial plantations to a new operational plan focusing on conservation and protection of the forest area. This government-endorsed change to the RKU will be the foundation for the project's implementation. Community-based Forest and fire patrol teams will be established to conduct regular patrols to monitor, report and help prevent illegal activities, control fires, and prevent agricultural encroachment into the forest areas. To support these activities, forest patrol guard posts and base camps will be constructed within the project area.

3) Agricultural area expansion is controlled

²² The Rencana Kerja Usaha (RKU) or PT CMI ten-year management plan (2018-2027) (number 2/1/IUPHHK-HTI/PMDN/2015) was revised and formally approved on 5 of August 2023 by the Indonesian Minister of Environment and Forestry. In the revised RKU, all areas in the PT CMI concession are considered environmental service areas for carbon sequestration and carbon storage (*penyerapan dan penyimpanan karbon*).

The project team will also collaborate with communities and stakeholders to manage the expansion of agricultural crops within the project area. This will involve activities such as documenting existing crop locations, conducting training to increase community awareness on the negative impacts of uncontrolled oil palm expansion, and developing agricultural and agroforestry programs aimed at improving yields and economic returns from the current farming footprint in the project area.

Outcome 2. Increased trees cover in non-forested areas

The project will increase tree cover in non-forest areas by working with local communities to implement afforestation, reforestation and revegetation (ARR) activities to restore the non-forest areas using a broad diversity of native species ecologically adapted to local conditions. The ARR activities will be collaboratively designed by communities and agricultural specialists to ensure the development and implementation of climate-resilient systems that meet community needs. This outcome will be achieved and measured through the following actions:

1) Tree nurseries are established

The project team will construct a tree nursery to supply seedlings for implementing the ARR activities. Local community members will be hired to work in the nursery.

2) Degraded forest areas are restored

Over a 10-year period, the project team will restore 3,660 ha of degraded land. The nursery will grow multi-purpose native tree species such as *Shorea stenoptera* (light red meranti), *Durio zibethinus* (durian), *Shorea pinanga* (pinanga), *Artocarpus spp.* (breadfruit and jackfruit family) and *Pithecellobium jiringa* (blackbead or dog fruit). Seedling survival and growth rates will be monitored annually.

3) Agroforestry is developed in agricultural areas

The project team will develop an agroforestry scheme across 3.316 hectares of agricultural land over 10 years. This program will focus on cultivating fruit tree such as, coffee, cacao, pepper and durian, intercropped with paddy rice. The final selection of commodities will depend on the on-going agroforestry study results and input from the community. The seedlings, for the agroforestry system, will also come from the nursery. Best practices of integrated pest management will be implemented to reduce the occurrence of pests and disease. The survival and growth rate of the planted seedlings will be monitored annually.

Outcome 3. Reduced fires and water pollution

A comprehensive forest conservation shall include reduced risk regarding the impact of Fire and use of fertilizer. Currently, slash and burn are still a normal agricultural practice for land preparation, pesticides and fertilizers application contribute to the reduced water pollution. Small-scale illegal mining around the project area also contributes into the water pollution. To address these issues, the project will invest in efforts to minimize fire events and improve water quality. The following outputs will support the achievement of this outcome:

1) Land and forest fires are controlled and reduced

A fire monitoring system will be established, along with community-based forest and fire patrol teams to prevent land and forest fires during the dry season. The project team will be provided with the firefighting equipment and training sessions on “no burning farming”, an activity that will be introduced to the local farming communities.

2) Water quality is controlled and monitored

Regular water quality monitoring checks will be conducted to evaluate water indicator (e.g. contamination and sediment level) in local rivers and drainage systems in the project area. The project will monitor pesticides and fertilizer usage, as well as establish waste collection points in each village to reduce pollution sources in nearby streams and rivers.

Outcome 4. Enhanced population of threatened species and maintained habitat

The project aims to enhance the population of threatened species, such as *Presbytis chrysomelas*, *Manis javanica*, and *Hylobates muelleri abbotti*, while maintaining their habitats by implementing activities to conserve forest areas, increase the trees cover in non-forest areas, and reduce fires and water pollution. In addition, the following outputs will contribute to achieve this outcome:

1) The movement and population dynamics of threatened species are monitored

The project team will regularly monitor threatened species using standardized monitoring protocols. Permanent transect plots for flora and fauna monitoring will be established within the project area and field staff will be trained in flora and fauna identification, monitoring standards, and data analysis. The project team will also establish Memorandums of Understanding (MoU) with universities and/or research institutes for collaborative research within the project area.

2) Hunting of threatened species is controlled

In parallel, local communities will be consulted in designing and implementing hunting restrictions mainly focused on the protection of threatened species. Patrol teams, consisting of field staff and community members will enforce these agreed-upon rules. Additionally, training sessions will be conducted to increase community awareness of threatened species and their importance within the local ecosystem. To formalize this effort, it is expected that the project will cooperate with tribes leaders to create Tribe Law(s) in banning endangered species hunting activities.

In addition, all the forest conservation and restoration efforts described in outcomes 1, 2, and 3 are designed to contribute to outcome 4.

Outcome 5. Improved community livelihoods

Limited economic opportunities and financial resources often compel the community to convert forest lands for expanded cropping areas, engage in illegal logging, and practice illegal hunting. Additionally, the collection of timber and non-timber forest products (NTFPs) is a vital source of livelihood alternatives, though its sustainability remains uncertain. The project will therefore invest to improve community livelihood outcomes and strengthen community capacity to sustainably manage forest lands. The following outputs will achieve this outcome:

1) Alternative livelihoods for the local communities are developed through a localized 'green economy'

The project team will collaborate with the local communities to design, develop, and implement alternative livelihoods strategies, as a substitute for the expansion of oil palm plantations and agricultural crops. Through market analysis, participatory rural appraisal, and participatory land use planning, alternative livelihood options for each village will be

- formulated. Current options discussed with communities include n, managed livestock production, ecotourism and the sustainable management of NTFPs.
- 2) The communities' involvement in agroforestry is maintained
Similarly, the project team will involve the local communities to design, develop, and implement agroforestry systems within existing agricultural areas. This will involve identifying suitable land and agricultural systems for the introduction of tree crops, forming community agricultural groups for each village, and market and value chain analysis for key tree and crops species within the agroforestry systems.
 - 3) The communities' capacity to manage natural resources is improved
The project team will conduct a series of training programs to enhance community capacity in natural resource management. These programs will include agroforestry, non-timber forest product (NTFP) management, and adaptation to changing climate regimes. The training aims to increase awareness of the importance of forests, sustainable land use practices, and their impacts on climate change and its derivatives. The ultimate goal is to strengthen community capacity for sustainable natural resource management that aligns with climate-resilient and biodiversity-friendly practices. Additionally, the project team will build the capacity of village-owned enterprises (BUMDES) to manage and equitably distribute financial resources from the project through village development funds.

Outcome 6. Increased community wellbeing

Limited environmental knowledge and low education levels are key contributors to land and forest degradation. Furthermore, limited access to healthcare services leads to poor health outcomes. To address these issues and enhance overall community wellbeing, the project will invest in the following areas:

- 1) Educational access is improved
The project team will support the education of local communities by providing “Kejar Paket B” and “Kejar Paket C” education programs (junior and senior high school education equivalency programs). Environmental education workshops will be provided for students, and partial scholarships will be made available to community members within the project zone to attend state universities.
- 2) Healthcare access is enhanced
Health care access will be improved by the project through the implementation of the program “doctor goes to village”, bringing health professionals to supplement existing auxiliary community health centres (*Pustu*). In addition, the project team will conduct workshops on traditional medicines sourced from forest resources to preserve and disseminate this knowledge.
- 3) Local culture and wisdom are preserved
Efforts to preserve and strengthen local culture and wisdom will focus on empowering youth and women in the local communities. This will include documenting culturally significant sites and local wisdom, as well as establishing conservation measures to conserve and protect them. Traditional competitions, such as *sumpit* (blowgun) contests will also be organized to raise community awareness about their local culture, local wisdom and heritage.
- 4) Clean water access is improved

The project will upgrade and maintain existing piping systems while protecting and restoring forest cover in upstream areas. This dual approach not only enhances natural water filtration but also stabilizes flows and reduces erosion.

5) Flood early warning system is available and ready for operationalization

The project zone outside the project area faces an increased risk of flooding, posing a threat to local communities. To mitigate significant damage caused by floods, the project plans to install a flood early warning system. This system will provide timely information to relevant stakeholders, enabling them to take necessary measures to be conducted prior to floods reach residential area.

2.1.18 Sustainable Development Contributions (VCS, 3.17)

The project activities are expected to contribute to several of the United Nations Sustainable Development Goals (SDGs). The relevant SDG targets and indicators, as well as their alignment with national sustainable development priorities²³ are described in Table 7. The project contribution to SDGs

Table 7. The project contribution to SDGs

SDG	Project Contribution
SDG 1: No poverty	The project will contribute to the achievement of SDG 1 to end poverty in all its forms everywhere. The project aims to build alternative livelihood pathways that are financially and culturally sustainable as well as employment opportunities for communities living in the Project Zone. The project activities will directly contribute to the national sustainable development priority to reduce extreme poverty as well as to reduce the number of citizens living below the national poverty line (Indonesian national SDG 1 ²⁴).
SDG 3: Good health and well-being	The project will contribute to the achievement of SDG 3 by building healthy lifestyle pathways and promoting well-being for all, at all ages. The project will supplement the existing auxiliary community health centres (<i>Pustu</i>), through “doctor goes to village” programs. In addition, the project team will conduct workshops to share knowledge on traditional medication from forest resources. The project is committed to providing essential health services and supporting the national sustainable development priority to reduce maternal and infant mortality rates (Indonesian national SDG 3 ²⁵).

²³ [BERANDA - SDGs Indonesia \(bappenas.go.id\)](https://bappenas.go.id/beranda)

²⁴ [GOAL 1 - SDGs Indonesia \(bappenas.go.id\)](https://bappenas.go.id/goal-1)

²⁵ [GOAL 3 - SDGs Indonesia \(bappenas.go.id\)](https://bappenas.go.id/goal-3)

SDG	Project Contribution
SDG 4: Quality education	The project will contribute to the achievement of SDG 4 by ensuring inclusive and equitable access to quality education and the promotion of lifelong learning opportunities for all. The project will support the education facilities for the local communities by providing Kejar Paket B” and “Kejar Paket C” programs (junior and senior high school education equivalency programs), as well as by conducting environmental education workshops for students, and providing partial scholarship for community members within the project zone to attend state universities. In this aspect, the project supports the national sustainable development priority to increase the number of citizens with quality education and to improve the proportion of citizens attending universities (Indonesian national SDG 4. ²⁶).
SDG 5: Gender equality	The project will actively contribute to the achievement of SDG 5, promoting gender equality and empowering women and girls within the project zone. All project activities will encourage the participation of all residents in the targeted villages, ensuring that key decisions affecting communities, households and individuals are inclusive. Special processes will be established to ensure the safe and respectful representation and inclusion of women, youth, and marginalized groups within these discussions. Additionally, the project will provide equal opportunities for women, girls and other vulnerable groups to be involved, engaged and actively participate in all community activities. This includes fair, open and transparent hiring practices for both community and project staff. The project is committed to provide enabling conditions for gender equality and women empowerment which aligns to the national sustainable development priority under SDG 5. ²⁷ .
SDG 8: Decent work and economic growth	The project will contribute to the achievement of SDG 8 by developing sustainable and inclusive economic opportunities, full and productive employment and decent work for all. The project aims to provide alternative livelihoods and employment opportunities for communities in the project zone. These activities will support the national sustainable development priority to increase per capita gross domestic product within the project zone. In addition the project’s safety culture to implement OHS regulations also aligns with the national sustainable development priority under SDG 8. ²⁸ .

²⁶ [GOAL 4 - SDGs Indonesia \(bappenas.go.id\)](https://bappenas.go.id)

²⁷ [GOAL 5 - SDGs Indonesia \(bappenas.go.id\)](https://bappenas.go.id)

²⁸ [GOAL 8 - SDGs Indonesia \(bappenas.go.id\)](https://bappenas.go.id)

SDG	Project Contribution
SDG 13: Climate action	The project will contribute to the achievement of SDG 13 by taking urgent action to combat climate change and enhance climate resilience for local communities. All project activities will address climate change impacts by reducing deforestation, restoring degraded ecosystems, and reducing GHG emissions. Here, the project directly addresses the national sustainable development priority SDG 13. ²⁹ by reducing GHG emissions within the project area.
SDG 15: Life on land	The project will contribute to the achievement of SDG 15 by protecting, restoring and promoting the sustainable use of terrestrial ecosystems and tropical forests. It aims to halt and reverse land degradation and prevent biodiversity loss. This includes conserving threatened species, supporting biodiversity, and facilitating ecosystem restoration within the project area. The project is aligned with the national priority to increase the forest cover as well as to reduce the illegal activities linked with biodiversity loss (Indonesian national SDG 15. ³⁰).

Particularly during monitoring stage, the project shall conduct appropriate measure to ensure and monitor the achievement of each of the identified SDGs indicator.

2.1.19 Implementation Schedule (CCB, G1.9)

Key dates and milestones in the project's development and implementation are presented in Table 8 below.

Table 8. Project development milestones

Date	Milestone(s) in the project's development and implementation
2023	- March 2023, CarbonX acquired PT CMI - May – November 2023, CarbonX conducted boundary delineation activities to establish clear boundary of the concession area and provide legal clarity. - July – November 2023, CarbonX adjusted PT CMI management plan (<i>Rencana Kerja Usaha</i> or RKU)). The revised RKU was approved on 9th of November. - By November 2023, four full-time field staff are employed - By December 2023, a social and economic assessment was completed with local communities

²⁹ [GOAL 13 - SDGs Indonesia \(bappenas.go.id\)](https://bappenas.go.id)

³⁰ [GOAL 15 - SDGs Indonesia \(bappenas.go.id\)](https://bappenas.go.id)

	By December 2023, biodiversity and carbon stock assessment completed with the participation of local communities.
2024	- March 2024, nursery's construction was started - March 2024, agroforestry demonstration plots and a farmer group in Raut Muara Village were formed - By April 2024, community engagement was completed - On May 2024, stakeholder consultations and FPIC process were conducted - Project design and drafting of the VCS and CCB Project description were underway until the end of 2024 - December 2024, Agroforestry and development document prepared by PT Re.mark Asia funded by USAID Segar program was completed.
2024-2025	Third-party VCS and CCB Validation
2024 onwards	Continue the implementation of project activities.
2025	Anticipated first verification
2026 onwards	Ongoing verification at least bi-yearly

2.1.20 Risks to the Project (CCB, G1.10)

Risks to the expected climate, community, and biodiversity outcomes are identified and mitigation measures to address these risks are summarized in Table 9 below.

Table 9. Risk to the project

Identified Risk	The potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Economic risk: Demand and/or prices fall in the voluntary carbon market	The project's financial resources may be insufficient if demand and/or prices for VCU declines.	The project will diversify its revenue streams to mitigate risk, including by the sale of agroforestry and actively looking for additional funding from grants.
Economic risk: Agricultural commodity price fluctuation	Fluctuations in the price for agroforestry crops can affect the economic stability of the community involved.	Species of cash crops will be diversified targeted by the project at the start of the project and can change depending on future market demand and pricing conditions.

Identified Risk	The potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Social risk: Land tenure issues	The project proponent has a legal right to control and operate the project in the project area according to prevailing forestry regulations in Indonesia. That said, indigenous people have been using the project area for their agricultural activities for many years, in particularly in non-forest area in Raut Muara village. Currently the project has consent to develop the project. However, the change in community expectations could lead into land tenure issues. Such issues could significantly impact all project activities, particularly those focussed on agricultural areas.	The project will: - Conduct detail mapping to identify the exact locations of community groups prone to land use conflicts, including indigenous populations and agricultural users. - Engage stakeholders through consultations to understand their land use rights and expectations. - Actively involve communities in key activities, such as agroforestry design, species selection, benefit sharing, and the establishment of grievance mechanism. - Develop MoUs in collaboration with land users before conducting agroforestry activities within the community claimed lands.

Technical risk: Failure of agroforestry activities	• Failure to provide alternative income pathways for communities and the risk of reverting to agricultural expansion into forest areas • Failure to achieve climate benefits and emission removal outcomes of the ARR activities.	All agroforestry activities will be collaboratively designed and implemented by communities and agricultural specialists to ensure the development and implementation of climate-resilient agroforestry systems meet community needs. This process will include: - Selecting suitable sites, choosing multipurpose tree species. The project aims to intensify land use and generate multiple revenue streams alongside paddy cultivation. This strategy includes cultivating cash crops that could yield monthly income, high-value commodities such as cocoa or coffee, and seasonal fruits. The goal is to achieve an optimal balance that enhances environmental benefits without jeopardizing economic returns. - Building silvicultural knowledge and practices. Recognizing that local communities may rely on traditional methods such as slash-and-burn agriculture, the project will engage agronomist and farmer/village coordinators to accompany and monitor sustainable agroforestry practices. Targeted training programs and technical support will be provided to empower local communities with the skills necessary for effective agroforestry production. - Identifying markets for agroforestry products. Recognizing that local communities often grow commodities for which nearby market access and reliable buyers are crucial, the project
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Identified Risk	The potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
		will ensure that any recommended commodity is supported by a robust market strategy. To achieve this, an agro-business development specialist will be hired to establish and maintain market linkages, ensuring consistent access to buyers for agroforestry products. - Establishing community monitoring practices for all plantation and agroforestry activities. This is to ensure ongoing oversight of plantation and agroforestry activities, facilitating adaptive management and sustained economic benefits.

Identified Risk	The potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Stakeholder risk: Communities engaged in agricultural cultivation may oppose the project if the economic benefit below their business as usual. Additionally, conflicting stakeholder interests could emerge: while tribal leaders might prioritize preserving cultural heritage, agricultural users are likely to focus on securing economic benefits	All project activities that generate climate, community, and biodiversity will be hampered.	The project has engaged local communities and stakeholders in all activities since the project development stage. During project implementation, the project will ensure continued community involvement to assess and manage community needs and expectations, while enhancing their understanding of the project. In addition to involving communities in key activities such as benefit sharing and grievance mechanism, the project will also conduct evidence-based adjustments informed by stakeholder feedbacks and consultation records. Additionally, the project is designed to deliver better livelihood outcomes compared to 'business-as-usual' or current land use practices. To support successful transition, the project will provide incentives such as subsidies for agroforestry inputs and facilitate market access by securing offtakers or acting as an offtaker itself to ensure guaranteed purchasing and fair pricing for sustainable products.

Identified Risk	The potential impact of risk on climate, community and/or biodiversity benefits	Actions needed and designed to mitigate the risk
Natural Risk Natural hazard like floods. Recently the several villages experienced floods caused by the combination of increased precipitation and spread of smallholder palm oil plantation in the project zone.	Destruction of large areas of forest and agricultural lands as well as habitat destruction of endangered species.	The project will: 1. Protect forest lands on areas of high erosion risk and restore degraded forest areas including the area in river riparian to reduce the potential for flooding and landslides. If needed, the project will build river embankment on identified areas with high risk of floods. 2. The project will establish waste management activity to make sure the waste does not into the river. 3. Educate and support the community to transition from shifting cultivation and the use of fire to burn newly cleared areas into sustainable agricultural practice.

2.1.21 Benefit Permanence (CCB, G1.11)

The project is designed to address current threats to climate, community, and biodiversity, ensuring these benefits will remain beyond the project's initial crediting period of 50 years. The license can be extended by up to 90 years per the regulations of the Government of Indonesia³¹ and allowing for potential continuation if necessary.

While the license and project could be extended, the project's ultimate objective is to build community capacity for the sustainable management of natural resources in alignment with climate and biodiversity friendly practices. This will be achieved by multi-layer training program combined with years of hands-on implementation. By the end of the project, the community is expected to be capable of independently managing the project area sustainably, utilizing the skill and knowledge gained throughout the project's duration. Commitment will be reinforced by institutionalizing local management committees and formalizing agreements and partnerships with relevant government bodies enabling these committees to independently manage project area beyond the carbon project's duration.

2.1.22 Financial Sustainability (CCB, G1.12)

The project proponent, CarbonX is a financially secure, sophisticated company committed to catalyzing climate action while achieving high-impact social, environmental, and economic

³¹ Government regulation number 23/2021 regarding forestry management

outcomes. Established in 2021, the company is led by a highly experienced senior management team, with expertise in technical, commercial, community, and legal domains.

The project is expected to generate sufficient cash flow to sustain long-term implementation, ensuring the achievement of its climate, community, and biodiversity objectives. Additionally, the project proponent has secured sufficient investment to cover all project costs until it achieves positive cash flow. Supplemental funding, in the form of grants from USAID, has also been obtained to complete an agroforestry study in the project area.

Documents supporting these investments and outlining the project's financial feasibility, project returns, and breakeven points can be produced the project's independent validation and verification body (VVB) upon request (See Annex 9).

2.2 Without-project Land Use Scenario and Additionality

2.2.1 Conditions Prior to Project Initiation and Land Use Scenarios without the Project (VCS, 3.13; CCB, G2.1)

The Sanggala Corridor Project is located in the Sanggau and Landak Regencies of West Kalimantan Province, Indonesia. The forested areas where the Avoided Planned Deforestation (APD) project activities will be implemented, as well as the shrubland where Afforestation, Reforestation, and Revegetation (ARR) activities will take place, are characterized by lowland tropical forests. These areas predominantly feature species such as Meranti (*Shorea spp.*), Medang (*Litsea spp.*), Jambu-jambuan (*Syzygium spp.*), and other species from the *Dipterocarpaceae* family. In other ARR project areas, the land is primarily used for community agriculture, primarily for paddy fields.

The current condition of the APD project area is forest, which has been degraded due to historical logging activities conducted by Yasayan Maju Kerja during the 1980s and 1990s. The forest has been undergoing natural succession. In contrast, the ARR project area comprises non-forested land, formed mainly due to agricultural activities. This land includes shrubland that was previously used for community farming. Most of the local community are farmers practicing slash-and-burn agricultural techniques. Notably, the project has not yet contributed to generating greenhouse gas (GHG) emissions.

Before project initiation, the area was designated for conversion into an Industrial Timber Plantation (HTI). In 2015, the Government officially granted the permit for PT CMI to operate and develop Industry Timber Plantation in the area. This permit was officially gazetted on February 13, 2015, by the Ministry of Environment and Forestry (Decree No. 2/1/IUPHHK-HTI/PMDN/2015), granting a 60-year permit for the development of Industrial Timber Plantation.

Prior to field work, PT CMI undertook a series of preparatory works to develop an industrial timber plantation within its concession area. In 2018, PT CMI obtained government approval for the General Work Plan (*Rencana Kerja Usaha* or *RKU*), a 10 years business plan detailing activities such as land clearing, planting, and seedling preparation. This plan was approved by the government through Ministry of Forestry Decree No. 4693/Men-LHK-PHPL/UHP/HPL.1/7/2018. In 2019, PT CMI conducted forest inventory activities to assess the commercialize standing timber stocks that would result from land clearing activities.

In parallel with these technical preparation, PT CMI secured investment by finalizing a cooperation agreement with PT Indonesia Wood Investment (PT IWI) in 2019. The agreement grants exclusive authority to PT IWI to operate logging activities, including providing financing for the operation of such activities. PT IWI is entitled to the revenue from the timber, while PT CMI shall be entitled to an upfront payment and royalties from each timber harvest.

The COVID pandemic in 2020 led to an operational pause of PT CMI, delaying the development of the industrial timber plantation. However in 2021, PT CMI resumed operation, by adjusting the permit to align with the new regulatory framework of Omnibus Law on Job Creation (Law 11 of 2020). This adjustment converted the concession permit into a Forest Utilization Business License (PBPH - *Perizinan Berusaha Pemanfaatan Hutan*) under the Ministry of Environment and Forestry Decree 1039/Menlhk/Setjen/HPL.0/11/2021.

In 2022, PT CMI continued its efforts by submitting an annual work plan (*Rencana Kerja Tahunan* or *RKT*) to the relevant government system, emphasizing the land clearing plan.

These adjustments along with associated planning documents are critical for the "without-project" scenario, as the PBPH license provides PT CMI the legal authority to continue converting the standing forest into industrial timber plantations. Under the HTI permit, PT CMI's concession was designated for conversion to industrial timber plantations with *Anthocephalus cadamba* as the primary species and *Hevea brasiliensis* as the secondary species. Further, both the General Work Plan and Annual Work Plan of PT CMI specified the area on which deforestation would have occurred in the "without-project" scenario.

The year 2023 was the turning point for the management of PT CMI concession area, as CarbonX finalized took over the management of the company. This also marks the initiation of conservation (APD) and restoration (ARR) initiatives along with community development program in the concession area under the name of Sanggala Corridor Project.

Both the APD and ARR project areas are part of the same landscape. The project is located in a tropical rainforest climate, characterized by warm and humid conditions throughout the year. Several rivers flow through the project area, including the Landak River, and the site is situated within the Kapuas Watershed.

Topographically, the project area exhibits varied terrain, ranging from flat land to steep slopes. The soils in the project area comprise three major types: Ultisols, Oxisols, and Inceptisols. No Histosols are present within the project area. A more detailed description of the environmental parameters in the project zone, including the project area, can be found in Section 2.1.14.

In the absence of the project, the most likely scenario for land use would involve conversion into industrial timber plantations, without any efforts to reduce or remove GHG emissions. Additionally, several other land-use scenarios could occur in the project zone, including community agriculture, smallholder palm oil cultivation, and social forestry.

Currently CarbonX owns 99% of PT CMI shares and is the project proponent. Under CarbonX's management, the project area will be protected and restored.

2.2.2 Most-Likely Scenario Justification (CCB, G2.1)

In the absence of the project, the most likely land use scenario for the project area would have been the conversion of forest land into an industrial timber plantation, primarily featuring of *Anthocephalus cadamba* as the main plant and *Hevea brasiliensis* as the subsistence plant. It becomes the baseline scenario of the project, and the justification for this most likely land use scenario is elaborated further in Section 3.1.4.

Given the legal framework, the most probable scenario in the project area would be the conversion of the current standing forest into an industrial timber plantation. The Ministry of Forestry Decree No 733/Menhut-II/2014 categorizes the project area as a production forest and the Director General of Forestry Planning Letter Number 2.90/VII-WP3H/2014 categorizes all of PT CMI working area as permanent production forest, which provide legal framework for industrial timber plantation to be developed.

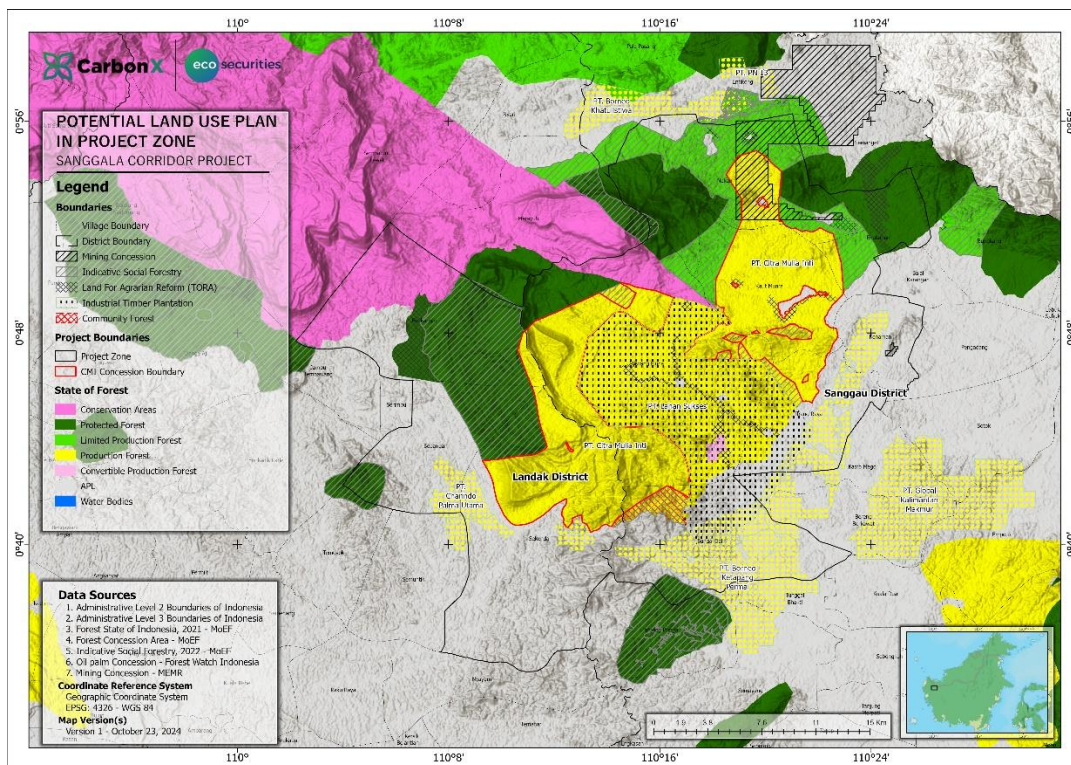
PT CMI has held a 60-year industrial timber plantation permit since February 13, 2015 (Ministry of Environment and Forestry Decree No. 2/1/IUPHHK-HTI/PMDN/2015) covering a total area of 20,645 hectares. This permit was later adjusted to a Forest Utilization Business License (PBPH) under Ministry of Environment and Forestry Decree 1039/Menlhk/Setjen/HPL.0/11/2021, which continues to legally authorize PT CMI to convert the forest into an industrial timber plantation. On top of the permit to develop industrial timber plantation, PT CMI has the general work plan approved by the Ministry of Environment and Forestry through Ministry of Environment of Forestry Decree 4693/MenLHK-PHPL/UHP/HPL.1/7/2018 dated July 6th 2018. This general work plan clearly states the plan to conduct land clearing for the existing forest stand in the concession area.

Considering the project has been granted an industrial timber plantation permit and the presence of approved general work plan by the ministry, the project area would have converted into an industrial timber plantation. The operating industrial timber plantation which adjacent to PT CMI further demonstrates the likeability of this scenario to take place.

In the project zone, several land use scenarios could happen. The project area is surrounded by the area that given a function as other utilization area (*Area Penggunaan Lain/ APL*) by Indonesia Government. This kind of land zonation functions could be developed for non-forestry businesses activity such as development of palm oil plantation and agricultural activities.

Adjacent to the southern boundary of the concession where the project area is situated, PT Charindo Palma Oetama, a palm oil company, holds a license to operate and develop a palm oil plantation. Furthermore, southeast of the concession boundary, another palm oil company, PT Borneo Ketapang Permai, has obtained authorization to convert the area into a palm oil plantation. Other utilization areas, which are not currently under the ownership of palm oil companies, are utilized by local communities for agricultural purposes and housing. In the future, the areas that currently utilized for agricultural purposes may either remain as community agricultural lands or be converted into palm oil plantations by the communities themselves, motivated by the potential for increased economic returns from palm oil cultivation. It is common for communities to receive palm oil seedlings from the companies, and once the palms bear fruit, they typically sell the produce back to nearby companies. The concession area of PT CMI consists of two distinct blocks, separated by a permanent production forest area, a zonation where PT

CMI also located. Within this area, PT Lahan Sukses has been granted a permit to operate and convert the land into an industrial plantation forest. On the western to northwestern side of PT CMI's concession lies a protected forest. According to indicative social forestry data from the Ministry of Environment and Forestry (KLHK), a portion of this protected forest has been identified as suitable for social forestry activities. These activities may include the sustainable utilization of non-timber forest products and the provision of environmental services, in accordance with the zoning regulations applicable to this protected forest area. In addition, the indicative social forestry area extends into the production forest zone located on the northern side of Block II of PT CMI's concession. Given that this area is designated as a limited production forest, the activities permitted include the sustainable harvesting of timber and non-timber forest products, along with the provision of environmental services. PT CMI's concession also shares a boundary with the Gunung Nyiut Nature Reserve. As nature reserves are generally preserved and not subject to conversion for other land uses, it is anticipated that this area will remain unchanged in the future. Furthermore, within the project zoning and the project area, several regions have been identified as suitable for development into Land for Agrarian Reform Objects (TORA). TORA is a government initiative aimed at accommodating "inadvertent occupation," a term used in forestry and land tenure to describe the occupation or use of state land by individuals or groups without formal authorization. Following is overlaid land use development plan within the project zone.



Map 14. Land use plan overlaid with the project zone

2.2.3 Community and Biodiversity Additionality (CCB, G2.2)

In the without-project scenario, the positive impacts provided by the project would not occur. The most likely baseline scenario suggests the conversion of the area into an industrial timber

plantation, with little evidence to support that this conversion would improve community livelihoods or contribute to biodiversity conservation. Indeed, there are activities in the without-project scenario that could potentially benefit the communities, but they are minimal, such as providing jobs mainly for harvesting and plantation maintenance roles, as well as income from rubber plantation development. However, this would also create conflicts due to unequal distribution. The establishment of rubber plantations for communities only takes place in two villages, Raut Muara and Nekan, while other areas receive no benefits.

Meanwhile, under the project scenario, various benefits are planned to be utilized by the communities, including the following benefits:

- **Continuation of ecosystem services:** In the project scenario, ecosystem services within the area are maintained or enhanced, ensuring that communities continue to benefit from these services under a regime of permanent forest protection. The project is designed to conserve and safeguard high conservation value areas, which provide essential ecosystem services to the communities within the project zone. Conversely, in the absence of the project, the area will be converted into monoculture plantations, leading to the loss or severe degradation of multiple ecosystem services. Traditional sources of food and medicinal plants are either lost or significantly reduced. Additionally, there is a decline in water quality and quantity, accompanied by increased erosion and sedimentation of streams and other water resources. Furthermore, the conversion activities compromise areas that naturally mitigate flooding and landslides, increasing the risk and frequency of such events.

According to Indonesian regulations, a High Conservation Value (HCV) assessment is not mandatory for the development of Industrial Timber Plantations. Only companies or concession holders aiming to be certified under the Forest Stewardship Council (FSC) standard are required to conduct such assessments. However, PT CMI's work plan does not include any intention to obtain this certification. While there is a government regulation requiring the protection of 20% of the total concession area, this does not guarantee the protection of all HCV areas within the project site.

- **Increase access to healthcare and education:** In the without-project scenario, based on PT CMI's work plan, there is no plan to improve access to healthcare and education services for the community. On the other hand, with the project, programs to enhance healthcare access will be implemented, such as doctor visits to villages. The project will also improve access to education through programs like Paket B and C (equivalency education programs), environmental education programs for high school students, scholarship opportunities, and various vocational training programs to help communities develop skills in natural resource management.
- **Employment Opportunities:** In the without-project scenario, PT CMI will provide employment opportunities to a limited number of villagers, primarily in roles such as harvesting and plantation maintenance. However, due to the restricted number of available positions, not all communities will benefit, and most may lose access to their agricultural lands and farming livelihoods as a result of the establishment of industrial tree plantations. In the project scenario, the company will offer employment opportunities that align with the educational background and skills of community

members applying. The project will also support the development of agroforestry systems in agricultural areas to enhance crop diversity, increase income streams, and establish more climate-resilient agricultural practices.

Additionally, other livelihood activities will be developed collaboratively with the communities to diversify and increase income sources through initiatives such as but not limited to livestock raising, honey production, and the sustainable collection and production of non-timber forest products (NTFPs).

In the biodiversity perspective, the conversion of the concession area into an Industrial Timber Plantation with monoculture crops will lead to habitat loss for various species living within the project area, with only 20% of the project area being protected as per existing regulations. In fact, the entire forested area within the project site serves as a habitat for diverse biodiversity, including threatened species such as *Hylobates muelleri abotti*, *Manis javanica*, and *Presbytis chrysomelas*.

Furthermore, the absence of prevailing customary laws protecting wildlife in the area increases the risks human-wildlife conflict. This risk stems from the fact that local communities have preexisting custom of wildlife hunting and the displacement of wildlife as their natural forest habitat is lost. Since local communities often adhere more closely to the customary laws than formal government regulations, the lack of relevant customary laws could further threaten biodiversity population in the project area.

While in the project scenarios, all forests within the concession area will be protected through various activities, such as community awareness programs, forest patrols, and training sessions aimed at reducing pressures on the forest ecosystem and its biodiversity.

The project will also conduct forest restoration by planting native species to rehabilitate degraded ecosystems. This activity is expected to expand habitats for biodiversity. Additionally, the project will implement agroforestry practices using native forest species and high-value, non-invasive cash crops. Agroforestry serves as a win-win solution, enhancing biodiversity while also improving the local economy and reducing emissions.

Under the Industrial Timber Plantation (HTI) model, the concession was primarily managed for timber production, with no structured focus on biodiversity conservation or community livelihood programs. Financially, forest restoration and conservation require substantial investment without immediate revenue, whereas HTI operations prioritize commercially attractive activities, often in direct contrast to conservation efforts, such as timber logging, including within forested areas. Additionally, implementing biodiversity conservation and community development programs requires specialized skills that differ significantly from the conventional expertise of HTI management, which has traditionally focused on timber production and resource extraction.

The identified without project scenario for both community and biodiversity is described in more detail in Section 4.1.4 and 5.1.3, respectively.

2.2.4 Benefits to be used as Offsets (CCB, G2.2)

The project does not intend to generate any community or biodiversity benefits for use as offsets.

2.3 Safeguards and Stakeholder Engagement

2.3.1 Stakeholder Identification (VCS, 3.18, 3.19; CCB G1.5)

The project team employed various methods and strategies to identify stakeholders within the project zone.

Local community stakeholders were identified through four main approaches: 1) spatial identification of villages boundaries overlapping with the project area boundary; 2) consultations to identify key community stakeholders; 3) interviews to confirm key community stakeholders and to identify other key community stakeholders ('snowballing' sampling); and 4) Participatory Rural Appraisal (PRA) to obtain information about the existence of customary communities and village institutions that have close ties to the forest.

Other stakeholders were identified through 1) spatial identification of all land, forest users and other land allocation plans surrounding the project area; 2) assessing the legal status of existing concessions surrounding the project area; and 3) consulting with the communities during the PRA process as mentioned above.

2.3.2 Stakeholder Descriptions (VCS, 3.18, 3.19; CCB, G1.6, G1.13)

The stakeholders identified as per the process outlined in Section 2.3.1 are described in Table 10 below.

Table 10. Stakeholder list and description.

Stakeholder	Rights, interest, and overall relevance to the project
Residents from the 20 sub-villages of the 10 identified villages living within the project area	Rights: Participation in project planning, implementation, and benefit-sharing, with direct access to training, resources, and other benefits from the project Interest: Improved livelihoods, sustainable agriculture, better resource access. Overall relevance: All of 20 sub-villages within the project area consists of indigenous communities from 5 customary groups. The communities will be directly impacted by the project due to they lives in the project area for considerable time and rely on land and ecological services provided by the forest for their livelihoods. They conduct basic activities such as hunting-gathering and shifting cultivation, while implementing customary rules to ensure sustainable management of forest resources. These communities are significantly impacted by the proposed carbon project within the PT CMI boundaries. Their main interest to the project is mainly to maintain security of their access to land and natural resources to maintain their livelihood, and to get additional benefit from the project. The existence of the proposed carbon project within PT CMI boundaries is expected to benefit the communities, as the project will include community participation in its programs as well as benefit sharing from the programs. The project is directly relevant to these communities due to the project area is in their customary areas, where they use land and forest for their livelihood. The project will use these areas for the conservation and agroforestry activities, and the communities will receive a portion of the revenue from carbon trading. These communities have a high interest in the project and are also likely to have high influence on the success of the project, therefore, they are considered to be ‘promoters’ and critically important to the project.

Stakeholder	Rights, interest, and overall relevance to the project
Residents from the 30 sub-villages of the 10 identified villages living outside the project area, but in the project zone.	Rights: Participation in project planning, implementation, and benefit-sharing, with direct access to training, resources, and other benefits from the project. Interest: Improved livelihoods, sustainable agriculture, better resource access. Overall relevance: Communities of 30 additional sub-villages outside the project area but belonging to the same 10 villages and five customary groups are closely tied to the project. Although situated outside the project area, their historical settlements and customary land rights create a strong connection to the land and communities who lives within the project area. These communities are still likely to have influence on the project because they are closely connected to the communities living within the project area and their village boundaries fall outside of the project area. PT CMI's activities and the carbon project will likely have a significant direct and indirect impact on these communities. They are relevant to the project as the project will provide local employment, health extension services as well as project awareness and environmental education.
Head of sub-village	Rights: Representation of the sub-village in project decisions, ensuring community interests are considered, especially in land use and benefit-sharing. Interest: Ensuring fair benefit distribution, and community welfare. Overall relevance: The head of sub-village (Kepala Dusun) are responsible for the implementation and administration of development programs at the sub-village level. Kepala Dusun at the project area is considered to be influential, have high interest and significantly relevant to the project, especially during the project implementation due to its role in monitoring the development programs at sub-district level. In addition, together with customary leader at the sub-village level, head of sub-village may represent voices of the community who lives inside the project area.

Stakeholder	Rights, interest, and overall relevance to the project
Customary leader and customary head at a village level and sub-village levels	Rights: Consultation rights to ensure project activities respect cultural practices, traditions, and sustainable land management approaches. Interest: Protection of traditional practices, cultural heritage, sustainable resource use and aligning project with sub-village custom and livelihoods Overall relevance: The customary head at the sub-village level (Kepala Adat) govern and enforce customary laws. In the event of a conflict, the Kepala Adat is the first authority to attempt a resolution. The Temenggung (head of the customary group at the village level) are responsible for governing and implementing customary laws at the village level. If a conflict cannot be resolved at the sub-village level by Kepala Adat, it shall be brought to the Temenggung for resolution. Decisions are made through deliberation and are often settled harmoniously. Due to their responsibility to the implementation of customary laws, Temenggung and Kepala Adat are very influential and also have high interest to the project.
Head of village	Rights: Authority to represent the village in project meetings, ensure benefit distribution aligns with community needs, and participate in monitoring and feedback. Interest: Village development, improved livelihoods, and sustainable land use. Overall relevance: The head of village (Kepala Desa) works closely with the Temenggung to jointly lead community welfare and village development. Chosen through popular vote by the villagers, the Kepala Desa is a representation of provincial government in managing village's budget, administration and various development programs. Even though the influence of the head of village is lower, the interest to the project is high due to possible contribution of the project to community welfare and enhancing development programs is significant.

Stakeholder	Rights, interest, and overall relevance to the project
Village Council (BPD)	Rights: Oversight of project transparency, participation in consultations, and ensuring that benefits align with community development plans. Interest: Fair benefit distribution, transparency, and community inclusion. Overall relevance: The village council jointly with the village government, creates regulation and oversee development programs at the village level. BPD interest and influence to the project is similar to the head of village.
Village-Owned Enterprises (BUMDES)	Rights: Right to develop sustainable enterprises aligned with the project activities, ensuring economic benefits reach the local community. Interest: Expanding income streams through sustainable enterprises. Overall relevance: BUMDES are social and commercial enterprises that are formed by the village government. They aim to strengthen the village economy through the development of community enterprises. BUMDES interest and influence to the project is similar to the head of village.
Credit Union (CU)	Rights: Do not have formal rights to the project, but could help on provision of financial services to the village community, enhancing access to resources for agroforestry and sustainable land use. Interest: Support the project through enhanced financial literacy. Overall relevance: CU plays an important role in the community economy by providing cooperative services, such as savings and loans that are driven by and tailored to the needs of the community. However, CU is less influential and considered less interest to the project.

Stakeholder	Rights, interest, and overall relevance to the project
Tourism Awareness Group (POKDARWIS) in Sekendal Village	Rights: Participation in the project activity to develop and promote eco-tourism linked to the project, benefiting from increased visitor interest and income opportunities. Interest: Eco-tourism development, job creation, and community income growth. Overall relevance: POKDARWIS conserves tourism areas and are responsible for developing the ecotourism potential in the villages. This group has high interest and high influential to the project due to their interest in establishing infrastructure for ecotourism in the project area but also interest to conserve forest around the waterfall as tourist destination.
Family Welfare Empowerment (PKK)	Rights: Participation in project activities that support family welfare, women involvement, and community capacity-building efforts. Interest: Women empowerment, family welfare, sustainable livelihoods Overall relevance: This women's organization plays a significant role in facilitating actions to achieve family welfare, especially the welfare of women and children. PKK is relevant to the project, especially inside the project area. They have interest in the programs for improving community welfare.
Youth Organization (Karang Taruna)	Rights: Participation in project activities that support youth capacity development Interest: Youth empowerment, livelihood opportunities, social issues, tradition and customs. Overall relevance: This youth organization educates young people to improve awareness of religion, livelihood opportunities, social issues, tradition and custom. Karang Taruna is relevant to the project, especially inside the project area. They have an interest in the programs for improving community welfare.

Stakeholder	Rights, interest, and overall relevance to the project
Landak and Sanggau Regent's Offices	Rights: Engagement in regional project coordination, ensuring alignment with development plans and participating in oversight activities. Interest: Regional economic growth, biodiversity conservation, regulatory compliance. Overall relevance: These offices have authority in all matters relating to land and community development in Landak and Sanggau Regencies. These regent offices are relevant to the project due to their interest to the land use and community development even though has lower influences at the project level.
Sub-district Offices of Entikong, Beduai, Air Besar, and Sekayam	Rights: Local administration support, coordination, and participation in monitoring to ensure alignment with sub-district development goals. Interest: Local sustainable development, community benefits, and climate resilience. Overall relevance: These offices are responsible for expanding central and regional government policies. These sub-district offices are relevant to the project due to their interest in monitoring land uses and development in their area, even though they have lower influences at the project level.

Stakeholder	Rights, interest, and overall relevance to the project
Directorate General of Sustainable Forest Management (DG SFM) of Ministry of Environment and Forestry (MoEF) and Bureau of Sustainable Forest Management unit VIII Pontianak.	Rights: Policy oversight, technical guidance, and support for project compliance with national sustainable forest management goals. Interest: National compliance, sustainable forest management, climate goals. Overall relevance: The DG SFM of MoEF is the main partner of PT CMI due to its authority over policy and governance for production forest area in Indonesia. Coordination on license and workplan are directly with DG SFM, while technical oversee of the concession management are conducted by Bureau of Sustainable Forest Management unit VIII Pontianak. These institutions have high interest to the project due to their authorities in relation to the project compliances with Indonesian laws, however, they have limited influences at the project level.
Environment and Forestry Service of West Kalimantan (Dinas LHK Kalimantan Barat) and Forest Management Unit Unit III (KPH Unit III) of Sanggau Barat	Rights: Provision of local support, monitoring, and technical assistance for sustainable forest management and conservation activities. Interest: Alignment with regional conservation strategies, restoration efforts. Overall relevance: These institutions are responsible for managing forest functions, especially KPH Unit III Sanggau Barat which is relevant to the project due to their role in monitoring and control the forest licenses. They have high interest to the project due to their authorities in relation to the project compliances with Indonesian laws, however, they have limited influences at the project level.

Stakeholder	Rights, interest, and overall relevance to the project
The MoEF Forest Planning and Environmental Governance (BAPLAN), BPKH III Pontianak, and National Land Agency (BPN);	Rights: Facilitation of land tenure clarity and land use planning, ensuring legal compliance and supporting land rights for the project. Interest: Legal compliance, clear land tenure, and effective forest management. Overall relevance: BAPLAN is responsible for forest planning, including reviewing the existing forest estate and boundary demarcation. The BAPLAN office in West Kalimantan is BPKH III Pontianak, and along with BPN may consider realizing land for agrarian reform from the forest estate, which includes some areas within the project area
Directorate General of Climate Change Control (Ditjen PPI)	Rights: Ensuring alignment with national climate targets and emissions reductions. Interest: National climate and emission reduction goals Overall relevance: Ditjen PPI serves as the national focal point for formulating and implementing policies aimed at controlling climate change. It oversees actions related to climate change through integrated low-carbon sustainable development plans. As a key regulator of climate change policies and carbon projects, Ditjen PPI is an important stakeholder. Ditjen PPI has high interest to the project due to their authorities in relation to the project compliances with Indonesian laws, however, it has limited influences at the project level.

Stakeholder	Rights, interest, and overall relevance to the project
The MoEF Directorate General of Nature Resources and Ecosystem Conservation (Ditjen KSDAE) and BKSDA West Kalimantan;	Rights: Oversight of biodiversity protection measures, ensuring conservation outcomes align with national strategies. Interest: Biodiversity protection, ecosystem restoration, sustainable land use. Overall relevance: Ditjen KSDAE and its regional office, BKSDA West Kalimantan, has authority to manage conservation areas, including Gunung Nyiut Nature Reserve. The nature reserve shares boundaries with the project area. The project will inevitably impact this nature reserve and vice versa. They have high interest to the project due to their authorities in relation to the project compliances with Indonesian laws, however, they have limited influences at the project level.
The MoEF Directorate General Social Forestry and Environmental Partnership (Ditjen PSKL) and Center for Social Forestry and Environmental Partnership (BPSKL) Kalimantan.	Rights: Facilitating community-based management and sustainable land use. Interest: Community-based management, sustainable forest use, livelihood improvement. Overall relevance: Ditjen PSKL and its regional office, BPSKL Kalimantan, in collaboration with KPH Sanggau Barat Unit III and DLHK West Kalimantan, regulate and implement social forestry and environmental partnership programs. These stakeholders support and assist communities to apply for social forestry licenses and monitor their operations. In Sungai Ilai Village, a community forest plantation managed by Koperasi Karyawan Rimba Jaya shares boundaries with the project zone. Additionally, there are some 'indicative' social forestry areas within the project zone that will provide licenses for community use of the forest, though this will only indirectly affect the project.

Stakeholder	Rights, interest, and overall relevance to the project
Social forestry license holders	Rights: Do not have formal right to the project, but could have indirect right of engagement in discussions to prevent conflicts Interest: Reduced conflict and learning how the project could maximize benefit from sustainable management and income generation Overall relevance: Koperasi Karyawan Rimba Jaya has a license for community forest plantation (Hutan Tanaman Rakyat - HTR) in the southern part of the project area. It is likely to have some influence on local communities in the nearby area.

Concession holders around the project area	Rights: Do not have formal right to the project, but could have indirect right of engagement in discussions to prevent conflicts. Interest: Reduced conflict Overall relevance: The PT CMI concession area is surrounded by other concession holders. PT Lahan Sukses: Block I and Block II of PT CMI concession are separated by the PT Lahan Sukses industrial timber plantation concession. This stakeholder has a significant impact in the project's success as they operate in the same villages as PT CMI, notably the villages of Mawang Muda, Sungai Ilai, and Thang Raya. This company has lower interest and influence to the project, however, the project should monitor the activity of this company due to it might has direct affect to the project area. PT Fajar Cemerlang Indomulya: A mining concession has been allocated in the northern part of PT CMI concession. This concession managed by PT Fajar Cemerlang Indomulya overlaps with Block I of the PT CMI concession. Mining operations have been temporarily halted but may start again soon. In addition, in order to be fully operational at the project area, this concession will need to obtain permission to borrow and use forest areas from the MoEF. This company has great interest in and influence to the project due to the activities are inside the project area. It could contribute to the negative impact to the project. This company has not obtained land use agreement (<i>Izin pinjam pakai Kawasan hutan</i>) PT Borneo Katulistiwa Palma, PT Borneo Ketapang Permai, and PT Charindo Palma Oetama: These three oil palm concessions are situated close to the PT CMI concession. They are expected to influence local community decisions on land use in the project zone as they continue to encourage palm oil development. This company has lower interest and influence to the project, however, the project should monitor the activity of this company due to it might have direct affect to the project area.
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Stakeholder	Rights, interest, and overall relevance to the project
Agriculture middlemen	Rights: As part of the broader community living in the project zone. The agriculture middlemen has rights to participate in project planning, implementation, and benefit-sharing, with direct access to training, resources, and other benefits from the project Interest: Improved livelihoods, sustainable agriculture, better resource access. Overall relevance: Individuals or groups of villagers around the project are making a living from purchasing agricultural products (e.g. palm oil, peppers, rubber, and durian) from farmers and on-selling these products at a higher price at regional markets. Due to the activities of the project will also addressing market of agricultural products from the project area, these middlemen will be highly impacted. Their interest and influence on the project is significant.
Taxi bike operators	Rights: As part of the broader community living in the project zone. The taxi bike operator has rights to participate in project planning, implementation, and benefit-sharing, with direct access to training, resources, and other benefits from the project Interest: Improved livelihoods, sustainable agriculture, better resource access. Overall relevance: Individuals or groups of villagers around the project make a living by providing taxi bike transportation services and moving agricultural products such as oil palm, peppers, rubber, and durian from remote areas to collection points. Most of these taxi bike drivers are based in the villages of Engkangin, Sekendal, Sungai Ilai, Raut Muara, and Mawang Muda. Due to the activities of the project will also limiting commodities that might damaging the forest, such as oil palm, demand on transportation services for these commodities in the project area will be lower, therefore, these middlemen will be highly impacted. Their interest and influence to the project is significant.

Stakeholder	Rights, interest, and overall relevance to the project
Local miners	Rights: As part of the broader community living in the project zone. The taxi bike operator has rights to participate in project planning, implementation, and benefit-sharing, with direct access to training, resources, and other benefits from the project Interest: Improved livelihoods, sustainable agriculture, better resource access. Overall relevance: Individuals or groups of villagers around the project area make a living by gold panning in rivers close to Sekendal Village, Raut Muara Village, and in Nekan Village. This activity is considered illegal as they don't have the correct licenses to conduct these mining operations. Such illegal activity was also reported to occur within the PT CMI concession area. The project activities might prevent the local miners from polluting the rivers, therefore, their interest and influence to the project is high.
Non-Governmental Organization (NGO)	Rights: Do not have formal right to the project Interest: Low interest but could have interest in participation in project monitoring, advocacy for transparency, and ensuring adherence to social and environmental standards. Overall relevance: Several local NGOs work in the project zone, collaborate with communities in the project area and project zone, especially in relation to indigenous rights advocacy, such as AMAN, PPSDAK-PK and LBBT. These NGOs has little interest to the project, however, they have direct influence to community for supporting and/or not supporting the project. On addition, there are several NGOs who worked in the project zone, such as Yayasan Planet Indonesia for conservation in Gunung Nyiut Nature Reserve and Yayasan Solidaridad Network Indonesia for small holder oil palm plantations. These NGOs have less interest and are influential to the project.

2.3.3 Stakeholder Access to Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

To provide stakeholders access to project documents, the field office will maintain a full, printed version of the project description (PD) document in English for public viewing. The executive summary of the PD will be translated to Bahasa Indonesia and given to community leaders to make information available to the community in surrounding villages. The documents related to the project are also available virtually on Verra and the project website (to be developed by the Project Proponent).

2.3.4 Dissemination of Summary Project Documents (VCS, 3.18, 3.19; CCB, G3.1)

The project information summary has been disseminated to local stakeholders during the informational meetings in FPIC processes (see Annex 4 for official report of informational meeting). For all stakeholders, the summary of project documents will be translated into Bahasa Indonesia, and access will be provided through the project offices, village information boards, and online resources through Verra and the project website.

In the subsequent years, the summary of monitoring results will be disseminated regularly to the communities through relevant channels and key stakeholders (e.g. village office, customary hall, tribal leaders, etc.). Prior to the dissemination of said results, Focus Group Discussions (FGD) will be conducted to confirm findings of the monitoring activities.

2.3.5 Informational Meetings with Stakeholders (VCS, 3.18, 3.19; CCB, G3.1)

The project began conducting informational meetings with stakeholders during the project development stage. The purpose of these meetings was to develop trust, build relationships with project stakeholders and engage in further efforts to understand the conditions of the local community as well as the potential community concerns. The stakeholders were informed at least three days before the meeting via letter or phone. The list of stakeholder engagements conducted is provided in Table 11 below.

Table 11. The list of stakeholder engagement

Date	Type of Engagement	Description
8 -11 Nov 2023	Focus Group Discussions (FGD) with the local teams	The local teams were recruited from the villages in the project zone. The FGD was used to inform the local teams about the project objectives and to plan the field work.

Date	Type of Engagement	Description
13 Nov 2023	A workshop attended by 10 village representatives; 4 people from each village were invited including the head of each village, head of the customary group, women, and other community leader	An FGD was conducted to explain the project area, the project objectives, and the field work plan. The participants were consulted on the schedule for the field work and key stakeholders at the village level were engaged during the field work. Information was collected and added to a village map regarding access to the villages as well as additional information that may be useful for geographical orientation, such as rivers and mountains. Information was collected and added to a village map regarding access to the villages and additional information that may be useful for geographical orientation, such as rivers and mountains.
14-19 Nov 2023	FGD, workshop, PRA and social transect in Raut Muara Village	The workshop was held to explain the project area, objectives, and field work plan. During the workshop, participants who consented filled out socio-economic questionnaires. Additionally, a Participatory Rural Appraisal (PRA) was conducted to gather information on community stakeholders, land use practices, occurrence of important species, culturally significant locations, and essential resources for local communities' basic needs. Social transects were carried out by visiting various locations to verify culturally important sites and resources crucial for meeting the communities' basic needs.
19-23 Nov 2023	FGD, workshop, PRA and social transect in Enkangin Village	Same as above.
23-27 Nov 2023	FGD, workshop, PRA and social transect in Sepangah Village	Same as above.

Date	Type of Engagement	Description
27-30 Nov 2023	FGD, workshop, PRA and social transect in Sekendal Village	Same as above.
1-5 Dec 2023	FGD, workshop, PRA and social transect in Mawang Muda Village	Same as above.
6-8 Dec 2023	FGD, workshop, PRA and social transect in Sungai Ilai Village	Same as above.
9-11 Dec 2023	FGD, workshop, PRA and social transect in Thang Raya Village	Same as above.
12-15 Dec 2023	FGD, workshop, PRA and social transect in Kenaman Village	Same as above.
16-18 Dec 2023	FGD, workshop, PRA and social transect in Nekan Village	Same as above.
19-21 Dec 2023	FGD, workshop, PRA and social transect in Engkahan Village	Same as above.
21 November – 6 December 2023	Informal discussion with 10 village officials	Informal discussion was conducted to introduce the project and explore the community's acceptance of the project. The discussion was also focusing on the potential community involvement in the project lifetime.
27 November 2023	Discussion with Beduai sub-district official	Courtesy visit and notification of CMI activities and the company's purpose to conduct consultations and FPIC (Free, Prior, and Informed Consent) with the village community.
29 November 2023	Discussion with Sekayam sub-district official	Courtesy visit and notification of CMI activities and the company's purpose to conduct consultations and FPIC (Free, Prior, and Informed Consent) with the village community.
29 November 2023	Discussion with Entikong sub-district official	Courtesy visit and notification of CMI activities and the company's purpose to conduct consultations and FPIC (Free, Prior, and Informed Consent) with the village community.

Date	Type of Engagement	Description
30 November 2023	Discussion with Air Besar sub-district official	Courtesy visit and notification of CMI activities and the company's purpose to conduct consultations and FPIC (Free, Prior, and Informed Consent) with the village community.
24 February 2024	Informal discussion with Raut Muara Village Official	Discuss the agroforestry pilot group and ask for the potential land to create a demonstration plot for agroforestry. The output of this meeting is the project obtained 1 hectare of land for a demonstration plot and 0.7 hectares for a nursery, which were voluntarily provided by the villagers of Raut Muara.
5 March 2024	Workshop	The workshop was conducted to socialize the nursery, agroforestry demonstration plot and development of pilot agroforestry group.
29 – 30 April 2024	Workshop	Informational meeting with community from 10 villages representatives to inform the project plan and make sure the community understand the project benefit, cost and risk.
28 May 2024	FGD with Forest Management Unit (KPH) Sanggau Barat Unit III	Courtesy visit and notification of CMI activities and discussion about the procedure for reporting illegal activities within the concession.

Additionally, project description summary - such as Flyer, poster, Theory of change - has been put on the board of every village office in the project zone. These steps were carried out as part of the Free and Prior Informed Consent (see Section 2.5.7).

2.3.6 Risks from the Project and No Net Harm (VCS, 3.18, 3.19)

Table 12 below identifies risks from the Project and their potential impact, and the mitigation or preventive measures put in place by the project.

Table 12. Identified risk, impact, and mitigation or preventive measures put in place by the project

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Natural and human induced risks to stakeholders' wellbeing	Crop failure in agroforestry	Loss of income	All agroforestry activities will be jointly designed between communities and agricultural specialists to ensure appropriate agroforestry systems are designed and implemented to meet community needs, while climate resilience is included. This approach will involve working with communities to selection of suitable sites, choosing the preferable multipurpose tree species to be planted, as well as building silvicultural knowledge and practices. Design, marketing, and standard operating procedures of agroforestry practice was prepared in collaboration with technical expert (USAID) and will be implemented in accordance with technical feasibility and community's approval.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Risks to stakeholder participation	Lack of engagement	Reduced support, leading to potential conflict with stakeholder	The project will conduct regular community meetings to maintain engagement. Where appropriate and when necessary, specific consultation processes will be established to target women and other marginalized groups in the community to ensure their engagement in project's decision making processes.
Working conditions	No risk identified	NA	The project will develop occupational health and safety (OH&S) regulations and Standard Operating Procedures (SOPs). It will also ensure fair wages and conduct training programs on all OH&S policies and regulations. These measures aim to ensure that working conditions do not negatively impact stakeholders, the ecosystem and biodiversity.

Safety of women and girls	Participation in physical activities such as planting or forest monitoring may expose women and girls to physical strain or unsafe working condition, especially if safety protocol are not enforced. And presence of outside workers or changes in community dynamic due to project activities might increase the risk of harassment or violence against women and girls	If women and girls experience physical strain or unsafe condition, they may become discouraged from participating in the project activities and could result of decline in community involvement in the project Due to the presence of harassment will develop conflict within the communities and the project and could result of decline in community involvement in the project	The project goal is to develop gender equity among the project staff and the community. The project will ensure women's participation in all stages of the project and provide equal access to resources and training. The project will conduct community engagement and awareness program for gender sensitive awareness campaign to engage both men and women in understanding of the importance of women's safety and equality Training and activities that promote the role and importance of woman and girls will be developed and implemented in the project. Moreover, the project has develop grievance mechanism that available for all community member including women and girls to express concern and provide confidential reporting channels for any incidents of harassment or violence. The project will take actions for any harassment and violence that occur in all the community
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	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
			member, including women and girls. Additionally, several document will be provided by the project, including: • Sexual harassment SOP • Additional standard and policy on how safety of woman could be ensured in workplace • Enforce health and safety protocol

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Safety of minority and marginalized groups, including children	Discrimination and exclusion of marginalized group	Deeper social and economic marginalization and mistrust and reduced engagement	The project will conduct regular stakeholder consultations in each village and make sure all sub-groups, minorities and marginalized groups, actively participate and are included in the project. While children will not take part in the stakeholder consultations and project activities, access to education and health care facilities will be provided by the company to ensure they also have a positive impact from project development. The project will make sure to Implement inclusive participation strategies, ensuring all voices are heard, particularly of those from vulnerable groups, during project planning and execution.

Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)	Pollution and improper waste handling	Health issue and environmental degradation	The project will identify and categorize all waste products into organic (fruit residues, paddy straw, pruning residues, etc), plastic (nursery containers, irrigation components, mulch film, etc), and other types of waste classification to ensure the correct separation of recyclable and non-recyclable. The project believes that understanding the composition of waste will enable it to tailor strategies and implement the project to tailor management practices for effective waste management. The project will adopt the principles of Reduce, Reuse and Recycle. Organic waste will be separated and
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	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
			either composted or disposed of through burial. Inorganic waste will be sorted into recyclable components, which will be processed by local partners, while residual inorganic waste will be removed from the site and disposed of at the nearest government-run waste disposal facilities. The project will train farmers on safe handling and disposal of agrochemicals.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Discrimination	Discrimination in workplace	Reduced engagement and social conflict	The project team and individuals involved will be held to the highest standards of integrity, to prevent any form of discrimination. This includes discrimination based on gender, race, religion, ethnicity, social class, or any other aspects of human individuality. The project will adhere to all relevant laws and regulations. To ensure compliance, a grievance system will be established, providing project staff, local community members, and other stakeholders with a formal system of report and address any instances of discriminatory actions.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Sexual harassment	Sexual harassment in remote working environments	Injury, mental health issues and legal consequence	The project has strict policies regarding sexual harassment. Training on sexual harassment had been provided to existing staff, while new staff members will be briefed on this matter during induction process, SOP on sexual harassment is already in place and ready to be enforced on every party associated with the project development.
Equal pay for equal work	Payment discrimination	Gender-gap payment	The project will make sure the any remuneration is transparent, and fair towards the services that the employee had provided. The project will make sure there are no gender pay gap payments or other gap based on race, religion, ethnicity, social class, or any other aspect of human individuality

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Gender equity in labor and work	Work discrimination	Reduced engagement and social conflict	The project team and all individuals involved will be held to the highest standards of integrity, to ensure both women and men will be treated appropriately, according to the company regulation. The project will comply with all relevant laws and regulations. To ensure adherence, a grievance system will be established, providing project staff, local community members, and other stakeholders with a formal system to report and address any discriminatory actions that may occur.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Forced labor ³²	No risk identified	NA	The project enforces stringent policies against illegal activities, including the use of forced labor. All newly hired staff will undergo training to ensure they are fully aware of these policies. Additionally, the project will implement rigorous hiring practices and work exclusively with verified contractors to ensure that neither contractors nor community staff members engage in forced labor in any project or project-related activities
Child labor	No risk identified	NA	The project will not employ children or individuals under the age of 15, in accordance with the Indonesia's minimum age for employment regulations.

³² The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking must be inclusive of staff and contracted workers employed by third parties.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Human trafficking	No risk identified	NA	The project has strict policies on illegal activities, including human trafficking. Training will be provided for newly hired staff to ensure they understand these policies. Additionally, the project will implement rigorous hiring practices and only collaborate with verified contractors to ensure that neither contractor nor community staff members are involved in human trafficking.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Recognition of, respect of, and promotion of the rights to IPs, LCs and customary rights holders	No risk identified	NA	The project activities do not involve the removal or relocation of property rights or territories belonging to local communities and indigenous peoples. A customary area assessment has been conducted across the project zone and recognized all the project zone as customary lands. The project proponents have committed to protecting and patrolling both project and customary boundaries in accordance with the approved governance and monitoring activities.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Preserving and protecting cultural heritage	No risk identified	NA	One of the project objectives is to preserve and protect cultural heritage, for this reason community will have continuing access to all culturally identified areas and it will be implementing programs to promote cultural knowledge and the exchange of this knowledge between generations.
Protecting and preserving property rights, customary rights, or protecting legal or customary tenure/access rights to territories, property, and resources, including collective and/or conflicting rights	No risk identified	NA	The project has conducted customary area assessments across the project zone and acknowledged and recognizes the project zone as customary lands. Additionally, FPIC process has been conducted 10 villages, signifying the consent of communities to the general design of the project. During the project lifetime, project will conduct subsequent similar activities to obtain communities' consent on each of the specific activities proposed by the project

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems	No risk identified	NA	The project will conduct conservation and protection activities in the project area. The lands with degradation will be restored through native species. Agroforestry practices will only be conducted in previously established agricultural areas. Additionally, no invasive species will be introduced. Hunting restrictions and a ban on hunting threatened species will be established and enforced by the project team and forest patrol teams.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Soil degradation and soil erosion	No risk identified	NA	The project will restore degraded lands and implement agroforestry practices to improve soil fertility. Organic fertilizers will also be used to improve soil nutrients, in addition to planting legume species able to fixate nitrogen directly from the atmosphere. Areas with high erosion profiles are in the project area, but the risk is low due to the current forest cover. The project will protect and conserve the forest, and the multilayered nature of agroforestry regime would be able to prevent further degradation of soil condition.
Water consumption and stress	No risk identified	NA	The project area has high annual rainfall (2500-3000 mm/year). This condition will bring sufficient water supply for on-going community needs and for the development of agroforestry systems without increasing water availability stress.

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Habitats (and areas needed for habitat connectivity) for rare, threatened, and endangered species	No risk identified	NA	The project will conserve the remaining forest area and will restore degraded area, as well as establish agroforestry systems to enhance biodiversity and habitat connectivity.
Areas needed for habitat connectivity	No risk identified	NA	The project will conserve the remaining forest area and will restore degraded areas, as well as establish agroforestry systems to enhance biodiversity and habitat connectivity.
Invasive species	No risk identified	NA	No invasive species will be introduced, and the project will regularly monitor biodiversity and forest condition to identify dispersal of any invasive species

	Identified risk(s)	Potential impact of risk on stakeholders, ecosystem health, and biodiversity	Mitigation or preventative measure(s) taken
Ecosystem conversion	No risk identified	NA	The project will conduct conservation and protection activities in the project area. When a degraded land is identified, the ecosystem will be restored through the planting of native species. Agroforestry practices will only be conducted in previous agricultural areas, where coffee and cacao will be introduced alongside other native species.

2.3.7 Community Costs, Risks, and Benefits (CCB, G3.2)

The project has conducted stakeholder consultations, described further in Section 2.3.10. Information on potential costs, risks, and benefits to communities is further outlined in section 4.2.1

2.3.8 Information to Stakeholders on Validation and Verification Process (VCS, 3.18.6, 3.19; CCB, G3.3)

The project will inform the local communities about the CCB and VCS validation and verification process. Prior to these processes, the project team will conduct site visits to provide information to village leaders and hold village meetings. All communities and residents will be informed if their participation is required during any of these processes.

2.3.9 Site Visit Information and Opportunities to Communicate with Auditor (VCS, 3.18.6; CCB, G3.3)

Project staff will inform relevant stakeholders in advance of all audit processes and help to prepare and support them for the auditor's site visit, ensuring they have sufficient opportunity to communicate with the auditor. The project will also arrange meetings with relevant stakeholders where the auditor can freely and openly ask them questions.

2.3.10 Stakeholder Consultations (VCS, 3.18; CCB, G3.4)

Table 13 below details stakeholder consultations that have been conducted by the project.

Table 13. Stakeholder consultation information

Date of stakeholder consultation	29 April 2024 – 30 April 2024
Stakeholder engagement process	Project consultations have occurred in each of the 10 villages in the project zone (see Section 2.3.5). One week before each meeting, community representatives, including women and young leaders were to choose their representative. The language used in these meetings was Bahasa Indonesia. These consultation meetings were conducted as informational workshops before moving into the formal consent process. During the workshops, the following information was presented and discussed: • Explanation on the PT CMI multi-forestry business permit and an outline on the various regulatory and legal issues in establishing a forest carbon project as part of the multi-business forestry permit. • Survey outcomes on current biodiversity conditions, High Conservation Values, and the identified socio-economic conditions of the local communities. • Results of the initial assessment on possible economic, social, cultural and environmental impacts including potential risks, and the fair and equitable distribution of benefits. • Future activities to be implemented by the project, considering the outcomes of the studies above. This included how project activities are to be implemented, including involve workers and the role of local communities and indigenous peoples, company staff, research institutions, NGOs and government institutions. • Proposed benefits, risks and costs to the community as well as the positive and negative impacts of carbon trading activities under a multi-business forestry permit. • Proposed benefit that discussed with community: - Additional and alternative income from agroforestry activity and involvement in restoration activity

- The benefits from carbon trading will be channeled to the community in the form of various empowerment programs (health, education, and training), the creation of new sustainable businesses, initial capital for revolving funds, and other forms that may be further determined.
- The enhancement of knowledge through various training programs that will be provided as the project progresses, such as agroforestry training, land preparation without burning, organic fertilizer production training, and non-timber forest products training.
- Woman involvement in the project activity
- Alternative job opportunities such as participation in planting activities, forest monitoring, tree inventory, and biodiversity monitoring.
- Increase access to education and health as the project progresses
- Identified risk of the project:
 - The limited use of wood is restricted to the community's basic needs to ensure that the remaining forests are preserved.
 - The limitation of mining activities, which prohibit the use of cyanide or any substances that could pollute water, is the prevention of water contamination.
 - The limitation on hunting, which only allows the hunting of animals that are not endangered, endemic, or protected.
 - The limitation on the expansion of new agriculture and oil palm plantations is due to the optimization of existing land and the avoidance of clearing the remaining forests.
- The project will finance all the cost of the project
- The project theory of changes.
- The proposed complaints procedures, complaint handling and the grievance redress mechanism.
- Cooperation plans between PT CMI, the project team and the community.

	• Agreement on a draft consent letter for the project proposal. After the workshop, posters and brochures summarizing the project were distributed to the community representatives to further inform all community members in all villages about the project's aims and benefits. To document all meeting outcomes, the project team and community representatives jointly drafted an official meeting report before asking the community to sign the report as a fair and consistent interpretation of the meeting. The official report of the informational meeting can be seen in Annex 4.
Consultation outcome	The community representatives presented at the consultation meetings have a well-developed understanding of the project aims, project design, anticipated benefits and potential risks to the community. This is evidenced through the agreement and signing of the official meeting report. The community representatives requested some time to consult widely with all residents of their village and to inform the entire village community of the project's risks, costs, and benefits. The draft consent agreement was also widely distributed, discussed and debated. The outcome of this process is all villagers of all 10 villages provides their free, prior and informed consent (FPIC) for the project to commence (see Section 2.5.7).
Stakeholder input	The project team received substantial feedback during the workshop regarding the risks, costs, and benefits for the community arising from project implementation. No changes were made to the project design following these consultation meetings, as the project had already been co-designed with input from the community and other stakeholders.

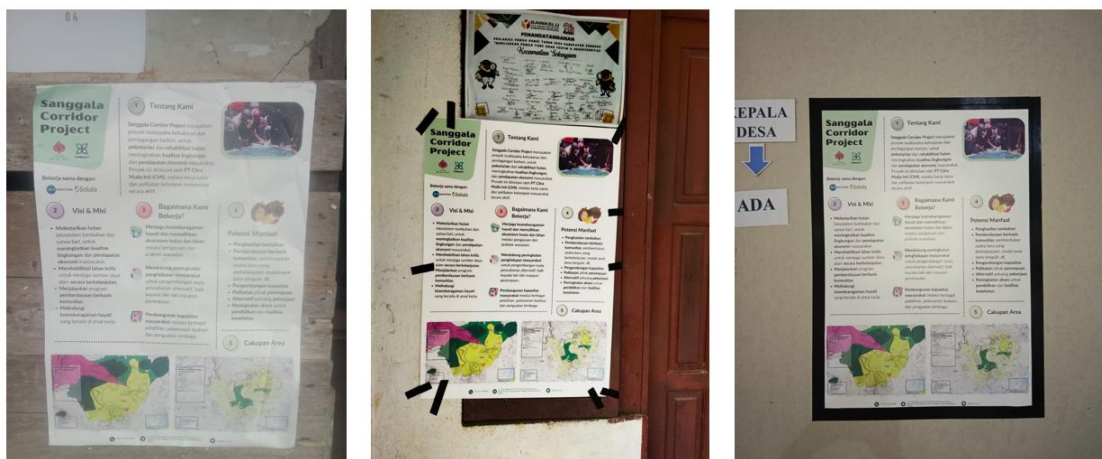


Figure 4. The project posters that were distributed have been put up by village representatives in their respective villages. (Left: project poster at the traditional hall of Bungkang Hamlet, Mawang Muda Village, center: project poster at the Raut Muara Village office, right: project poster at the Sekendal Village office).

The summary of comments and action taken by the project are described in Table 14 below. The official report of the meetings can be seen in Annex 4 and the meeting notes can be seen in Annex 5.

Table 14. Summary of comments and action taken by the project.

Summary of comment received	When comment was received	Actions taken
The community does not really understand about carbon projects because it is a new concept for them.	29 April 2024	The project has explained what a carbon project is to the community.
The community asked what species the project would like to plant and suggested planting fruit trees so the community could harvest them.	29 April 2024	The project, with the community, will include fruit trees such as durian trees and hardwood trees in restoration and agroforestry activities. This will allow the community to harvest fruits from these trees. There are no changes to the project design because this suggestion is already included in the project design.
The community suggested the project to involve the community through employment opportunities.	29 April 2024	The project design includes collaboration with the community and has already involved them in various activities, such as biodiversity surveys and nursery establishment and maintenance

		This community input aligns with the project's goals, and no changes to the design are necessary.
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2.3.11 Continued Consultation and Adaptive Management (VCS, 3.18; CCB, G3.4)

The project team is deeply committed to establishing broad communication and consultation processes, with the aim of ensuring that all stakeholders are informed on all project activities and that feedback is heard and acted upon.

Throughout the project's lifetime, the project team will maintain a direct line of communication with community leaders, representatives, members and relevant stakeholders. Each channel of communication will be established to meet solely the needs of the stakeholders and not of the project staff.

The project field staff will consult and engage with local community members and relevant stakeholders through regular formal and informal in-person meetings. Staff will actively listen to recommendations and adjust their approach as necessary. Employing local community members within the project team and for project activities will further integrate communication channels into the project structure.

An adaptive management plan is in place to ensure the project evolves effectively as it progresses. This plan will be strengthened through a rigorous monitoring, reporting, and evaluation process. The project will periodically review all plans to incorporate lessons learned, new technologies, new scientific findings, and community feedback.

2.3.12 Stakeholder Consultation Channels (CCB, G3.5)

The project will inform, engage and consult with all community leaders, community representatives and other stakeholders throughout its life. Multiple channels will be used to facilitate such as informal meetings, Semesterly formal meetings with community leaders and community representatives, ad-hoc project reporting and updates (with information translated into Bahasa Indonesia), information distribution through village notice boards and other locally important mechanisms, and consultation on key project milestones and activities such as project validation and verification processes. Stakeholder consultation processes are further described in Section 2.3.10.

2.3.13 Stakeholder Participation in Decision-Making and Implementation (VCS, 3.18, 3.19; CCB, G3.6)

Project will provide the platform which would allow Local communities to be actively engaged in the decision-making processes that will impact their livelihood due to intervention initiated by Sanggala Corridor Project. Meetings related to project design and implementation will be open to all community members and organized by village leaders in accordance with local customs.

To ensure inclusivity, the meetings will be structured to encourage the participation of women and other marginalized groups. If necessary, the project team will facilitate separate meetings

led by women for women, ensuring that their voices and perspectives are clearly heard in the decision-making process.

2.3.14 Anti-Discrimination Assurance (VCS 3.19; CCB, G3.7)

The project team maintains a high standard of integrity for all individuals involved in the project design and implementation, ensuring that no one is involved in or complicit in any form of discrimination or sexual harassment. This includes discrimination based on gender, race, religion, ethnicity, social class, or any other aspects of human individuality.

The project complies with all laws and regulations governing anti-discrimination, including the:

- Constitution 1945 (UUD 1945) especially section 27
- Law No. 40/2008 on the elimination of racial and ethnic discrimination
- Law No. 1/2023 on the criminal law code

In addition, the project will comply with any regulations on manpower and worker safety including Law No. 1/1970 on the Occupational Safety (Occupational Safety Law).

The company will implement best practices in recruiting, including a zero-tolerance policy for sexual harassment and discrimination based on gender, religion, ethnicity, or sexual orientation

To ensure compliance with company policy and national laws and regulations, the project will establish a grievance redress mechanism. This system will allow all staff, local community members, and other stakeholders to formally report any discriminatory actions and ensure that appropriate measures are taken to resolve issues promptly.

2.3.15 Feedback and Grievance Redress Procedure (VCS, 3.18.4; CCB, G3.8)

The development and design of the grievance redress procedure intended to resolve any conflicts which may arise between the project proponent and stakeholders are described in Table 15 below.

Table 15. Project Grievance Procedures

Development process	The project's grievance procedure encompasses the following steps: grievance registration, verification, identification of possible solutions with the complainant, taking action to resolve grievances, reporting and monitoring the result. All grievances and disputes are to be resolved within a set timeframe (generally within 10 working days), in a transparent and accountable manner.
Grievance redress procedure	• The stakeholders may submit a complaint to the project manager, any team member, head of village, or nearby village office at any time (the head of village or village office will be required to pass all complaints onto the CarbonX project manager). Complaints can be made by directly calling a team

	member, visiting a project field office, or submitting an email. The report will need to include: 1) the name and contact details of the complainant, 2) any request for confidentiality or anonymity, and 3) a description of the dispute or grievance. • The dispute or complaint will be registered and documented in the Grievance Register, and the complaint person or parties will be notified of any follow-up action required to address the complaint. • The project manager will appoint an independent and senior project team member to investigate the complaint and conduct a follow-up action within 10 business days of receiving the complaint. The investigator will develop an action, implement it, monitor the actions, evaluate and communicate the outcomes back to the complaint. • If grievance, complaint or dispute is found unsubstantiated, the investigator will provide an explanation and if appropriate, meet with the complainant to explain their reasoning and discuss any follow-up actions needed to address the grievance or complaint correctly. In the case the investigation team requires more time to investigate the dispute or complaint, they will inform the aggrieved or complainant directly or via email, text messages or WhatsApp, to let them know the time that will be required to conduct and resolve the complaint. • If appropriate the complaint will be jointly investigated by the project team and the community, <i>with the aim of reaching a mutual deliberation wherever possible</i>. If the person or parties are not satisfied with the resolution, they have the right to not accept it by notifying the CarbonX project manager and explaining the reasons within 15 business days. The reasons can be directly reported by a formal written report, email, text message, or WhatsApp message to the project manager. • Although it is not expected, any dispute or complaint that is not resolved through mediation shall be referred by the project manager to a neutral third party for either a) arbitration, to the extent allowed by the laws of the relevant jurisdiction, or b) competent courts in the relevant jurisdiction. This does not preclude any party's right to submit the complaint to a competent supranational adjudicatory body. • The outcomes and resolution of any arbitration or court ruling shall be clearly communicated to all parties, including the reasons for the decisions. The introduction and/or implementation of outcomes and resolutions from the arbitration of court process will be overseen, if applicable, by
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	the CarbonX project manager or other independent third-party entity. The above procedure may be updated from time to time to reflect any new procedural steps or contact information, e.g., hotline number, new dedicated email, website or social media account, etc. The above procedure may be updated occasionally to reflect any new procedural steps or contact information, e.g., hotline number, new dedicated email, website or social media account, etc.
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2.3.16 Accessibility of the Feedback and Grievance Redress Procedure (VCS, 3.19; CCB, G3.8)

The grievance redress mechanism (GRM) will be clearly communicated to all local communities through the communication channels outlined in Section 2.3.12. The grievance form can be seen in Annex 6. The GRM mechanism was initially developed by project proponent, and communicated to communities through several meetings leading to signing of the FPIC document. The project proponent has also incorporated input regarding the GRM, such as on how the communities could lodge their input to project proponent.

The project will encourage local community members and relevant stakeholders to visit the project office to discuss any concerns directly with the project staff. Verbally reported grievances will be documented by the project staff and submitted into the formal system for resolution. Grievance submission forms will be available at the project office and can be submitted directly to the project staff. Additionally, the community liaison officer's phone number and email will be distributed to community leaders, and they will be available to meet with stakeholders to receive reported grievances. To ensure transparency, the project will maintain a record book with all received and resolved grievances, which can be viewed upon request by auditors.

2.3.17 Worker Training (VCS, 3.19; CCB, G3.9)

The project will require all staff to complete an induction of orientation program where they will be provided information on the project's policies and Standard Operating Procedures (SOPs), with a particular emphasis on health and safety. This orientation program will be updated yearly, or as needed, with all staff required to complete this revised orientation program.

The project staff will also receive technical training based on their roles and to expand their prospects for future employment. Training will be provided in the following areas: agroforestry design and management, biodiversity monitoring and assessment, carbon stock assessment, community facilitation and development, Monitoring, Reporting, and Verification (MRV), and other project related training needs.

The project will also offer community-based training that complements and aligns with the staff-based training. Identified training needs for the community include agroforestry, biodiversity, forest inventory, and sustainable agricultural practices.

To ensure knowledge retention and transfer, the project will maintain records of all training materials, manuals, and procedures, and will create both digital and physical repositories of

project knowledge and resources. Community training programs for trainers will also be developed to facilitate the transfer of local knowledge to other communities.

2.3.18 Community Employment Opportunities (VCS, 3.19.13; CCB, G3.10)

The project will actively recruit local community members in project-based employment opportunities. This may include roles in:

- Nursery establishment and management.
- Forest restoration and agroforestry planting.
- Forest and fire patrolling.
- Biodiversity monitoring.
- Community development and community facilitation.
- Ecotourism ventures (upon completion of the ecotourism plan and viability of this plan).

The project is committed to providing equal employment opportunities. Where possible, and in line with their work capacity and capability, the community can fill middle management positions. Employment will be based on merit, but this will also be balanced according with the project's commitment to enhance the livelihood opportunities for women and other marginalized community groups through the project.



Figure 5. Job opportunities announcement in the village office

2.3.19 Occupational Safety Assessment (VCS, 3.19; CCB, G3.12)

The project is committed to ensuring the health and safety for all project staff and will follow all relevant laws and regulations on this matter.

Project staff are exposed to several risks that include:

- Tips, falls and accidents related to strenuous activities under wet, humid, and hot conditions,
- Encounters with dangerous wildlife, including snakes, wild boars and possibly bear attacks
- Insect and spider bites
- Exposure to mosquito-borne sickness.
- Getting lost during field exercises and patrolling activities
- Exposure to smoke and fire during fire management activities.
- Harm and injury by confronting illegal poachers, hunters, loggers or other conducting illegal activities in the project zone.

The project has developed a comprehensive Occupational Health and Safety (OH&S) Plan (see Annex 7), which will be communicated to all staff to inform them of work-related risks and safety practices for risk mitigation. A hard copy of the relevant project regulations will be kept at the project office for reference by all project staff.

During the staff induction process, all new staff will be briefed on the OH&S Plan and basic first aid procedures. Prior to any field activities, all staff and community members involved will meet with the project leader to discuss the risks and mitigation measures associated with the field exercise. These measures will be aligned with local norms and cultural practices to ensure they are accepted and integrated into the work plans by community members. For example, the project will provide navigation tools and markings on clothing, work in a buddy system, require the use of personal protective equipment and be provided with first aid kits.

All incidents will be reported through a project Incident reporting form. This form will be submitted to the Project Team for actions to be taken, relative to the severity of the incident, and for mitigation measure to be put in place to prevent future incidents arising from a similar risk.

2.4 Management Capacity

2.4.1 Project Governance Structures (CCB, G4.1)

PT CarbonX Bumi Harmoni (CarbonX) is the project proponent and main entity to lead the project design, implementation, monitoring and reporting on the project. CarbonX will be supported by other entities as described in Section 2.4.4.

The project will be managed by CarbonX. The organization structure of the project team, highlighting the different roles between CarbonX and PT CMI is shown in Figure 6. CarbonX roles will focus on the efficient management and administration of the project while PT CMI roles will focus on field operations and project implementation.

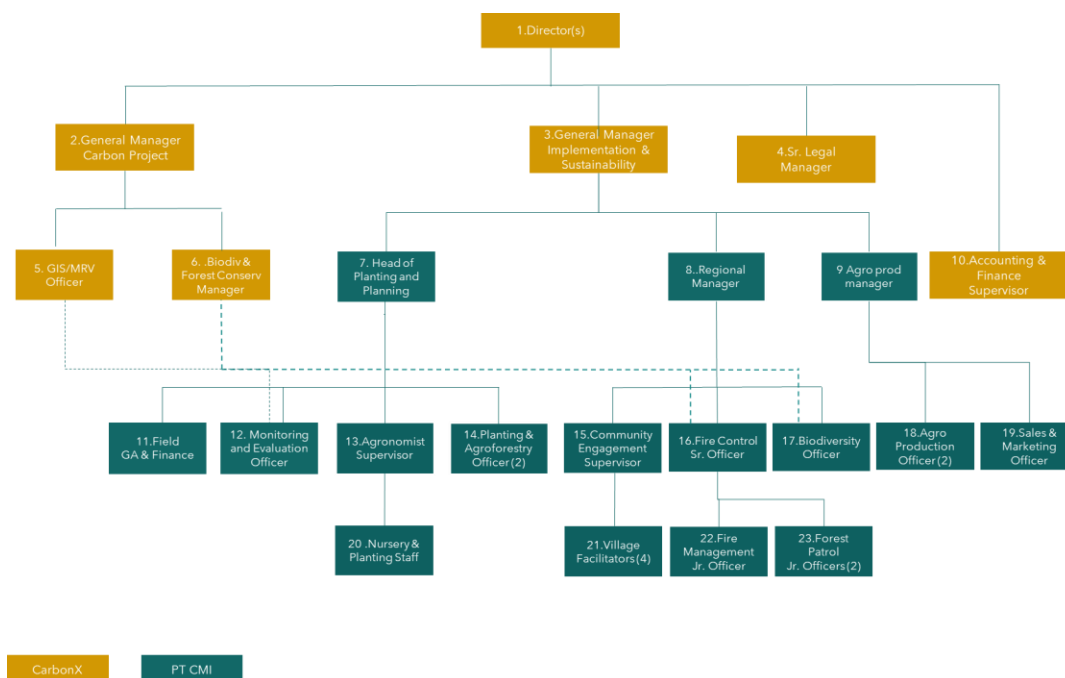


Figure 6. The schematic diagram of the project structure

2.4.2 Required Technical Skills (VCS, 3.19; CCB, G4.2)

The key technical skills required to implement the project are outlined in Table 16. These skills cover numerous project activities, including forest conservation, forest restoration, carbon accounting, biodiversity conservation, agriculture and agroforestry development community engagement, and community development.

Table 16. Required technical skills for the project

Technical Skill	Description	Role in Project
Agroforestry Practices	Knowledge of the integration of trees and shrubs into agricultural systems. Knowledge of silvicultural management and tree crop production within agricultural systems	Designing and implementing agroforestry systems, improving soil fertility, and increasing biodiversity.
Soil Science	Understanding of soil properties, nutrient management, and conservation techniques.	Assessing soil health, implementing soil conservation measures, and managing soil fertility.
Forest Conservation	Techniques and strategies for protecting and managing forest ecosystems.	Preventing deforestation, managing protected areas, and restoring degraded forests.
Restoration Ecology	Principles and methods for restoring ecosystems to their natural state.	Planning and executing restoration activities in degraded areas.

Technical Skill	Description	Role in Project
Sustainable Agriculture	Practices that promote environmentally friendly and economically viable farming.	Implementing sustainable farming techniques, reducing environmental impact, and enhancing productivity.
Community Engagement	Skills in working with local communities, participatory planning, and conflict resolution.	Collaborating with local communities, ensuring their involvement and support, and addressing concerns.
Biodiversity Conservation	Knowledge of protecting and promoting biodiversity within ecosystems.	Enhancing habitat connectivity, protecting endangered species, and promoting ecosystem health.
Water Management	Techniques for efficient water use, irrigation, and watershed management.	Managing water resources sustainably, implementing efficient irrigation systems, and reducing water stress.
Carbon Accounting	Methods for measuring and calculating carbon emissions and sequestration.	Quantifying carbon reductions, monitoring project impact, and reporting on carbon credits.
Geographic Information Systems (GIS)	Using GIS software for mapping and spatial analysis.	Mapping project areas, monitoring land use changes, and planning restoration activities.
Project Management	Skills in planning, executing, and managing projects.	Coordinating project activities, managing timelines and budgets, and ensuring project goals are met.
Environmental Impact Assessment (EIA)	Assessing and mitigating the environmental impacts of projects.	Conducting EIAs, identifying potential impacts, and implementing mitigation measures.
Monitoring and Evaluation (M&E)	Techniques for tracking project progress and evaluating outcomes.	Developing M&E plans, collecting data, and assessing project effectiveness.
Data Analysis	Analysing data to inform decision-making and reporting results.	Interpreting project data, generating reports, and supporting adaptive management.
Fundraising and Grant Writing	Skills in securing funding and writing grant proposals.	Obtaining financial support for the project, writing proposals, and managing grants.

2.4.3 Management Team Experience (VCS, 3.19; CCB, G4.2)

The management team's expertise and prior experience implementing land management and carbon projects are described below.

Dessi Yuliana – Director

Dessi is the Director at CarbonX. She is an established professional with over 20 years of experience in business, finance, and investment, including six years as a green investment specialist for the international organization, the Global Green Growth Institute. Previously, Dessi

led a successful four-year Indonesia Green Climate Fund readiness program in Indonesia, where she developed sustainable business models and enhanced climate change project proposals to secure international climate financing. She holds a bachelor's degree in accounting from the University of Indonesia and a master's degree in business and information technology from the University of Melbourne.

Rusman Sudaya – General Manager for Carbon Projects

Rusman is CarbonX's General Manager for Carbon Projects with responsibility for managing all technical aspects of nature-based solutions projects throughout the project life cycles. Rusman is skilled in developing GHG quantification models and analyzing project data at CarbonX. For over three years, he was the Sustainability Director at PT Provident Agro Tbk and has led several projects across Southeast Asia in the development of GHG emission reduction certifications. Rusman earned his bachelor's degree of environmental engineering from Bandung Institute of Technology, and his MSc degree from Kungliga Tekniska Högskolan, Stockholm, Sweden.

Sigid Ariyanto – General Manager for Implementation and Sustainability

Sigid is managing the field team and various partners to ensure project milestones and compliance with international and national standards and regulations. He leads the design of community benefit sharing mechanisms and planning the rigorous Free Prior Informed Consent (FPIC) process. Sigid has over a decade of experience as a project manager for biodiversity, carbon, and community projects in international organization such as World Wildlife Fund, Conservation International, the Nature Conservancy, Orangutan Foundation International, and Hatfield Consultant. He holds a Master degree in Social and Political Science from Gadjah Mada University.

Adhi Wardana – Legal Senior Manager

Adhi is CarbonX's Legal Senior Manager. He leads legal affairs and advises CarbonX's management on all legal changes in government regulations and international standards. He is a qualified Indonesian lawyer with over ten years of experience. He has been involved in a wide range of legal transactions, including capital markets, mergers and acquisitions, foreign direct investment, and technology. In addition to his legal experience, Adhi has also worked for an Indonesian non-governmental organization focused on climate change, forestry, and palm oil governance. He earned his law degree from Universitas Padjadjaran.

Stefanus Agustino Sitor – Carbon Project Senior Officer

Stefanus Agustino Sitor is the Carbon Project Senior Officer in the CarbonX team. He supports the team's operational work, particularly on issues related to carbon, forestry, and nature conservation. Prior to CarbonX, he worked at an independent media outlet and at GGGI, an international organization supporting Indonesia actors in developing sustainable business models and enhancing their capacity in writing proposals for international climate finances. He obtained his bachelor's degree in forestry from Universitas Gadjah Mada.

Rizki Atthoriq Hidayat – GIS Officer

Rizki is GIS officer leveraging cutting-edged Geographical Data Science and remote sensing to uncover environmental insights and promote sustainability. He applies his expertise in GIS technologies to assess projects and has collaborated with NGOs and government agencies on

conservation efforts. His research has been published in leading journals. Rizki hold a Bachelor in Geography Science from Padang State University and is a certified in GIS and Data Science.

Dalyono – Field Manager

Dalyono is responsible for ensuring that project implementation activities adhere to relevant regulations. He also plays a role in building and fostering communication and cooperation both with government agencies such as Forest Management Unit (FMU), Regencies Government, as well as with the community in the concession area. Further, he is deeply involved in the preparation and subsequently coordinate the forest conservation and restoration activities according to the Project Design. Dalyono has more than 1 decade of experience as the Head of the Division in the field of industrial forest plantation. He holds a Bachelor of Forestry from Bogor Agricultural University.

Muh. Asnawi – Senior Fire Control

Muh. Asnawi is responsible for planning, coordinating, and implementing fire control and forest protection plans throughout the concession area. He leads the implementation of community programs intended as fire prevention measures such as the formation of *Masyarakat Peduli Api* (MPA) and patrolling activities to prevent fires and illegal activities in the concession area. As a former active soldier, his experience is instrumental in creating strategic plans and cooperation amongst relevant stakeholders. He completed his military education at the SECABA level.

Deddy Dwi Firwanta – Community Engagement Supervisor

Deddy plays an essential role in building relationships and fostering partnership with the communities, traditional leaders, and village leaders in the project zone area. He conveyed project's program and policy to relevant community stakeholders to foster collaboration and streamline coordination between project and communities' relevant stakeholders. In addition to the work related to community development, he supports the implementation and monitoring of project's biodiversity aspect. Deddy has more than 1 decade of experience in the field of mapping and surveying. He holds a bachelor's degree in forestry from Universitas Tanjungpura.

Yeni Pramita – Agronomist Officer

Yeni has the responsibility in planning, supervising, and evaluating the implementation of activities related to agroforestry aspect of the project. This ranges from study activities, design the technical aspects of community-based agroforestry, water and pest management, as well as farmer capacity building programs. She has 5 years of experience in community-based agriculture commodity production companies and at Earthworm Foundation, an international non-profit organization focusing on regenerative agriculture practices. Yeni holds a master's degree in Extension and Development Communication from Universitas Brawijaya.

2.4.4 Project Management Partnerships and Team Development (VCS, 3.19; CCB, G4.2)

The project proponent recognizes that hiring and retaining technically skilled and experienced staff will significantly influence the project's success. As outlined in Section 2.4.3, the project proponent's team consists of highly capable staff with appropriate skills, knowledge, and educational backgrounds. However additional knowledge and skills will be required to

complement the project proponent's expertise. Table 17 lists the institutions and areas for cooperation to ensure success.

Table 17. The list of institutions and respective subjects of cooperation

Institution	Subject of cooperation	Status
USAID SEGAR	Provide grants and expert to conduct research to determine culturally and economically viable agroforestry systems.	MoU established
SEBUMI	Review the ecotourism potential across the project zone and map out resource requirements and investment needs.	MoU established
Universitas Tanjung Pura (UNTAN)	Technical expertise in forestry including biodiversity, carbon stock and agroforestry.	MoU established
Solidaridad Network Indonesia	Collaboration to provide community support related to agroforestry, forest conservation and restoration.	MoU planned
PT Forestwise Wild Keepers	Cooperation for the supply and sale of illipe nuts as a key strategy to enhance alternative livelihoods by providing market access to forest-based commodities in the project zone area.	Cooperation Agreement established.

2.4.5 Financial Health of Implementing Organization(s) (CCB, G4.3)

The project proponent, CarbonX and PT CMI, is a financially secure, and sophisticated company with adequate financial resources committed to the project throughout its entire lifespan (see Section 2.1.22 for further information). CarbonX and PT CMI financial statements are annually audited by Binder Dijker Otte (BDO), an international assurance, tax and financial advisory services firm. Such audited financial statements will be provided to the project's independent validation and verification body (VVB) upon request.

2.4.6 Avoidance of Corruption and Other Unethical Behavior (VCS, 3.19; CCB, G4.3)

The project team is committed to upholding a high level of integrity and professionalism throughout all aspects of project design, implementation, monitoring and reporting. The project proponent has a zero-tolerance policy towards corruption and unethical behavior, and will not tolerate any form of corruption such as bribery, embezzlement, fraud, extortion, and collusion. The project is compliant with the US Foreign Corruption Practice Act (FCPA).

2.4.7 Commercially Sensitive Information (VCS, 3.5.2 – 3.5.4; CCB Rules, 3.5.13 – 3.5.14)

Our project's commercially sensitive information that is not publicly available includes:

- Project implementation plans and financial plans – Including detailed organizational project implementation plans, as well as initial and long-term financial plans for the project's duration.
- Original data, computer code, and GIS information – Including project boundary, land-use, biomass inventories, models assessing carbon calculations, and social assessments in the project area.
- Any documentations related to acquisition process of the concession between previous owner and PT Carbonx Bumi Harmoni as current majority shareholder of PT CMI
- Agreements and contracts between project proponents, other project entities, and relevant stakeholders – regarding the implementation of project activities, including consent from the community.
- Benefit sharing mechanisms plan

Specific information on these aspects can be provided upon request from the project validators.

2.5 Legal Status and Property Rights

2.5.1 National and Local Laws (VCS, 3.1, 3.6, 3.7, 3.14, 3.18, 3.19; CCB, G5.6)

The project is designed to be fully compliant with relevant national, regional, and local laws of the Republic of Indonesia on forest conservation projects. All relevant approvals have been obtained by the project in line with these laws and regulations. Following are relevant national and local laws for Sanggala Corridor Project

- Law No. 6 of 1994 concerning ratified Indonesia's commitment to the United Nations Framework Convention on Climate Change (UNFCCC)
- Law No. 41 of 1999 concerning Forestry
- Law No. 32 of 2009 concerning Environmental Protection and Management
- Law No. 16 of 2016 ratifies Indonesia's commitment to the Paris Agreement under the United Nations Framework Convention on Climate Change (UNFCCC)
- Government Regulation No. 22 of 2021 concerning Environmental Protection and Management
- Government Regulation No. 23 of 2021 concerning Forestry Management
- Presidential Regulation No. 98 of 2021 concerning Carbon Pricing
- Ministry of Environment and Forestry Regulation No. P32/Menlhk/Setjen/Kum.1/3/2016 concerning Forest and Land Fire Control

- Ministry of Environment and Forestry Regulation No. 8/2021 concerning Forest Governance and the Preparation of Forest Management and Utilization Plans in Protection and Production Forests
- Ministry of Environment and Forestry Regulation No. 21/2022 concerning Guidance for the implementation of Carbon Economic Value (NEK).
- Ministry of Environment and Forestry No.7/2023 concerning Carbon Trading Procedures in Forestry Sector.
- Ministry of Environment and Forestry No.12/2024 concerning Implementation of Nationally Determined Contributions in Addressing Climate Change

2.5.2 Relevant Laws and Regulations Related to Worker's Rights (VCS, 3.18.2; CCB, G3.11)

Indonesian employment law is governed by Law No.13/2003 regarding Manpower (Labour Law), as amended by Law No. 11/2020 and as amended by *Perpu* (Government Regulation in lieu of Law) No.2/2022 regarding Job Creation. The government has issued the following regulation to implement the Labour Law:

- Government Regulation No. 35/2021 concerning Fixed-Term Employment Agreements, Outsourcing, Working Hours and Rest Hours, and Termination of Employment.
- Government Regulation No. 36/2021 concerning wages

The other main statutes are:

- Law No.2/2004 concerning Industrial Relations Dispute Settlement
- Law No.21/2000 concerning Labour Unions.

The project will follow all the labor regulations above. By adhering to these labor provisions, the project ensures legal compliance and supports fair and ethical employment practices, enhancing worker protection and aligning with Indonesian labor laws.

In addition to the above laws and regulations, employers and employees are also subject to all relevant company regulations and the provisions of their employment agreements. Workers will be informed about their rights during the recruitment and induction process.

2.5.3 Human Rights (VCS, 3.19)

The project acknowledges and upholds the protection of the rights of Indigenous Peoples (IPs), Local Communities (LC), and customary rights holders in accordance with relevant international human rights laws. This includes adherence to the United Nations Declaration on the Rights of Indigenous Peoples and the International Labour Organization (ILO) Convention 169 on Indigenous and Tribal People. In this regard, the project:

- **Upholds Free, Prior and Informed Consent (FPIC):** Before commencing project activities on customary lands, the project have secured the FPIC from Indigenous Peoples IPs, Indigenous Peoples LCs, and customary rights holders. This involves engaging multiple stakeholders

through meaningful and transparent consultation processes and fair negotiation, prior to any implementation.

- **Recognizes Land and Resources Rights:** The project acknowledges and respects the land and resources rights of Indigenous Peoples IPs, Local Communities LCs, and customary rights holders. It ensures that their traditional territories and resources are not infringed upon by the forest conservation and restoration activities. This includes recognizing and protecting their rights to land, territories, and natural resources.
- **Encourages Participation and Inclusion:** The project actively involves Indigenous Peoples IPs, Local Communities LCs, and customary rights holders in all decision-making processes. Their traditional knowledge and expertise are highly valued and will be integrated into the project design, implementation, monitoring, and reporting to ensure their meaningful participation and inclusion.
- **Safeguards Cultural Heritage:** The project will take all measures to protect the cultural heritage and traditional practices of Indigenous Peoples IPs, Indigenous Peoples LCs, and customary rights holders. It ensures that all project activities do not disrupt or diminish their cultural identity, traditions, and customs.
- **Builds Capacity and Shares Benefit Equitably:** The project supports initiatives for capacity building and ensures that Indigenous Peoples IPs, Indigenous Peoples LCs, and customary rights holders fairly, equitably and transparently benefit from the project, both economically and socially. This includes providing training for all, employment opportunities for women and youth, and implementing a strong, positive and integrated community development plan.

2.5.4 Indigenous Peoples and Cultural Heritage (VCS, 3.18, 3.19)

The project will preserve and protect cultural heritage through all its activities. The project proponent views direct engagement with local communities in the region as essential. This engagement is crucial for fostering partnerships to protect and restore the natural ecosystems within the project area and for exchanging knowledge on climate-related issues. It is critical for understanding the primary social or environmental needs of Indigenous Peoples (IPs) and Local Communities (LCs) that could be addressed or alleviated through potential income-generating activities and revenue from the sale of Verified Carbon Units (VCUs). All forms of communication will be based on the project proponent's ethical principles and formal policies, and in recognition of the history, ancestral heritage, and intrinsic culture of the indigenous people living within the project zone.

2.5.5 Statutory and Customary Property Rights (VCS, 3.18, 3.19; CCB, G5.1)

There are no outside statutory claims in the project area as the project has officially received all statutory rights to manage the area under the license as described in Section 2.5.12.

However, based on the project community assessment¹⁰⁶ and HCV assessment¹⁰⁷, the project area and zone can be defined as customary areas of the Dayak community. This is proven by the existence of zones created based on the distribution of specific cultural values or archaeological sites, ritual activities, and biological natural resources to fulfill cultural needs.

Due to this, the project recognized the project zone as a customary area of the community. The project will continuously conduct stakeholder consultation during the implementation of project activities. The project has undertaken FPIC and obtained consent from the community to perform the project as described in section 2.5.7.

2.5.6 Recognition of Property Rights (VCS, 3.7, 3.18, 3.19; CCB, G5.1)

The project activities do not lead to involuntary removal or relocation of property rights holders from their lands or territories and do not force rights holders to relocate activities important to their culture or livelihood.

2.5.7 Free, Prior and Informed Consent (VCS, 3.18; CCB, G5.2)

The outcome of the FPIC process as part of the stakeholder consultation process is described in Table 18 below.

Table 18. Free, Prior, Informed and Consent process

Description of process for obtaining consent	The FPIC process conducted for the Sanggala Corridor Project followed the following seven steps: 1) Identification: Identification of Indigenous Peoples and Local Communities (IPs and LCs) who have rights (tenure, access, use, cultural rights) to the forest area, those who have a potential 'stake' in the land or resources or those groups who may be impacted by the proposed project. Indigenous Peoples or Local Communities are defined as a community groups with a historical connection to a residence on customary lands for generations. These communities are governed by customary laws and institutions that regulate the socio-cultural life and maintain sovereignty over land and natural resources, which is crucial for their sustainability. The community share a common cultural identity, including unique language, spirituality, values, attitudes, and behavior, and possess traditional values and knowledge system³³. Based on this definition, the project identified the existence of community groups with a common history, customary laws, cultural assets and/or customary objects, ownership' of traditional lands, and traditional government institutions or system³⁴. Prior to the fieldwork, the project team asked personnel from NGOs, such as the Samdhana Institute, AMAN network and PPSDAK—Pancur Kasih, whether customary communities lived in the area. The feedback confirmed that the
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³³ [Aliansi Masyarakat Adat Nusantara - AMAN | Mengenal Siapa Itu Masyarakat Adat](#)

³⁴ Customary Law Community definition according to Head of Geospatial Information Agency Regulation No. 12/2017 regarding Guidance on Mapping Territory of Customary Law Community.

entire project area is considered customary land of six Dayak sub-tribes.

- 2) **Preparation:** The project prepared information on the aims, objectives and methods of all assessment work needed for the project. The community was then consulted on the work needed to be completed and consent from the communities was sought to conduct the forest inventory assessment, the biodiversity assessment and participatory rural appraisal (PRA) and community mapping work needed for the socio-economic survey.

Another objective at this stage was to prepare for further engagement and seek agreement from the community on the overall scope of the FPIC process. The project team then developed a Standard Operating Procedure (SOP) for the FPIC process and established a local team from villages in the project area (8-11 November 2023) to help with the assessment work. Focus Group Discussions (FGDs) were conducted with the head of Sepangah Village, Engkangin Village, and the head of Air Besar Sub-district (10 Nov 2023) to identify key stakeholders, collect existing community information through a participatory mapping process, and other secondary data. From this consultative process, a pathway was developed to broadly engage the community and initiate the FPIC process.

The first workshop was conducted in Crediting Union Pancur Kasih Building in Balai Karangan Village on 13 November 2023. The workshop was attended by representatives from all 10 villages including 4 people from each village including the head of each village, heads of the customary community, women, and other public figures. In total, 46 participants from the 10 villages attended the workshop, including five women. The workshop aimed to explain the project area, project objectives and the fieldwork required. The participants were also consulted on the fieldwork schedule.

During the field work, key stakeholders at the village level were engaged and various information necessary for participatory mapping exercise, such as access to the villages and location of rivers and mountains, was gathered. In addition, the project team requested for consent from the community representatives to complete the necessary fieldwork for the social and forest inventory surveys. The consent letters included for this stage are included in annex 18.

- 3) **Research:** From November 2023 until the end of December 2023, village-level workshops were conducted in Raut Muara Village, Engkangin Village, Sepangah Village, Sekendal Village,

Mawang Muda Village, Sungai Ilai Village, Thang Raya Village, Kenaman Village, Nekan Village, and Engkahan Village.

During these village-level workshops, the project team conducted PRA exercises, community mapping of land and resource use, social transects, and a socio-economic survey to gather baseline information on the community's current condition and assess the potential impact (both positive and negative) of the project on the community. Other information collected included confirmation of community stakeholders (using a Diagram Venn tool), the mapping of land use by the communities, occurrence of important species, culturally important locations as well as areas or resources essential for local communities to meet their basic needs.

The forest inventory team conducted field surveys from 14 November 2023 until 22 December 2023, then again from January to April 2024

- 4) **Information and consultation:** From 29-30 April 2024 (see Section 2.3.10) the project team conducted an information sharing workshop with all community representatives. In addition to informing the communities regarding the findings of the social and forest inventory activities, it is also intended to verify the results of the study. Additionally, project's theory of change were laid out to participants of the workshop. From this point, the project's long-term objectives, impacts, outcomes, outputs, planned activities, potential community benefits and risks were again discussed and debated by the community.

A draft of the project agreement was also openly discussed, debated and jointly drafted with the community representatives.

During the workshop, all information, project clarifications and answers to every questions were presented transparently, accessible, and culturally appropriate to the community representative attending the meetings.

The community representatives who attended the workshop then freely signed an "official report" (*berita acara*) declaring that they received and understood the project information that was presented to them. The official report of the workshop can be seen in Annex 4.

The community representatives were then provided with ample time and resources to consult with their broader communities on the project benefits, risks and impacts to the community, as

well as the draft project agreement (see Section 2.3.10 for further details on this process).

- 5) **Negotiation:** The consultation process designed by the project team and agreed upon by the community (see step 2 of the process) included provisions for the community to review and negotiate the project agreement. This also encompassed a plan for participatory monitoring, conflict resolution, and a grievance redress mechanism

From 1 May 2024 to 11 June 2024, the community representatives who attended the workshop returned to their villages to discuss the project with their communities and gathered any concerns raised. A WhatsApp group was created during this period to facilitate discussion and clarify concerns from the community.

Agreement: At the end of the consultation and negotiation process, final workshops were held for community representatives. In those events, communities' comments and concerns on the draft project agreement were formally presented to representative of project proponent. This included all conditions that needed to be met for the community to provide their consent for the project to commence. During this workshop the time frame for the agreement was actively discussed. It was ultimately agreed that the final timeframe of the agreement is following with the period of the concession.

At the end of the workshop, a near final draft of the project agreement was prepared. The community representatives then took this final draft back to their communities for further consideration. Posters and brochures detailing the project agreement were distributed within the villages, such as the village's head office, and the WhatsApp.

Consultation at the community level continued until each village openly and freely provided their consent to the project agreement. This was formalized through final workshops held in each village:

- Kenaman Village on 15 May 2024.
- Raut Muara Village on 16 May 2024.
- Engkahan Village on 16 May 2024.
- Nekan Village on 17 May 2024.
- Sungai Ilai Village on 17 May 2024.
- Mawang Muda Village on 20 May 2024.
- Thang Raya Village on 20 May 2024.

	• Sepangah Village on 21 May 2024. • Engkangin Village on 3 June 2024. • Sekendal Village on 4 June 2024. 6) Implementation and monitoring: All conditions agreed upon by the community will now be implemented, based on The project's Theory of Change, which was already discussed with the communities in step 4. At this step, the project's monitoring and evaluation process is being discussed and formulated with the communities. It has also been agreed that when the agroforestry activities commence with potential impacts on private and communal lands, a more detailed community mapping and cooperation agreements discussions will be carried out. The project's feedback and grievance redress mechanism (see Section 2.3.15) will also be implemented ensuring that the community can raise concerns or objections regarding project implementation, the distribution of benefits or other project activities. This is an important part of the ongoing FPIC process and an important mechanism for consent to be withdrawn.
Outcome of FPIC process	All 10 village communities within the Sanggala Corridor Project have granted their FPIC for the project to commence and have all signed a formal agreement for the implementation of the project. All processes have been documented, and the signed consent document is detailed in Appendix 19. The project will not encroach on land and doesn't plan to relocate people within the project zone. Instead, it will work with the communities to ensure the continuity of the project activity. The community will benefit from the project through training, education, healthcare access, employment, opportunities, and more.

The attendance of the information workshop conducted on 29 April 2024 is depicted in Figure 7.



Figure 7. Communities and the project proponents during the information-sharing workshop on 29 April 2024.

2.5.8 Benefit Sharing Mechanisms (VCS, 3.18, 3.19;)

The benefit sharing mechanisms are outlined in Table 19 below.

Table 19. Benefit Sharing Mechanism

Process used to design the benefit sharing plan	The general benefit-sharing plan has been designed to incentivize behavioral change, ensure project permanence and provide an equitable and transparent distribution of both financial and non-financial benefits to local communities. The plan was communicated to the community during the informational workshop. From this activity, the community agreed on a general profit-sharing plan. However, profit sharing needs to be detailed further through a Memorandum of Understanding MoU at the hamlet level or with the current land users. Benefit-sharing arrangements will continue to be refined throughout the project period at the hamlet user level.
Summary of the benefit sharing plan	The following are the general benefit sharing plan with the communities: - From agroforestry and restoration activities, the communities shall obtain annual wages for the planting of seedlings in the first year and maintenance in the subsequent years - From agroforestry activities, the company shall be the main off taker for the agricultural commodities produced if and when an

	agreement is reached regarding the selection of agricultural commodities. • A percentage of proceeds from the sales of VCU shall be allocated and managed independently by community through established community funds and or agreed-upon activities based on community aspirations, financed by the sales of VCU. • In other activities related to the project (e.g. forest patrol, biodiversity monitoring, etc.), the company shall collaborate with local communities and provide daily wages for their contribution to the project.
Approval and dissemination of benefit sharing plan	The general benefit-sharing plan has been disclosed to and shared with local communities during the informational workshop (See Appendix 4). The community has given the consent on the project and the general benefit sharing plan of the project, but further subsequent detailed plan must be developed and agreed at the hamlets level or with the current land users.

2.5.9 Property Rights Protection (VCS, 3.18, 3.19; CCB, G5.3)

The project proponents have agreed that the project boundaries will be protected and patrolled according to approved governance and monitoring activities. The project activities do not involve the removal or relocation of property rights from the territories belonging to local communities.

2.5.10 Illegal Activity Identification (VCS, 3.19; CCB, G5.4)

The community has traditionally converted small patches of forest to paddy rice fields or other crops. This activity has occurred in a production forest area, making it illegal. According to the Forestry Law No.41/1999 and No. 19/2004, the Job Creation Law No.11/2020, and the Ministry of Forestry Decree No. P31/2014, agricultural activities are not legally permissible in areas designated as production forests.

Additionally, the community engage in activities such as smallholder palm oil plantations, wildlife hunting and illegal mining (see Figure 8). Generally, palm oil cultivation is prohibited in forest area as the species is not considered forest vegetation and it contributes to deforestation, habitat degradation and reduction of vital forest ecosystem services, such as water catchment. The region has faced pressure from palm oil expansion due to easy market access surrounding the project zone which has been growing since early 2000s. As of 2023 approximately 1.341 hectares have been converted to palm oil, which has been excluded in the crediting area. Wildlife hunting without permits is also illegal particularly when it involves protected species. Illegal traditional gold mining is another prohibited activity as the communities does not have necessary permits to conduct mining activities in the project area.

A comprehensive strategy to address these illegal activities should integrate economic incentives, law enforcement, and community engagement. Proposed measures include:

- Agroforestry schemes. Develop agroforestry systems with strong economic benefits offering viable alternatives to illegal practices and curbing the expansion of palm oil cultivation.
- Alternative livelihoods. Promote opportunities such as ecotourism and the commercialization of sustainable forest-based commodities. As an example, the project has signed an agreement with PT Forestwise to supply illipe nuts.
- Multi-Stakeholder forums. Establish a forum that brings together community members, local authorities, and other stakeholders to coordinate efforts and curb illegal activities.
- Customary law enforcement. Enforce customary laws that prohibit the hunting of protected wildlife species and other illegal activities.
- Community-based Forest patrols. Establish and empower community-based patrols to monitor and prevent illegal activities. During patrols, engage directly with community members using persuasive dialogue and awareness-raising measures to promote sustainable practices and discourage illegal activities.
- Awareness and sustainable practices. Implement a series of awareness-raising initiatives to educate communities about the detrimental effects of monoculture plantations, promote sustainable agriculture practices, and emphasize the importance of wildlife protection.

Additionally, the project will actively engage individuals involved in illegal activities, inviting them to participate in the program and become part of the solution.



Figure 8. A villager conducting traditional gold mining in the Riam Dait waterfall area, Sekendal Village

2.5.11 Ongoing Disputes (VCS, 3.18, 3.19; CCB, G5.5)

There are no ongoing disputes or previously unresolved conflicts inside the project area. The project will continue to engage with the community and conduct regular community meetings to

consult on project progress and implementation activities, identify any possible conflicts or disputes, and put in place actions to prevent all disputes and conflicts.

The project team will be staffed by dedicated community development specialists (see Section 2.4.1) experienced in conflict management and mediation.

If a dispute or conflict does arise, the grievance redress mechanism detailed in Section 2.3.15 will be used to resolve any dispute that arise.

2.5.12 Approvals (CCB, G5.7)

The project has received approval from the Ministry of Environment and Forestry to manage the project area. The approved documents from The Government of Indonesia are listed below in Table 20.

Table 20. List of approved documents that the project received from the Government of Indonesia

No	Indonesian Terms	English Terms
1	Surat Keputusan Menteri Lingkungan Hidup dan Kehutanan Republik Indonesia Nomor: 2/1/IUPHHK-HTI/PMDN/2015 tentang Pemberian Izin Usaha Pemanfaatan Hasil Hutan Kayu Pada Hutan Tanaman Industri Kepada PT. Citra Mulia Inti Atas Hutan Produksi Seluas 20,645 (Dua Puluh Ribu Enam Ratus Empat Puluh Lima) Hektar Di Kabupaten Sanggau dan Kabupaten Landak, Provinsi Kalimantan Barat.	MoEF Decree Number 2/1/IUPHHK-HTI/PMDN/2015 Regarding Granting of Business Permits for The Utilization of Wood Forest Products in Industrial Plant Forest to PT. Citra Mulia Inti for a Production Forest Area of 20,645 (Twenty Thousand Six Hundred Forty-Five) Hectares in Sanggau District and Landak District, West Kalimantan Province
2	Surat Keputusan Menteri Lingkungan Hidup dan Kehutanan Nomor: SK. 4693/MenLHK-PHPL/UHP/HPL.1/7/2018 tentang Persetujuan Rencana Kerja Usaha Pemanfaatan Hasil Kayu Hutan Tanaman Industri (RKUPHHK-HTI) Periode Tahun 2018-2027 Atas Nama PT. Citra Mulia Inti Di Provinsi Kalimantan Barat.	MoEF Decree Number SK. 4693/MenLHK-PHPL/UHP/HPL.1/7/2018 Regarding the Approval of the Work Plan for the Utilization of Industrial Timber Products (RKUPHHK-HTI) for the 2018-2027 Period on Behalf of PT. Citra Mulia Inti in West Kalimantan Province.
3	Surat Keputusan Menteri Lingkungan Hidup dan Kehutanan Republik Indonesia Nomor: SK.1039/MenLhk/Setjen/HPL.0/11/2021 Tentang Perubahan Atas Keputusan Kepala Badan Koordinasi Penanaman Modal Atas Nama Menteri Lingkungan Hidup dan Kehutanan Nomor 2/1/IUPHHK-	MoEF Decree Number SK.1039/MenLhk/Setjen/HPL.0/11/2021 Regarding Amendment to Decisions of the Head of the Investment Coordinating Board on behalf of the Minister of the Environment and Forestry Number 2/1/IUPHHK-HTI/PMDN/2015

No	Indonesian Terms	English Terms
	HTI/PMDN/2015 Tanggal 13 Februari 2015 tentang Pemberian IZin Usaha Pemanfaatan Hasil Hutan Kayu Pada Hutan Tanaman Industri Kepada PT.Citra Mulia Inti Atas Areal Hutan Produksi Seluas 20.645 (Dua Puluh Ribu Enam Ratus Empat Puluh Lima) Hektar Di Kabupaten Sanggau dan Kabupaten Landak, Provinsi Kalimantan Barat	on February 13, 2015 regarding the Granting of Business Permits for the Utilization of Timber Forest Products in Industrial Plantation Forests to PT. Citra Mulia Inti Over a Production Forest Area of 20,645 (Twenty Thousand Six Hundred Forty Five) Hectare in Sanggau District and Landak District, West Kalimantan Province
4	Surat Keputusan Menteri Lingkungan Hidup dan Kehutanan Nomor: SK.11965/MenLHK-PHL/PUPH/HPL.1/11/2023 Tentang Persetujuan Perubahan Rencana Kerja Usaha Pemanfaatan Hutan Perizinan Berusaha Pemanfaatan Hutan Periode Tahun 2018-2027 Atas Nama PT Citra Mulia Inti Di Provinsi Kalimantan Barat	Decree of the Minister of Environment and Forestry Number: SK.11965/MenLHK-PHL/PUPH/HPL.1/11/2023 Concerning the Approval of Changes to the Work Plan for Forest Utilization Business, Forest Utilization Business Licensing for the Period 2018-2027 on behalf of PT Citra Mulia Inti in West Kalimantan Province

Moreover, the project has signed consent from the community in the project zone to conduct the project, as outlined in Section 2.5.7.

2.5.13 Double Counting and Participation under Other GHG Programs (VCS, 3.23; CCB G5.9)

2.5.13.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program, or any other form of community, social, or biodiversity unit or credit?

☐ Yes ☒ No

2.5.13.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

2.5.13.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

2.5.14 Double Claiming, Other Forms of Credit, and Scope 3 Emissions (VCS, 3.24)

2.5.14.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

2.5.14.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

2.5.14.3 Supply Chain (Scope 3) Emissions

Do the project activities affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, "Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."?

☐ Yes ☐ No

2.6 Additional Information Relevant to the Project

2.6.1 Leakage Management (VCS, 3.11, 3.15)

For the Avoided Planned Deforestation (APD) project, two leakage modules are applied: activity-shifting leakage and market leakage. Both leakages have been assessed and calculated as zero.

The activity-shifting leakage is determined to be zero because PT CMI, the baseline entity for deforestation activities, does not have other operational areas. This makes the shifting of deforestation activities unfeasible. Presently, 99% of PT CMI's shares are owned by CarbonX, a conservation-focused company that has no intention or plan to engage in deforestation activities within the project scenario. Similarly, market leakage is also calculated as zero, as PT CMI's baseline scenario involves the export of timber. Consequently, there is no specific management required for either leakage source under the APD project.

For the Afforestation, Reforestation and Revegetation project, leakage is assessed as zero. As outlined in section 3.2.3.2, the project will not displace current agricultural commodities. The project identified paddy and palm oil production as possible foregone. No displacement of paddy fields inside the project area as the agroforestry scheme allocates 20% of agroforestry project area for paddy cultivation. Additionally, ARR project activities do not account the already cultivated palm oil area,

Moreover, displacement of fuel wood production is unlikely. Currently, fuel wood is produced as a commodity production but is collected from natural residues, such as fallen trunks and branches, without any dedicated cultivation activity. On the contrary, the ARR planting activities expected to provide the community with abundant supply of fuel wood. Therefore, there should be no displacement of fuel wood production within the project area.

Since no displacements of any commodities expected to occur during project scenario, ARR leakage management plan is no longer required. Nevertheless, a leakage monitoring plan will be implemented to count any leakages in the case of occurrence. An adaptive management approach will be adopted, allowing periodic reviews and adjustments based on monitoring data (section 3.3.3).

2.6.2 Further Information

Not applicable. All information related to the project is already in this document

3 CLIMATE

3.1 Application of Methodology

3.1.1 Title and Reference of Methodology (VCS, 3.1)

The project uses two VCS methodologies: a.) VM0007 REDD+ Methodology Framework v.1.8, which is used to calculate the emission reductions from avoided planned deforestation, and b.) VM0047 Afforestation, Reforestation, and Revegetation v.1.0, which is utilized to estimate the emission removals from the restoration and agroforestry activities. The associated tools and modules for each methodology have also been applied as outlined in Table 21 below.

Table 21. The title and reference of methodology.

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF)	1.8
Methodology	VM0047	Afforestation, Reforestation, and Revegetation	1.0
Module	VMD0001	Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools (CP-AB)	1.2
Module	VMD0005	Estimation of carbon stocks in the long-term wood products pool (CP-W)	1.2
Module	VMD0006	Estimation of baseline carbon stock changes and greenhouse gas emissions from planned deforestation/forest degradation and planned wetland degradation (BL-PL)	1.4

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Module	VMD0009	Estimation of emissions from activity shifting leakage for avoiding planned deforestation/forest degradation and avoiding planned wetland degradation (LK-ASP)	1.3
Module	VMD0011	Estimation of emissions from market-effects (LK-ME)	1.2
Module	VMD0013	Estimation of GHG emissions from biomass and peat burning (E-BPB)	1.3
Module	VMD0015	Methods for monitoring of greenhouse gas emissions and removals in REDD and CIW projects (M-REDD)	2.3
Module	VMD0016	Methods for stratification of the project area (X-STR)	1.3
Module	VMD0017	Estimation of uncertainty for REDD+ project activities (X-UNC)	2.2
Module	VMD0054	Module for estimation leakage from ARR activities	1.0
Tool	VT0001	Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities	3.0
Tool	VCS AFOLU NPR Tool	AFOLU Non-Permanence Risk Tool	4.2

Type (methodology, tool, module)	Reference ID (if applicable)	Title	Version
Tool	A/R Methodological Tool	Tool for Estimation of Direct Nitrous Oxide Emission from Nitrogen	1.0
Tool	A/R CDM Project Activities	Tool for testing significance of GHG emissions in A/R CDM project activities	1.0

3.1.2 Applicability of Methodology (VCS, 3.1)

All the applicability conditions for the methodologies, tools, and modules are met by the project. These conditions for each of these components are detailed in Table 22.

Table 22. The applicability conditions of the methodologies, tools, and modules

Reference ID/Title	Applicability condition	Justification of conformance
VM0007 – REDD-MF – Section 4.1.2	Land in the project area has qualified as forest for at least the 10 years prior to the project start date	Condition satisfied. The land in the project area has been a forest for over 10 years prior to the project start date covering the period from 2013 to 2023. The land use and land cover change have been assessed using high-resolution satellite imagery (SPOT 6).
VM0007 – REDD-MF – Section 4.1.2	If land within the project area is peatland and emission from the soil carbon pool are deemed significant, the relevant WRC modules must be applied alongside other relevant modules	Not applicable. The project area is not situated on peatland (see sub section 2.1.14.3 Soil). Thus, WRC is not applicable.
VM0007 – REDD-MF – Section 4.1.2	Baseline deforestation and forest degradation in the project area fall within on or more the following categories: Unplanned deforestation (VCS category AUDD) Planned deforestation/degradation (VCS category APD)	Condition satisfied. Baseline deforestation is planned deforestation due to the presence of legally sanctioned plan to conduct land clearing in the concession area of PT CMI.

Reference ID/Title	Applicability condition	Justification of conformance
VM0007 – REDD-MF – Section 4.1.2	Leakage avoidance activities must not include: • Flooding agricultural lands to increase production (e.g., rice paddies); and/or • Intensifying livestock production through use of feed-lots and/or manure	Condition satisfied. The project does not include agriculture on flooded land or intensification of livestock production.
VM0007 – REDD-MF – Section 4.2.3	Avoiding planned deforestation activities are applicable under the following condition: • Where conversion of forest lands to a deforested condition is legally permitted	Condition satisfied. Conversion of forest lands to Industrial Timber Plantation is legally allowed by the government, and the specific entity of deforestation has a permit to deforest the area. Moreover, the entity of deforestation also has a Management Plan (RKU) ³⁵ and an Annual Work Plan (RKT) ³⁶ which are the primary references in carrying out deforestation activities to conduct land clearing, and ultimately convert the forest area to an industrial timber plantation.
VM0047 – Afforestation, Reforestation, Revegetation v1.0	The project activities increase vegetative cover	Condition satisfied. The project scenario includes the planting of 722 trees per hectare through two (2) interventions: a.) agroforestry in agriculture areas, and b.) forest restoration in shrub areas.

³⁵ Management plan for industrial timber plantation (RKU) is a work plan covering the entire working area of an industrial timber plantation license for 10 years, including forest sustainability, business sustainability, environmental stability and socio-economic development of the local community. – Ministry of Environment and Forestry regulation no. P-11/MENLHK/SETJEN/KUM.1/3/2019

³⁶ Annual workplan for industrial timber plantation (RKT – Rencana Kerja Tahunan) is a work plan for a one-year period that is prepared based on the management plan (RKU) – Ministry of Environment and Forestry regulation no. P-11/MENLHK/SETJEN/KUM.1/3/2019

Reference ID/Title	Applicability condition	Justification of conformance
VM0047 – Afforestation, Reforestation, Revegetation v1.0	Area based, census based, or a combination of the two quantification approaches may be used provided approach-specific applicability conditions are met. Approaches must be selected at the project start date and used for the entire project credit period. Where the two approaches are used together, they must be applied in non-overlapping areas defined at the project start	Condition satisfied. The project will use the area-based approach to quantify emission removal from ARR activity.
VM0047 – Afforestation, Reforestation, Revegetation v1.0	This methodology is not applicable if Project activities involve mechanical removal offsite or burning of significant stocks pre-existing dead wood (e.g., for site preparation). Where project site preparation includes chipping, mastication or machine piling, all material must remain onsite within the project.	Condition satisfied. The project may involve mechanical removal but it is limited, and all material will be remain onsite within the project boundary
VM0047 – Afforestation, Reforestation, Revegetation v1.0	This methodology is not applicable under the following condition: if Project activities take place in tidal wetland (e.g., mangroves, salt marshes)	Condition satisfied. The project activities do not occur on wetland. The project site is neither flooded nor saturated with water for any of the year to a degree that would classify it as wetland.
VM0047 – Afforestation, Reforestation, Revegetation v1.0	This methodology is not applicable under the following condition: if Project activities that occur on organic soils or wetland and result in a manipulation of the water table.	Condition satisfied. The project area is not on organic soil or wetland and the activities would not result in manipulation of the water table.

Reference ID/Title	Applicability condition	Justification of conformance
VM0047 – Afforestation, Reforestation, Revegetation v1.0	For Area-based approach Project activities produce continuous tree/or shrub cover on any contiguous area exceeding one hectare	Condition satisfied. The planting will take place in several areas, each larger than one (1) hectare (ha), with a total area of 6,936 Ha. The ARR activity is clearly delineated spatially, and the area in hectares is calculated using GIS.
VM0047 – Afforestation, Reforestation, Revegetation v1.0	For Area-based approach Project may include direct (e.g., manual planting, broadcast seeding) and indirect activities (e.g., activities that permit or facilitate natural regeneration, like herbivory enclosures.	Condition satisfied. The trees will be planted using a direct planting method with seedlings from the nursery. The project will also implement activities that stimulate natural regeneration.
VMD0001 – CP-AB	This module is applicable to all forest types and age classes. Inclusion of the aboveground tree biomass pool as part of the project boundary is mandatory as per the framework module VM0007 REDD-MF	Condition satisfied. The project area for REDD APD is a forested area. The aboveground biomass of trees is included in the project emissions estimations.
VMD0005 – CP-W	This module is applicable to all cases where wood is harvested for conversion to wood products for commercial markets, for all forest types and age classes	Condition satisfied. The project considered harvested wood at the time of deforestation to produce the wood timber forest product.
VMD0006 – BL-PL	This module is applicable for estimating the baseline emission on forest land (usually privately or government owned) that are legally authorized and documented to be converted to non-forest land	Condition satisfied. The project areas are forest concessions legally authorized and documented to be converted to Industrial Timber Plantation permitted by the Ministry of Environment and Forestry under Decree No. 2/1/UPHHK-HTI/PMDN/2015.

Reference ID/Title	Applicability condition	Justification of conformance
VMD0009 – LK-ASP	This module is applicable for estimating the leakage emission due to activity shifting from forest land that are legally authorized and documented to be converted to non-forest land, including activity shifting to forested wetland that is drained or degraded as a consequence of project implementation. This tool must be used for project areas where planned deforestation happens on forested wetland, regardless of the absence of wetland within the project boundaries	Condition satisfied. The project area is forest concession, legally authorized and planned to be converted to Industrial Timber Plantation.
VMD0011 – LK-ME	This module is applicable for calculating market – effects leakage from projects that are anticipated to reduce levels of wood substantially and permanently.	This module is applicable. The project will reduce the level of harvested wood permanently as the project aims to conserve the forest. this project would then affect to the wood production in the country.
VMD0013 – E-BPB	This module is applicable to REDD project activities with emission from biomass burning	This module is applicable as the one of the project activities is REDD project. Although, the burning woody biomass is not considered in the baseline and project scenario but forest fire may occur during project implementation
VMD0015 – Methods for Monitoring Greenhouse Gas Emission and Removals – M-REDD	The module is mandatory for REDD, CIW REDD, RWE REDD and stand-alone CIW project activities.	Condition satisfied. The project is avoided planned deforestation which is a REDD project activity.

Reference ID/Title	Applicability condition	Justification of conformance
VMD0016 – X-STR	This module is fully mandatory for REDD project activities based on VM0007 MF V1.8	Applicable. The project uses a single forest stratum that represents the areas where pre-deforestation activities in the baseline scenario will occur.
VMD0017 – X-UNC	This module is applicable for estimating the uncertainty of estimates of emissions and removal of CO₂-e generated from REDD and WRC project activities. The module focuses on the following sources of uncertainty: • Determination of rates of deforestation and degradation • Uncertainty associated with estimation of stocks in carbon pools and changes in carbon stocks • Uncertainty associated with estimation of peat emissions • Uncertainty in assessment of project emissions.	Condition satisfied. The project is REDD Project.

Reference ID/Title	Applicability condition	Justification of conformance
VT0001 - T-ADD	- AFOLU activities the same or similar to the proposed project activity on the land within the proposed project boundary performed with or without being registered as the VCS AFOLU project shall not lead to violation of any applicable law even if the law is not enforced - Project proponent(s) proposing new baseline methodologies shall ensure consistency between determination of a baseline scenario and the determination of additionality of a project activity.	Condition satisfied. The project activities will not lead to a violation of any applicable law because the project boundary is legally owned and managed by PT CMI. Moreover, there is no potential to violate regulations when carrying out the carbon project activities within the project area because forest protection and reforestation in production areas can be implemented at the discretion of the forest concession holder. This project does not propose a new baseline methodology.
A/R Methodological Tool – Estimation of Direct Nitrous Oxide Emission from Nitrogen Fertilization	This tool allows for estimating direct nitrous oxide emission from applying nitrogenous fertilizer within the project boundary, for both ex-ante and ex post estimation. This tool applicable when: - The project activities are implemented on non-wetlands - The project activities are implemented not in irrigated or floodplain areas	Conditions satisfied. The project area is non-wetland area.

Reference ID/Title	Applicability condition	Justification of conformance
A/R CDM Project Activities - Tool for testing significance of GHG emissions in A/R CDM project activities	This tool shall be used for project activity: - To determine which decreases in carbon pools, and increases in emissions of the greenhouse gases measured in CO₂e that results from the implementation of the project activity, are insignificant and can be neglected - To ensure that is valid to neglect decreases in carbon pools and increases in GHG emissions by sources stated as being insignificant in the applicability conditions of the methodology	This tool was used to justify the omission of carbon pools and GHG sources within the project boundaries as well as to justify the omission of GHG emissions resulting from leakage mitigation activities.
VCS AFOLU Non-Permanence Risk Tool v4.2		This tool is fully mandatory for the given project activity and must be used to determine the number of buffer credits that shall be deposited into the AFOLU pooled buffer account

3.1.3 Project Boundary (VCS, 3.12)

3.1.3.1 Project Boundary Determination

To determine the eligible project area for REDD+ and ARR project activities, a geospatial analysis considering the application VM0007, VM0047, and VMD0006 was conducted. The Land Use and Land Cover (LULC) over a 10-year historical reference period (2013-2023) was analyzed based on SPOT 6 satellite imagery datasets. Spot 6 combined multi-temporal satellites imagery which was derived from several acquisition dates to minimize cloud cover. The spatial resolution and acquisition dates are provided in Table 23.

Table 23. Satellite imagery used for LULC analysis

Year	Spatial Resolution	Acquisition Date	Source
2013	1.5 Meters Panchromatic + 6 Meters Multispectral	May 22	Airbus, DS
		June 17	
		July 25	
2022		May 11	
July 15			
2023		March 14	
		April 9	
		April 14	

LULC analysis, conducted for 2013 and 2023, preliminary consisted of nine land use and land cover classes³⁷: Medium Density Forest, High Density Forest, Built-up, Shrubs, Agriculture, and Water Bodies. This analysis was conducted using a supervised object-based image segmentation approach and a random forest algorithm. It captures the dynamic changes between the 10-year period of historical conditions in t-10 (2013) and t0 (2023), incorporating water bodies and built-up areas.

The forest classes were further analysed using T-test to determine significant differences between the means of two samples AGB from medium-density and high-density classes collected from forest inventory data. The figure below describes the T-test result between medium-density and high-density classes.

	<i>Variable 1</i>	<i>Variable 2</i>
Mean	314.6728	292.181636
Variance	6373.992	12854.69923
Observations	25	41
Hypothesized Mean Difference	0	
df	63	
t Stat	0.943304	
P(T<=t) one-tail	0.174564	
t Critical one-tail	1.669402	
P(T<=t) two-tail	0.349129	
t Critical two-tail	1.998341	

Figure 9. T-test results for high-density (Variable 1) and medium-density (variable 2) classes

The T-test result indicates no statistically significant difference between the means of Variable 1 (high density forest) and Variable 2 (medium density forest). Both the one-tailed and two-tailed tests result in p-values greater than 0.05, suggesting that any observed difference in means could be due to random chance rather than a true underlying difference. Given that there is no significant difference between the two variables, both strata can be combined into a single stratum, classified as forest. The detailed results of the t-test can be seen in Annex 20.

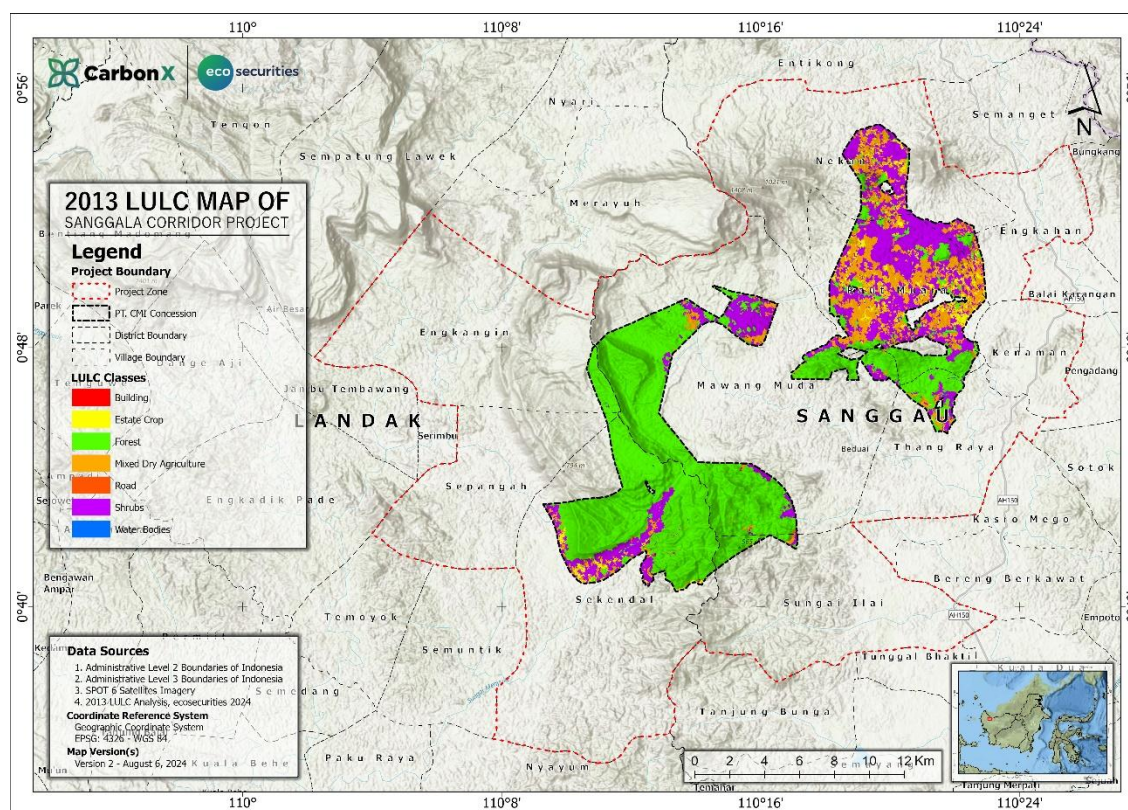
³⁷ Badan Standarisasi Nasional 2014. Klasifikasi penutup lahan - Bagian 1: Skala kecil dan menengah. [SNI 7645-1:2014](#). Jakarta

The 2013 LULC analysis showed that forest (11,979.91 ha) is the dominant LULC class within the project area, closely followed by shrubs (5,062.45 ha), mixed dry agriculture (2,696.68 ha), and estate crop (825.25 ha). Meanwhile, built-up areas, which are roads and buildings, take up only 28.17 ha., while water bodies cover 13.06 ha within the project area. The results of 2013 LULC analysis in detail can be seen in Table 24 and Map 15.

Table 24. The contemporary result of 2013 LULC classification

2013 LULC Class	Area (ha)
Built-up	28.17
Estate Crop	825.25
Forest	11,979.91
Mixed Dry Agriculture	2,696.68
Shrubs	5,062.45
Water Bodies	13.06
Grand Total	20,605.53

Source: SPOT 6 Satellite Imagery and LULC analysis



Map 15. The 2013 LULC map within project area

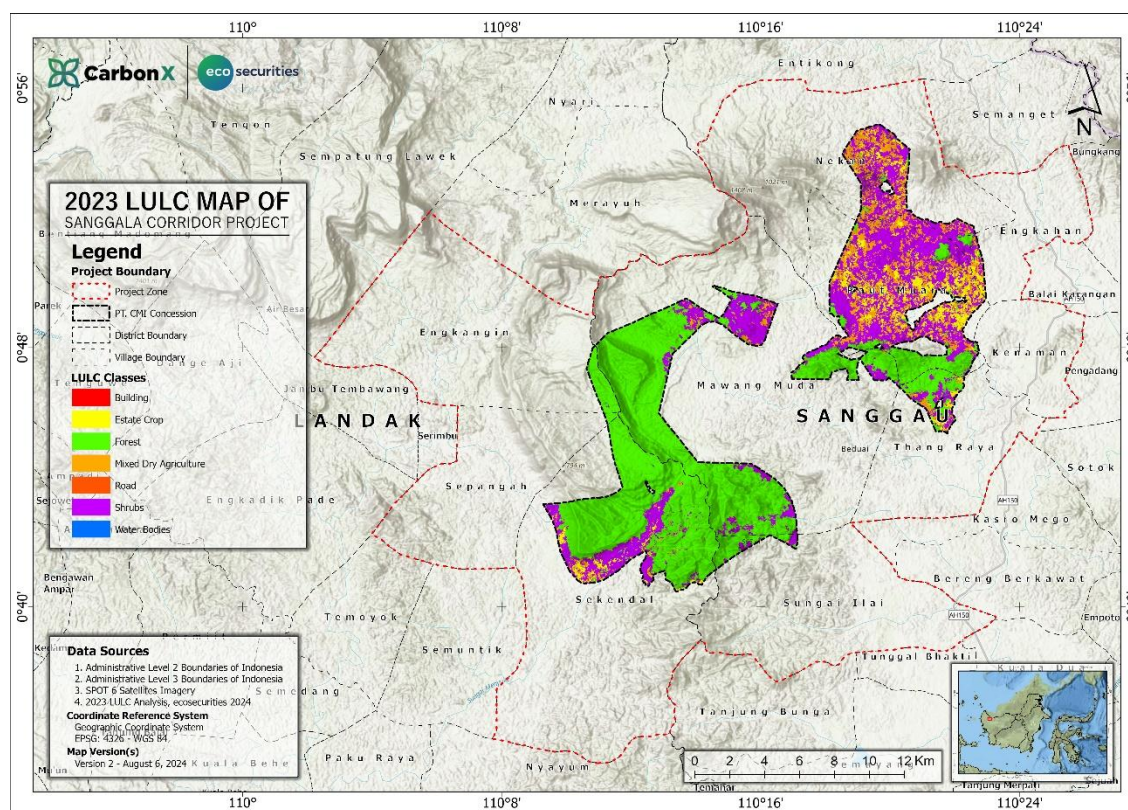
The calculated change of LULC within the 10-year period is shown in Table 25 and Map 16, detailing the 2023 results. Deforestation occurrence is detected from the conversion of a total 637.05 ha (5.32%) forest into shrubs. Otherwise, the portion of 1,479.75 ha (24.72%) shrubs in

2023 emerged from areas previously classified as mixed dry agriculture in 2013, which indicates vegetation regrowth in the project area. Furthermore, the 2023 LULC analysis also shows that there is an expansion of cultivated land, with the transition of shrubs in 2013 to mixed-dry agriculture (966.23 ha) and estate crop (218.90 ha) in 2023.

Table 25. The contemporary result of 2023 LULC classification

2023 LULC Class	Area (ha)
Built-up	38.39
Estate Crop	1,342.56
Forest	11,159.70
Mixed Dry Agriculture	2,065.85
Shrubs	5,986.47
Water Bodies	12.56
Grand Total	20,605.53

Source: SPOT 6 Satellite Imagery and LULC analysis



Map 16. The 2023 LULC map within the project area

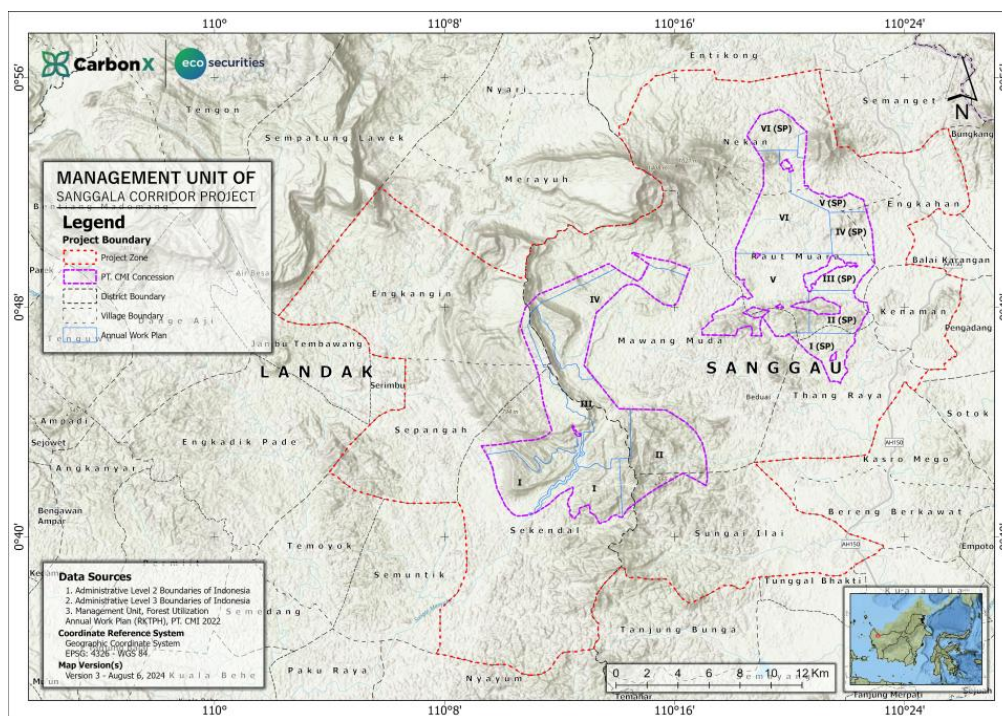
Using the contemporary 2023 LULC condition, an overlay analysis was conducted to identify the ten-year historical changes over the decade between 2013 and 2023 datasets. The area in hectares calculated from each LULC change is summarized in Table 26 below. The numbers highlighted in green in the table represent the area in hectares of stable classes.

Table 26. The result of LULC change analysis of 2013-2023

		2023						
	LULC	Built-up	Estate Crop	Forest	Mixed Dry Agriculture	Shrubs	Water Bodies	Grand Total
2013	Built-up	28.12	0.02			0.03		28.17
	Estate Crop		825.13	0.02	0.03	0.06		825.25
	Forest	2.55	33.50	11,159.62	147.18	637.05	0.01	11,979.91
	Mixed Dry Agriculture	0.02	264.76	0.02	952.13	1,479.75		2,696.68
	Shrubs	7.69	218.90	0.05	966.23	3,869.59		5,062.45
	Water Bodies		0.24		0.27		12.56	13.06
	Grand Total	38.39	1,342.56	11,159.70	2,065.85	5,986.47	12.56	20,605.53

Source: SPOT 6 Satellite Imagery and LULC analysis

The result of the LULC change overlay analysis is used to stratify the project area, incorporating various ancillary data sources³⁸ (i.e., NDVI³⁹, DEM, Landsat 8 images). The PT CMI annual workplan used to calculate the deforestation rate and slope percentage for determining ARR eligible area, is described in the succeeding sections.



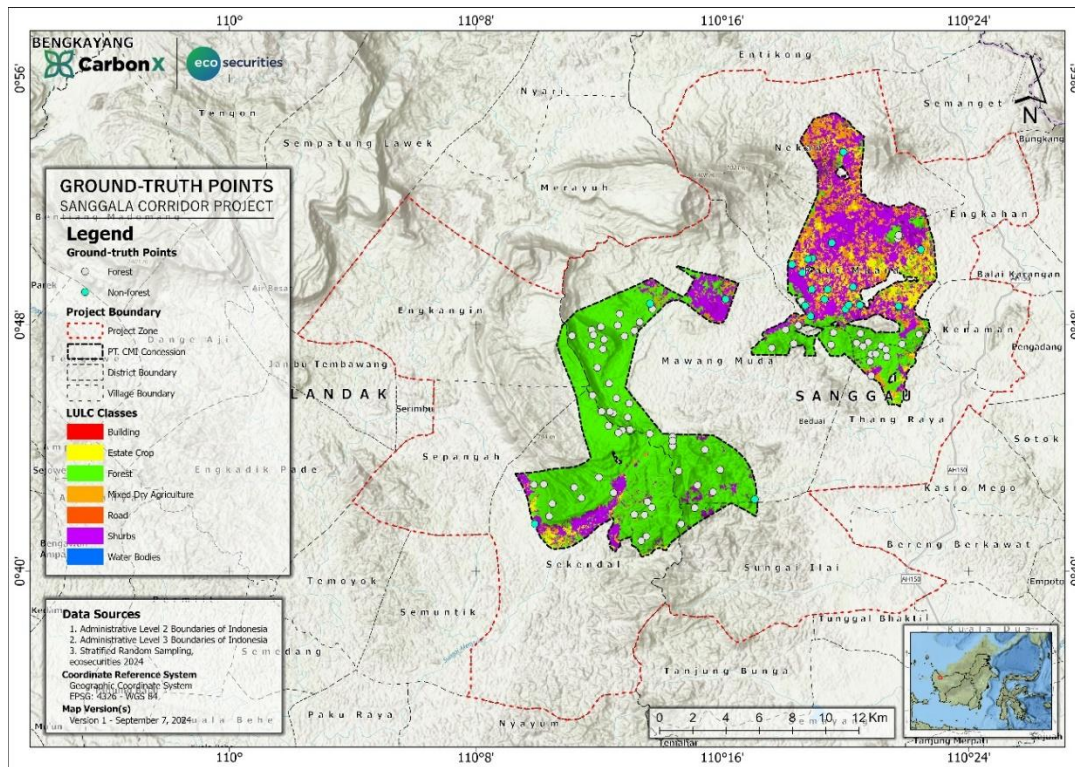
Map 17. PT CMI management plan map

³⁸ VMD0016 Methods for stratification of the project area (X-STR), V1.3

³⁹ A Zaitunah et al. (2018) IOP Conf. Ser.: Earth Environ. Sci. **126** 012112

Following module VMD0015 (M-REDD) from the VCS methodology VM0007, the forest/non-forest stratification map must achieve at least 90% of overall accuracy⁴⁰. This accuracy was assessed by comparing 94 points of ground-truth data (Map 18) from forest inventory result with the 2023 LULC classification, applying the equation below:

$$\text{Overall Accuracy} = \frac{\text{Number of Points Classified Correctly}}{\text{Total Number of Classified Points}}$$



Map 18. Ground-truth points for accuracy assessment

The calculation was satisfactory and indicates that the map reaches 97.9% of overall accuracy. Detailed confusion matrix, which includes user and producer accuracy and showed percentage of correctly classified points for each stratum, can be seen in Table 27.

Table 27. Accuracy Assessment Confusion Matrix

		Ground-truth		Total	User Accuracy (%)	Overall Accuracy (%)
Classes		Forest	Non-forest			
LULC Classification	Forest	72	1	73	98.63	
	Non-forest	1	20	21	95.23	

⁴⁰ VMD0015 Methods for monitoring of GHG emissions and removals in REDD and CIW projects (M-REDD), V2.3

		Ground-truth		Total	User Accuracy (%)	Overall Accuracy (%)
	Classes	Forest	Non-forest			
Total (%)		73	21	94		
Producer Accuracy (%)		98.63	95.23			
Overall Accuracy (%)						97.9

3.1.3.2 Project Boundary for APD

Geographical Boundaries

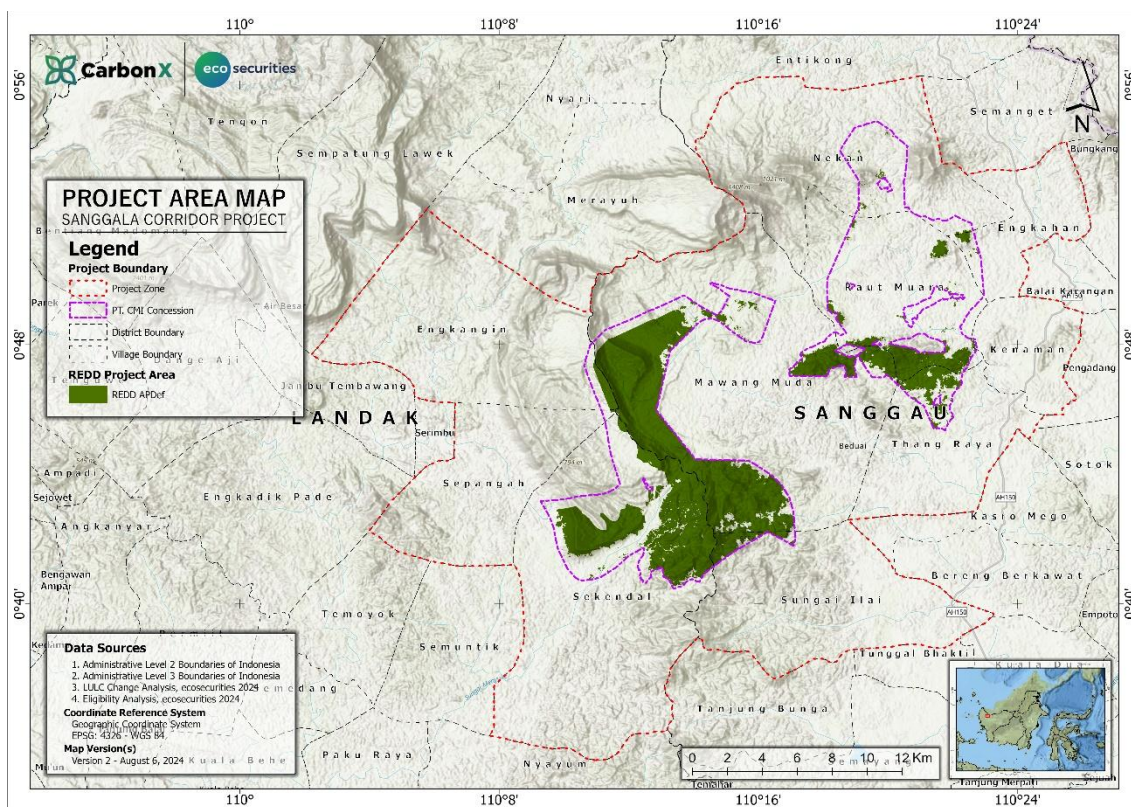
The project area is located in the Sanggau and Landak District of West Kalimantan Province, Indonesia, approximately 238 kilometers from the provincial capital, Pontianak. The boundaries are two separated blocks located at coordinates 00° 49' 53.3" N, 110° 20' 20.08" E and 00° 44' 23.95" N, 110° 13' 8.31" E.

The APD project area covers 9,309.46 ha within the PT. CMI concession boundaries, with a total of 20,605.53 ha based on the entire coverage of PT. CMI business work plan boundary. The eligible APD project area is forest remaining forest for 10 years. Village boundaries are included in order to identify the villages affected by the project as part of the project zone. It is important to highlight that all these villages have their administrative areas overlapping with the project area.

Table 28. APD Eligibility Area Calculation

2013 LULC	2023 LULC	Annual Block	Eligibility	Area (Ha)
Forest	Forest	2023	Eligible	2,047.92
Forest	Forest	2024	Eligible	2,496.01
Forest	Forest	2025	Eligible	2,030.38
Forest	Forest	2026	Eligible	1,737.49
Forest	Forest	2027	Eligible	960.59
Forest	Forest	2028	Eligible	37.06
Forest	Forest	Protected Areas	Not Eligible	1,848.26
Total Eligible				9,309.46
Total Not Eligible				1,850.16
Grand Total				11,159.62

From the geospatial analysis, project boundaries are illustrated below.



Map 19. The project area map of REDD APD

Table 29. The boundary details of PT CMI

Boundary Details	Concession 1	Concession 2
Name	Block – 1	Block - 2
Parcel ID	B-1	B-2
Landowner	PT. Citra Mulia Inti	PT. Citra Mulia Inti
Permit No.	- Ministry of Environment and Forestry Decree No. 2/1/UPHHK-HTI/PMDN/2015 - Ministry of Environment and Forestry Decree 1039/Menlhk/Setjen/HPL.0/11 /2021	- Ministry of Environment and Forestry Decree No. 2/1/UPHHK-HTI/PMDN/2015 - Ministry of Environment and Forestry Decree 1039/Menlhk/Setjen/HPL.0/11 /2021
Legal Classification	Industrial Timber Plantation	Industrial Timber Plantation
Geographic Coordinates	Latitude: 00° 49' 53.3" N Longitude: 110° 20' 20.08" E	Latitude: 00° 44' 23.95" N Longitude: 110° 13' 8.31" E

Boundary Details	Concession 1	Concession 2
Total Land Area	9,094.29 Ha	11,511.23 Ha
REDD APD Area	1,879.81 Ha	7,429.64 Ha

Temporal Boundaries

The project uses the temporal boundaries described in Table 30 below.

Table 30. The temporal boundaries of the project

Temporal Boundary	Start Date	End Date
Historical Reference Period	22 May 2013	14 April 2023
Project Crediting Period	16 March 2023	15 March 2073

Carbon Pool

Table 31 below illustrates the relevant carbon pools included in this project according to the VM0007 v1.8.

Table 31. Carbon Pools for APD

Carbon Pools	Included?	Justification/Explanation
Aboveground tree biomass	Included	Mandatory inclusion for the REDD activity following the requirement in VM0007 v1.8.
Aboveground non-tree biomass	Excluded	Excluded as conservative approach applied.
Belowground tree biomass	Included	Included in relation to the aboveground biomass to completely represent the whole tree (aboveground and belowground) biomass
Belowground carbon stocks in non-tree vegetation	Excluded	Excluded as conservative approach applied.
Deadwood	Excluded	Excluded as conservative approach applied.
Litter	Excluded	Excluded as conservative approach applied.
Harvested Wood Product	Excluded	Conservative to exclude and this pool is deemed insignificant analyzed by T-SIG (see Annex 16).
Soil organic carbon	Excluded	Excluded as conservative approach applied.

Sources of GHG Emissions

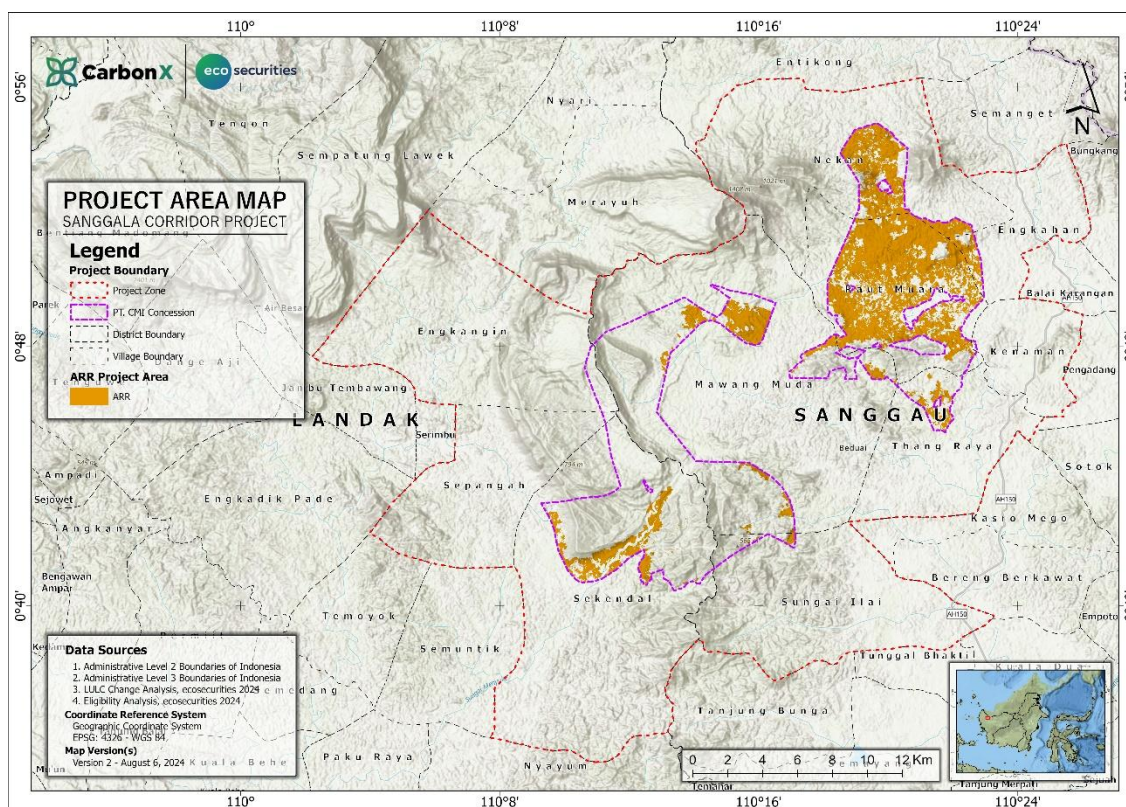
There are no sources of GHG emission sources for APD as justified in Table 32 below.

Table 32. GHG Emissions Sources for APD

Source		Gas	Included?	Justification/explanation
Baseline	Burning of woody biomass	CO ₂	Excluded	The use of fire to burn woody biomass is not considered in the baseline scenario because the Indonesian government prohibits deforestation activities using fire to burn trees. It is regulated in the Forestry Law no. 41 of 1999.
		CH ₄	Excluded	
		N ₂ O	Excluded	
	Combustion of fossil fuels	CO ₂	Excluded	It is conservative to exclude.
		CH ₄	Excluded	Potential emissions are negligible
		N ₂ O	Excluded	Potential emissions are negligible
	Use of fertilizer	CO ₂	Excluded	Based on the management plan, the use of fertilizer is needed to develop the industrial timber plantation. However, the N ₂ O emission is excluded because it is insignificant based on the T-SIG calculation. Hence, this GHG emission source is excluded.
		CH ₄	Excluded	
		N ₂ O	Excluded	
Project	Burning of woody biomass	CO ₂	Included	Burning woody biomass is not considered in the project scenario and is therefore excluded as an emission source, except in the case of forest fire occurrence
		CH ₄	Included	
		N ₂ O	Included	
	Combustion of fossil fuels	CO ₂	Excluded	Neglected because this emission source is excluded from baseline accounting.
		CH ₄	Excluded	Potential emissions are negligible
		N ₂ O	Excluded	Potential emissions are negligible
	Use of fertilizer	CO ₂	Excluded	Potential emissions are negligible
		CH ₄	Excluded	Potential emissions are negligible
		N ₂ O	Excluded	Emissions from the use of fertilizers are conservatively excluded in the project scenario because the protection of the standing forest does not need fertilizer inputs.

3.1.3.3 Project Boundary for ARR

The ARR activities for this project use an area-based approach to quantify emissions removal. LULC analysis was conducted to determine the project area for ARR by following the requirements of VM0047 v1.0. This analytical process estimated a total ARR project area of 6,976.22 Ha. The locations of the project area and management activities for ARR is depicted in Map 20 below.



Map 20. The project area map of ARR

Selected carbon pools included within the project boundary for both the area-based baseline and project scenarios are listed in Table 33 below.

Table 33. Selected Carbon Pools in the Project Boundary Using the Area-Based Approach

Carbon Pools	Included?	Justification/Explanation
Aboveground woody biomass	Yes	Major carbon pool.
Aboveground non-woody biomass	Yes	Included. The ex-ante calculation of aboveground non-woody biomass showed insignificant results based on the relative contribution estimation (see Annex 16). Despite these insignificant findings, this carbon pool is likely to increase with the implementation of project activity.
Belowground woody biomass	Yes	Major carbon pool.
Belowground non-woody biomass	Yes	Included. The ex-ante calculation of belowground non-woody biomass showed insignificant results based on the relative contribution estimation (see Annex 16). Despite these insignificant findings, this carbon pool is likely to increase with the implementation of project activity.
Dead wood	Yes	Included as carbon stock in this pool may increase due to implementation of the project activity

Carbon Pools	Included?	Justification/Explanation
Litter	Yes	Included. The Ex-Ante estimation for litter showed insignificant results based on the relative contribution estimation (see Annex 16). However, this carbon pool will be monitored in the implementation of project activity and estimated in ex-post calculation.
Harvested Wood Product	No	Excluded. The project activities are planting and conservation focused. The planted forest will not be managed for timber products. Conservative to exclude.
Soil organic carbon	No	Excluded as conservative approach applied. The project activity also does not involve soil inversion to a depth exceeding 25 cm or site preparation occurs more than once during the project crediting period

Selected GHG sources included in the project boundary in the area-based baseline and project scenarios according to VM0047 v1.0 are detailed in Table 34 below.

Table 34. GHG Sources Included in or Excluded from the Project Boundary

Source		Gas	Included?	Justification/explanation
Baseline	Burning of biomass (Either by natural or anthropogenic causes)	CO ₂	No	Conservative to exclude
		CH ₄	No	Conservative to exclude
		N ₂ O	No	Conservative to exclude
	Emissions from nitrogen fertilizer	CO ₂	No	Conservative to exclude
		CH ₄	No	Conservative to exclude
		N ₂ O	No	Conservative to exclude
	Burning of fossil fuels	CO ₂	No	Conservative to exclude
		CH ₄	No	Conservative to exclude
		N ₂ O	No	Conservative to exclude
Project	Burning of biomass (Either by natural or anthropogenic causes)	CO ₂	No	Carbon stock decreases due to the fact that burning is accounted as a carbon stock change
		CH ₄	Yes	May be a significant source.
		N ₂ O	Yes	May be a significant source.
	Emissions from nitrogen fertilizer	CO ₂	No	Conservative to exclude
		CH ₄	No	Conservative to exclude
		N ₂ O	Yes	The project uses synthetic fertilizers. Emissions from nitrogen fertilizer will be monitored. Initially, these emissions are considered as de minimis
		CO ₂	No	De minimis

Source		Gas	Included?	Justification/explanation
	<i>Burning of fossil fuels</i>	<i>CH₄</i>	No	De minimis
		<i>N₂O</i>	No	De minimis

3.1.4 Baseline Scenario (VCS, 3.13)

3.1.4.1 Baseline Scenario for APD

Determination of the most plausible baseline scenario builds on the outcome of the additionality analysis and must be consistent with the description of the conditions prior to the project start date. We use the stepwise approach from VM0007 section 6 to determine the most plausible baseline scenario. Following is the description of the stepwise approach:

Step 1: Reuse the plausible alternative land scenarios to the REDD project activity that have been listed as an outcome of sub-step 1b of the additionality tool VT0001.

The plausible alternative land use scenarios listed as the outcome of sub-step 1b are the following:

1. Continuation of pre-project land use scenario - Industrial Timber Plantation *Anthocephalus cadamba*.
2. Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project: Protected Forest without being registered as the VCS AFOLU.
3. Community forest/Social Forest.

Step 2a: Where the VT0001 barrier analysis is used to demonstrate additionality, apply the decision tree in Figure below to the list of all alternative land use scenarios from Step 1 that are not prevented by any barrier.

Since barrier analysis is used to demonstrate additionality, the decision tree in Figure below is applied to the result of step 1.

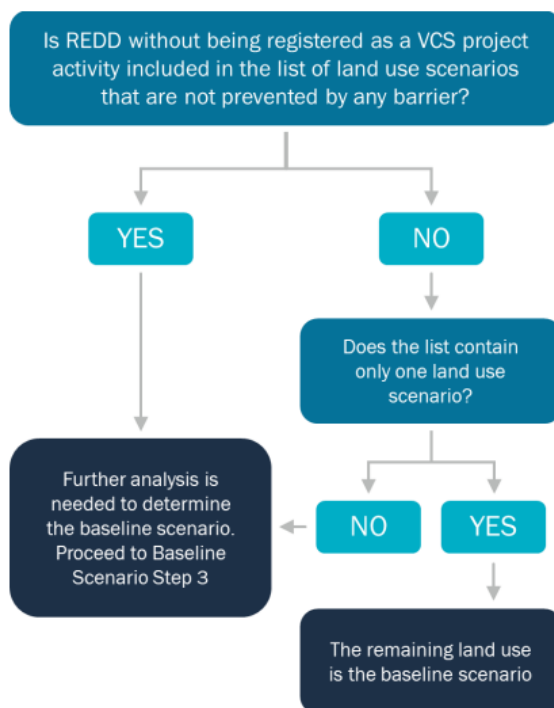


Figure 10. Barrier analysis decision tree

1. Is REDD without being registered as a VCS project activity included in the list of land use scenarios that are not prevented by any barriers?

Alternative land use scenario of project activity on the land within the project boundary performed without being registered as the VCS AFOLU project: Protected Forest without being registered as the VCS AFOLU project is included in the list that is prevented by investment and institutional barriers (Table 39).

2. Does the list contain only one land use scenarios?

Alternative land use scenarios (outcome of step 1)	NOT prevented by barrier?
1. Continuation of pre-project land use scenario - Industrial Timber Plantation Anthocephalus cadamba.	YES
2. Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project: Protected Forest without being registered as the VCS AFOLU.	NO
3. Community forest/Social Forest.	NO

The list contains only one land use scenario that is not prevented by any barriers which is alternative 1. Continuation of pre-project land use scenario: Industrial Timber Plantation *Anthocephalus cadamba*.

3. The remaining land use scenario is the baseline scenario.

Alternative 1 Continuation of pre-project land use scenario: Industrial Timber Plantation Anthocephalus cadamba is the remaining land use scenario that is not prevented by any barriers. Therefore, this alternative is the remaining land use scenario that is considered as the baseline scenario.

3.1.4.2 Baseline scenario for ARR

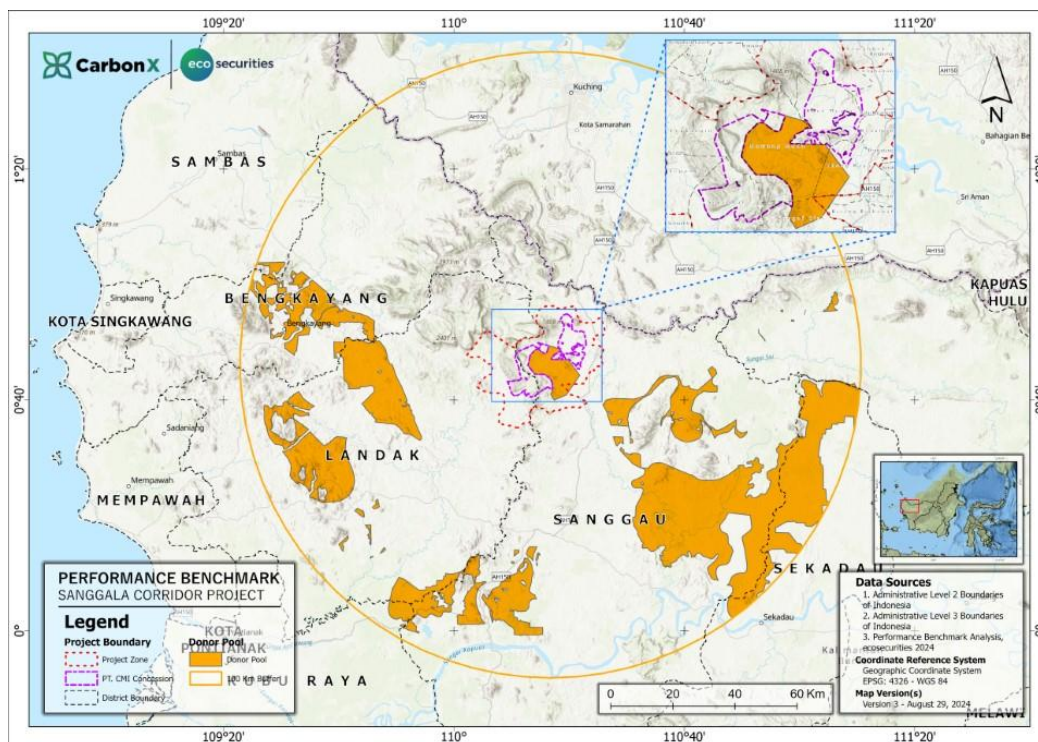
According to the applied VM0047 v1.0, the baseline scenario has been established using the performance benchmark approach outlined in Appendix 1 of the methodology. This baseline scenario reflects the expected business-as-usual growth of new vegetative stocking, as determined through observations conducted outside the project area in designated representative sample plots (control plots) with comparable land use/land cover conditions. The baseline vegetation growth in these control plots is assessed remotely.

Performance Benchmark

The performance benchmark is defined in VCS Program Definition v4.5 as a benchmark against which the performance of individual projects is assessed to determine additionality and/or the crediting baseline. The performance benchmark is set equal to the average cumulative increase in estimated vegetative stocks from their initial state, observed in designated control plots, relative to the project area. At this stage, the project estimated the performance benchmark through ex-ante estimation to assess additionality. An ex-ante estimate of the performance benchmark is calculated by referencing a value of $\Delta SI_{control, t}$ and $\Delta SI_{wp, t}$ using the stocking indices for the historic period $t=-10$, $t=-7$, $t=-3$ and $t=0$ for the selected control plots and project plots. The performance benchmark comparing the baseline scenario to the project scenario provides a clear and quantifiable measure of the project's impact on net GHG emissions.

Calculation of Performance Benchmark

Thirty (30) project plots (60 m x 60 m) for the non-forest area stratum were selected while control plots were determined by conducting a spatial analysis in the donor pool area. The donor pool area was determined by following the requirements of VM0047 v1.0. Donor pool area was used to estimate the control plots with conditions that the control plot is non-overlapping units with a size not exceeding ± 20 percent of the mean size of the project plots. Given that the mean size of the project plots is 0.36023 hectares, the control plots for this project should have an area of 0.288384 hectares. This can be achieved by dividing the donor pool area into a grid with dimensions of 53.68 m x 53.68 m.



Map 21. Donor Pool Analysis Based on 100 km Buffer

The process continued to evaluate project plots and conducted matching analysis between control plots and project plots. The project used stocking index data from the remote sensing metric DVI (Difference Vegetation Index)⁴¹ that derived from the use of Landsat satellite imagery. DVI estimated by using the equation as follows:

$$DVI = NIR - RED$$

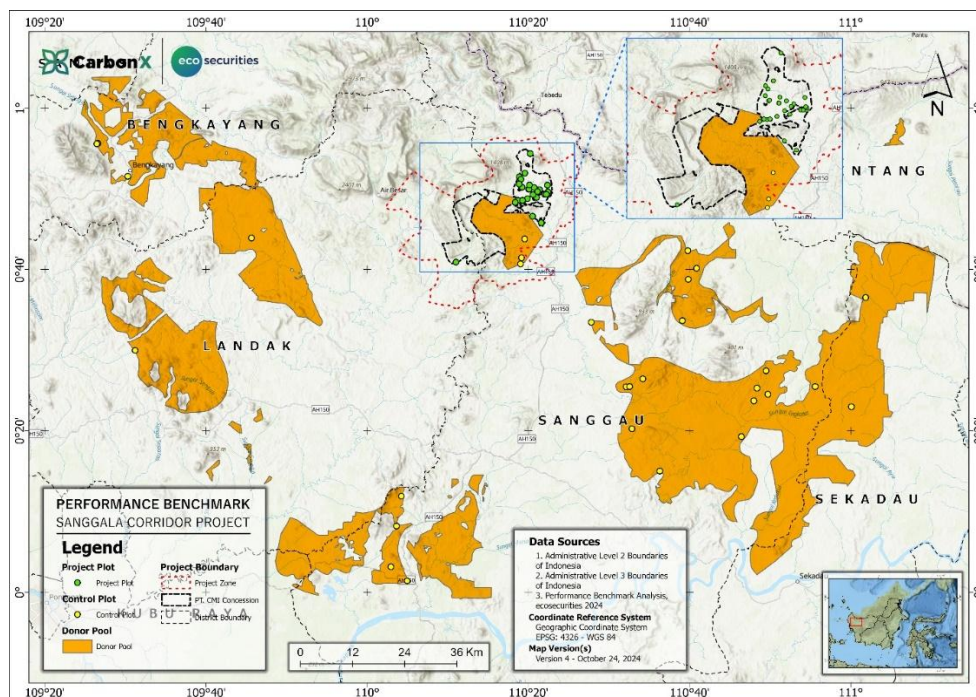
Where:

- DVI = Difference vegetation index
- NIR = Light reflected in the near-infrared spectrum from Landsat satellite imagery
- RED = Light reflected in the red range of the spectrum from Landsat satellite imagery

The stocking index was computed historically at $t=-10$, $t=-7$, $t=-3$ and $t=0$. Moreover, to match control plots and project plots a multivariate Euclidean distance was derived from control plots.

Based on those datasets, matching analysis was conducted using the K-nearest neighborhood approach (see Annex 8) to select the control plots that matched the project plots. The analysis generated 30 control plots that matched with project plots (see Map 22).

⁴¹ Purnamasari, E., Kamal, M., & Wicaksono, P. (2021). Comparison of vegetation indices for estimating above-ground mangrove carbon stocks using PlanetScope image. *Regional Studies in Marine Science*, 44, 101730. doi:10.1016/j.rsma.2021.101730



Map 22. Control plots and project plot locations

For matched control plots and project plots, the quality of the matching was evaluated using the Standardized Difference of Means (SDM). The SDM estimated by using equation A2 in VM0047 v.1.0 appendix 1, as follows:

$$SDM = ABS(\bar{X}_{wp,x} - \bar{X}_{bsl,x}) / \sqrt{\frac{\sigma^2_{wp,x} + \sigma^2_{bsl,x}}{2}}$$

Where:

- SDM = Standardized difference of means
- $\bar{X}_{wp,x}$ = Mean value of covariate x in the population of projects plots
- $\bar{X}_{bsl,x}$ = Mean value of weighted sums of covariate x in the population of matched sets of control plots
- $\sigma^2_{wp,x}$ = Variance of covariate x in the population of project plots
- $\sigma^2_{bsl,x}$ = Variance of covariate x in the population of control plots

Note that the methodology erroneously labels these sigma-squared values the “standard deviation” rather than variance. If standard deviation were to be used, it would also satisfy the necessary condition in all cases that the variance does.

Table 35 below shows the values for each of the four covariates used SI t=-10, SI t=-7, SI t=-3 and SI t=0.

Table 35. Standardized Difference of Means Calculations

	2,013	2,016	2,020	2,023
Xbsl,x	0.22	0.24	0.28	0.29
Xwp,x	0.21	0.24	0.28	0.30
Variance of Project Plots	0.001	0.002	0.002	0.000
Variance of Control Plots	0.002	0.003	0.002	0.000
SDM	0.12	0.01	0.01	0.17
Overall SDM	0.08			

This project generated a mean of SDM value of 0.08 from the matched control plots and project plots. The overall match results are deemed valid because the SDM value for each covariate is less than or equal to 0.25 as set by Step 3 of Appendix 1 VM0047.

For ex-ante estimation, The $\Delta SI_{control}$ and ΔSI_{wp} are estimated based on the selection of control and project plots in historical years. Both values were calculated using a linear regression model that generated the slopes based on the historical stocking index of project and control plots (please see Annex 8 for detailed explanation of estimating the $\Delta SI_{control}$ and ΔSI_{wp}). The process generated a $\Delta SI_{control}$ of 0.00529 and ΔSI_{wp} of 0.00757 over the past 10 years before the project start date.

Table 36. Values of $\Delta SI_{control}$ and ΔSI_{wp} and the Standard Errors Based on Linear Regression

Variables	Values
$\Delta SI_{control}$	0.00529
ΔSI_{wp}	0.00757
SE ΔSI_{wp}	0.0006852
SE $\Delta SI_{control}$	0.00074

The significance of the difference between $\Delta SI_{control}$ and ΔSI_{wp} is evaluated with a Z test as follows:

$$Z = \frac{\Delta SI_{wp,t} - \Delta SI_{control,t}}{\sqrt{SE_{\Delta SI_{wp,t}}^2 - SE_{\Delta SI_{control,t}}^2}}$$

Where:

- Z = Z value (unitless)
- $\Delta SI_{control,t}$ = Average annual increase (slope) in stocking index SI in control plots through time t
- $\Delta SI_{wp,t}$ = Average annual increase (slope) in stocking index SI in project plots through time t
- $SE_{\Delta SI_{control,t}}^2$ = Squared standard error of the average annual increase (slope) in stocking index SI in control plots through time t
- $SE_{\Delta SI_{wp,t}}^2$ = Squared standard error of the average annual increase (slope) in stocking index SI in project plots through time t

$t = 1, 2, 3, \dots, t$ years elapsed since the project start date

Where the parameters $\Delta SI_{control}$ and ΔSI_{wp} are not deemed significantly different ($|Z| < 1.96$), PB_t is set equal to 1.

Where the parameters $\Delta SI_{control}$ and ΔSI_{wp} are deemed significantly different ($|Z| \geq 1.96$), the project will calculate performance benchmark as the ratio of average change in SI in control plots to average change in SI in the project plots. Where the slope coefficient of the control plots, $\Delta SI_{control}$ is insignificant ($P > 0.05$) or less than zero, $\Delta SI_{control}$ is set equal to zero. The performance benchmark estimated following the equation from VM0047 v1.0 as follows:

$$PB_t = \Delta SI_{control,t} \times \frac{1}{\Delta SI_{wp,t}}$$

Where:

PB_t = Performance benchmark for the monitoring interval ending at year t , dimensionless

$\Delta SI_{control,t}$ = Average annual increase (slope) in stocking index SI in control plots through time t

$\Delta SI_{wp,t}$ = Average annual increase (slope) in stocking index SI in project plots through time t

$t = 1, 2, 3, \dots, t$ years elapsed since the project start date

$\Delta SI_{control}$ and ΔSI_{wp} are deemed significant based on the Z-Test result of 2.27. Accordingly, the performance benchmark calculations in the baseline scenario are equivalent to 0.70 based on the calculations using the formula above. Annex 8 will explain in detail how to estimate SDM, $\Delta SI_{control}$, ΔSI_{wp} , and other parameters to calculate the performance benchmark.

Baseline Scenario

The realistic and credible alternative scenario that would have occurred in the absence of the reforestation project activity, under the VCS standard for the current project area, is the business-as-usual changes in above-ground biomass in control plots. The most credible alternative scenario is the continuation of pre-project activities, as this has been the typical practice in the project area. In accordance with VM0047, the project assesses the baseline scenario by considering a historical performance benchmark that represents the business-as-usual change in vegetation cover before the project start date.

Over the 10 years prior to the project start date, the land conditions revealed a performance benchmark of 0.70 with $\Delta SI_{control}$, specifically 0.00529, and SI_{wp} , 0.00757. These values reflect the baseline conditions in both the project area and the control area as a result of the business-as-usual scenario over the past decade in non-forest areas.

Based on these results, the pre-project condition likely persists in the area targeted by the project (non-forest areas) according to performance benchmark results. Therefore, without ARR activity, the project-targeted areas would remain non-forest areas with sparse biomass.

3.1.5 Additionality (VCS, 3.14)

3.1.5.1 Regulatory Surplus (VCS, 3.14)

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☒ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☒ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☐ No

3.1.5.2 [Additionality Methods](#) (VCS, 3.14)

Additionality was assessed using VT0001 – Tool for the Demonstration and Assessment of Additionality in VCS Agriculture, Forestry and Other Land Use (AFOLU) Project Activities, v3.0. VT0001 is used as a tool for demonstrating Additionality for Avoid Planned Deforestation. Meanwhile, for Afforestation, Reforestation and Revegetation applies VM0047 for additionality assessments.

Below is an explanation of the additional assessment results for Avoid Planned Deforestation and Afforestation, Reforestation and Revegetation.

3.1.5.2.1 **Avoid Planned Deforestation**

VM0007 requires VT0001 to demonstrate the additionality for all project activities, with the exception of WRC in tidal wetlands. This additionality test is applicable to the PT CMI concession project areas, where the avoided planned deforestation project activities will be implemented. There are no tidal wetlands within the concession area.

The four steps applied in the additionality test required in VT0001 are as follows:

1. Step 1. Identification of alternative land use scenarios to the AFOLU project activity's
2. Step 2. Investment analysis to determine that the proposed project activity is not the most economically or financially attractive of the identified land use scenarios; or
3. Step 3. Barrier analysis; and
4. Step 4. Common practice analysis.

Step 1. Identification of alternative land use scenarios to the proposed VCS AFOLU project activity

Sub-step 1a. Identified credible alternative land-use scenarios to the proposed project activity.

Identification of realistic and credible land-use scenarios that would have occurred on the land within the proposed project boundary in the absence of the AFOLU project activity under the VCS are described in Table 37 below. These alternative scenarios are feasible for the project area, considering relevant national and/or sectoral policies and circumstances, such as historical land uses, practices and economic trends. Providing the justification, the alternative scenarios are deemed realistic and credible.

Table 37. List of alternative land use scenario and justification

Application of sub-step 1a.	
Alternative Land Use Scenario	Justification
Continuation of pre-project land use scenario - Industrial Timber Plantation <i>Anthocephalus cadamba</i>	The acquisition of PT. CMI by CarbonX occurred on 16 Mar 2023. Prior to this event, there has been series of important event as follows: • 2015, the government officially granted PT CMI the 60 years permit to develop Industry Timber Plantation⁴². • 2018, the General Work Plan (<i>Rencana Kerja Usaha</i> or <i>RKU</i>), detailing the 10 years business plan obtained⁴³. • 2020, the 2019 detail work plan (<i>Rencana Kerja Tahunan</i> 2019) was approved by the government. • Furthermore, in 2019, the investment agreement between PT. CMI and PT Indonesia Wood Investment (PT IWI) officially signed. These events confirmed that the pre-project land use scenario was designated for Industrial timber plantation <i>Anthrocehalus cadamba</i>. Moreover, since the area is classified as production forests, its designated for timber plantation development remains unchanged. If PT CMI does not proceed with the plantation, the government will reallocate the license to another company to ensure the land continues to be used for industrial timber production as originally intended.
Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project:	The central government is the one with the authority to designate forest area status. Forest areas designated as Production Forests can legally be converted into protected areas under government decree. However, this is a rare occurrence because it needs solid political support, that goes from a village up to a national level. The legal process of converting the Production Forest to a Protected Forest is at the discretion and responsibility of the Ministry of Forestry, as the central government has the authority to designate forest area states.
Protected Forest without being registered as the VCS AFOLU project	One example of this rare occurrence is parts of Berbak-Sembilang National Parks, were converted into national parks⁴⁴ managed by Directorate General of Natural Resources and Ecosystem Conservation of Ministry Environment and Forestry (MoEF).

⁴² Decree No. 2/1/IUPHHK-HTI/PMDN/2015

⁴³ Ministry of Forestry Decree No. 4693/Men-LHK-PHPL/UHP/HPL.1/7/2018.

⁴⁴ Silvius, M. J., & Suryadiputra, N. (2005). Review of policies and practices in tropical peat swamp forest management in Indonesia. *Wetlands International*.

Application of sub-step 1a.	
Alternative Land Use Scenario	Justification
Smallholder agriculture	The community members who live both within the project area and project zone mostly work as farmers. They practice mixed dry agriculture and oil palm plantation covering a total hectareage of 3,408.41 Ha (see table 25) inside the project boundary. In the absence of VCS AFOLU project activity, the expansion of small holder agriculture would likely have continued.
Community forest/ Social Forest	Social forestry is a positive trend that is promoted by the Indonesian government to provide opportunities for community who live around forests to be benefited from the forest where they live. Converting Production Forest status into Social Forestry is possible option in some cases. Currently, social forestry permits in West Kalimantan province have reached a total of 378,305 hectares, which makes this province as the one with the highest number of social forestry permits in Indonesia. ⁴⁵ .

Outcome of sub-step 1a: List of credible alternative land use scenarios that could have occurred on the land within the project boundary of the VCS AFOLU project.

Alternative 1: Continuation of pre-project land use scenario - Industrial Timber Plantation *Anthocephalus cadamba*.

Alternative 2: Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project: Protected Forest without being registered as the VCS AFOLU project.

Alternative 3: Smallholder agriculture.

Alternative 4: Community forest/ Social Forest.

Sub-step 1b. Consistency of credible land use scenarios with enforced mandatory applicable laws and regulations

The four (4) major land use scenarios identified under Sub-step 1a are next considered in the context of enforced mandatory laws and regulations in Indonesia. The results of this analysis are shown in Table 38.

⁴⁵ [PPID | Kementerian Lingkungan Hidup dan Kehutanan | Kalimantan Barat Paling Banyak Terima Izin Hutan Sosial \(menlhk.go.id\)](#)

Table 38. Consistency of credible land use with enforced mandatory applicable laws and regulations.

Alternative Land Use Scenario	Consistency analysis
Alternative 1: Continuation of pre-project land use scenario - Industrial Timber Plantation <i>Anthocephalus cadamba</i>	The Government Regulation No. 7/1990 concerning Industrial Plantation Forest Concession Rights (<i>Peraturan Pemerintah No.7/2019 tentang Hak Pengusaha Hutan Tanaman Industri</i>) and Ministry of Environment and Forestry Decree Regulation No. P.62/MENLHK/SETJEN/KUM.1/10/2019 concerning Development of Industrial Timber Plantation (<i>Peraturan Menteri Lingkungan Hidup dan Kehutanan No. P.62/MENLHK/SETJEN/KUM.1/10/2019 tentang Pembangunan Hutan Tanaman Industri</i>) legally allow the development of Industrial Timber Plantation <i>Anthocephalus cadamba</i>. As the project area is designated as production forest. hence this alternative land use scenario is consistent with enforced mandatory applicable laws on “Land Function” in Indonesia.
Alternative 2: Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project: Protected Forest without being registered as the VCS AFOLU project	Land designated as production forest can legally be converted to protected forest or conservation forest. This conversion is permitted under applicable laws and regulations, including the Forestry Law No. 41/1999, No. 19/2004, the Ministry of Forestry Regulation No. 31/2014, and the latest Ministry of Environment and Forestry Regulation No. 7/2021. A PBPH License for ecosystem restoration could potentially be obtained by a non-profit or for-profit entity engaged in forest protection. Hence this alternative land use scenario is consistent with enforced mandatory applicable laws and regulations in Indonesia
Alternative 3: Small holder agriculture	According to the Forestry Law No.41/1999 (<i>Undang-undang No.41/1999 tentang Kehutanan</i>) and No. 19/2004, Job Creation Law No. 11/2020 (<i>Undang-Undang Cipta Kerja No.11/2020</i>), and Ministry of Forestry Decree No. P31/2014, all agricultural activities are not legally permissible in areas designated as Production Forests. Hence this alternative land use scenario is NOT consistent with enforced mandatory applicable laws and regulations in Indonesia.

Alternative Land Use Scenario	Consistency analysis
Alternative 4: Community forest/Social Forest	The Ministry of Environment and Forestry Decree No. 9/2021 concerning the Management of Social Forestry (<i>Peraturan Menteri Lingkungan Hidup dan Kehutanan No. 9/2021 tentang Pengelolaan Perhutanan Sosial</i>) legally allows social forestry in Production Forest land function/status. Hence this alternative land use scenario is consistent with enforced mandatory applicable laws on “Land Function” regulation in Indonesia.

Outcome of sub-step 1b. List of plausible alternative land use scenarios to the VCS AFOLU project activity that are in compliance with mandatory legislation and regulations taking into account their enforcement:

Alternative 1: Continuation of pre-project land use scenario - Industrial Timber Plantation *Anthocephalus cadamba*.

Alternative 2: Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project: Protected Forest without being registered as the VCS AFOLU.

Alternative 4: Community forest/Social Forest.

As a result of the legal analysis, aligned with the prevailing laws and regulations in Indonesia, smallholder agriculture is not considered credible alternative land-use scenarios in the project area as it is not legally permissible.

Sub-step 1c. Selection of the baseline scenario

This step is described in section 3.1.4.

Step 2. Investment analysis

An investment analysis has not been conducted. VT0001 provides the option of either conducting an investment or barrier analysis. If the barrier analysis is satisfied, then the investment analysis is not necessary.

Step 3. Barrier Analysis.

The three remaining major land use scenarios identified in step 1 were then considered in the context of a barrier analysis.

Sub step 3a. Identify barriers that would prevent the implementation of the type of proposed project activity.

The identified barriers for proposed project activity are described in Table 39.

Table 39. Identified barriers for proposed project activity

Alternative Land Use Scenario	Barrier analysis
Alternative 2: Project activity on the land within the project boundary performed without being registered as the VCS AFOLU project: Protected Forest without being registered as the VCS AFOLU project	<u>Investment Barriers:</u> Financing for conservation projects in West Kalimantan exists through international and local NGOs; However, this typically supports only small-scale areas and time-bound conservation projects. The creation of large-scale protected forest within the project area would not have local political support as it would require the local government to use existing budgets to cover the cost of forest protection. The government prioritizes land use that generates tax revenue and local jobs. Currently, the average state budget for environmental protection of forest areas at the provincial level is below 1% of the total state budget ⁴⁶. <u>Institutional barriers.</u> The land is legally designated for production, which provides economic benefit for the state and country. In 2023, production forest area generated export value from forest products amounting \$13,17 billion and increasing \$0.25 billion every year⁴⁷. The forestry sector in Indonesia only contributes 0.66% to Indonesia's GDP and the government aims to continuously increase GDP from the forestry sector⁴⁸. Moreover, Ministry of Environment and Forestry aims to increase the area of Industrial timber plantations from 3,86 million ha to 4.57 million ha in 2045⁴⁹. Due to this, seeking protection status for the forest without carbon financing would be politically unpopular, as the area would not generate tax revenue and economic benefits for the state and local communities.

Sub-step 3b. Show that the identified barriers would not prevent the implementation of at least one of the alternative land use scenarios (except the proposed project activity)

⁴⁶ Orchidea, Meidiza & Mulatsih, Sri & Purnamadewi, Yeti. (2016). Effectiveness of Deconcentration Budget Policy toward Environmental Quality Improvement. Journal of Natural Resources and Environmental Management. 6. 200-210. 10.19081/jpsl.2016.6.2.200.

⁴⁷ <https://phl.menlhk.go.id/publikasi/lapkin>

⁴⁸ [Tingkatkan Kontribusi Hutan pada PDB, Pemerintah Dorong Pemanfaatan Hutan Lestari - Forest Insights Indonesia](#)

⁴⁹ Roadmap Forest Production Development of Indonesia - [Road Map Pembangunan Hutan Produksi Tahun 2019 -2045 - Asosiasi Pengusaha Hutan Indonesia \(rimbawan.com\)](#)

In table 40, the identified barriers for the proposed project activity (investment and institutional barriers) are applied to the remaining alternative land use scenarios

Table 40. Identified barriers applied to the alternative land use scenarios

Alternative Land Use Scenario	Barrier analysis
Alternative 1: Continuation of pre-project land use scenario - Industrial Timber Plantation <i>Anthocephalus cadamba</i>.	<u>Investment Barrier:</u> There are no investment barriers to this land use scenario. Investment agreement with PT Indonesia Wood Investments shows that investment would have been disbursed in the case that project activity does not occur. Full agreement between PT CMI and PT Indonesia Wood Investment can be seen in Annex 12 <u>Institutional Barrier:</u> There are no institutional barriers to this land use. The project area is legally eligible for Industrial Timber Plantation development. Under the Ministry of Forestry Decree No 733/Menhut-II/2014, the area where the project located is designated as Production Forest. In 2015, PT CMI was granted an Industrial Timber Plantation license under the Ministry of Environment and Forestry Decree Number 2/1/IUPHHK-HTI/PMDN/2015.
Alternative 4: Community forest/Social Forest	<u>Investment Barriers:</u> Community forests or social forestry are usually implemented under the assistance from the non-government organization grants, which generally only cover a small project area. In Indonesia, the expansion average area of social forestry is only 676 hectares/year⁵⁰. Currently, only limited funding is available to implement social forestry applied for large areas. <u>Institutional barriers:</u> According to the Ministry of Environment and Forestry under Decree No. 9/2021 concerning the Management of Social Forestry, social forestry permits can only be granted to individuals, farmer groups, and community cooperatives. However, the project area is already granted to PT CMI as a concession holder, which makes it not eligible for community forest or social forest.

In conclusion, the identified barriers (investment and institutional barriers) would not prevent the implementation of alternative 1: Continuation of pre-project land use scenario - Industrial Timber Plantation *Anthocephalus cadamba*. Alternative 2: Project activity on the land within the project

⁵⁰ Pambudi, Andi. (2023). CAPAIAN, TANTANGAN IMPLEMENTASI, DAN REKOMENDASI KEBIJAKAN PERHUTANAN SOSIAL DI INDONESIA. Jurnal Kebijakan Pemerintahan. 6. 74-94. 10.33701/jkp.v6i2.3551.⁵¹ <https://hutanharapan.id/en/hutan-harapan-human-rights-policy-ham/>

boundary performed without being registered as the VCS AFOLU project and alternative 4: Community forest/Social Forest, both face either investment or institutional barriers.

Step 4. Common practice analysis

No similar activities identified to have been already diffused in the geographical area of the proposed VCS AFOLU project activity. Moreover, establishing protected forest on land legally designated for forest production is not common practice in Indonesia. Outside of legally designated protected areas, and without the prospect of revenues from carbon finance, few examples exist. Those that do tend to be small projects backed by stable philanthropic donors.,

In Indonesia, there is only one large-scale project in a production forest area aimed at protecting and restoring the forest. The project, called Hutan Harapan⁵¹, located in the provinces of Jambi and South Sumatra. It is funded by several philanthropic donors to support all its activities. An example of a project aimed at protecting production forest areas is the corridor conservation initiative between Wasur National Park, Bian Wildlife Reserve, and Bupul Nature Reserve, implemented by WWF Indonesia's Papua Program since 2017⁵². This project is also supported by philanthropic donors.

These examples illustrate that forest protection efforts within production forest designated areas in Indonesia are uncommon without philanthropic funding support.

It can be concluded that no similar activity observed to the proposed VCS AFOLU project activity and VCS AFOLU project activity is not the baseline scenario. Hence, it is additional.

3.1.5.2.2 Afforestation, Reforestation and Revegetation

According to VM0047 v1.0 and VCS Methodology requirements v4.4, the project is required to apply the performance method for demonstration of additionality. This method requires the following steps to be applied:

- 1) Step 1. Regulatory surplus
- 2) Step 2. Performance benchmark
- 3) Step 3. Investment barrier

Project must apply step 3 only when there are revenues or financial incentives other than from the sales of carbon credits.

Step 1 Regulatory Surplus

This step refers to section 3.1.5.1. The project areas located in West Kalimantan, Indonesia which included on the list of UNFCCC Non-Annex 1 country⁵³. Any law, statue or other regulatory framework in Indonesia do not mandate ARR Project activities.

The list in table below demonstrates the regulatory surplus. It is the exhaustive list of law, statutes and other regulatory frameworks related to climate change mitigation under the Government of Indonesia regulatory framework that do not mandate the implementation of such project activity.

⁵¹ <https://hutanharapan.id/en/hutan-harapan-human-rights-policy-ham/>

⁵² KORIDOR WABUBI Report. WWF Indonesia Papua Program 2017

⁵³ https://unfccc.int/process/parties-non-party-stakeholders/parties-convention-and-observerstates?field_national_communications_target_id%5B514%5D=514

Taking into account the land function of Industrial Forest and the regulatory surplus demonstrated in table 4-12, it is found that there is no such obligation for the private sector (and also the government agencies) to implement an ARR project.

No.	Regulatory framework	Mandating the implementation of project activity?	
		Yes	No
1	Law No. 6/1994 on Ratification of the United Nations Framework Convention on Climate Change		☒
2	Law No. 41/1999 on Forestry		☒
3	Law No. 25/2004 on the National Development Planning System		☒
4	Law No. 17/2007 on Long-Term Development Plan 2005 – 2025		☒
5	Law No. 32/2009 on Environmental Protection and Management		☒
6	Law No. 16/2016 concerning Ratification of the Paris Agreement to the United Nations Framework Convention on Climate Change		☒
7	Law No. 11/2020 on Job Creation amended by Perpu (Government Regulation in lieu of Law) No. 2/2022 dated 30 December 2022 regarding Job Creation (Job Creation Law)		☒
8	Government Regulation No. 46/2017 concerning Environmental Economic Instruments		☒
9	Government Regulation No. 22/2021 on Environmental Protection and Management		☒
10	Government Regulation No. 23/2021 concerning Forestry Management		☒
11	Presidential Regulation No. 98/2021 on Carbon Pricing		☒
12	Minister of Environment and Forestry Regulation No. P.32/Menlhk/Setjen/Kum.1/3/2016 concerning Forest and Land Fire Control		☒
13	Minister of Environment and Forestry Regulation No. P.33/Menlhk/Setjen/Kum.1/3/2016 concerning Guidelines for Formulating Climate Change Adaptation Actions		☒
14	Minister of Environment and Forestry Regulation No. P.70/Menlhk/Setjen/Kum.1/12/2017 concerning Procedures for Implementing Reducing Emissions from Deforestation and Forest		☒
15	Degradation, Role of Conservation, Sustainable management of forest and enhancement of forest carbon stocks		☒
16	Minister of Environment and Forestry Regulation No. P.71/Menlhk/Setjen/Kum.1/12/2017 concerning Implementation of a national registration system for climate change control		☒

No.	Regulatory framework	Mandating the implementation of project activity?	
		Yes	No
17	Minister of Environment and Forestry Regulation No. P.72/Menlhk/Setjen/Kum.1/12/2017 concerning Guidelines for the Implementation of Measurement, Reporting and Verification of Climate Change Controlling Actions and Resources		☒
18	Minister of Environment and Forestry Regulation No. P.73/Menlhk/Setjen/Kum.1/12/2017 concerning Guidelines for the Implementation and Reporting of the National Greenhouse Gases Inventory		☒
19	Minister of Environment and Forestry Regulation No. P.7/Menlhk/Setjen/Kum.1/2/2018 concerning Guidelines for the Study of Vulnerability, Risk, and Impact of Climate Change		☒
20	Minister of Environment and Forestry Regulation No. P.8/Menlhk/Setjen/Kum.1/2/2018 concerning Technical Criteria for Alert and Emergency Status of Forest and Land Fires		☒
21	Minister of Environment and Forestry Regulation No. P.9/Menlhk/Setjen/Kum.1/2/2018 concerning Fixed Procedures for Field Checking Hotspot Information and/or Forest and Land Fire Information		☒
22	Decree of the Minister of Environment and Forestry No. SK.679/Menlhk/Setjen/Kum.1/12/2015 concerning Monitoring the Implementation of Nationally Determined Contribution (NDC)		☒
23	Minister of Environment and Forestry Regulation No. 8 year 2021 regarding Forest Governance and Preparation of Forest Management and Utilization Plans in Protection Forest and Production Forest		☒
24	Minister of Environment and Forestry Regulation No. P.71/Menlhk/Setjen/Kum.1/12/2017 concerning Implementation of a national registration system for climate change control		☒
25	Minister of Environment and Forestry Regulation No. P.72/Menlhk/Setjen/Kum.1/12/2017 concerning Guidelines for the Implementation of Measurement, Reporting and Verification of Climate Change Controlling Actions and Resources		☒
26	Minister of Environment and Forestry Regulation No. P.8/2021 concerning Tata Hutan dan Penyusunan Rencana Pengelolaan Hutan, serta Pemanfaatan Hutan di Hutan Lindung dan Hutan Produksi		☒

Step 2. Performance Benchmark

A performance benchmark is used to set the crediting baseline. Procedures to define the performance benchmark are provided in Appendix 1 of the VM0047 v1.0, which is explained in section 3.1.4 of the baseline scenario. Based on the analysis, this project is additional due to it generating a Z value of 2.27 and which led to performance benchmark calculation result less than 1 (0.70).

Step 3. Investment Barrier

The project demonstrates through its financial model (available to the auditor at validation) that the project faces capital or investment return constraints that can be overcome by the additional revenues associated with the sale of GHG carbon credits. The ARR project requires significant upfront capital investment in land preparation, planting and maintenance. To demonstrate this, the project was using benchmark analysis while considering another revenue from the sale of agroforestry products. Following the VCS Methodology Requirement V4.4, the financial benchmark should be consistent with the Weighted Average Cost of Capital (WACC) or the cost of equity commonly applicable to the country, sector and type of project.

As the project has potential revenue from agroforestry commodity sales, the financial analysis assesses the investment viability under assumption that the project activities rely solely on agroforestry revenue. The model shows a significantly low Internal Rate of Return (IRR) compared to WACC for the agriculture sector in Indonesia, with prolonged negative cash flow exceeding 15 years. In such scenario, it becomes evident that without the funding mechanisms provided by carbon credits sales, the project's financial viability would be untenable. Based on that, the ARR project is additional as the project does not meet the required financial benchmark without carbon credit revenues. The project financial analysis spreadsheet can be found in Annex 9 and it is only available for validation and verification needs conducted by the Validation and Verification Bodies.

3.1.6 Methodology Deviations (VCS, 3.20)

There are no methodological deviations in this project.

3.2 Quantification of Estimated GHG Emission Reductions and Removals

3.2.1 Baseline Emissions (VCS, 3.15)

3.2.1.1 Avoided Planned Deforestation (APD)

In this section, the annual area of land to be deforested were determined using the criteria set in VMD0006 v1.4.

Part 1. Calculating annual area of land deforested

Identify the agent of planned deforestation in each baseline stratum i

The company PT Citra Mulia Inti (PT CMI) is the only agent of planned deforestation in the baseline scenario. PT CMI is an Indonesian company operating in forest management, and the

industrial timber plantation sector. The company currently holds an Industrial Timber Plantation permit, which grants them the authority to manage and convert the entire area of 20,645 ha of standing forest into an Industrial Timber Plantation.

Area of Deforestation

1. Legal permissibility for deforestation

Under the Ministry of Forestry Decree No 733/Menhut-II/2014, the project area is legally classified as a Production Forest. This designation allows for the development of plantations. According to Ministry of Environment and Forestry (MoEF) regulation No. 8/2021⁵⁴, production forest is designated to produce forest production of forest products such as sawn wood. The State of Indonesia's Forest 2022 document⁵⁵ report further defines Production Forests as areas characterized by gentler slopes, less sensitive soils, and lower precipitation intensity, making them suitable for selective logging or intensive management through plantation forestry. Accordingly, this forest can be legally utilized for developing forest plantations. Based on Permit No 2/1/IUPHHK-HTI/PMDN/2015, all areas managed by PT Citra Mulia Inti fall within a permanent production forest, which are legally approved for conversion into industrial timber plantation.

2. Suitability of project area for conversion to alternative non-forest land use.

PT Citra Mulia Inti has a work plan to develop an Industrial Timber Plantation with, *Anthocephalus cadamba* (A. Cadamba) as the primary species. *A. cadamba* is a resilient and fast-growing tree that thrives in the tropical climates and requires minimal special care. It can tolerate temperatures ranging from 32 to 42°C and grows optimally in areas situated at 300 to 800 meters above the sea level. The project area, located on Kalimantan Island, Indonesia, falls within a tropical and equatorial zone. Additionally, the entire project area is situated in lowland regions, with no land exceeding 1,000 meters in elevation. According to the technical feasibility report prepared by PT Citra Mulia Inti, *A. cadamba* is highly suitable and viable for cultivation within this project area.

3. Government approval for deforestation to occur.

PT Citra Mulia Inti has received legal approval to convert the project area into an Industrial Timber Plantation from the Ministry of Environment and Forestry under Decree No. 2/1/UPHHK-HTI/PMDN/2015. This decree grants PT Citra Multi Inti a business permit for the utilization of timber forest plantations within production forest areas, covering a total 20,645 ha in the Sanggau and Landak Regencies, West Kalimantan Province. The permit authorizes PT Citra Mulia Inti to operate in the project area for a duration of 60 years.

4. Intention to deforest.

PT CMI has a granted a permit for developing Industrial Timber Plantation, demonstrating a clear intent to deforest the project area. This intent is supported by key documents, including the government-approved land use management plans (Rencana Kerja Usaha/RKU) and the annual work plan (Rencana Kerja Tahunan/RKT)⁵⁶.

⁵⁴ [P.8 TAHUN 2021.pdf \(menlhk.go.id\)](https://phl.menlhk.go.id/static/file/publikasi/1664941652-Digital_SoIF0%202022_09.25.22.pdf)

⁵⁵ https://phl.menlhk.go.id/static/file/publikasi/1664941652-Digital_SoIF0%202022_09.25.22.pdf

⁵⁶ Provided in annex 10

The RKU also is sufficient to demonstrate the intent to deforest; however, the following sequence of events and supporting documents further reinforce this intention.

After PT. CMI was granted a government license in 2015, PT CMI undertook preparatory works for an industrial timber. In 2018, PT CMI obtained government approval for its 10-year General Work Plan (Rencana Kerja Usaha or RKU), detailing activities such as land clearing, planting, and seedling preparation. This plan was approved by the government through Ministry of Forestry Decree No. 4693/Men-LHK-PHPL/UHP/HPL.1/7/2018. By 2019, PT CMI conducted forest inventory activities to assess the commercialize standing timber stocks that would result from land clearing activities⁵⁷.

In parallel with these technical preparation, PT CMI secured investment by finalizing a cooperation agreement with PT Indonesia Wood Investment (PT IWI) in 2019, granting PT IWI exclusive right in exchange of an upfront payment and royalties on harvested timber.

Although, the COVID pandemic in 2020 led to an operational pause, delaying the development of the industrial timber plantation. PT CMI resumed operation in 2021, by adjusting the permit to align with the new regulatory framework of Omnibus Law on Job Creation (Law 11 of 2020). This adjustment converted the concession permit into a Forest Utilization Business License (PBPH - Perizinan Berusaha Pemanfaatan Hutan) under the Ministry of Environment and Forestry Decree 1039/Menlhk/Setjen/HPL.0/11/2021.

In 2022, PT CMI continued its efforts by submitting an annual work plan (Rencana Kerja Tahunan or RKT) to the relevant government system, emphasizing the land clearing plan.

This documented sequence of events confirms that prior to the project start date in 16 Mar 2023, PT CMI shows a clear intention to deforest.

Rate of deforestation ($D\%_{planned,i,t}$)

The deforestation rate ($D\%_{planned,i,t}$) for this project is derived from the management plan (RKU) of PT Citra Mulia Inti which was authorized by the Ministry of Environment and Forestry under the Decree No. 4693/MenLHK-PHPL/UHP/HPL.1/7/2018. The total area planned for the Industrial Timber Plantation in the management plan is 9,309.46 Ha. The annual activities are divided into six (6) phases, as outlined in the annual work plan (RKT), covering the period from 2023 to 2028 (see Table 41).

Table 41. Deforestation Area Based on PT Citra Mulia Inti Management Plan

RKT	Deforestation Area		Post-Deforestation	
	Year	AAplanned,i,t (ha)	A. cadamba (ha)	Rubber (ha)
1	2023	2,047.92	1,684.94	362.98
2	2024	2,496.01	2,073.92	422.09
3	2025	2,030.38	2,015.45	14.94
4	2026	1,737.49	1,655.34	82.16
5	2027	960.59	923.45	37.14
6	2028	37.07	30.26	6.81
		9,309.46		

⁵⁷ Laporan Hasil Cruising, Mei 2019

The deforestation rates each year is calculated according to the annual workplan (RKT) and is converted to a deforestation rate for each year in Table 42.

Likelihood of Deforestation ($L - D_i$) and Annual Area of Deforestation ($AA_{planned,i,t}$)

The likelihood of deforestation is assumed to be 100%, as all Industrial Timber Plantations in Indonesia are zoned for deforestation and fall under the legal control of the concessionaire for the duration of the concession license. This assumption of complete deforestation likelihood is applied in calculating the total annual area of planned deforestation.

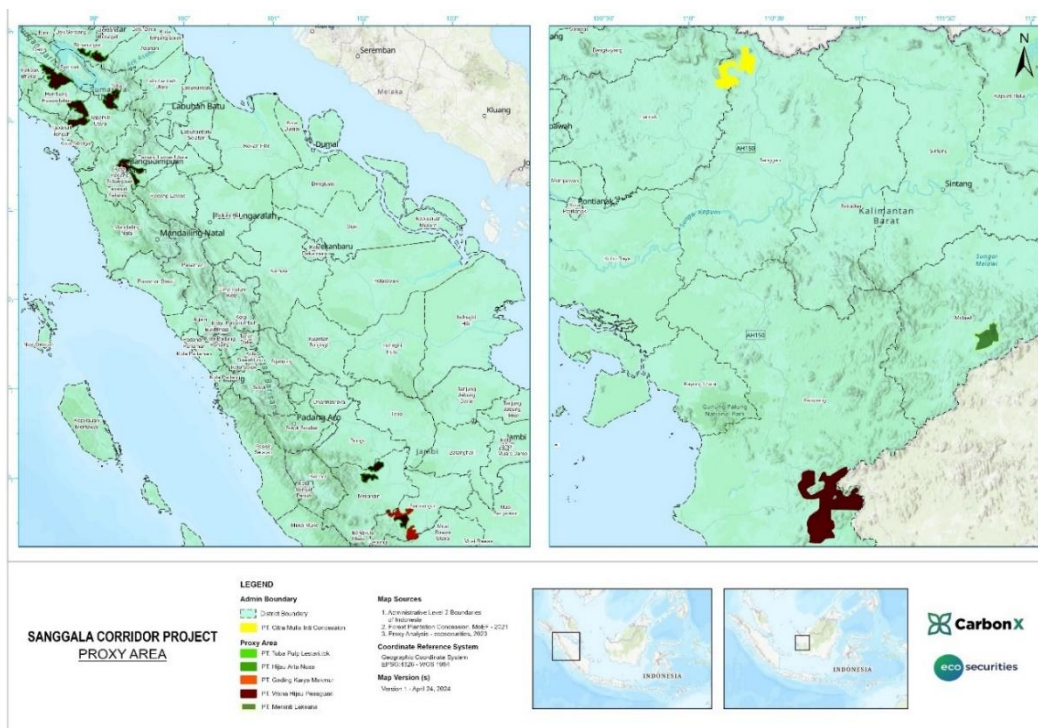
Table 42. Rate Deforestation based on RKT

RKT	Deforestation rate				
	Year	Aplanned,i (ha)	D%	L-Di	AAplanned,i,t (ha)
1	2023		22.00%	100%	2,047.92
2	2024		26.81%	100%	2,496.01
3	2025		21.81%	100%	2,030.38
4	2026		18.66%	100%	1,737.49
5	2027		10.32%	100%	960.59
6	2028		0.40%	100%	37.07
		9,309.46	100%		9,309.46

Risk of Abandonment

Five (5) areas were identified from a similar class of deforestation agent to PT CMI, i.e., other forest concession holders that develop industrial timber plantations. These proxy areas have similar conditions, i.e., forest types, soil type, and elevation, with the project area. The proxy areas were spatially analyzed based on the geographic political boundaries at the province level, i.e., West Kalimantan, Jambi and North Sumatera (see Map 23). These proxy areas are as follows:

1. PT. Meranti Laksana (West Kalimantan Province)
2. PT. Wana Hijau Pesaguan (West Kalimantan Province)
3. PT. Hijau Arta Nusa (Jambi Province)
4. PT. Gading Karya Makmur (Jambi Province)
5. PT. Toba Pulp Lestari (North Sumatera Province)



Map 23. Distribution of 5 Proxy Areas Within in Sumatera and Kalimantan Island

1. Forest Types in Project Area and Proxy Areas

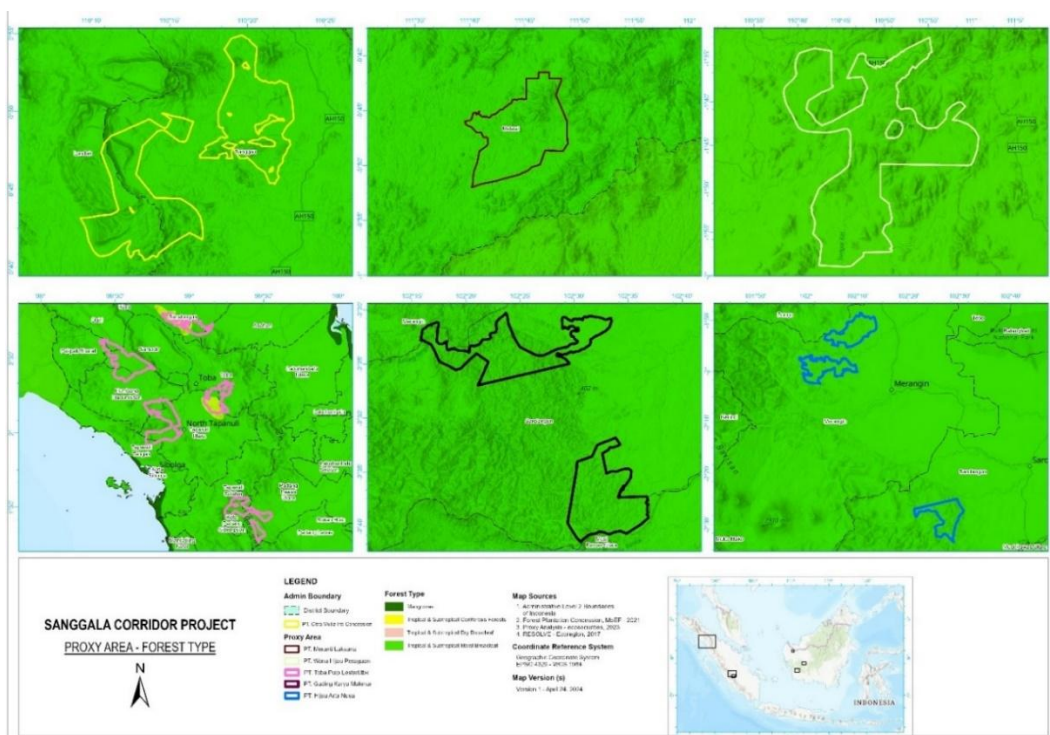
The proxy areas have the same forest type condition and proportions prior to deforestation with the project area ($\pm 20\%$). The forest type was classified based on LULC analysis, and Resolve Ecoregions, 2017.⁵⁸ The details of these similarities in forest type and proportion are shown in Table 43 and Map 24 below.

Table 43. Forest type in the project area and proxy areas

Concession	Area Identifier	Province	Forest Type	Proportion
PT Citra Mulia Inti	Project Area	West Kalimantan	Tropical & Subtropical Moist Broadleaf Forest	100%
PT. Meranti Laksana	Proxy Area 1	West Kalimantan	Tropical & Subtropical Moist Broadleaf Forest	100%
PT. Wana Hijau Pesaguan	Proxy Area 2	West Kalimantan	Tropical & Subtropical Moist Broadleaf Forest	100%
PT. Hijau Arta Nusa	Proxy Area 3	Jambi	Tropical & Subtropical Moist Broadleaf Forest	100%
PT. Gading Karya Makmur	Proxy Area 4	Jambi	Tropical & Subtropical Moist Broadleaf Forest	100%

⁵⁸ Dinerstein, Eric & Olson, David & Joshi, Anup & Vynne, Carly & Burgess, Neil & Wikramanayake, Eric & Hahn, Nathan & Palminteri, Suzanne & Hedao, Prashant & Noss, Reed & Hansen, Matt & Locke, Harvey & Ellis, Erle & Jones, Benjamin & Barber, Charles & Hayes, Randy & Kormos, Cyril & Martin, Vance & Crist, Eileen & Saleem, Muhammad. (2017). An Ecoregion-Based Approach to Protecting Half the Terrestrial Realm. BioScience. 1. 1-12. 10.1093/biosci/bix014.

Concession	Area Identifier	Province	Forest Type	Proportion
PT. Toba Pulp Lestari	Proxy Area 5	North Sumatera	Tropical & Subtropical Moist Broadleaf Forest	87%
			Tropical & Subtropical Coniferous Forest	13%



Map 24. Forest Types at Project Area and Proxy Areas

2. Types of Soil in the Project Area and Proxy Areas

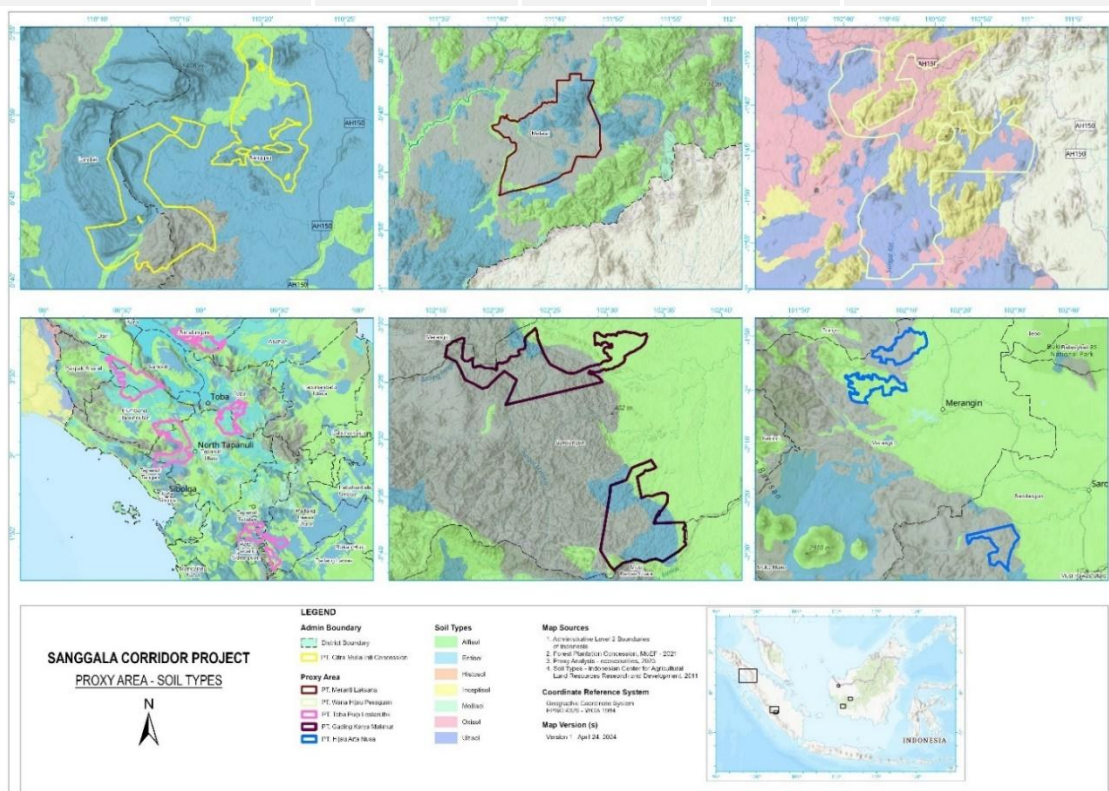
Soil types were identified using data from the Indonesian Center for Agricultural Land Resources Research and Development.⁵⁹ The result of the identification showed that the soil types found within the project area, and the proxy areas are entisol, inceptisol, oxisol and ultisol. The project area is covered in inceptisol, oxisol and ultisol, while the proxy areas are entisol, inceptisol, oxisol and ultisol. Table 44 and Map 25 below provides an overview of the soil classes in a project area and various proxy areas. See annex 11 for the complete proportion.

Table 44. Soil Types in The Project Area and Proxy Areas

Plantation Forest Concession	Area	Soil Types	%	% Proportion From Project Area
PT CITRA MULIA INTI	Project Area	Inceptisol	15.04%	
		Oxisol	19.65%	
		Ultisol	65.32%	
PT MERANTI LAKSANA	Proxy Area 1	Inceptisol	2.98%	2.67%

⁵⁹ <https://sdip-bsip-ppid.pertanian.go.id/>

Plantation Forest Concession	Area	Soil Types	%	% Proportion From Project Area
PT GADING KARYA MAKMUR	Proxy Area 2	Oxisol	62.54%	56.04%
		Ultisol	34.48%	30.90%
		inceptisol	20.42%	27.04%
		oxisol	52.74%	69.83%
		ultisol	26.85%	35.55%
PT HIJAU ARTHA NUSA	Proxy Area 3	inceptisol	36.94%	57.71%
		oxisol	63.06%	98.54%
PT TOBA PULP LESTARI Tbk	Proxy Area 4	entisol	43.70%	356%
		inceptisol	33.38%	272%
		oxisol	9.84%	80%
		ultisol	12.95%	106%
PT WANA HIJAU PESAGUAN	Proxy Area 5	Inceptisol	23.20%	90%
		Oxisol	29.41%	114%
		Ultisol	46.86%	182%



Map 25. Proxy Area Types of Soil

All the soil types are suitable for industrial timber plantation as the land-use practice. These soil types are suitable for planting *Paraserianthes falcataria*, *Anthocephalus cadamba*, *Acacia spp.*, etc. as species commonly used as main plants for industrial plantation forests. Following the requirements of proxy analysis, the proportion of soil types in the proxy areas is aligned with the project area as the proportion has exceeded the required 20% for each soil type.

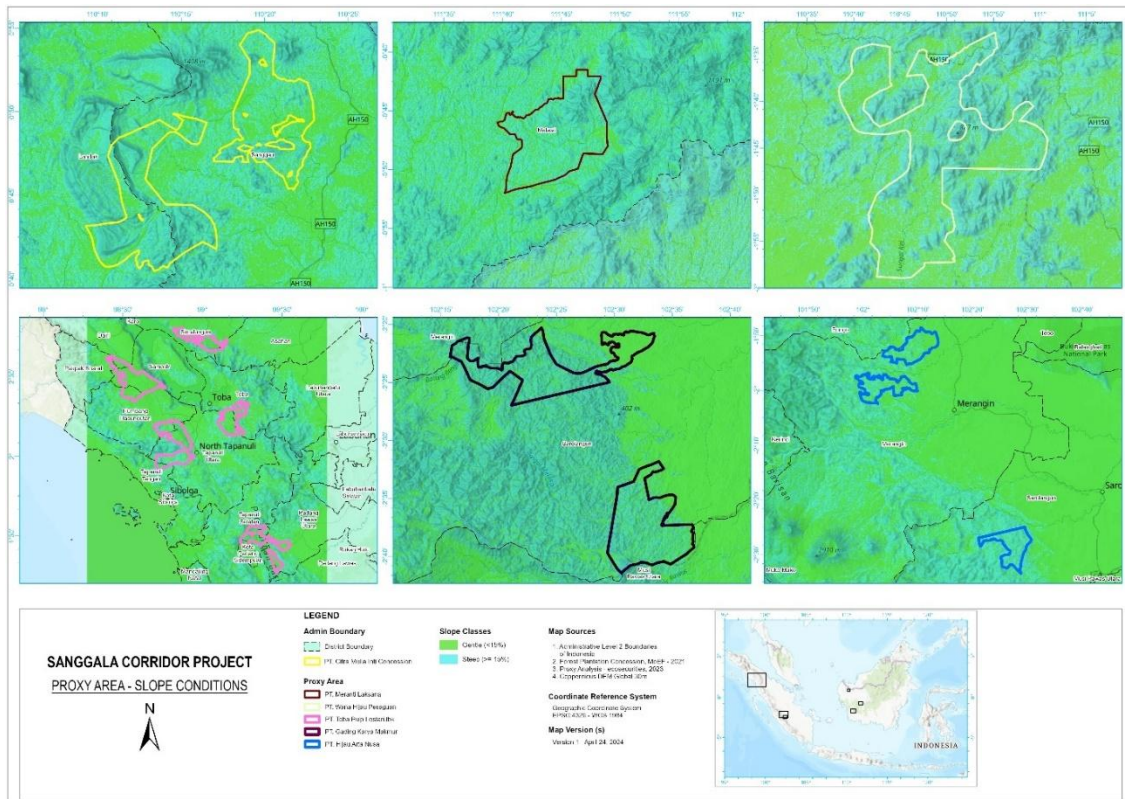
3. Slope Classes in The Project Area and Proxy Areas

According to VMD0006, slope classes are divided into 2 classes, gentle (< 15%) and steep (≥ 15%). Data from Copernicus DEM Global 30m were used to identify slope conditions within the project and proxy areas. Based on the slope analysis, the proportion of slope classes in the project area is 36% gentle and 64% steep. while the proxy area had an average of 50% gentle and 49.99% steep. Table 45 and Map 26 provide the information of slope classes for project area and proxy areas.

Table 45. Slope Classes in The Project Area and Proxy Areas

Plantation Forest Concession	Area	Soil Types	%	% Proportion From Project Area
PT CITRA MULIA INTI	Project Area	Gentle (< 15%)	35.75%	
		Steep (≥ 15%)	64.33%	
PT MERANTI LAKSANA	Proxy Area 1	Gentle (< 15%)	34.38%	30.76%
		Steep (≥ 15%)	65.62%	58.73%
PT GADING KARYA MAKMUR	Proxy Area 2	Gentle (< 15%)	42.94%	56.86%
		Steep (≥ 15%)	57.06%	75.55%
PT HIJAU ARTHA NUSA	Proxy Area 3	Gentle (< 15%)	50.86%	79.47%
		Steep (≥ 15%)	49.14%	76.78%
PT TOBA PULP LESTARI Tbk	Proxy Area 4	Gentle (< 15%)	78.74%	642.30%
		Steep (≥ 15%)	21.26%	173.40%
PT WANA HIJAU PESAGUAN	Proxy Area 5	Gentle (< 15%)	46.90%	182.48%
		Steep (≥ 15%)	52.70%	205.01%

Following the requirement of the proxy areas, the selected proxy areas have met the requirement of equal proportions for each slope class with a project area of ±20%.



Map 26. Slope Ranges Identified in Percentage within Project Area and Proxy Areas

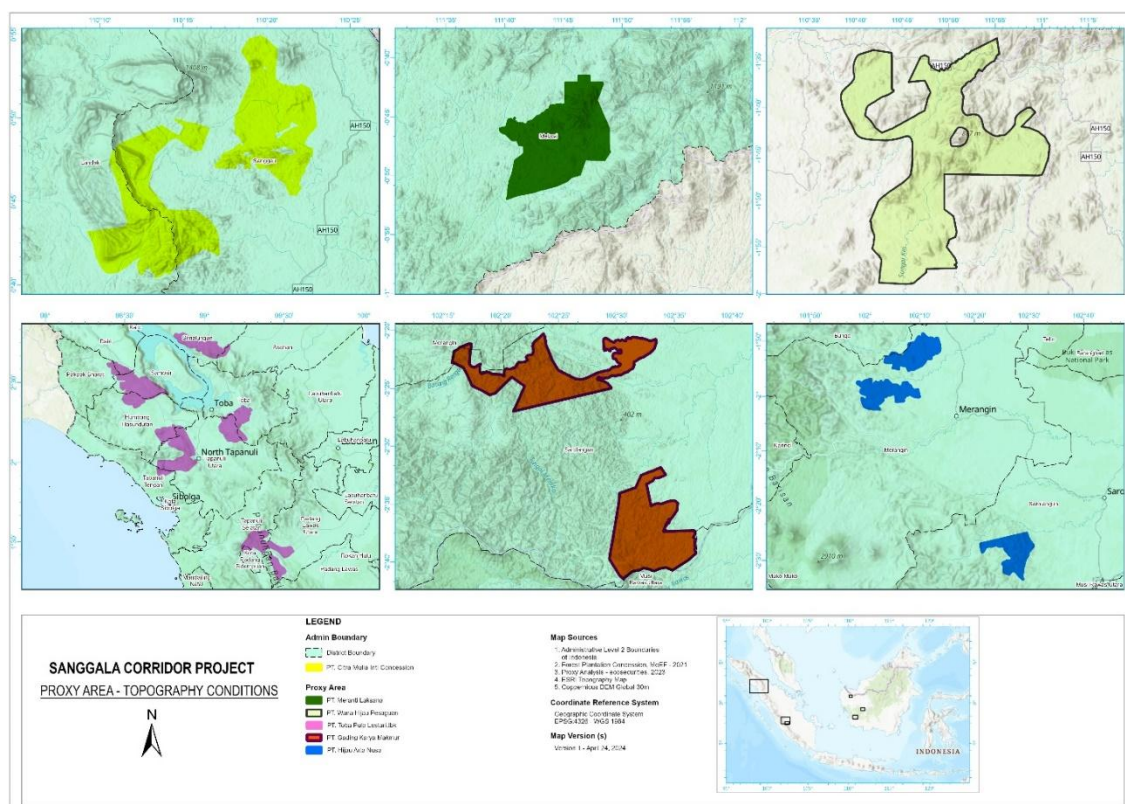
4. Elevation Classes in The Project Area and Proxy Areas

The Copernicus DEM Global 30m dataset was used to identify elevation conditions within project area and proxy areas. The requirements for elevation conditions from the methodology are the elevation class calculated for the project area and proxy areas must be under 500 meters above sea level (MASL), with the proportion of the proxy area to the project area equal to $\pm 20\%$. According to the results shown in Table 46 and Map 27 below, the proxy areas are aligned and have the same proportions of elevation classes as the project area.

Table 46. Elevation Conditions in Project Area and Proxy Areas

Plantation Forest Concession	Area	Elevation	%	% Proportion From Project Area
PT CITRA MULIA INTI	Project Area	≤ 500	98.29%	
		> 500	1.71%	
PT MERANTI LAKSANA	Proxy Area 1	≤ 500	97.44%	87.19%
		> 500	3.19%	2.86%
PT GADING KARYA MAKMUR	Proxy Area 2	≤ 500	99.51%	131.76%
		> 500	1.30%	1.73%
PT HIJAU ARTHA NUSA	Proxy Area 3	≤ 500	100.00%	157.04%
		> 500	0.00%	0.40%
PT TOBA PULP LESTARI Tbk	Proxy Area 4	≤ 500	12.25%	0.99%
		> 500	87.75%	721.78%

Plantation Forest Concession	Area	Elevation	%	% Proportion From Project Area
PT WANA HIJAU PESAGUAN	Proxy Area 5	≤ 500	98.45%	381.49%
		> 500	1.55%	7.88%

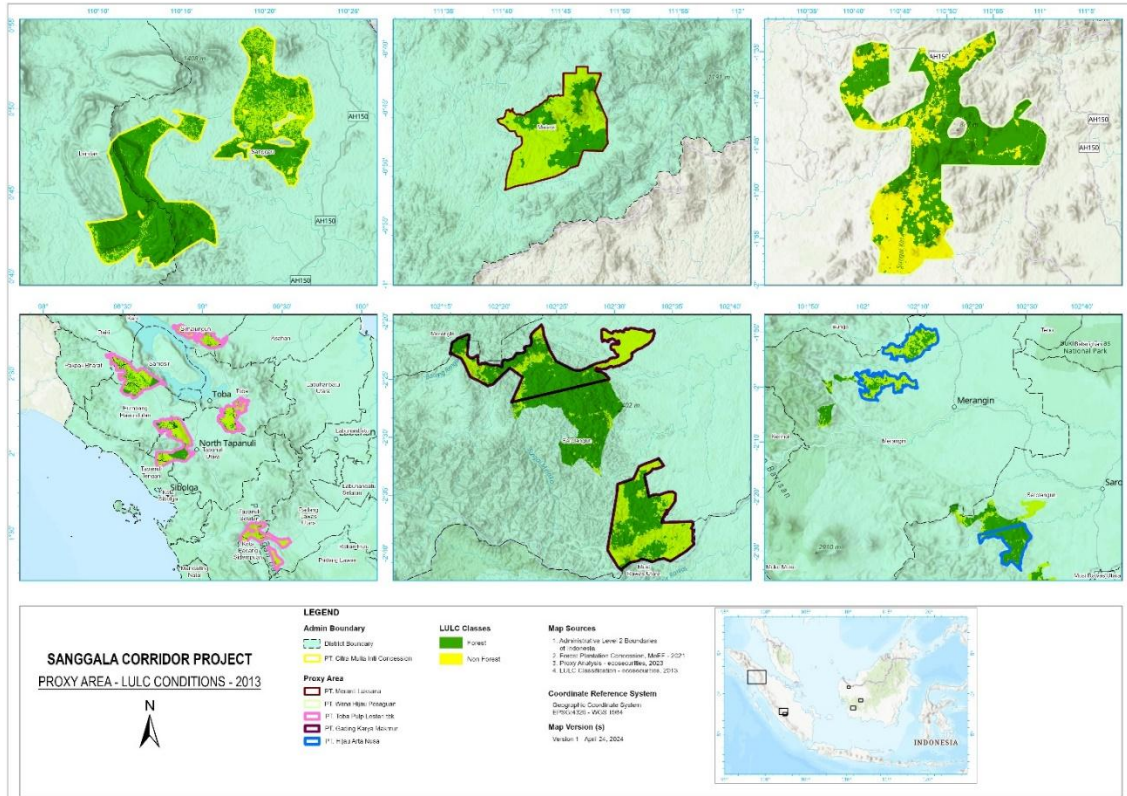


Map 27. Elevation Conditions within Project Area and Proxy Areas

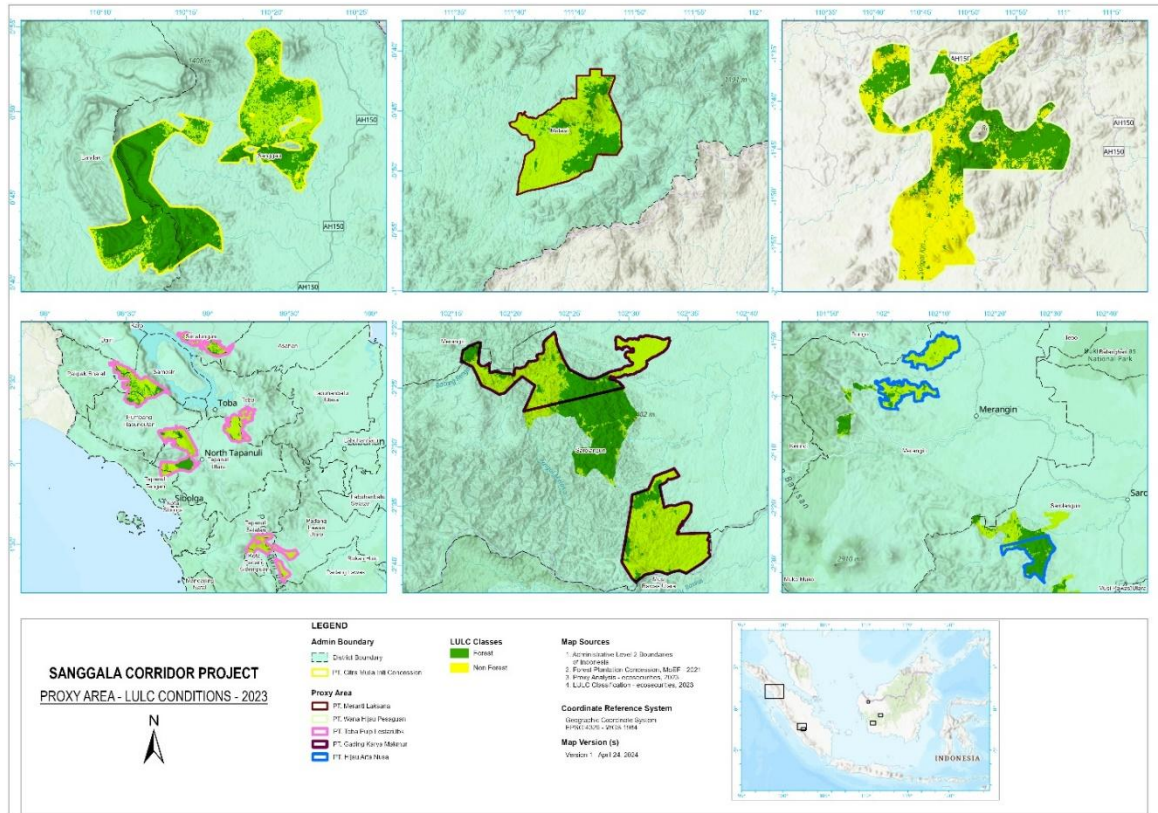
Furthermore, risk abandonment analysis was conducted to detect if forest regrowth occurred within the proxy areas managed by an industrial timber plantation company, which was identified as an agent of deforestation. The result confirmed that proxy areas were not abandoned, as deforestation was detected. Table 47 below provides the detailed deforestation within the proxy areas from 2013 – 2023.

Table 47. LULC Change Analysis for The Proxy Areas from 2013 - 2023

Proxy Areas	LULC Change Analysis				
	2013		2023		Rate Deforestation
	Forest	Non-Forest	Forest	Non-Forest	
PT MERANTI LAKSANA	6,047.28	12,537.36	4,701.69	13,882.95	33%
PT GADING KARYA MAKMUR	13,954.86	13,531.14	5,320.71	22,165.29	62%
PT HIJAU ARTHA NUSA	16,924.95	15,503.22	11,807.37	20,620.80	30%
PT TOBA PULP LESTARI Tbk	43,776.36	125,108.55	40,783.77	128,101.14	26%
PT WANA HIJAU PESAGUAN	49,009.95	31,401.81	32,797.44	47,614.32	37%



Map 27. Proxy Areas Forest Cover in 2013



Map 28. Proxy Areas Forest Cover in 2023

Based on Table 48, it was confirmed that the proxy areas were not abandoned due to deforestation within those areas. Over the 10 years prior to the project's start, deforestation did occur within the proxy areas, with the most significant activity observed in PT Gading Karya Makmur's concession area, where the deforestation rate reached 62%.

Annual Area of Deforestation

The annual area of deforestation is calculated by the following equation:

$$\square AA_{planned,i,t} = (A_{planned,i,t} * D\%_{planned,i,t}) * L - Di$$

Where:

- $AA_{planned,i,t}$ = Annual area of baseline planned deforestation for stratum i at time t ; ha
- $D\%_{planned,i,t}$ = Projected annual proportion of land that will be deforested in stratum i during year t . If actual annual proportion is known and documented (e.g. 25% per year for 4 years), set to proportion
- $A_{planned,i,t}$ = Total area of planned deforestation over the baseline period for stratum i ; ha
- $L - Di$ = Likelihood of deforestation for stratum i ; %

Part 2. Baseline carbon stock change

The baseline emissions for planned deforestation were estimated in accordance with VMD0006 BL-PL, as mandated by VCS methodology VM0007.

The baseline carbon stock change is estimated by annual emissions from deforestation due to the conversion of the natural forest into *A. cadamba* and *H. brasiliensis* plantations by PT CMI as the specific deforestation agent. The annual rate of deforestation is detailed in Section 3.1.4 Baseline Scenario.

As per BL-PL module (VMD0006), the above ground and below ground net carbon stock changes in the baseline are equal to pre-deforestation carbon stocks minus the post deforestation carbon stock, as defined in the following equation:

$$\Delta C_{AB_tree,i} = C_{AB_tree,bsl,i} - C_{AB_tree,post,i}$$

$$\Delta C_{BB_tree,i} = C_{BB_tree,bsl,i} - C_{BB_tree,post,i}$$

Where:

- $\Delta C_{AB_tree,i}$ = Baseline carbon stock change in above ground tree biomass in stratum *i*; tCO_{2e}/ha
- $\Delta C_{BB_tree,i}$ = Baseline carbon stock change in below ground tree biomass in stratum *i*; tCO_{2e}/ha
- $C_{AB_tree,bsl,i}$ = Pre deforestation forest carbon stock in above ground tree biomass in stratum *i*; tCO_{2e}/ha
- $C_{BB_tree,bsl,i}$ = Pre deforestation forest carbon stock in below ground tree biomass in stratum *i*; tCO_{2e}/ha
- $C_{AB_tree,post,i}$ = Post deforestation Forest carbon stock in above ground tree biomass in stratum *i*; tCO_{2e}/ha
- $C_{BB_tree,post,i}$ = Post deforestation Forest carbon stock in below ground tree biomass in stratum *i*; tCO_{2e}/ha

The forest inventory result that was conducted on November 2023 – February 2024 is used as pre-deforestation carbon stock⁶⁰, while the post-deforestation carbon stock is calculated by using long-term average carbon stock from the biomass dynamic of *A. cadamba* and *H. brasiliensis* following the VCS AFOLU guidance⁶¹.

The pre-deforestation carbon stock is estimated using an allometric equation from a peer-reviewed study, following the methodology outlined by Krisnawati et al (2012)⁶². This project utilized approach-2⁶³ out of the eight approaches provided. Additionally, 10 regional models are available

⁶⁰ Forest inventory report

⁶¹ AFOLU Guidance: example for calculation Long Term Average Carbon Stock for ARR project with harvesting

⁶² Krisnawati, Haruni & adinugroho, Wahyu & Imanuddin, Rinaldi. (2012). Monograph Allometric Models for Estimating Tree Biomass at Various Forest Ecosystem Types in Indonesia.

⁶³ Approach02 is used when the tree biomass allometric model for a species or ecosystem type to be estimated is not available at the site, but the biomass allometric model for the species or ecosystem type has been developed for another site, then the allometric model for a tree species or ecosystem type that has been developed for another site can be used as long as the diameter range of tree from forest inventory is in the diameter range of the sample tree used to develop the tree biomass allometric model.

from Manuri et al. (2017)⁶⁴. The allometric equation was developed using 1,300 tree samples from lowland tropical forests characterized by mineral soils, with 758 of these samples obtained from Kalimantan Island. This makes the equation well-suited to the location and ecosystem type of the project area in Kalimantan.

From the 10 models, we chose to use the most comprehensive model that uses two traditional factors (diameter and wood density) called Model DG1, which was recommended by Manuri et al. (2017) as the influence of region is trivial for the accuracy of AGB estimates.

$$AGB = 0.171D^{2.564}G^{0.909}$$

Where:

AGB = Above ground biomass; kg.d.m
D = Diameter at breast height (cm)
G = Wood density (grammes cm⁻³)

Below ground biomass will estimate depending on the AGB stocks and would be multiplied by root-to-shoot ratio from 2nd FREL of Indonesia (0.29)⁶⁵.

To determine the biomass dynamic, we estimate the growth of *A. cadamba* and *H. brasiliensis* by using allometric equation for each species. The allometric equation can be seen in Table 48.

Table 48. The allometric equation used for the analysis

Species	Equation	Author
<i>A. cadamba</i>	$Y = ((1.954)(2.773) + 24.354 X^{1.242}) / 2.773 + X^{1.242}$	Wahyudi & Pamungkas (2013)
<i>A. cadamba</i>	$H = 1.3 + b_0 \exp(-b_1 \exp(-b_2 D))$	Krisnawati et al. (2011)
<i>A. cadamba</i>	$0.014 * (D)^{2.958}$	Indrajaya & Siarudin (2014)
<i>H. brasiliensis</i>	$DBH = 42.8505 / (1 + \exp(2.8662 - 0.162773 * A))$	Qi et al. (2019)
<i>H. brasiliensis</i>	$H = -0.00726 + 4.15 * \ln(D)$	Oyebade & Onyambo (2011)
<i>H. brasiliensis</i>	$AGB = 0.0419 * D^{2.316} * H^{0.478}$	Yang et al. (2017)

A. cadamba plantation long-term average carbon stock is estimated from the biomass dynamic of *A. cadamba* in plantation with the rotation of 6 years. For *H. brasiliensis* the long-term average carbon stock is estimated from the biomass dynamic of *H. brasiliensis* in plantation with rotation of 20 years following the information in PT CMI's RKU. The following table shows the carbon stock used in the calculation of APD activity.

⁶⁴ Manuri, Solichin & Brack, Cris & Rusolono, Teddy & noor'an, Rahimahyuni & Verchot, Louis & Maulana, Sandhi & adinugroho, Wahyu & Kurniawan, Hery & Sukisno, Dian & Kusuma, Gita & Budiman, Arif & Anggono, Rahmad & Siregar, Chairil & Onrizal, Onrizal & Yuniati, Dhany & Soraya, Emma. (2017). Effect of species grouping and site variables on aboveground biomass models for lowland tropical forests of the Indo-Malay region. Annals of Forest Science.

⁶⁵ https://redd.unfccc.int/media/2nd_frl_indonesia_final_submit.pdf

Table 49. The carbon stock used in the calculation of APD activity

Landuse	AGC (tCO ₂ e/ha)	BGC (tCO ₂ e/ha)	Notes
Forest	518.68	150.42	Pre deforestation carbon stock
<i>A. cadamba</i> Plantations	83.46	24.20	Post deforestation carbon stock
<i>H. brasiliensis</i> Plantations	35.10	10.18	

Furthermore, the carbon stock changes above ground accounted for at the time of deforestation and the carbon stock changes in below ground biomass are estimated to be emitted at an annual rate of 1/10 of the stock change for 10-years. The sum of carbon stock changes above ground and below ground is estimated using the following equation.

$$\Delta C_{BSL,it} = AA_{planned,i,t} * (\Delta C_{ABtree,i} - \Delta C_{WP,i}) + \left(\sum_{t=10}^t A_{planned,i,t} \right) * (\Delta C_{BB,tree,i}) * \left(\frac{1}{10} \right) + \left(\sum_{t=10}^t AA_{unplanned,i,t} \right) * (C_{WP100,i} + \Delta C_{SOC,i}) * \left(\frac{1}{20} \right)$$

$\Delta C_{BSL,it}$	=	Sum of the baseline carbon stock change in all terrestrial pools in stratum i in year t, t CO ₂ -e
$AA_{planned,i,t}$	=	Annual area of baseline planned deforestation for stratum i in year t; ha
$\Delta C_{ABtree,i}$	=	Post-deforestation carbon stock in aboveground tree biomass in stratum i; t CO ₂ -e ha ⁻¹
$\Delta C_{WP,i}$	=	Baseline carbon stock change in wood products in stratum i; t CO ₂ -e ha ⁻¹
$\Delta C_{BB,tree,i}$	=	Baseline carbon stock change in belowground tree biomass in stratum i; t CO ₂ -e ha ⁻¹
$C_{WP100,i}$	=	Carbon stock entering the wood products pool at the time of deforestation that is expected to be emitted over 100-years from stratum i; t CO ₂ -e ha ⁻¹
$\Delta C_{SOC,i}$	=	Baseline carbon stock change in soil organic carbon in stratum i; t CO ₂ -e ha ⁻¹
i	=	1, 2, 3, ... M strata
t	=	1, 2, 3, ... t* years elapsed since the projected start of the project activity

In the baseline scenario, the project excluded Soil Organic Carbon (SOC) from quantification as a conservative approach. While the wood product pool was initially considered as outlined in the management plan, it was ultimately excluded from the baseline emissions quantification after a significance test showed insignificant results. Thus, the carbon pool from wood product can be omitted.

According to the management plan, the baseline scenario included the use of nitrogen fertilizer for establishing forest plantations. The project calculated the greenhouse gas (GHG) emissions associated with this fertilizer use. However, after conducting a significance test following T-SIG procedures, the results indicated that the GHG emissions from fertilizer use were insignificant and negligible. Consequently, these emissions were set to zero and excluded from the baseline scenario estimation.

Other CO₂e emissions from fossil fuel combustion ($E_{FC,i,t}$) and non-CO₂ emissions due to biomass burning ($E_{BiomassBurn,i,t}$) are conservatively excluded in emissions baseline quantification. Emissions

due to biomass burning are conservatively excluded as Indonesian government prohibits deforestation activities by use of fire to burn trees. It is regulated in forestry law no. 41 of 1999.

The baseline net GHG emissions for planned deforestation are determined according to VMD0006 (BL-PL) equation 1 presented at the beginning of section 4.1 (baseline emissions). As net GHG emissions are conservatively set to 0 in the baseline scenario, then it will be equal to net carbon stock changes in all pools.

$$\Delta C_{BSL, planned} = \sum_{t=1}^{t^*} \Delta C_{BSL,i,t}$$

- $\Delta C_{BSL, planned}$ = Net greenhouse gas emissions in the baseline from planned deforestation up to year t^* ; t CO₂-e
- $\Delta C_{BSL, i,t}$ = Net carbon stock changes in all pools in the baseline stratum i in year t; t CO₂-e
- t = 1,2,3,... t^* years elapsed since the projected start of the project activity

Table 50 below presents the values for the net GHG emissions in the baseline from planned deforestation up to year 50 is calculated for the project area.

Table 50. Total Net GHG Emissions in the Baseline Scenario

Year	AGB (tCO ₂ e)	BGB (tCO ₂ e)	Total (tCO ₂ e/ha)
0	0	0	0
1	908,023	26,333	934,356
2	1,105,722	58,399	1,164,121
3	883,582	84,023	967,605
4	759,475	106,047	865,522
5	419,483	118,212	537,695
6	16,446	118,689	135,135
7	0	118,689	118,689
8	0	118,689	118,689
9	0	118,689	118,689
10	0	118,689	118,689
11	0	92,357	92,357
12	0	60,291	60,291
13	0	34,667	34,667
14	0	12,642	12,642
15	0	477	477
16	0	0	0
17	0	0	0
18	0	0	0
19	0	0	0
20	0	0	0

Year	AGB (tCO2e)	BGB (tCO2e)	Total (tCO2e/ha)
21	0	0	0
22	0	0	0
23	0	0	0
24	0	0	0
25	0	0	0
26	0	0	0
27	0	0	0
28	0	0	0
29	0	0	0
30	0	0	0
31	0	0	0
32	0	0	0
33	0	0	0
34	0	0	0
35	0	0	0
36	0	0	0
37	0	0	0
38	0	0	0
39	0	0	0
40	0	0	0
41	0	0	0
42	0	0	0
43	0	0	0
44	0	0	0
45	0	0	0
46	0	0	0
47	0	0	0
48	0	0	0
49	0	0	0
50	0	0	0

Part 3: Greenhouse gas emissions

The GHG emissions in the baseline within the project boundary must be estimated as:

$$GHG_{BSL,E,i,t} = EFC_{i,t} + E_{BiomassBurn,i,t} + N2O_{direct-N,i,t}$$

Where:

$GHG_{BSL,E,i,t}$ = Greenhouse gas emissions as a result deforestation activities within the project boundary in the stratum i in year t; t CO2-e

$EFC_{i,t}$	=	Net CO ₂ e emission from fossil fuel combustion in stratum i in year t; t CO ₂ -e
$E_{BiomassBurn,i,t}$	=	Non-CO ₂ emissions due to biomass burning in stratum i in year t; t CO ₂ -e
$N2O_{direct-N,i,t}$	=	Direct N ₂ O emission as a result of nitrogen application on the alternative land use within the project boundary in stratum i in year t; t CO ₂ -e
i	=	1, 2, 3, ... M strata
t	=	years elapsed since the projected start of the REDD VCS project activity

In the ex-ante calculation, these three components are considered as zero and excluded from GHG Emissions Sources for APD (Table 32).

Part 4. Frequency of baseline renewal

In accordance with section 3.2.5 VCS Standard, the baseline must be revised every ten years for ongoing planned deforestation where the agent of deforestation is known. PT. CMI prior to the acquisition by CarbonX is identified as the agent of deforestation.

3.2.1.2 Afforestation, Reforestation and Revegetation (ARR)

The carbon stock changes in the baseline scenario are accounted for by applying the crediting baseline performance benchmark value to the estimation of carbon dioxide removals quantification (see section 3.1.4). The performance benchmark, defined as the business-as usual increase in vegetative stocking relative to the project, is set based on data from representative control plots outside the project area through remote sensing and spatial analysis.

To measure future baseline emissions, statistical methods will be used to compare changes in the stocking index between the project area and a control plot. A detailed performance benchmark is explained in section 3.1.4.2.

Ex ante Estimation for baseline scenario

As described in section 3.1.4.2, the project plots were defined using a 60 x 60 m grid, resulting in an average plot size of 0.36023 hectares. For the control plots, a 53.68 x 53.68 m grid was used. Both the control and project plots will be combined with DVI datasets to calculate the stocking index for each plot. This stocking index will be used to estimate the ex-ante performance benchmark, based on the values of $\Delta SI_{control}$ and ΔSI_{wp} .

In the ex-ante estimation process, the performance benchmark must be estimated 10 years after the project begins, in accordance with VM0047, and then extrapolated to cover the 50 year crediting period and lifetime project estimation. According to VM0047, deriving ΔSI_{wp} in the ex-ante estimation requires the project to model the relationship between the stocking index and tC in biomass using a regression model. For the control plots, the project must use a regression model based on the historical performance of the stocking index to generate $\Delta SI_{control}$.

Based on the ex-ante estimation, the performance benchmark from t₀ to t₅₀ is shown in the table below.

Table 51. The Results of Ex-ante Estimation for Performance Benchmark

Years	T	PB
2024 – 2033	1 – 10	0.39
2034 – 2043	11 – 20	0.32
2044 – 2053	21 – 30	0.31
2054 – 2063	31 – 40	0.33
2063 – 2073	41 – 50	0.37

The detailed explanation of ex-ante estimation for performance benchmark is explained on Annex 8

3.2.2 Project Emissions (VCS, 3.15)

3.2.2.1 Avoided Planned Deforestation

Project GHG emissions will be considered according to VMD0015 (M-REDD). The net GHG emissions in the project scenario are equal to the sum of stock changes due to deforestation and forest degradation plus the total GHG emissions, according to VMD0015, equation 01:

$$\Delta C_{WPS-REDD} = \sum_{t=1}^{t^*} \sum_{i=1}^M (\Delta C_{P,DefPA,i,t} + \Delta C_{P,Deg,i,t} + \Delta C_{DistPA,i,t} + GHG_{p-E,i,t})$$

Where:

$\Delta C_{WPS-REDD}$	=	Net GHG emissions in the REDD project scenario up to year t^* ; t CO2-e)
$\Delta C_{P,DefPA,i,t}$	=	Net carbon stock change because of deforestation in the project area in the project case in stratum i in year t ; t CO2-e
$\Delta C_{P,Deg,i,t}$	=	Net carbon stock change because of degradation in the project area in the project case in stratum i in year t ; t CO2-e
$\Delta C_{DistPA,i,t}$	=	Net carbon stock change because of natural disturbance in the project area in the project case in stratum i in year t ; t CO2-e
$GHG_{p-E,i,t}$	=	Greenhouse gas emissions because of deforestation and degradation activities within the project area in the project case in stratum i in year t ; t CO2-e
i	=	1, 2, 3, ...M strata
t	=	1, 2, 3, ... t^* years elapsed since the start of the project activity

The parameters that must be considered are discussed in the sections below.

Deforestation in the Project Scenario

Ex-ante deforestation projections in the project scenario, assumes that no deforestation will occur as the project scenario aims to conserve the forest. The project proponent is committed to not undertake any deforestation on the project area to the detriment of developing conservation projects to generate carbon credits (See Annex 3 explained management plan to conserve the project area). Accordingly, the deforestation in the project scenario will set to zero. However, the

deforestation may occur during the ex-post scenario. In that case, the net carbon stock change as a result of deforestation will be estimated.

$$\Delta C_{P,defPA,i,t} = \sum_{u=1}^U (A_{DefPA,u,i,t} \times \Delta C_{pools,P,Def,u,i,t})$$

Where:

$\Delta C_{P,defPA,i,t}$	=	Net carbon stock change as a result of deforestation in the project case in the project area in stratum I in year t; t CO ₂ -e
$A_{DefPA,u,i,t}$	=	Area of recorded deforestation in the project area stratum i converted to land use u in year t; ha
$\Delta C_{pools,P,Def,u,i,t}$	=	Net carbon stock changes in all pools in the project case in land use u in stratum i in year t; t CO ₂ -e ha ⁻¹
u	=	1,2,3,....., U post-deforestation land uses
i	=	1,2,3,...., M strata in the project scenario
t	=	1,2,3,...., t* years elapsed since the start of the project activity

The emission per unit area is equal to the difference between the stock before and after deforestation. **Degradation in Project Scenario**

GHG emissions related to forest degradation are considered in relation to timber, fuelwood and charcoal collection activities in the project area in the project case.

For ex-ante estimates, forest degradation projections were made as a function of the initial participatory rural appraisal (PRA) results. According to the PRA, all households living inside and around the project area utilize the forest area for timber and fuel wood. These activities are considered low impact, driven by subsistence purposes. No timber or fuelwood economic activities were identified as a result of the PRA activity. Hence, emissions resulting from wood extraction for timber or firewood are assumed to be zero in the ex-ante estimation.

For ex-post estimates, the PRA shall be conducted and repeated every 2 years to identify the potential where trees may be extracted to generate timber products or firewood for community use.

Selective Logging in the Project Scenario

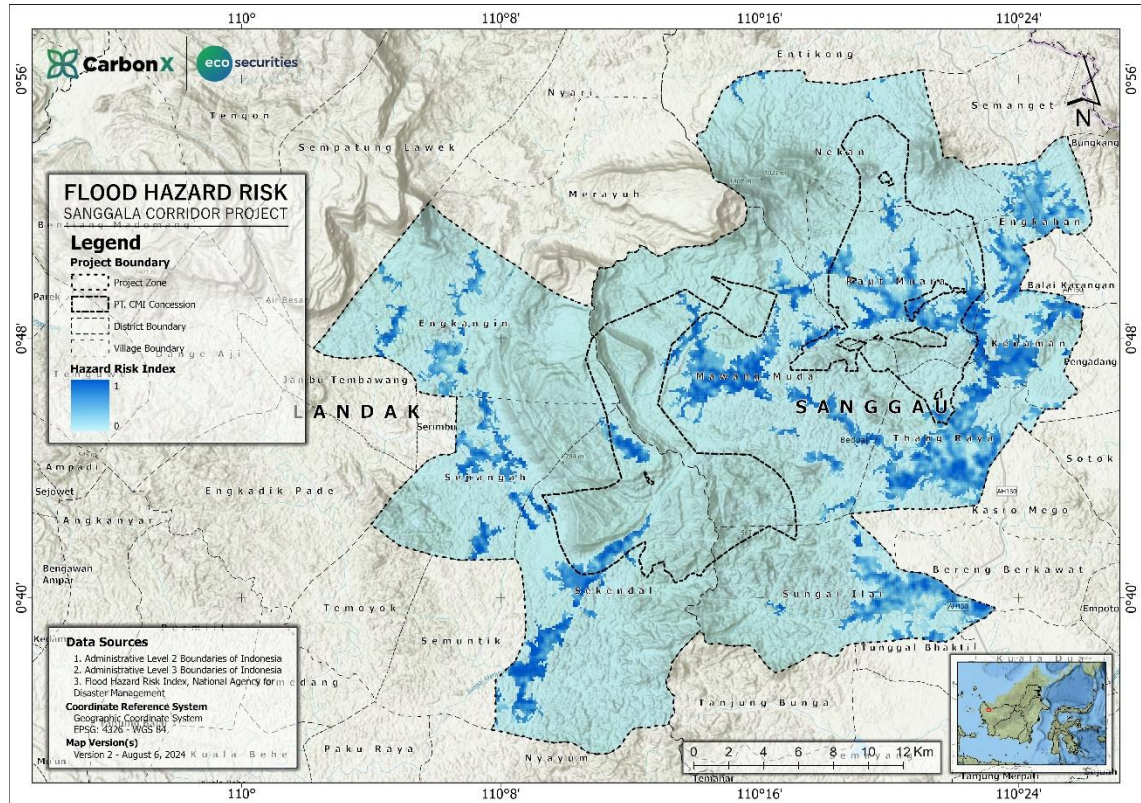
No potential selective logging is planned in the project scenario due to the project's aims to conserve the forest.

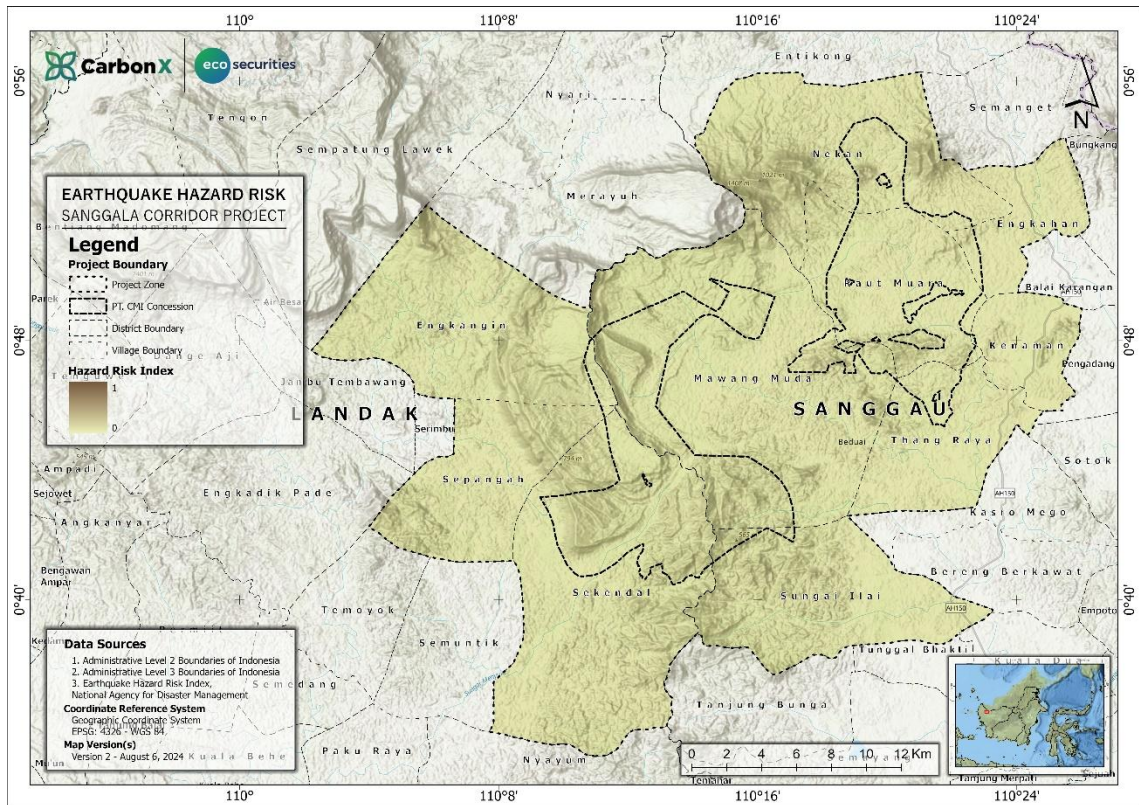
Natural Disturbance in the Project Scenario

There are no potential natural disturbances, that would affect the viability of the project the generate carbon credits. According to InaRisk geoportal⁶⁶ from National Agency for Disaster Management of Indonesia, most of the project zone falls within the Low-Risk classification for potential flood hazards, with the index range of 0 - 1. Moreover, the project area is also classified as low risk for potential earthquake with the index of 0, and wildfires, with an index range of 0 - 0.56. The flood, earthquake, and wildfire hazard risks for the project area shown in Map 30, Map 31, and Map 32.

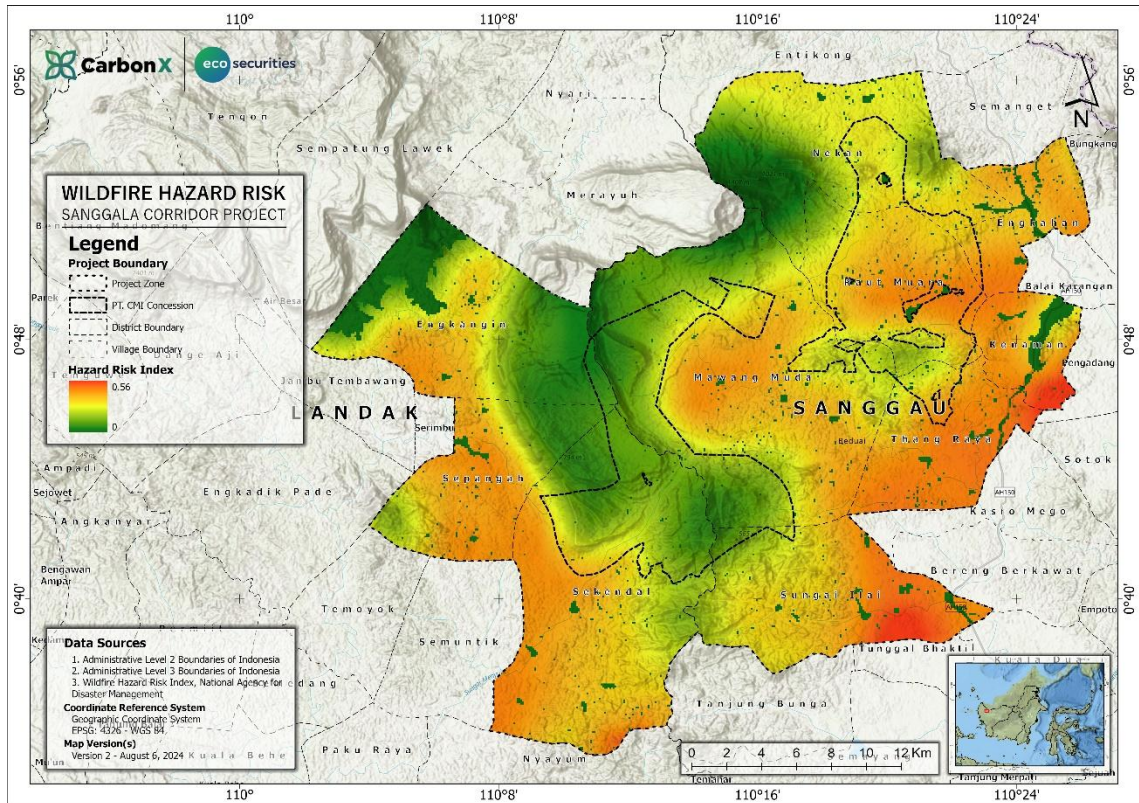
⁶⁶ InaRisk is GIS-based information system provides disaster risk assessment (hazard, capacity, vulnerability, and risk) and monitoring of disaster risk index reduction in Indonesia.

Therefore, in the ex-ante calculation net carbon stock change due to natural disturbance is assumed to be zero. In ex-post calculation, the determination of project emission coming from natural disturbance will be determined using remote sensing imagery combined by ground verification or GPS coordinates.





Map 30. Earthquake hazard risk within the project area



Map 31. Wildfire hazard risk within the project area

Project Emissions

The project emissions are defined as emission resulted from biomass burning, fossil fuels combustion and fertilizer use.

According to the Indonesian regulations, there are no potential emissions from biomass burning because the government prohibits the burning activity in the forest areas. Accordingly, the emissions from biomass burning are conservatively set to zero in ex-ante condition. However, forest fire incident may occur during the implementation of project activities. In that case, the emissions from biomass burning will be considered and monitored accordingly.

Meanwhile, the emissions from fossil fuel combustion were conservatively excluded in the baseline scenario and will also be excluded in the project scenario in both ex-ante and ex-post scenario (Table 32).

The emission resulting from nitrogen applications were considered in the baseline scenario, but the result based on the T-SIG application shows that the emission from nitrogen applications is insignificant (see Annex 16). Hence as a conservative measure, they were excluded in the baseline scenario. Meanwhile, the project scenario aims only to conserve the forest areas that do not involve fertilization activities. This condition leads into no emission from nitrogen application in the project scenario. Therefore, both in ex-ante and ex-post scenario the emissions from nitrogen applications are considered as zero.

Based on the explanation above, ex-ante estimation for net GHG emissions in project scenarios ($\Delta C_{WPS-REDD}$) generates a value of 0. In ex-post scenario, emissions from deforestation, degradation and natural disturbance will be monitored.

3.2.2.2 Afforestation, Reforestation and Revegetation (ARR)

The project will conduct reforestation activities on eligible areas under methodology VM0047 for ARR projects through forest restoration and agroforestry development. Agroforestry is defined as a dynamic, ecologically based natural resource management system that, through the integration of trees on farms and in agricultural landscapes or through the production of agricultural products in forests, diversifies and sustains production for increased economic, social, and environmental benefits for land users⁶⁷. Meanwhile, forest restoration refers to activities aimed at increasing forest cover, structure, function, and/or species composition⁶⁸.

Agroforestry will develop on mixed dry agriculture strata and shrubland, which was mixed dry agriculture 10 years ago. Forest restoration activities will be conducted on the stable shrub for 10 years. Table 52 and Table 53 below provide the information on the Agroforestry and Forest restoration planting design for this project

⁶⁷ Borelli et al. Agroforestry module, FAO, Forestry Department.

⁶⁸ Julia Jones, David Ellison, Silvio Ferraz, Antonio Lara, Xiaohua Wei, Zhiqiang Zhang, Forest restoration and hydrology, Forest Ecology and Management, Volume 520, 2022, 120342, ISSN 0378-1127,

Table 52. Agroforestry Planting Design

Agroforestry				
Tree Species	Space (m)		%	Number of Seedlings
Vegetation			25	123
<i>Shorea stenoptera</i>	5	5	15	60
<i>Macaranga .sp</i>	4	4	10	63
Paddy field			20	50,000
<i>Oryza sativa</i>	0.2	0.2	20	50,000
Cash crop			10	111
<i>Piper nigrum</i>	3	3	10	111
Plantation tree			15	167
<i>Theobroma cacao</i>	3	3	15	167
Fruits			30	94
<i>Durio zibethinus</i>	8	8	20	31
<i>Pithecellobium jiringa</i>	4	4	10	63

Table 53. Forest Restoration Planting Design

Restoration				
Tree Species	Space (m)		%	Number of Seedlings per ha
<i>Shorea stenoptera</i>	5	5	10	40
<i>Durio zibethinus</i>	8	8	20	31
<i>Shorea pinanga</i>	5	5	10	40
<i>Pithecellobium jiringa</i>	3	3	10	111
Natural Regeneration			50%	

Agroforestry integrates various types of vegetation, including paddy fields, plantation trees, cash crop plants and fruit trees, to accommodate the native species and social culture^{69, 70, 71} within the project area. The amount of stocking density in the agroforestry model is 384. Implementation of forest restoration activities combines planting activities with the amount of stocking density of 222 and assisted natural regeneration. For any species used in ex-ante, estimation is tentative, and the implementation process will adjust to the field condition, community consensus and availability of seedlings in the field.

ARR activities are divided into 10 phases of planting within both mixed dry agriculture and shrub areas, encompassing a total eligible area of ARR of 6,976.22 ha. Detailed information on phase planting for ARR activities was provided on Table 54 below.

⁶⁹ Muslim, Nahdzatul & Abdul Majid, Amin Malik Shah. (2010). *Pithecellobium Jiringa*: A Traditional Medicinal Herb. 1.

⁷⁰ MORFOLOGI VEGETATIF JENIS POHON TENGGAWANG (*Shorea* spp) DI DESA MENSIAU KECAMATAN BATANG LUPAR KABUPATEN KAPUAS HULU

⁷¹ Siregar, Mustaid. (2006). Species Diversity of Local Fruit Trees in Kalimantan: Problems of Conservation and Its Development. Biodiversitas, Journal of Biological Diversity.

Table 54. Total Area Planting Area for ARR activities

Year	Planting Area		Total (ha)
	Agroforestry (ha)	Restoration (ha)	
1	50.00	50.00	100.00
2	200.00	100.00	300.00
3	400.00	200.00	600.00
4	380.91	472.84	853.75
5	380.91	472.84	853.75
6	380.91	472.84	853.75
7	380.91	472.84	853.75
8	380.91	472.84	853.75
9	380.91	472.84	853.75
10	380.91	472.84	853.75
	Total Planting Area:		6,976.22

The procedure for the quantification of carbon stock changes and project emissions for ARR activities area calculated based on the agroforestry development and following the area-based quantification approach following the methodology VM0047. The project emissions from the agroforestry model for this project are estimated as follows:

$$\Delta C_{WP,t} = (\Delta C_{WP-biomass,t} + \Delta C_{WP-SOC,t}) \times \frac{44}{12}$$

Where:

- $\Delta C_{WP,t}$ = Project carbon stock change in year t (t CO₂e)
- $\Delta C_{WP-biomass,t}$ = Change in carbon stock in biomass carbon pools in the project scenario through year t (t C)
- $\Delta C_{WP-SOC,t}$ = Change in carbon stock in SOC in the project scenario through year t (t C)
- $44/12$ = Ratio of molecular weight of carbon dioxide to carbon (unitless)
- t = 1, 2, 3, ..., t years elapsed since the project start date

Change in carbon stock in biomass carbon pools in the project scenario are estimated as follows:

$$\Delta C_{wp-biomass,t} = \Delta C_{WP-woody,t} + \Delta C_{WP-herb,t} + \Delta C_{WP-DW,t} + \Delta C_{WP-LI,t}$$

Where:

- $\Delta C_{wp-biomass,t}$ = Change in carbon stock in biomass carbon pools in the project scenario through year t (t C)
- $\Delta C_{WP-woody,t}$ = Change in carbon stock in woody biomass in the project scenario through year t (t C)
- $\Delta C_{WP-herb,t}$ = Change in carbon stock in non-woody biomass in the project scenario through year t (t C)
- $\Delta C_{WP-DW,t}$ = Change in carbon stock in dead wood in the project scenario through year t (t C)

$\Delta C_{WP-LI,t}$ = Change in carbon stock in litter in the project scenario through year t (t C)
t = 1,2,3,..., t years elapsed since the project start date

In ex ante estimation, woody biomass is estimated based on the growth of trees that will be planted. In the meantime, the others carbon pools such as non-woody biomass, deadwood and litter are estimated based on the stock change approach.

Woody Biomass

The carbon stock change in tree woody biomass will be calculated using the equation below:

$$\Delta C_{WP-woody,t} = A \times (C_{WP-woody,t} - C_{WP-woody,t=0})$$

Where:

$\Delta C_{WP-woody,t}$ = Change in carbon stock in woody biomass in the project scenario through year t (t C)
A = Area (ha)
 $C_{WP-woody,t}$ = Average carbon stock in woody biomass in the project scenario in year t (t C/ha)
t = 1,2,3, ..., t years elapsed since the project start date

$$C_{WP-woody,t} = C_{WP-woody-AB,t} \times (1 + R)$$

Where:

$\Delta C_{WP-woody,t}$ = Average carbon stock in woody biomass in the project scenario in year t (t C/ha)
R = Root to shoot ratio (t root d.m./t shoot d.m.)
 $C_{WP-woody-AB,t}$ = Average carbon stock in aboveground woody biomass in the project scenario in year t (t C/ha)
t = 1,2,3, ..., t years elapsed since the project start date

Pre-existing woody biomass was not measured at t = 0, immediately before initiation of the project activity. The initial stocks will be measured within the next 10-year period and before verification, pre-existing woody biomass will be equal to the initial stock measurement.

Non-Woody Biomass

According to VCS Methodology Requirements v4.4, GHG emissions from the removal of non-woody biomass in ARR project may be deemed de minimis without formal evaluation. However, VM0047 indicates the pools need to be evaluated for de minimis as it may be increase or decrease in project scenario. Accordingly, the non-woody biomass can be estimated as follows:

$$\Delta C_{WP-herb,t} = A \times (C_{WP-herb,t} - C_{WP-herb,t=0})$$

$$C_{WP-herb,t} = DM_{WP-herb,t} \times CF$$

Where:

$\Delta C_{WP-herb,t}$	=	Change in carbon stock in non-woody biomass in the project scenario through year t (t C)
A	=	Area (ha)
$C_{WP-herb,t}$	=	Average carbon stock in non-woody biomass in the project scenario in year t (t C ha ⁻¹)
$DM_{WP-herb,t}$	=	Average non-woody biomass in the project scenario in year t (t.d.m. ha ⁻¹)
CF	=	Carbon fraction of dry biomass (t C/t.d.m)
t	=	1,2,3,..., t years elapsed since the project start date

Dead Wood

The net carbon stock change in dead wood in the project scenario is estimated by using equation below:

$$\Delta C_{WP-DW,t} = A \times (C_{WP-DW,t} - C_{WP-DW,t=0})$$

Where:

$\Delta C_{WP-DW,t}$	=	Change in carbon stock in dead wood in the project scenario through year t (t C)
A	=	Area (ha)
$C_{WP-DW,t}$	=	Average carbon stock in dead wood in year t (t C/ha)
t	=	1,2,3,..., t years elapsed since the project start date

Dead wood may comprise two components: standing dead wood that is fully dead (i.e., absence of green leaves and green cambium) and lying dead wood.

$$C_{WP-DW,t} = (B_{SDW,t} + B_{LDW,t}) \times CF$$

Where:

$C_{WP-DW,t}$	=	Average carbon stock of dead wood in year t (t C/ha)
$B_{SDW,t}$	=	Average biomass of standing dead wood in year t (t.d.m/ha)
$B_{LDW,t}$	=	Average biomass of lying deadwood in year t (t.d.m/ha)
CF	=	Carbon fraction of dry biomass (t C/t.d.m)
t	=	1,2,3,..., t years elapsed since the project start date

Based on ex-ante estimation, carbon stock from dead wood is deemed insignificant as the result of relative contributions less than 5 percent (1.5%). However, it may increase in project scenarios and need to be evaluated in ex-post calculation. Based on that, this carbon pool will be monitored during implementation of project activity.

Litter

Carbon stock of litter will be estimated from direct measurement in project scenario. The net carbon stock change in litter is estimated as follows:

$$\Delta C_{WP-LI,t} = A \times (C_{WP-LI,t} - C_{WP-LI,t=0})$$

$$C_{WP-LI,t} = DM_{WP-LI,t} \times CF$$

Where:

$\Delta C_{WP-LI,t}$	=	Change in carbon stock in litter in the project scenario through year t (t C)
A	=	Area (ha)
$C_{WP-LI,t}$	=	Average carbon stock in litter in the project scenario in year t (t C/ha)
$DM_{WP-LI,t}$	=	Average litter dry mass per ha
CF	=	Carbon fraction of dry biomass (t C/t.d.m)
t	=	1,2,3,..., t years elapsed since the project start date

Based on ex-ante estimation, carbon stock from litter is deemed insignificant as the result of relative contributions less than 5 percent (0.30%). However, it may increase in project scenarios and need to be evaluated in ex-post calculation.

Project Emissions

Project emissions will be estimated using the next equation:

$$PE_t = PE_{bburn,t} + PE_{fert,t}$$

Where:

PE_t	=	Project emissions from biomass burning and fertilizer in year t (t CO ₂ e)
$PE_{bburn,t}$	=	Project emissions due to biomass burning in year t (t CO ₂ e)
$PE_{fert,t}$	=	Project emissions from nitrogen fertilizer in year t (t CO ₂ e)

The project activities do not involve any burning of biomass (not even for site preparation) therefore, the project emissions resulting from biomass burning will be set to zero. However, burning of biomass from natural causes and community agricultural activity may occur during the crediting period. In that case, the emissions from biomass burning will be estimated according to the equations below:

$$PE_{bburn,t} = A_{burn,t} \times \sum_{g=1}^G (GWP_g \times EF_g \times B_{WP,t} \times COMF \times 10^{-3})$$

Where:

PE_{bburn}	=	Project emissions due to biomass burning in year t (t CO ₂ e)
$A_{burn,t}$	=	Area burned in the monitoring interval ending in year t (ha)
GWP_g	=	Global warming potential for gas g (dimensionless)
EF_g	=	Emission factor for gas g (kg gas/t.d.m. burned)
$B_{WP,t}$	=	Average aboveground biomass stock subject to burning in the project scenario in the monitoring interval ending in year t (t.d.m/ha)
$COMF$	=	Combustion factor (dimensionless)
t	=	1,2,3, ..., t years elapsed since the project start date
G	=	1, ..., G greenhouse gases (methane and nitrous oxide) (dimensionless)
10^{-3}	=	Conversion of kg CO ₂ e to tCO ₂ e

The average aboveground biomass stock subject to burning is estimated as follows:

$$B_{WP,t} = (C_{WP-woody-AB,t-\Delta t} + C_{WP-herb,t-\Delta t} + C_{WP-DW,t-\Delta t} + C_{WP-LI,t-\Delta t}) \times (1/CF)$$

Where:

$B_{wp,t}$	=	Average aboveground biomass stock subject to burning in the project scenario in the monitoring interval ending in year t (t d.m./ha)
$C_{WP-woody-AB,t-\Delta t}$	=	Average carbon stock in aboveground woody biomass in the project scenario in year t-Δt (t C/ha)
$C_{WP-herb,t-\Delta t}$	=	Average carbon stock in non-woody biomass in the project scenario in year t-Δt (t C/ha)
$C_{WP-DW,t-\Delta t}$	=	Average carbon stock in dead wood in the project scenario in year t-Δt (t C/ha)
$C_{WP-LI,t-\Delta t}$	=	Average carbon stock in litter in the project scenario in year t-Δt (t C/ha)
CF	=	Carbon fraction of dry biomass (t C/t d.m)
T	=	1,2,3, ..., t years elapsed since the project start date
Δt	=	Length of monitoring interval ending in year t (years)

The project will also consider the use of fertilizer for project activities. Where nitrogen fertilizer is applied due to the project activities, nitrous oxide emissions are calculated as:

$$PE_{fert,t} = PE_{Ndirect,t} + PE_{Nindirect,t}$$

Where:

$PE_{fert,t}$	=	Project emissions from nitrogen fertilizer in year t (t CO _{2e})
$PE_{Ndirect,t}$	=	Direct nitrous oxide emissions due to fertilizer use in the project scenario in year t (t CO _{2e})
$PE_{Nindirect,t}$	=	Indirect nitrous oxide emissions due to fertilizer use in the project scenario in monitoring interval ending in year t (t CO _{2e})
t	=	1,2, 3, ..., t years elapsed since the project start date

An organic fertilizer was proposed to be used in the project implementation. To estimate direct nitrous oxide emissions due to fertilizer use in project scenario using the equation as follows:

$$PE_{N,direct,t} = (F_{wp,SN,t} + F_{wp,ON,t}) \times EF_{Ndirect} \times \frac{44}{28} \times GWP_g$$

Where:

$PE_{N,direct,t}$	=	Direct nitrous oxide emissions due to fertilizer use in the project scenario in year t (t CO _{2e})
$F_{wp,SN,t}$	=	Synthetic N fertilizer applied in the project scenario in year t (t N)
$F_{wp,ON,t}$	=	Organic N fertilizer applied in the project scenario in year t (t N)
$EF_{Ndirect}$	=	Emission factor for nitrous oxide emissions from N additions due to synthetic fertilizers, organic amendments and crop residues (t N ₂ O-N/t N applied)
44/28	=	Ratio of molecular weight of N ₂ O to molecular weight of N (applied to convert N ₂ O-N emissions to N ₂ O emissions) (unitless)
GWP_g	=	Global warming potential for gas g (here, nitrous oxide) (dimensionless)
t	=	1,2, 3, ..., t years elapsed since the project start date

The synthetic N fertilizer that applied in the project scenario can be estimate using following equation below.

$$F_{wp,SN,t} = M_{wp,SF,t} \times NC_{wp,SF,t}$$

Where:

- $F_{wp,SN,t}$ = Synthetic N fertilizer applied in the project scenario in year t (t N)
 $M_{wp,SF,t}$ = Mass of N-containing synthetic fertilizer applied in the project scenario in the monitoring interval ending in year t (t fertilizer)
 $NC_{wp,SF,t}$ = N content of synthetic fertilizer applied in the project scenario in year t (t N/t fertilizer)
t = 1,2,3,...., t years elapsed since the project start date

Meanwhile, for organic N fertilizer applied in the project scenario can be estimate using following equation below.

$$F_{wp,ON,t} = M_{wp,OF,t} \times NC_{wp,OF,t}$$

Where:

- $F_{wp,ON,t}$ = Organic N fertilizer applied in the project scenario in year t (t N)
 $M_{wp,OF,t}$ = Mass of N-containing organic fertilizer applied in the project scenario in the monitoring interval ending in year t (t fertilizer)
 $NC_{wp,OF,t}$ = N content of organic fertilizer applied in the project scenario in year t (t N/t fertilizer)
t = 1,2,3,...., t years elapsed since the project start date

To estimate indirect nitrous oxide emissions due to fertilizer use in project scenario using the equation as follows:

$$PE_{N,indirect,t} = Nfert_{wp,volat,t} + Nfert_{wp,leach,t}$$

Where:

- $PE_{N,indirect,t}$ = Indirect nitrous oxide emissions due to fertilizer use in the project scenario in monitoring interval ending in year t (t CO₂e)
 $Nfert_{wp,volat,t}$ = Indirect nitrous oxide emissions produced from atmospheric deposition of N Volatilized due to nitrogen fertilizer use in year t (t CO₂e)
 $Nfert_{wp,leach,t}$ = Indirect nitrous oxide emissions produced from leaching and runoff of N, in regions where leaching and runoff occurs, due to nitrogen fertilizer use in year t (t CO₂e)
t = 1,2,3,...., t years elapsed since the project start date

Indirect nitrous oxide emissions produced from atmospheric deposition of N Volatilized due to fertilizer use can be estimate using following equation below.

$$Nfert_{wp,volat,t} = [(F_{wp,SN,t} \times Frac_{GASF}) + (F_{wp,ON,t} \times Frac_{GASM})] \times EF_{Nvolat} \times \frac{44}{28} \times GWP_g$$

Where:

- $Nfert_{wp,volat,t}$ = Indirect nitrous oxide emissions produced from atmospheric deposition of N Volatilized due to nitrogen fertilizer use in year t (t CO₂e)
 $F_{wp,SN,t}$ = Synthetic N fertilizer applied in the project scenario in year t (t N)

$Frac_{GASF}$	=	Fraction of all synthetic N added to soils that volatilized as NH_3 and NO_x (dimensionless)
$F_{wp,ON,t}$	=	Organic N fertilizer applied in the project scenario in year t (t N)
$Frac_{GASM}$	=	Fraction of all organic N added to soils that volatilizes as NH_3 and NO_x (dimensionless)
EF_{Nvolat}	=	Emission factor for nitrous oxide emissions from atmospheric deposition of N on soils and water surfaces (t N_2O -N/(t NH_3 -N + NO_x -N volatilized))
44/28	=	Ratio of molecular weight of N_2O to molecular weight of N (applied to convert N_2O -N emissions to N_2O emissions) (unitless)
GWP_g	=	Global warming potential for gas g (here, nitrous oxide) (dimensionless)
t	=	1,2,3,....., t years elapsed since the project start date

Indirect nitrous oxide emissions produced from leaching and runoff of N in region due to fertilizer use can be estimate using following equation below.

$$Nfert_{wp,leach,t} = (F_{wp,SN,t} + F_{wp,ON,t}) \times Frac_{LEACH} \times EF_{Nleach} \times \frac{44}{28} \times GWP_g$$

Where:

$Nfert_{wp,leach,t}$	=	Indirect nitrous oxide emissions produced from leaching and runoff of N, in regions where leaching and runoff occurs, due to nitrogen fertilizer use in year t (t CO_2e)
$F_{wp,SN,t}$	=	Synthetic N fertilizer applied in the project scenario in year t (t N)
$F_{wp,ON,t}$	=	Organic N fertilizer applied in the project scenario in year t (t N)
$Frac_{LEACH}$	=	Fraction of synthetic or organic N added to soils that is lost through leaching and runoff, in regions where leaching and runoff occurs (dimensionless)
EF_{Nleach}	=	Emission factor for nitrous oxide emissions from leaching and runoff (t N_2O -N/t N leached and runoff)
44/28	=	Ratio of molecular weight of N_2O to molecular weight of N (applied to convert N_2O -N emissions to N_2O emissions) (unitless)
GWP_g	=	Global warming potential for gas g (here, nitrous oxide) (dimensionless)
t	=	1,2,3,....., t years elapsed since the project start date

At this stage, the emissions from fertilizer used are estimated based on the 10-years planting plan for restoration and agroforestry schemes. According to ex-ante estimation performed, the project may generate emissions due to fertilizer use of 4.494.21 t CO_2e over 10 years.

Table 55. Ex-Ante estimation of the emissions from fertilizer use

Year	Restoration		PEfert,t (t CO_2e)	Agroforestry		PEfert,t (t CO_2e)	Total PEfert,t (t CO_2e)
	PEN indirect	PE _{n,direct,t}		PEN indirect	PE _{n,direct,t}		
1	10.56	28.23	38.79	6.79	18.16	24.95	63.74
2	21.12	56.47	77.59	27.16	72.63	99.79	177.37
3	42.24	112.94	155.17	54.32	145.25	199.57	354.75

Year	Restoration		PEfert,t (tCO2e)	Agroforestry		PEfert,t (tCO2e)	Total PEfert,t (tCO2e)
	PEN indirect	PE _{n,direct,t}		PEN indirect	PE _{n,direct,t}		
4	99.86	267.00	366.86	51.73	138.32	190.05	556.91
5	99.86	267.00	366.86	51.73	138.32	190.05	556.91
6	99.86	267.00	366.86	51.73	138.32	190.05	556.91
7	99.86	267.00	366.86	51.73	138.32	190.05	556.91
8	99.86	267.00	366.86	51.73	138.32	190.05	556.91
9	99.86	267.00	366.86	51.73	138.32	190.05	556.91
10	99.86	267.00	366.86	51.73	138.32	190.05	556.91
Total							4,494.21

According to Appendix 2 of VM0047, the relative contribution of these emissions was found to be less than 5% (0.07%), and therefore, these emissions are considered de minimis (see Annex 16). All parameters and factors used in these estimations are reported in section 3.3.1.

Ex-ante Estimation

a. Woody Biomass

According to the VM0047, the project description must include an ex-ante estimation of emissions removals. This requires making estimates for the 10 years from the time of validation t to t+10. However, in this project, estimates were extended to cover the entire crediting period of period (50 Years). For the ex-ante estimation, several parameters were considered, including DBH, wood density, AGB, BGB and $\Delta C_{biomass}$ (tCO₂e) to calculate emissions removal from ARR activities.

Estimated carbon stock from those parameters in this project, used several growth models from scientific papers, the 2nd FREL of Indonesia and IPCC. The detailed growth model used for this project can be seen in Table 56 below.

Table 56. Growth Parameters Used for Ex-ante Estimation

Parameters	Equation/Value	Sources
Dbh Dipterocarpaceae (cm)	$\ln Dbh = 5.36641 - 6.31351 \cdot (1/A)^{0.38024}$	Bustomi et al. (2009)
Dbh Non Dipterocarpaceae (Cm)	$Dbh = A \cdot (1 - (e^{-0.035t})^{1/(1-0.40)})$	Luo et al. (2018) - Richard Model
AGB (kg/tree)	$0.088(D^2GH)^{0.954}$	Manuri et al. (2017)
AGC (tC/ha)	$AGC = MAX \cdot (1 - EXP(-k \cdot Age))^{1/(1-m)}$	Chapman-Richards
Root to Shoot Ratio	0.29	2nd FREL of Indonesia (2022)
Wood Density for all Species	0.528	Manuri et al. (2017)
Carbon fraction	0.47	IPCC (2006)
C to CO ₂	3.67	IPCC (2006)

According to the table above, the growth model to estimate DBH is divided based on the family of trees such as *Dipterocarpaceae* and *non-Dipterocarpaceae*. At the same time, other parameters used a similar growth model for all species, including wood density. The estimation used the average wood density in Indonesia from Manuri et al. (2017) for all species used due to a lack of information on wood density for each species in Indonesia.

To perform ex-ante estimation for biomass growth, there are several steps need to be done:

- Estimate DBH growth for each species by using a selected allometric equation
- Calculation of above-ground biomass per tree based on DBH growth and using the selected allometric equation
- Calculations of above-ground biomass (t CO₂e) per ha based on tree density for each scheme
- Calculations of below-ground biomass (t CO₂e) per ha based on the AGB value per ha and root-to-shoot ratio in the absence of below-ground biomass equations.
- Calculations of the mean biomass (t CO₂e) per ha.
- Calculations of change in carbon stock in tree biomass per ha

Figure 11 and Figure 12 below explains the total biomass growth per ha for each scheme (agroforestry and Restoration) using equations and parameters provided in Table 56.

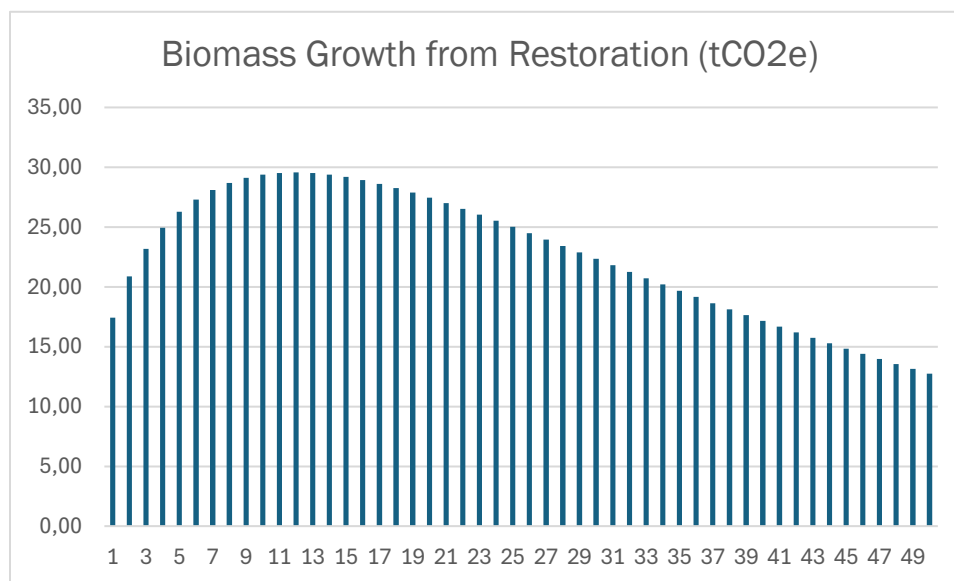


Figure 11. Biomass Growth from Restoration Activity

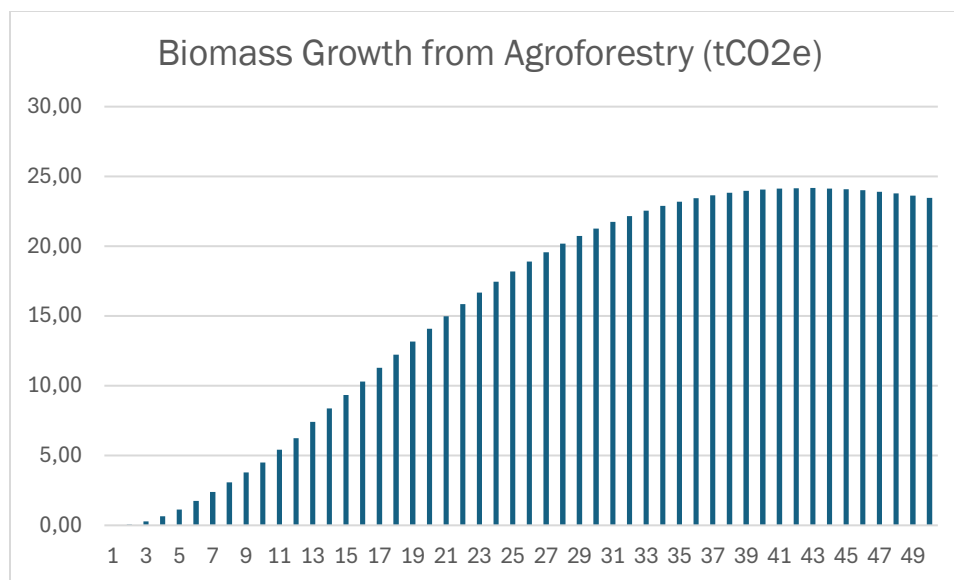


Figure 12. Biomass Growth from Agroforestry Activity

Biomass growth based on the figures above, is used to estimate emission removal based on the eligible area for restoration and agroforestry activities.

b. Non-Woody Biomass

The project includes non-woody biomass as a carbon pool, which is estimated using the stock difference approach according to VM0047. Non-woody species such as paddy and *Piper nigrum* are considered in the agroforestry areas, while understorey vegetation is included for the restoration areas. Additionally, the calculation includes the replanting cycle of *Piper nigrum* every five years, in accordance with the IPCC guidelines for perennial crops. The parameters used in these calculations are detailed in the table below.

Table 57. Parameters used for ex-ante calculation of non-woody biomass

Parameters	Value	Unit	Source
Paddy Field (Total Biomass)	12.36	tdm/ha	2nd FREL
carbon fraction	0.47		IPCC
C to CO2	3.67		IPCC
Understorey	2.70%		2nd FREL
Forest establishment	20	Years	IPCC
<i>Pipper nigrum</i> (Carbon Stock)	21.28	tdm/ha	IPCC
Harvest period	5	Years	IPCC

The non-woody biomass calculation follows a 10-phase planting plan over 10 years. It also considers forest establishment over 20 years, following IPCC guidelines. After 20 years, it is assumed that an equilibrium is reached, and no further changes in the non-woody biomass stock will occur.

Based on the calculations, non-woody species contributed to the removal of 55,848.26 t CO₂-e through agroforestry and restoration activities. The figure below presents a detailed breakdown of the emissions removal by non-woody species.

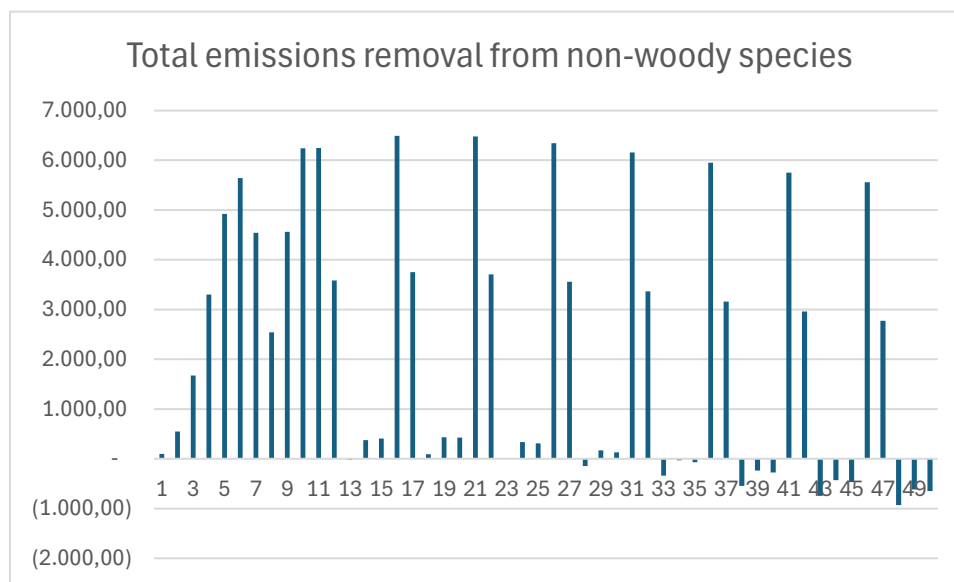


Figure 13. Total emissions removal from non-woody species

Following Appendix 2 in VM0047, the project shall estimate the relative contribution to determine insignificant emissions sources and/or changes in carbon pools. Relative contribution can be estimated following the equation below:

$$RC_{Es} = \frac{E_s}{\sum_{s=1}^S E_s}$$

Where:

- RC_{Es} = Relative contribution of each source s to the sum of project and leakage GHG emissions
- E_s = GHG project emissions, leakage emissions and decreases in carbon pools s
- s = 1,2,3,..., S sources of the project and leakage GHG emissions and decreases in carbon pools

The results of the relative contribution analysis indicate that the emissions removal from non-woody species, including both above-ground biomass (AGB) and below-ground biomass (BGB), is less than 5 percent (0.85%), and thus can be considered insignificant. However, it is important to monitor this carbon pool throughout the project implementation, as it could exceed the ex-ante estimates. The detailed of non-woody biomass and also relative contribution estimation can be seen in Annex 16.

c. Deadwood

The project used a stock difference approach to estimate carbon stock change in dead wood for ex-ante estimation. The estimation follows the conditions from IPCC that consider the equilibrium conditions for forest establishment in 20 years which means after 20 years no change in carbon stock is expected for dead wood. In addition to using stock difference approaches, the project also

considers the dead wood from mortality rate for the species that will be planted. The agroforestry scheme only accounts for dead wood resulting from the mortality rate of planted species, as the land used for agroforestry is agricultural and is assumed to contain an absence of vegetation. several parameters considered in the calculation are shown in the table below:

Table 58. Parameters used for ex-ante calculation of dead wood

Parameters	Value	Unit	Source
Shrub	11.72	tdm/ha	Hidayat et al
Tropical Forest	18.2	tdm/ha	IPCC
carbon fraction	0.47		IPCC
C to CO2	3.666666667		IPCC
Forest establishment	20	Years	IPCC

Based on the parameters above, carbon stock change in deadwood is estimated by considering the planting phase for 10 years for restoration and agroforestry. Accordingly, the calculation generates the total carbon from dead wood in ex-ante estimation of 99,010.60 t CO₂-e.

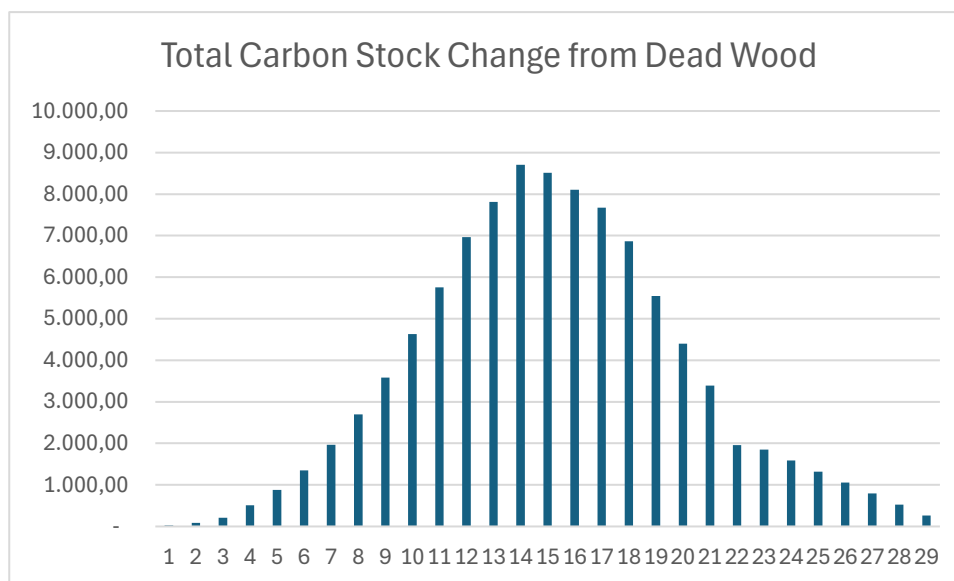


Figure 14. Total Carbon from Deadwood

Following Appendix 2 in VM0047, the project shall estimate the relative contribution to determine insignificant emissions sources and/or changes in carbon pools. Based on relative contribution estimation, total carbon from dead wood is deemed insignificant as the result of a relative contribution of less than 5% (1.50%). For detailed explanation regarding this calculation can be seen in Annex 16.

d. Litter

Litter carbon stock estimated following the provision of VM0047 that compare stock differences from t1 and t+1. the calculation process for this carbon source also considers forest establishment for up to 20 years and several other parameters. Table below describes several parameters used for the calculation process.

Table 59. Parameters used for ex-ante calculation of litter

Parameters	Value	Unit	Source
Shrub	2.84	tdm/ha	Hidayat et al
Tropical Forest	4.47	tdm/ha	IPCC
carbon fraction	0.47		IPCC
C to CO2	3.666666667		IPCC
Forest establishment	20	Years	IPCC

The estimation of litter carbon stock follows a 10-phase planting plan over 10 years. Based on the calculations, litter contributed to the removal of 19,573.41 t CO₂-e through agroforestry and restoration activities. However, the relative contribution analysis indicates that litter's contribution is insignificant, with a relative value of less than 5 percent (0.30). Despite this, continuous monitoring of this carbon pool during project implementation is necessary, as its contribution may exceed the ex-ante estimates.

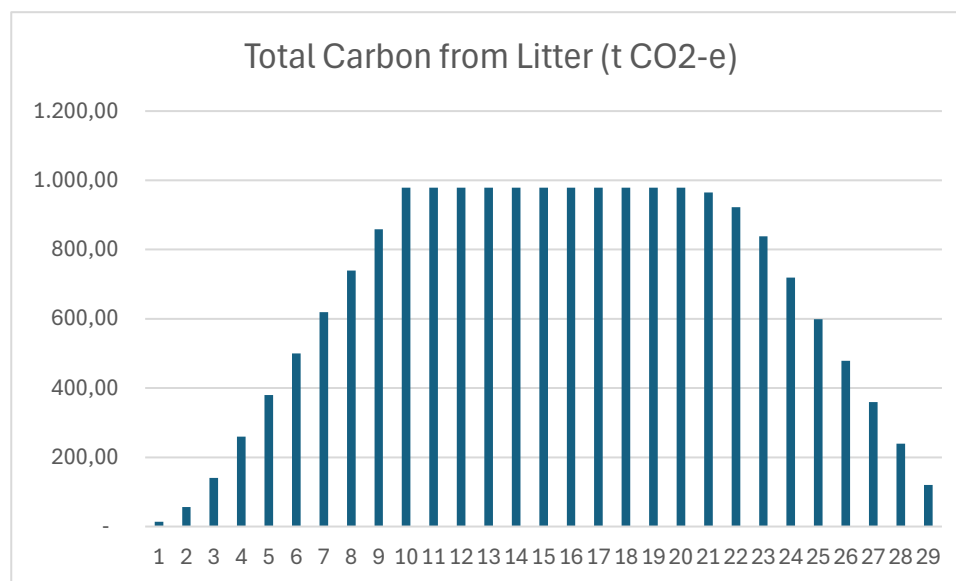


Figure 15. Total carbon from litter

3.2.3 Leakage Emissions (VCS 2.5, 3.2, 3.6, 3.15, 4.3)

3.2.3.1 Leakage Emission from Avoided Planned Deforestation

GHG emissions due to leakage are based on activity shifting and market-effects leakage throughout the project lifetime. Leakage emissions are determined according to the VM0007 equation below:

$$\Delta C_{LK-REDD} = \Delta C_{LK-AS,planned} + \Delta C_{LK-ME}$$

Where:

$\Delta C_{LK-REDD}$ = Net GHG emissions due to leakage from the REDD project activity up to year t*(tCO₂e)

$$\Delta C_{LK-AS,planned} = \text{Net GHG emissions due to activity shifting leakage for projects preventing planned deforestation up to year } t^* \text{ - from Module LK-ASP (tCO}_2\text{e)}$$

$$\Delta C_{LK-ME} = \text{Net GHG emissions due to market-effects leakage up to year } t^* \text{ - from module LK-ME (tCO}_2\text{e)}$$

Activity Shifting Leakage

GHG leakage emissions due to activity shifting from lands legally authorized for conversion to non-forest land are estimated following VMD0009 (LK-ASP) criteria. However, as explained in subsection 3.1.4.1 and according to VMD0009 (LK-ASP), “where the specific deforestation agent can be identified, leakage does not need to be considered where it can be demonstrated that the management plans and/or land-use designations of the deforestation agent’s other lands (which must be identified by location) have not materially changed as a result of the project”. In this project, PT, Citra Mulia Inti is identified as the deforestation agent and does not have other legally designates areas for timber concession beyond the project area. Therefore, in the case of this project the GHG leakage emissions due to activity shifting from forest lands ($\Delta C_{LK-AS,planned}$) will be set to zero.

Market Effects Leakage

According to management plan, the deforestation agent will convert forest area into industrial timber plantation, producing commercial timber products such as plywood. An agreement between Citra Mulia Inti as a deforestation agent and PT Indonesia Wood Investment in 2019 (please see annex 12), indicates that the sawn wood will be exported globally. Module VMD0011 (LK-ME) is used to calculate market-effects leakage for projects expected to substantially and permanently reduce wood harvest levels. However, it only accounts for market effects in the local market, as the VCS Methodology Requirements state that market leakage should be assessed at the national level.⁷² Therefore, Module VMD0011 (LK-ME) is not relevant in this case, as it does not account for the impacts of reduced wood supply on the local market.

Additionally, the performance report on sustainable forest management from MoEF Indonesia.⁷³(2021 – 2023), shows that timber production exceeded the targets over the past 3-years. This indicates that the project’s existence and activities will not decrease overall timber production in Indonesia. Therefore, market leakage is assumed to be zero.

Net GHG Emissions due to leakage from the REDD APD Project Activity

According to explanations above for activity shifting and market effects leakage considering deforestation agent and project activity, thus the net GHG emissions due to leakage ($\Delta C_{LK-REDD}$) for the REDD APD project will be set to zero.

3.2.3.2 Leakage Emission from Afforestation, Reforestation and Revegetation (ARR)

Emissions from leakage are accounted by applying VMD0054 V1.0 Module for Estimating Leakage from ARR activities. This module accounts for leakage related to the displacement of

⁷² <https://verra.org/wp-content/uploads/2023/08/VCS-Methodology-Requirements-v4.4.pdf>

⁷³ <https://phl.menlhk.go.id/publikasi/lapkin>

pre-project agricultural activities. The module proposes a stepwise methodology to determine the leakage that are summarized in the next steps:

- a. Step 1: Determine foregone production in Project Area
- b. Step 2: Determine the impact of leakage mitigation activities
- c. Step 3: Determine amount of new land brought into production
- d. Step 4: Determine change in carbon stocks in new lands brought into production
- e. Step 5: Determine leakage emissions

Step 1: Determine Foregone Production in Project Area

The baseline of agricultural and fuelwood commodity production in the project area must be demonstrated using historical data production records for each commodity during the reference period. Based on field visit assessment and LULC analysis, there are 2 (two) major commodities cultivated by communities, which are paddy and palm oil. Meanwhile, fuel wood commodity production is not found.

Inside the project area, community in remote hamlets, where access to government-subsidized LPG gas is limited, rely on fuelwood as the main source household energy. However, fuelwood is not cultivated and commoditized; rather, it is from the leftover, fallen trunk and existing tree branch. Since there is no formal fuelwood production, no displacement of fuelwood is expected. Additionally, the planting activity under ARR scheme will provide the community with abundant fuelwood supply, further ensuring no disruptions to community energy needs from fuelwood.

Paddy fields serve as their primary source of subsistence crops, with the yield used solely to meet their daily sustenance. In the project scenario, the ARR activity will maintain existing paddy field, with 20% of community-cultivated areas allocated for paddy production. As a result, no displacement of paddy field inside the project area is expected to occur.

The other agricultural commodity identified inside project area is palm oil, consisting of 1.341 ha. But since ARR project activity does not account the already cultivated palm oil area, the displacement of palm oil to the area outside of the project area is expected not to happen.

Result of step 1: No foregone production identified in project area.

Step 2: Determine the Impact of Leakage Mitigation Activities

The project proponent may implement activities outside the project area that reduce leakage in designated leakage mitigation areas. However, since the result of step 1 shows no foregone production identified in project area, step 2 is no longer applicable. Furthermore, leakage mitigation is optional.

Result of step 2: The impact of leakage mitigation activities is not accounted for.

Step 3: Determine Amount of New Land Brought into Production

Since the displaced production of paddy, as determined in Step 1, is zero, no new land is expected to be brought into production outside the project area.

Result of step 3: The amount of new land brought into production is zero.

Step 4: Determine Change in Carbon Stocks in New Lands Brought into Production

Since the amount of new land brought into production is zero (as determined in step 3), this step is no longer applicable. Consequently, the change in carbon stocks in the new land that is brought into production is also zero.

Result of step 4: Change in carbon stock in new land brought into production is zero.

Step 5: Determine Leakage Emissions

Since there is no new land brought into production, and no change in carbon stocks, leakage emissions are consequently also zero.

Result of step 5: Leakage emissions are zero.

In line with the ex-ante scenario, ARR leakage emissions are considered as zero demonstrated by the steps above. Nevertheless, the ARR leakage shall be monitored in the ex-post scenario to accommodate possible leakage occurrence. The detail ARR leakage monitoring plan is available in section 3.3.3.2.

3.2.4 Uncertainty

3.2.4.1 Uncertainty for Avoided Planned Deforestation (APD)

Uncertainty analysis for REDD APD is performed according to VMD0017. The uncertainty in baseline estimations was calculated by evaluating deforestation rates, the stocks in various carbon pools, and changes in carbon stock. Uncertainty in baseline rate of deforestation ($Uncertainty_{BSL,RATE,t}$) is set as zero, as the deforestation rates are based on actual deforestation plans, as per VMD0017 recommendations.

Uncertainty should be expressed as the 95% confidence interval half-width as a percentage of the mean. Uncertainty is first propagated across pools within strata. The uncertainty is calculated as follows:

$$U_{REDD-BSL,SS,i} = \frac{\sqrt{\sum_1^n (U_{REDD-BSL,SS,i,pool\#} \times E_{REDD-BSL,SS,i,pool\#})^2}}{\sum_1^n E_{REDD-BSL,SS,i,pool\#}}$$

Where:

$U_{REDD-BSL,SS,i}$	=	Percentage uncertainty in the combined carbon stocks and greenhouse gas sources in the REDD baseline scenario in stratum i (%)
$U_{REDD-BSL,SS,i,pool\#}$	=	Percentage uncertainty for carbon stocks and greenhouse gas sources in the REDD baseline scenario in stratum i (%)
$E_{REDD-BSL,SS,i,pool\#}$	=	Carbon stock or GHG sources in the REDD baseline scenario (tCO ₂ e)
i	=	1,2,3,... M strata (unitless)

Uncertainty across combined strata must be assessed as follows:

$$U_{REDD-BSL,SS} = \frac{\sqrt{\sum_{i=1}^M (U_{REDD-BSL,SS,i} \times E_{REDD-BSL,SS,i})^2}}{\sum_{i=1}^M E_{REDD-BSL,SS,i}}$$

- $U_{REDD-BSL,SS}$ = Total uncertainty in the combined carbon stocks and greenhouse gas sources in the REDD baseline scenario (%)
- $U_{REDD-BSL,SS,i}$ = Percentage uncertainty in the combined carbon stocks and greenhouse gas sources in stratum i in the REDD baseline scenario (%)
- $E_{REDD-BSL,SS,i}$ = Sum of combined carbon stocks and GHG sources in the REDD baseline scenario (t CO₂-e)
- i = 1,2,3,... M strata (unitless)

Total Uncertainty in REDD APD baseline scenario is estimated according to equation below:

$$Uncertainty_{REDD-BSL,t*} = \sqrt{Uncertainty_{BSL,RATE,t*}^2 + Uncertainty_{REDD-BSL,SS}^2}$$

Where:

- $Uncertainty_{REDD-BSL,t*}$ = Cumulative uncertainty in REDD baseline scenario up to year t* (%)
- $Uncertainty_{BSL,RATE,t*}$ = Cumulative uncertainty in the baseline rate of deforestation up to year t (%)
- $Uncertainty_{REDD-BSL,SS}$ = Total uncertainty in the combined carbon stocks and greenhouse gas sources in the REDD baseline scenario (%)
- t = 1,2,3,...,t* years elapsed since the project start of the project activity (years)

To estimate $U_{REDD-BSL,SS,i}$, this project use the equation from Winrock that expressed uncertainty by using the confidence interval as a percent of the mean.⁷⁴ The following equation is based on the Winrock Guidance to estimate the $U_{REDD-BSL,SS,i}$,

$$Uncertainty = \frac{CI}{\bar{X}}$$

Where:

- CI = Confidence interval at a specific confidence level often 95%
- $\bar{X}$ = Mean

The calculation of $U_{REDD-BSL,SS,i}$ results in a value of 8.33%. Furthermore, $U_{REDD-BSL,SS}$ is estimated based on $U_{REDD-BSL,SS,i}$ and $E_{REDD-BSL,SS,i}$. These result are used to determine $Uncertainty_{REDD-BSL,t*}$. The detailed calculation to estimate $Uncertainty_{REDD-BSL,t*}$ is explained in the table below:

Table 60. Detailed calculations of $Uncertainty_{REDD-BSL,t*}$

Stratum	Project Area	Average Biomass (tdm/ha)	CI 95%	$U_{REDD-BSL,SS,i}$	$E_{REDD-BSL,SS,i}$	$Uncertainty_{BSL,Rate,t}$	$U_{REDD-BSL,SS}$	$Uncertainty_{REDD-BSL,t}$
Forest	9,309.46	300.70	25.05	8.33%	4,828,623	0	8.33%	8.33%

⁷⁴ <https://winrock.org/wp-content/uploads/2018/08/Winrock-Guidance-on-calculating-carbon-stocks.pdf>

Following VMD0017, $Uncertainty_{BSL,Rate,t}$ value is set to zero as the project utilizes a deforestation rate based on the actual deforestation recorded in the management plan. The estimation for $Uncertainty_{REDD-BSL,t*}$ yields a value of 8.33%.

The uncertainty in REDD project scenario is estimated according to VMD0017 with the equation below:

$$U_{REDD-WPS,SS,i} = \frac{\sqrt{\sum_1^n (U_{REDD-WPS,SS,i,pool\#} \times E_{REDD-WPS,SS,i,pool\#})^2}}{\sum_1^n E_{REDD-WPS,SS,i,pool\#}}$$

Where:

- $U_{REDD-WPS,SS,i}$ = Percentage uncertainty in the combined carbon stocks and greenhouse gas sources in the REDD project scenario in stratum i (%)
- $U_{REDD-WPS,SS,i,pool\#}$ = Percentage uncertainty for carbon stocks and greenhouse gas sources in the REDD project scenario in stratum i (%)
- $E_{REDD-WPS,SS,i,pool\#}$ = Carbon stock or GHG sources in stratum i in the REDD project scenario (t CO₂)
- i = 1, 2, 3, ...M strata (unitless)

As no ex-post measurements of carbon pools or GHG sources have been made at this time, uncertainty from these sources is already included in $Uncertainty_{REDD-BSL,t*}$ and $Uncertainty_{REDD-WPS}$ is set to zero. Therefore, the total error in the REDD APD project activity is calculated by equation:

$$\begin{aligned} &NER_{REDD+ERROR} \\ &= \sqrt{\left(Uncertainty_{REDD_{BSL,t*}} \times \Delta C_{BSL-REDD,t*} \right)^2 + \left(Uncertainty_{REDD_{WPS}} \times \Delta C_{WPS-REDD,t*} \right)^2} \\ &\times \left(\frac{1}{\Delta C_{BSL-REDD,t*} + \Delta C_{WPS-REDD,t*}} \right) \end{aligned}$$

Where:

- $NER_{REDD+ERROR}$ = Cumulative uncertainty for the REDD+ (REDD and WRC) project activities up to year t* (%)
- $Uncertainty_{REDD_{BSL,t*}}$ = Cumulative uncertainty in REDD baseline scenario up to year t (%)
- $\Delta C_{BSL-REDD,t*}$ = Net GHG emissions in the REDD baseline scenario up to year t* (t CO₂e)
- $Uncertainty_{REDD_{WPS}}$ = Total uncertainty in the REDD project scenario (%)
- $\Delta C_{WPS-REDD,t*}$ = Net GHG emissions in the REDD project scenario up to year t* (t CO₂e)
- t = 1, 2, 3, ...t* time elapsed since the project start (year)

As the total error in the REDD APD activity shows the value of 8.33% or does not surpass 15% of the $(NER_{REDD+ERROR})$ at the 95% confidence level, then deductions of net emissions reductions are not applied, and Uncertainty deemed to be 0. Detail estimation of the uncertainty can be seen in Annex 16.

3.2.4.2 Uncertainty for Afforestation, Reforestation and Revegetation (ARR)

The uncertainty associated with sample error will be quantified and accounted for through application of quality assurance/quality control (QA/QC) procedures detailed in section 4.1.3. The performance benchmark is assumed to have zero uncertainty. Uncertainty will be calculated by propagating errors associated with estimates of included pools as:

$$UNC_t = MIN (100\%, MAX (0, (\sum_{p=1}^n (U_{p,t=0} \times C_{p,t=0})^2 + \sum_{p=1}^n (U_{p,t} \times C_{p,t})^2)^{\frac{1}{2}} \times (\frac{1}{\Delta C_{WP-biomass,t} + \Delta C_{WP-SOC,t}}) - 10\%))$$

Where:

UNC_t	=	Uncertainty in cumulative removals through year t(percent)
$U_{p,t}$	=	Percentage uncertainty (expressed as 90 percent confidence interval as a percentage of the mean) in carbon stock estimate of pool p (representing woody biomass, non-woody biomass, dead wood, litter and SOC) in the project scenario in year t (percent)
$C_{p,t=0}$	=	Carbon stock estimate of pool p (representing woody biomass, non-woody biomass, dead wood, litter and SOC) in the project scenario in year t (t CO ₂ e)
$\Delta C_{WP-biomass,t}$	=	Change in carbon stock in biomass carbon pools in the project scenario through year t (t C)
$\Delta C_{WP-SOC,t}$	=	Change in carbon stock in SOC in the project scenario through year t (t C)
t	=	1, 2, 3, ..., t years elapsed since the project start date

According to VM0047 v1.0, for ex-ante estimations the project shall use a minimum uncertainty deduction of 10 percent.

3.2.5 Estimated GHG Emission Reductions and Carbon Dioxide Removals (VCS, 3.15, 4.1)

3.2.5.1 Estimated GHG Emissions Reductions from APD

Net GHG emission reduction estimates are based on the equation below:

$$NER_{REDD} = \Delta C_{BSL-REDD} - \Delta C_{WPS-REDD} - \Delta C_{LK-REDD}$$

Where:

NER_{REDD}	=	Total net GHG emission reductions of the REDD project activity up to year t (t CO ₂ e)
$\Delta C_{BSL-REDD}$	=	Net GHG emissions in the REDD baseline scenario up to year t* (t CO ₂ e)
$\Delta C_{WPS-REDD}$	=	Net GHG emissions in the REDD project scenario up to year t* (t CO ₂ e)

$$\Delta C_{LK-REDD} = \text{Net GHG emissions due to leakage from the REDD project scenario up to year } t^* \text{ (t CO}_2\text{e)}$$

Specific procedures for the quantification of the net GHG emissions in the REDD APD baseline scenario, project scenario and GHG emission due to leakage can be found in section 3.2.1, 3.2.2, and 3.2.3, respectively. Results are shown in Table 61 below as the result of the calculation to estimate total net GHG emission reductions of the REDD APD project activity within the project area.

Table 61. Total net GHG emission reductions of the REDD APD project activity

Year	$\Delta C_{BSL-REDD}$	$\Delta C_{WPS-REDD}$	$\Delta C_{LK-REDD}$	NER REDD
0	-	-	-	-
1	934,356	-	-	934,356
2	1,164,121	-	-	1,164,121
3	967,605	-	-	967,605
4	865,522	-	-	865,522
5	537,695	-	-	537,695
6	135,135	-	-	135,135
7	118,689	-	-	118,689
8	118,689	-	-	118,689
9	118,689	-	-	118,689
10	118,689	-	-	118,689
11	92,357	-	-	92,357
12	60,291	-	-	60,291
13	34,667	-	-	34,667
14	12,642	-	-	12,642
15	477	-	-	477
16	-	-	-	-
17	-	-	-	-
18	-	-	-	-
19	-	-	-	-
20	-	-	-	-
21	-	-	-	-
22	-	-	-	-
23	-	-	-	-
24	-	-	-	-
25	-	-	-	-
26	-	-	-	-
27	-	-	-	-
28	-	-	-	-

Year	ΔCBSL-REDD	ΔCWPS-REDD	ΔCLK-REDD	NER REDD
29	-	-	-	-
30	-	-	-	-
31	-	-	-	-
32	-	-	-	-
33	-	-	-	-
34	-	-	-	-
35	-	-	-	-
36	-	-	-	-
37	-	-	-	-
38	-	-	-	-
39	-	-	-	-
40	-	-	-	-
41	-	-	-	-
42	-	-	-	-
43	-	-	-	-
44	-	-	-	-
45	-	-	-	-
46	-	-	-	-
47	-	-	-	-
48	-	-	-	-
49	-	-	-	-
50	-	-	-	-

3.2.5.2 Estimated GHG Emissions Removal From ARR

The procedure for quantifying net GHG emission reductions and removals during the monitoring period will use the area-based approach outlined in the VM0047.

$$CR_t = \left(\left(\Delta C_{WP,t} \times (1 - PB_t) \times (1 - UNC_t) \right) - LK_t \right) - PE_t - \left(\left(\Delta C_{WP,t-1} \times (1 - PB_{t-1}) \times (1 - UNC_{t-1}) \right) - PE_{t-1} - LK_{t-1} \right)$$

Where:

- CR_t = Carbon dioxide removals from the project activity in year t (t CO₂e)
- $\Delta C_{WP,t}$ = Project carbon stock change in year t (t CO₂e)
- PB_t = Performance benchmark for the monitoring interval ending in year t (percent)
- UNC_t = Uncertainty in cumulative removals through year t (percent)

LK_t = Leakage through year t (t CO₂e)
 PE_t = Project emissions from biomass burning and fertilizer in year t (t CO₂e)
 t = 1,2,3,... t years elapsed since the project start date

All process quantification of emission removal from ARR activities already explained in section 3.1.4, 3.2.1, 3.2.2, 3.2.3 and 3.2.4, respectively. Results are shown in Table 62 below as the result of the calculation to estimate total net GHG emission removal of the ARR project activity within the project area.

Table 62. Total net GHG emission reductions of the ARR project activity

Year	$\Delta C_{wp,t}$	PB_t	UNC_t	P_{et}	LK_t	CR_t
2024	1,064	0.31	0.1	0.00	0	663
2025	3,958	0.31	0.1	0.00	0	2,465
2026	10,419	0.31	0.1	0.00	0	6,489
2027	23,468	0.31	0.1	0.00	0	14,616
2028	38,719	0.31	0.1	0.00	0	24,115
2029	55,583	0.31	0.1	0.00	0	34,618
2030	73,695	0.31	0.1	0.00	0	45,898
2031	92,785	0.31	0.1	0.00	0	57,788
2032	112,641	0.31	0.1	0.00	0	70,154
2033	133,084	0.31	0.1	0.00	0	82,887
2034	144,262	0.27	0.1	0.00	0	94,517
2035	152,538	0.27	0.1	0.00	0	99,939
2036	158,872	0.27	0.1	0.00	0	104,089
2037	163,697	0.27	0.1	0.00	0	107,250
2038	167,293	0.27	0.1	0.00	0	109,606
2039	169,862	0.27	0.1	0.00	0	111,289
2040	171,563	0.27	0.1	0.00	0	112,404
2041	172,522	0.27	0.1	0.00	0	113,032
2042	172,843	0.27	0.1	0.00	0	113,242
2043	172,615	0.27	0.1	0.00	0	113,093
2044	171,913	0.32	0.1	0.00	0	105,741
2045	170,802	0.32	0.1	0.00	0	105,057
2046	169,337	0.32	0.1	0.00	0	104,156
2047	167,568	0.32	0.1	0.00	0	103,068
2048	165,537	0.32	0.1	0.00	0	101,819
2049	163,283	0.32	0.1	0.00	0	100,432
2050	160,838	0.32	0.1	0.00	0	98,928
2051	158,233	0.32	0.1	0.00	0	97,326
2052	155,493	0.32	0.1	0.00	0	95,641
2053	152,642	0.32	0.1	0.00	0	93,887

Year	$\Delta C_{wp,t}$	PBt	UNCt	Pet	LKt	CRt
2054	149,701	0.40	0.1	0.00	0	80,534
2055	146,687	0.40	0.1	0.00	0	78,912
2056	143,618	0.40	0.1	0.00	0	77,261
2057	140,508	0.40	0.1	0.00	0	75,588
2058	137,370	0.40	0.1	0.00	0	73,900
2059	134,215	0.40	0.1	0.00	0	72,203
2060	131,053	0.40	0.1	0.00	0	70,502
2061	127,895	0.40	0.1	0.00	0	68,803
2062	124,747	0.40	0.1	0.00	0	67,109
2063	121,617	0.40	0.1	0.00	0	65,425
2064	118,510	0.53	0.1	0.00	0	49,775
2065	115,433	0.53	0.1	0.00	0	48,483
2066	112,391	0.53	0.1	0.00	0	47,205
2067	109,386	0.53	0.1	0.00	0	45,943
2068	106,424	0.53	0.1	0.00	0	44,699
2069	103,506	0.53	0.1	0.00	0	43,474
2070	100,636	0.53	0.1	0.00	0	42,268
2071	97,816	0.53	0.1	0.00	0	41,084
2072	95,048	0.53	0.1	0.00	0	39,921
2073	92,333	0.53	0.1	0.00	0	38,780

3.2.5.3 Calculation of AFOLU Pooled Buffer Account Contributions

State the non-permanence risk rating (%)	12%
Has the non-permanence risk report been attached as either an appendix or a separate document?	☒ Yes ☐ No
For ARR and IFM projects with harvesting, state, in tCO ₂ e, the Long-term Average (LTA).	The project aims to conserve the forest and restore the deforested/degraded area by implementing forest restoration and agroforestry without harvesting activity.
Has the LTA been updated based on monitored data, if applicable?	☐ Yes ☐ No Not applicable
State, in tCO ₂ e, the expected total GHG benefit to date.	
Is the number of GHG credits issued below the LTA?	☐ Yes ☐ No Not applicable

The number of credits to be held in the AFOLU pooled buffer account is determined as a percentage of the total carbon stock benefits, excluding any emissions due to leakage. The project used the AFOLU Non-Permanence risk assessment tool v4.2 (Annex 15. AFOLU Non-Permanence Risk Assessment)

where the overall risk rating of 12% was determined (Table 63). Referring to Table 63, the total risk buffer based on the risk category does not reach 12%. However, according to the AFOLU Non-Permanence Risk Tool V4.2, the minimum risk buffer must be set at 12%, regardless of whether the calculated risk buffer falls below this threshold.

The total estimated buffer pool from the project (REDD APD and ARR) is 1,065,084 t CO₂e for 50 years by using buffer risk 12%.

Table 63. Risk rating from the AFOLU Non-Permanence Risk Assessment

Risk Category		Rating
a)	Internal risk	5.25
b)	External risk	3.00
c)	Natural Risk	1.77
Overall risk rating (a + b + c)		12

Table 64. The Non-Permanence Risk Buffer Withholding from the project.

Year	Buffer APD (t CO ₂ e)	Buffer ARR (t CO ₂ e)	Total Buffer (t CO ₂ e)
0	-	-	-
1	112,122.73	79.53	112,202
2	139,694.47	295.84	139,990
3	116,112.58	778.72	116,891
4	103,862.69	1,753.93	105,617
5	64,523.46	2,893.81	67,417
6	16,216.23	4,154.14	20,370
7	14,242.71	5,507.80	19,751
8	14,242.71	6,934.60	21,177
9	14,242.71	8,418.52	22,661
10	14,242.71	9,946.47	24,189
11	11,082.79	11,342.03	22,425
12	7,234.87	11,992.64	19,228
13	4,160.01	12,490.65	16,651
14	1,517.03	12,870.02	14,387
15	57.23	13,152.71	13,210
16	-	13,354.72	13,355
17	-	13,488.43	13,488
18	-	13,563.79	13,564
19	-	13,589.06	13,589
20	-	13,571.16	13,571
21	-	12,688.86	12,689

Year	Buffer APD (t CO2e)	Buffer ARR (t CO2e)	Total Buffer (t CO2e)
22	-	12,606.83	12,607
23	-	12,498.72	12,499
24	-	12,368.13	12,368
25	-	12,218.25	12,218
26	-	12,051.85	12,052
27	-	11,871.41	11,871
28	-	11,679.11	11,679
29	-	11,476.89	11,477
30	-	11,266.46	11,266
31	-	9,664.04	9,664
32	-	9,469.50	9,469
33	-	9,271.37	9,271
34	-	9,070.59	9,071
35	-	8,867.99	8,868
36	-	8,664.33	8,664
37	-	8,460.25	8,460
38	-	8,256.34	8,256
39	-	8,053.12	8,053
40	-	7,851.05	7,851
41	-	5,973.04	5,973
42	-	5,817.96	5,818
43	-	5,664.60	5,665
44	-	5,513.18	5,513
45	-	5,363.87	5,364
46	-	5,216.82	5,217
47	-	5,072.17	5,072
48	-	4,930.04	4,930
49	-	4,790.50	4,791
50	-	4,653.65	4,654

3.2.5.4 Calculation of Verified Carbon Units

VCUs are calculated by considering VCS non-permanence risk buffer and uncertainty (explained in section 3.2.4) from net emission reduction and removal for this project. This project also considered buffer released assuming that 15% of risk buffer pool credits used to deduct net GHG emissions reduction and removal will be returned to the project proponent every 5 years through risk mitigation

activities, following the conditions defined in the VCS Registration and Issuance Process v4.5 document⁷⁵

Through the consideration of VCS non-permanence risk buffer, uncertainty and buffer released, the total VCUs from REDD APD and ARR activities in 50 years of project crediting period is 7,970,382 t CO₂e with the average 159,408 t CO₂e and average per hectare is 9.79 t CO₂e. Detail VCUs estimation explained in Table 65 below.

Table 65. Total Estimated Verified Carbon Units (VCUs) from All Project Activities

Vintage Period	Estimated Net GHG Emission Reduction	Estimated Net GHG Emission Removal	Total Emission Reduction Removal	Total Risk Buffer 12%	Buffer Released	Estimated Net VCUs
	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e	t CO ₂ e
16 March 2023 to 15 March 2024	0	0	0	0		0
16 March 2024 to 15 March 2025	934,356	663	935,019	112,202		822,817
16 March 2025 to 15 March 2026	1,164,121	2,465	1,166,586	139,990		1,026,596
16 March 2026 to 15 March 2027	967,605	6,489	974,094	116,891		857,203
16 March 2027 to 15 March 2028	865,522	14,616	880,138	105,617		774,522
16 March 2028 to 15 March 2029	537,695	24,115	561,811	67,417	71,205.07	565,598
16 March 2029 to 15 March 2030	135,135	34,618	169,753	20,370		149,383
16 March 2030 to 15 March 2031	118,689	45,898	164,588	19,751		144,837
16 March 2031 to 15 March 2032	118,689	57,788	176,478	21,177		155,300
16 March 2032 to 15 March 2033	118,689	70,154	188,844	22,661		166,182
16 March 2033 to 15 March 2034	118,689	82,887	201,576	24,189	22,706.50	200,094
16 March 2034 to 15 March 2035	92,357	94,517	186,873	22,425		164,449
16 March 2035 to 15 March 2036	60,291	99,939	160,229	19,228		141,002
16 March 2036 to 15 March 2037	34,667	104,089	138,755	16,651		122,105
16 March 2037 to 15 March 2038	12,642	107,250	119,892	14,387		105,505
16 March 2038 to 15 March 2039	477	109,606	110,083	13,210	14,531.88	111,405
16 March 2039 to 15 March 2040	0	111,289	111,289	13,355		97,935
16 March 2040 to 15 March 2041	0	112,404	112,404	13,488		98,915
16 March 2041 to 15 March 2042	0	113,032	113,032	13,564		99,468
16 March 2042 to 15 March 2043	0	113,242	113,242	13,589		99,653
16 March 2043 to 15 March 2044	0	113,093	113,093	13,571	10,080.89	109,603
16 March 2044 to 15 March 2045	0	105,741	105,741	12,689		93,052
16 March 2045 to 15 March 2046	0	105,057	105,057	12,607		92,450
16 March 2046 to 15 March 2047	0	104,156	104,156	12,499		91,657
16 March 2047 to 15 March 2048	0	103,068	103,068	12,368		90,700

⁷⁵ [Registration and Issuance Process, v4.5 \(verra.org\)](#)

Vintage Period	Estimated Net GHG Emission Reduction	Estimated Net GHG Emission Removal	Total Emission Reduction Removal	Total Risk Buffer 12%	Buffer Released	Estimated Net VCUs
	t CO2 e	t CO2 e	t CO2 e	t CO2 e	t CO2 e	t CO2 e
16 March 2048 to 15 March 2049	0	101,819	101,819	12,218	9,560.06	99,161
16 March 2049 to 15 March 2050	0	100,432	100,432	12,052		88,380
16 March 2050 to 15 March 2051	0	98,928	98,928	11,871		87,057
16 March 2051 to 15 March 2052	0	97,326	97,326	11,679		85,647
16 March 2052 to 15 March 2053	0	95,641	95,641	11,477		84,164
16 March 2053 to 15 March 2054	0	93,887	93,887	11,266	8,894.63	91,515
16 March 2054 to 15 March 2055	0	80,534	80,534	9,664		70,870
16 March 2055 to 15 March 2056	0	78,912	78,912	9,469		69,443
16 March 2056 to 15 March 2057	0	77,261	77,261	9,271		67,990
16 March 2057 to 15 March 2058	0	75,588	75,588	9,071		66,518
16 March 2058 to 15 March 2059	0	73,900	73,900	8,868	7,311.29	72,343
16 March 2059 to 15 March 2060	0	72,203	72,203	8,664		63,538
16 March 2060 to 15 March 2061	0	70,502	70,502	8,460		62,042
16 March 2061 to 15 March 2062	0	68,803	68,803	8,256		60,547
16 March 2062 to 15 March 2063	0	67,109	67,109	8,053		59,056
16 March 2063 to 15 March 2064	0	65,425	65,425	7,851	6,345.31	63,920
16 March 2064 to 15 March 2065	0	49,775	49,775	5,973		43,802
16 March 2065 to 15 March 2066	0	48,483	48,483	5,818		42,665
16 March 2066 to 15 March 2067	0	47,205	47,205	5,665		41,540
16 March 2067 to 15 March 2068	0	45,943	45,943	5,513		40,430
16 March 2068 to 15 March 2069	0	44,699	44,699	5,364	4,622.97	43,958
16 March 2069 to 15 March 2070	0	43,474	43,474	5,217		38,257
16 March 2070 to 15 March 2071	0	42,268	42,268	5,072		37,196
16 March 2071 to 15 March 2072	0	41,084	41,084	4,930		36,154
16 March 2072 to 15 March 2073	0	39,921	39,921	4,791		35,130
16 March 2073 to 15 March 2074	0	38,780	38,780	4,654	4,504.06	38,631
Total						7,970,382
Average Annual ERs						159,408
Average Annual ERs per Ha						9.79

3.3 Monitoring

3.3.1 Data and Parameters Available at Validation (VCS, 3.16)

The table below show all data and parameters that are determined or available at validation and remain fixed throughout the project crediting period. Data and parameters monitored during the operation of the project are included in Section 3.3.2 (Data and Parameters Monitored) below.

3.3.1.1 Avoided Planned Deforestation (APD)

Data / parameter	Forest cover map
Data unit	ha
Description	The map showing the location of forest area within the project area 10 years prior project start date
Source of data	Remote sensing analysis
Value applied	9,307.62
Justification of choice of data or description of measurement methods and procedures applied	The map was produced from remote sensing analysis using high-resolution satellite imagery SPOT-6. Forest cover map is used to identify forest condition within the project area for baseline scenario.
Purpose of data	Identify the eligible areas for avoid planned deforestation
Comments	N/A

Data / parameter	$A_{planned,i}$
Data unit	Ha
Description	Total area of planned deforestation over the fixed baseline period for stratum i
Source of data	Management plan from PT Citra Mulia Inti
Value applied	9,307.62
Justification of choice of data or description of measurement methods and procedures applied	Determine the area of planned deforestation according to the management plan document of PT Citra Mulia Inti which has been authorized by Ministry of Environment and Forestry of

	Indonesia. By following this document, we used legal sources to determine planned deforestation area
Purpose of data	Determination of baseline scenario
Comments	N/A
Data / parameter	$D_{planned}$
Data unit	%
Description	Projected annual proportion of land that will be deforested in stratum i at year t
Source of data	Management Plan of PT Citra Mulia Inti
Value applied	Year 1 = 22% Year 2 = 26.81% Year 3 = 21.81% Year 4 = 18.67% Year 5 = 10.31% Year 6 = 0.40%
Justification of choice of data or description of measurement methods and procedures applied	Determine rate of deforestation according to the management plan document of PT Citra Mulia Inti which has been authorized by Ministry of Environment and Forestry of Indonesia. By following this document, we used legal sources to determine rate of planned deforestation in the project area
Purpose of data	Determination of baseline scenario
Comments	N/A
Data / parameter	$L - Di$
Data unit	%
Description	Likelihood of deforestation in stratum i
Source of data	VMD0006 v1.3
Value applied	1.00
Justification of choice of data or description of	For all areas not both under government control and zoned for deforestation, $L - Di$ must be equal to 1

measurement methods and procedures applied	
Purpose of data	Determination of baseline scenario
Comments	N/A
Data / parameter	$C_{AB,tree,bsl,i}$
Data unit	tCO ₂ /Ha
Description	Carbon stock in aboveground biomass in trees in the baseline in stratum i
Source of data	Forest Biomass Inventory
Value applied	518.21
Justification of choice of data or description of measurement methods and procedures applied	Forestry Inventory is performed according to VMD0001 v1.1
Purpose of data	Calculation of baseline emissions
Comments	N/A
Data / parameter	$C_{AB,tree,post,i}$
Data unit	tCO ₂ /Ha
Description	Carbon stock in aboveground biomass in trees in the post deforestation in stratum i
Source of data	Long Term Average (LTA) analysis
Value applied	Plantation Forest A. <i>cadamba</i> = 83.46 Plantation Forest H. <i>brasiliensis</i> = 35.10
Justification of choice of data or description of measurement methods and procedures applied	VCS AFOLU Guidance for Calculating the Long-term Average Carbon Stock for ARR With Harvesting ⁷⁶
Purpose of data	Calculation of baseline emissions

⁷⁶ [VCS-Guidance-Harvesting-Examples_0.pdf \(verra.org\)](#)

Comments	N/A
Data / parameter	$C_{BB,tree,bsl,i}$
Data unit	tCO ₂ /Ha
Description	Carbon stock in belowground biomass in trees in the post deforestation in stratum i
Source of data	Belowground biomass estimation by using Root-to-shoot ratio
Value applied	150.28
Justification of choice of data or description of measurement methods and procedures applied	Belowground biomass value from each stratum obtained from calculations between $C_{AB,tree,bsl,i}$ and root-to-shoot ratio
Purpose of data	Calculation of baseline emissions
Comments	N/A
Data / parameter	$C_{BB,tree,post,i}$
Data unit	tCO ₂ /Ha
Description	Carbon stock in belowground biomass in trees in the baseline in stratum i
Source of data	Belowground biomass estimation by using Root-to-shoot ratio
Value applied	Plantation Forest <i>A. cadamba</i> = 24.18 Plantation Forest <i>H. brasiliensis</i> = 10.17
Justification of choice of data or description of measurement methods and procedures applied	Belowground biomass value from each stratum obtained from calculations between $C_{AB,tree,post,i}$ and root-to-shoot ratio
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / parameter	<i>R</i>
Data unit	dimensionless
Description	Root to shoot ratio appropriate to forest type applied to calculate belowground biomass per unit area based on aboveground biomass
Source of data	2 nd FREL of Indonesia
Value applied	0.29
Justification of choice of data or description of measurement methods and procedures applied	The root to shoot ratio value used are national reference data available in Indonesia
Purpose of data	Calculation of baseline emissions and project emissions
Comments	N/A

Data / parameter	<i>N</i>
Data unit	dimensionless
Description	Number of sample plots
Source of data	Forest Biomass Inventory
Value applied	66
Justification of choice of data or description of measurement methods and procedures applied	Determining the number of sampling points in the field consist of several working stages as follows: - Stratification of the project area Stratification involves dividing the area into units with similar biophysical characteristics. Satellite imagery data analysis by using SPOT 6 imagery data from 2013 and 2023 was conducted to determine the stratification. Moreover, the historical information on concession permits in the project area was observed as well as comparing SPOT 6 images from the past 10 years and Landsat satellite imagery. - Number of plot determinations The number of sample plots was determined by using Winrock sample plot calculators that are suggested and used by CDM

	standards⁷⁷ and the Government of Indonesia⁷⁸. The determination of the sampling plot number requires information on the standard deviation of AGB value in each stratum. With no information regarding the standard deviation and carbon stock data, the calculation was done by using information from other resources. In this case the total number of sample plots was based on preliminary forest biomass inventory.
Purpose of data	Calculation of baseline emissions and project emissions
Comments	N/A

Data / parameter	$f_j(x,y,...)$
Data unit	t d.m. tree ⁻¹
Description	Allometric equation for species j linking measured tree variable(s) to aboveground biomass of living trees
Source of data	From peer reviewed literature (Manuri et al., 2017)
Value applied	$AGB = 0.171D^{2.564}G^{0.909}$
Justification of choice of data or description of measurement methods and procedures applied	This project used allometric equations from Manuri et al. (2017) based on the approach provides by Krisnawati et al. (2012). Detail explanation and justification for choosing the allometric equation from Manuri et al. (2017) are provided on section 3.2.1.1
Purpose of data	Calculation of baseline emissions
Comments	N/A

Data / parameter	CF
Data unit	t C/t d.m.
Description	Carbon fraction of dry biomass
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories

⁷⁷ [ar-am-tool-03-v2.1.0.pdf \(unfccc.int\)](#)

⁷⁸ Peraturan Menteri Lingkungan Hidup dan Kehutanan Nomor 8 Tahun 2021 tentang Tata Hutan dan Penyusunan Rencana Pengelolaan Hutan, Serta Pemanfaatan Hutan di Hutan Lindung dan Hutan Produksi suggested to use Winrock Terrestrial Sampling Calculator to determine the number of sampling plot.

Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	IPCC is a reputable source approved under the VCS
Purpose of data	Calculation of baseline emissions and project emissions
Comments	

Data / parameter	D_j
Data unit	t d.m. m ⁻³
Description	Wood density appropriate to species/genus/family
Source of data	Global Wood Density Database
Value applied	See Appendix 16 - ER model GHG calculations
Justification of choice of data or description of measurement methods and procedures applied	Wood density is an important parameter used in allometric equations that estimate tree biomass and carbon stocks from stem diameter. Using wood density can reduce uncertainty in the use of allometric biomass equations for predicting above-ground tree biomass.
Purpose of data	Calculation of baseline and project emissions
Comments	

Data / parameter	$COMF_t$
Data unit	dimensionless
Description	Combustion factor for stratum i (vegetation type)
Source of data	IPCC
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	Default values in Table 2.6 of IPCC, 2006 (Appendix 2). The combustion factor is a measure of the proportion of the fuel that is actually combusted, which varies as a function of the size and architecture of the fuel load (i.e., a smaller proportion of

	large, coarse fuel such as tree stems will be burnt compared to fine fuels, such as grass leaves), the moisture content of the fuel and the type of fire (i.e., intensity and rate of spread). Default values must be updated whenever new guidelines are produced by the IPCC
Purpose of data	Calculation of baseline and project emissions
Comments	

Data / parameter	G_{gi}
Data unit	g kg ⁻¹ dry matter burnt
Description	Emission factor for stratum i for gas g
Source of data	IPCC
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	Default can be found in Volume 4, chapter 2, of the IPCC 2006 inventory guidelines in table 2.5 (see Appendix 2: emission factors for various types of burning for CH₄ and N₂O). Default values must be updated whenever new guidelines are produced by the IPCC
Purpose of data	Calculation of baseline and project emissions
Comments	

Data / parameter	GWP_g
Data unit	Dimensionless
Description	Global warming potential for gas g
Source of data	Default factor from the latest IPCC assessment report
Value applied	N/A
Justification of choice of data or description of measurement methods and procedures applied	N/A

Purpose of data	Calculation of baseline and project emissions
Comments	

3.3.1.2 Afforestation, Reforestation and Revegetation (ARR)

Data / parameter	A
Data unit	Ha
Description	Project Area
Source of data	Calculated from Remote Sensing Analysis
Value applied	Project-specific
Justification of choice of data or description of measurement methods and procedures applied	The project area was delineated using remote sensing analysis, ground-truthing, supported by high-resolution satellite imager (SPOT 6) and GIS data coverage. All Remote sensing images and GIS dataset are correctly georeferenced.
Purpose of data	Calculation of project emissions using the area-based quantification approach
Comments	

Data / parameter	R
Data unit	dimensionless
Description	Root to shoot ratio (i.e., ratio of belowground (root) biomass to aboveground biomass, per tree species)
Source of data	2 nd FREL of Indonesia
Value applied	0.29
Justification of choice of data or description of measurement methods and procedures applied	The root to shoot ratio value used are national reference data available in Indonesia. The root-to-shoot ratio value used which is sourced from 2 nd FREL of Indonesia is derived based on National Forest Inventory (NFI) conducted from 1989 – 2014. The NFI process by considering 7 mayor island/regions in Indonesia, including Java, Sumatera, Kalimantan, Maluku, Nusa Tenggara,

	Papua and Sulawesi. Until 2014, there are 3,928 clusters/sampling plots distributed in seven major islands/regions. Kalimantan has the largest plot allocation, with 32.5% or 1,277 clusters. As mentioned above NFI conducted by considering 7 mayor island/regions in Indonesia and based on the explanation from 2nd FREL Indonesia, root-to-shoot ratio determined depending on the type of forest ecosystem in Indonesia, such as: • Dryland Forest • Swamp Forest • Mangrove Forest The root-shoot-ratio value from FREL was selected as it complies with requirement from VM0047 with consider region, and forest ecosystem. Moreover, mostly the cluster that measured is in Kalimantan Island where project area located.
Purpose of data	Calculation of project emissions using the area-based approach
Comments	
Data / parameter	CF
Data unit	t C/t d.m.
Description	Carbon fraction of dry biomass
Source of data	IPCC 2006 Guidelines for National Greenhouse Gas Inventories
Value applied	0.47
Justification of choice of data or description of measurement methods and procedures applied	IPCC is a reputable source approved under the VCS
Purpose of data	Calculation of Project emissions using the area-based approach
Comments	
Data / parameter	EFN_{direct}
Data unit	t N ₂ O-N/t N applied

Description	Emissions factor for direct nitrous oxide emissions from N additions due to synthetic fertilizer, organic amendments, and crop residues
Source of data	Table 11.1, Chapter 11 in Volume 4 of the 2019 Refinement to the 2006 <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.01
Justification of choice of data or description of measurement methods and procedures applied	IPCC is a reputable source approved under the VCS
Purpose of data	Calculation of Project emissions using the area-based approach
Comments	
Data / parameter	EF_{Nleach}
Data unit	t N ₂ O-N/t N leached and runoff
Description	Emissions factor for nitrous oxide emissions from leaching and runoff
Source of data	Table 11.1, Chapter 11 in Volume 4 of the 2019 Refinement to the 2006 <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.011
Justification of choice of data or description of measurement methods and procedures applied	IPCC is a reputable source approved under the VCS
Purpose of data	Calculation of Project emissions using the area-based approach
Comments	
Data / parameter	$Frac_{GASF}$
Data unit	Dimensionless

Description	Fraction of all synthetic N added to soils that volatilizes as NH ₃ and NO _x
Source of data	Table 11.3, Chapter 11 in Volume 4 of the 2019 Refinement to the 2006 <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.11
Justification of choice of data or description of measurement methods and procedures applied	IPCC is a reputable source approved under the VCS
Purpose of data	Calculation of Project emissions using the area-based approach
Comments	
Data / parameter	<i>Frac_{LEACH}</i>
Data unit	Dimensionless
Description	Fraction of synthetic or organic N added to soils that is lost through leaching and runoff
Source of data	Table 11.3, Chapter 11 in Volume 4 of the 2019 Refinement to the 2006 <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.24
Justification of choice of data or description of measurement methods and procedures applied	IPCC is a reputable source approved under the VCS
Purpose of data	Calculation of Project emissions using the area-based approach
Comments	
Data / parameter	<i>EF_{Nvolat}</i>
Data unit	t N ₂ O-N/(t NH ₃ -N + NO _x -N volatilized)

Description	Emissions factor for nitrous oxide emissions from atmospheric deposition of N on soils and water surfaces
Source of data	Table 11.3, Chapter 11 in Volume 4 of the 2019 Refinement to the 2006 <i>IPCC Guidelines for National Greenhouse Gas Inventories</i>
Value applied	0.01
Justification of choice of data or description of measurement methods and procedures applied	IPCC is a reputable source approved under the VCS
Purpose of data	Calculation of Project emissions using the area-based approach
Comments	
Data / parameter	GWP_g
Data unit	Dimensionless
Description	Global warming potential for gas g
Source of data	Default factor from the latest IPCC Assessment Report
Value applied	Most recent IPCC Assessment report
Justification of choice of data or description of measurement methods and procedures applied	IPCC is a reputable source approved under the VCS
Purpose of data	Calculation of Project emissions using the area-based approach
Comments	
Data / parameter	D_j
Data unit	t d.m. m ⁻³
Description	Wood density appropriate to species/genus/family
Source of data	Global Wood Density Database

Value applied	See Appendix 16 - ER model GHG calculations
Justification of choice of data or description of measurement methods and procedures applied	Wood density is an important parameter used in allometric equations that estimate tree biomass and carbon stocks from stem diameter and height. Using wood density can reduce uncertainty in the use of allometric biomass equations for predicting above-ground tree biomass.
Purpose of data	Calculation of Project emissions using the area-based approach
Comments	

Data / parameter	$p_{j,h}$
Data unit	Unit production, varies by commodity
Description	Production in the project area for commodity j in year h of the historical reference period
Equations	VMD0054 (1)
Source of data	Central Agency of Statistics of West Kalimantan
Value applied	See Appendix 16 - ER model GHG calculations
Justification of choice of data or description of measurement methods and procedures applied	Data from Central Agency of Statistics is the official data from West Kalimantan provincial government that is used for leakage estimation. It is used as historical production data for commodities is currently only available based on data from the Central Agency of Statistics, hence this project uses this data to calculate forgone production.
Purpose of data	Calculation of baseline production in the project area for commodity j in year t
Comments	

Data / parameter	H
Data unit	years
Description	Number of years within historical reference period used to determine baseline production within the project area

Equations	VMD0054 (1), (3)
Source of data	NA
Value applied	3
Justification of choice of data or description of measurement methods and procedures applied	A value between 3 and the number of years for one complete crop rotation where applicable, whichever is greater.
Purpose of data	Calculation of leakage
Comments	
Data / parameter	$OP_{j,h}$
Data unit	Unit of production, varies by commodity
Description	Production units of commodity j in the leakage mitigation area in year h of the historical reference period
Equations	VMD0054 (3)
Source of data	Central Agency of Statistics of West Kalimantan
Value applied	See Appendix 16 - ER model GHG calculations
Justification of choice of data or description of measurement methods and procedures applied	Data from Central Agency of Statistics is the official data from West Kalimantan provincial government used for leakage estimation. It is used as historical production data for commodities is currently only available based on data from the Central Agency of Statistics, hence this project uses this data to calculate forgone production.
Purpose of data	Calculation of baseline production in the leakage mitigation area for commodity j in year t
Comments	

3.3.2 Data and Parameters Monitored (VCS, 3.16)

3.3.2.1 Avoided Planned Deforestation (APD)

Data / parameter	Project Forest Cover Monitoring Map
Data unit	-
Description	Map showing the location of forest land within the project geographic area at the beginning of each monitoring period. If within the project geographic area some forest land is cleared, the benchmark map must show the deforested areas at each monitoring event.
Source of data	Remote sensing analysis in combination with GPS data collected during ground-truthing
Description of measurement methods and procedures to be applied	The minimum map accuracy must be 90% for the classification of forest/non-forest in the remote sensing imagery. If the classification accuracy is less than 90% then the map is not acceptable for further analysis. More remote sensing data and ground truthing data will be needed to produce a product that reaches the 90% minimum mapping accuracy
Frequency of monitoring/recording	Prior to every verification
Value applied	N/A
Monitoring equipment	N/A
QA/QC procedures to be applied	-
Purpose of data	Calculations of baseline and project emissions
Calculation method	-
Comments	Where forest land contains more than one forest class, the map must be stratified into forest classes using module X-STR
Data / parameter	DBH
Data unit	cm

Description	Diameter at breast height of tree
Source of data	Field measurements in sample plots
Description of measurement methods and procedures to be applied	Diameter shall be generally measured at 1.3 m above the ground (about breast height). DBH shall be measured on standing trees outside the bark. In trees with irregular shapes, such as protruding knots, swelling or forking, DBH should be taken at a different point where the irregularity does not affect the measurement, usually conservatively above breast height. DBH shall be measured in all trees within the plot. The detail about DBH and how to measure it, explained in Annex 13 Forest Inventory SOP
Frequency of monitoring/recording	Monitoring must occur at least every ten years for baseline renewal.
QA/QC procedures to be applied	Adopted QA/QC procedures from national forest monitoring
Purpose of data	Calculations of baseline and project emissions
Calculation method	N/A
Comments	

Data / parameter	N
Data unit	Dimensionless
Description	Number of sample plots
Source of data	Recording and archiving of number of sample plots
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	Monitoring must occur for at least ten years for baseline renewal.
QA/QC procedures to be applied	N/A

Purpose of data	Calculations of baseline and project emissions
Calculation method	N/A
Comments	

Data / parameter	$A_{DefPA,i,u,t}$
Data unit	Ha
Description	Area of recorded deforestation in the project area in stratum i converted to land use u in year t.
Source of data	Remote sensing analysis
Description of measurement methods and procedures to be applied	To estimate the change of forest cover within the project area. The benchmark map generated at the t=0 or previous monitoring event will be compared with the latest update of forest cover map. The update of forest cover will be based on remote sensing analysis and satellite imagery with 90% of the image to be free of cloud.
Frequency of monitoring/recording	Must be monitored every 2 years or prior to any verification event
Value applied	N/A
Monitoring equipment	N/A
QA/QC procedures to be applied	
Purpose of data	Calculations of project emissions
Calculation method	N/A
Comments	

Data / parameter	$A_{DegW,i,t}$
Data unit	Ha

Description	Area potentially impacted by degradation processes in stratum i
Source of data	Remote sensing analysis and ground-truthing (field tracking)
Description of measurement methods and procedures to be applied	Composed of a buffer from all access points (access buffer), such as roads and rivers or previously cleared areas. The width of the buffer shall be determined by the depth of degradation penetration as defined as a PRA and/or Limited Degradation Survey output.
Frequency of monitoring/recording	Must be repeated each time the PRA and Limited Degradation Survey indicates a potential for degradation
Value applied	N/A
Monitoring equipment	Global Positioning System tracking device
QA/QC procedures to be applied	N/A
Purpose of data	Calculations of $\Delta CP, Def, PA_{i,t}$ component in project emissions
Calculation method	-
Comments	
Data / parameter	$C_{DegW,i,t}$
Data unit	t CO ₂ -e
Description	Biomass carbon of trees cut and removed through degradation process from plots measured in stratum i at time t
Source of data	Field Measurement
Description of measurement methods and procedures to be applied	The diameter of all tree stumps in the designated plots will be measured and conservatively assumed to be the same as the DBH. If the stump is a large buttress, identify several individuals of the same species nearby and determine a ratio of the diameter at DBH to the diameter of buttress at the same height above ground as the measured stumps. This ratio will be applied to the measured stumps to estimate the likely DBH of the cut tree. The above and below ground carbon stock of each harvested tree must be estimated using the same allometric regression equation and root to shoot ratio used in the module for estimating the carbon

	pool in trees (CP-AB) in the baseline scenario. See detailed guidance in Module CP-AB for aboveground biomass estimation.
Frequency of monitoring/recording	Monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years, examination must occur prior to any verification event
Value applied	N/A
Monitoring equipment	
QA/QC procedures to be applied	N/A
Purpose of data	Calculations of $\Delta CP_{Deg,w,i,t}$ component in project emissions
Calculation method	-
Comments	-

Data / parameter	AP_i
Data unit	Ha
Description	Total area of degradation sample plots in stratum i
Source of data	Field Measurement
Description of measurement methods and procedures to be applied	The sampling plan must be designed using plots systematically placed over the buffer zone so that they sample at least 3% of the area of the buffer zone.
Frequency of monitoring/recording	Monitored at least every 5 years or if verification occurs on a frequency of less than every 5 years, examination must occur prior to any verification event
Value applied	N/A
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculations of $\Delta CP_{DegW,i,t}$ component of the project emissions

Calculation method	-
Comments	
Data / parameter	Degradation PRA Results
Data unit	-
Description	Degradation PRA Results
Source of data	PRA assessment
Description of measurement methods and procedures to be applied	The PRA will consist of semi-structured interviews / questionnaires. The PRA shall evaluate whether the following activities may be occurring in the project area: -harvesting of fuel wood -harvesting of wood for charcoal production -timber harvest If $\geq 10\%$ of those interviewed/surveyed believe that degradation may be occurring within the project boundary then the limited on the-ground degradation survey shall be triggered. An additional output of the PRA shall be a depth of penetration of degradation pressure. A maximum distance shall be recorded for penetration into the forest from access points (such as roads, rivers, already cleared areas) to harvest fuel wood, charcoal and/or timber. It is likely that differing distances shall exist for each degradation pressure. If multiple pressures exist in the same stratum the deepest depth of penetration shall be used to define Adeg.i.
Frequency of monitoring/recording	Monitored at least every 2 years or if verification occurs on a frequency of less than every 2 years, examination must occur prior to any verification event
Value applied	N/A
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculations of project emissions
Calculation method	-

Comments	
Data / parameter	Result of Limited Degradation Survey
Data unit	-
Description	
Source of data	PRA assessment
Description of measurement methods and procedures to be applied	Sampled by surveying several transect of known length and width across the access-buffer area (equal in area to at least 1% of $A_{DegW,i}$ to check whether new tree stumps are evident or not.
Frequency of monitoring/recording	Must be repeated each time the PRA indicates a potential for degradation
Value applied	N/A
Monitoring equipment	N/A
QA/QC procedures to be applied	N/A
Purpose of data	Calculations of project emissions
Calculation method	-
Comments	

Data / parameter	$A_{DistPA,q,i,t}$
Data unit	Ha
Description	Area impacted by natural disturbance in the project stratum i converted to natural disturbance stratum q in year t; ha
Source of data	Remote sensing analysis combined with ground-truthing
Description of measurement methods and procedures to be applied	Minimum monitoring unit shall be equal to a minimum of 11 Landsat pixels or one hectare.

Frequency of monitoring/recording	Must be monitored every 2 years or prior to any verification event
QA/QC procedures to be applied	.
Purpose of data	Calculations of project emissions
Calculation method	-
Comments	

Data / parameter	$A_{burn,i,t}$
Data unit	Ha
Description	Area burnt in stratum i in year t
Source of data	Remote sensing
Description of measurement methods and procedures to be applied	Best practices for remote fire area monitoring
Frequency of monitoring/recording	Prior to each verification event
QA/QC procedures to be applied	Best practices for remote fire area monitoring
Purpose of data	Calculation of baseline and project emissions
Calculation method	N/A
Comments	To be monitored when accounting of $E_{biomassburn,i,t}$ Is included in the project scenario

3.3.2.2 Afforestation, Reforestation and Revegetation (ARR)

Data / parameter	<i>Sl_{control,t} and Sl_{wp,t}</i>
Data unit	Unspecified
Description	Stocking index in scenario (control plot j or project plot i) at time t
Source of data	SI is an unspecified remote sensing metric that has demonstrated correlation with terrestrial aboveground carbon stocks (e.g., normalized difference fraction index¹⁵ from Landsat imagery, average canopy height derived from LiDAR or percentage canopy cover interpreted from aerial imagery). Variability due to seasonality must be minimized (e.g., by setting a target data collection period at the project start and collecting all monitoring imagery from within that period). Target period should coincide with minimal seasonal phenological variation, and where passive remote sensors are employed, this should coincide with months of lowest cloud cover. Spatial scale: Divide the entire project area into polygons from 0.09 hectares (30 × 30 m) to 10 hectares in area. Polygons must be of equal size with at least 75 percent of each polygon located within the project area boundary. project plots may be represented by individual pixels or aggregates of pixels.
Description of measurement methods and procedures to be applied	
Frequency of monitoring/recording	At least annually or prior to any verification event
Value applied	
Monitoring equipment	GIS, Remote Sensing
QA/QC procedures to be applied	The remote sensing metric applied must: 1) Have significant correlation with terrestrial carbon stocks at least with aboveground biomass that has been previously substantiated with published or peer-reviewed studies 2) Be validated with direct measurements from the project region (collected from within the project ecoregion; ecoregion defined at the biome level¹⁶). The processing and analysis of remote sensing data must apply established best practices, such as

	those found in: Global Forest Observations Initiative (2016). Integration of remote-sensing and ground-based observations for estimation of emissions and removals of greenhouse gases in forests: Methods and guidance from the Global Forest Observations Initiative, edition 2.0. U.N. Food and Agriculture Organization.17 Mitchell, A. L., Rosenqvist, A. & Mora, B. (2017). Current remote sensing approaches to monitoring forest degradation in support of countries measurement, reporting and verification (MRV) systems for REDD+. Carbon Balance and Management, 12, 9. 18
Purpose of data	Selection of control plots and derivation of performance benchmark for the area-based approach
Calculation method	VM0047 appendix 1
Comments	Note that SI may be derived using different remote sensing metrics for the selection of control plots and for monitoring the performance benchmark. It is expected that the same remote sensing technology may not be available for both the historical analysis (selection of control plots) and monitoring ex post. The same remote sensing metric must be used for monitoring SI ex post in both control plots and project sample plots. Where more accurate remote sensing metrics become available over time, the remote sensing metric used for monitoring SI ex post may be changed when: 1) The new metric offers equivalent or better accuracy (in terms of correlation with terrestrial carbon stocks); 2) It is possible to harmonize the new metric with the previous metric, applying procedures from peer-reviewed literature to ensure data continuity and remove sources of misalignment (e.g., geometric, radiometric and/or spectral artifacts) introduced by the new metric; 3) The procedure to harmonize the new metric incorporates temporally coincident observations of both (previous and new) remote sensing metrics from the project and control plots from within an overlap interval of not less than two years, or as prescribed by the procedure.
Data / parameter	CWP-woody-AB,t
Data unit	t C/ha

Description	Average aboveground woody biomass stocks in the project scenario in year t
Source of data	Field measurement
Description of measurement methods and procedures to be applied	Aboveground woody biomass of each sampled woody plant (e.g., tree, shrub) is estimated using published allometric equations applied to one or more measured attributes. Since project activity involves facilitated natural regeneration, equation(s) is chosen from the equations specific to the species. Attributes (e.g., diameter at breast height, total height) incorporated as independent variables in allometric equations must be directly measured in the field. Measurement protocols must be detailed in standard operating procedures.
Frequency of monitoring/recording	Must be monitored every 2 years or prior to any verification event
Value applied	Determined ex ante and monitored ex post
Monitoring equipment	
QA/QC procedures to be applied	The QA/QC procedures will be determined by project proponent and outlined in standard operating procedures governing field data collection
Purpose of data	Calculation of project emissions using the area-based quantification approach
Calculation method	Calculated as the average of sample measurements
Comments	

Data / parameter	$DM_{WP-herb,t}$
Data unit	t d.m./ha
Description	Average non-woody biomass in the project scenario in year t
Source of data	Field measurements
Description of measurement methods	Non-woody biomass is measured via plot-based sampling. Stratification may be employed to improve precision but is not

and procedures to be applied	required. Sample design need to be held constant across all monitoring and verification events. Sample measurement must: 1) Be demonstrated to be unbiased and derived from representative sampling; and 2) Ensure accuracy through adherence to best practice and quality assurance/quality control (QA/QC) procedures (to be determined by the project proponent and outlined in standard operating procedures governing field data collection). Aboveground non-woody biomass is defined as a pool that includes both living and dead non-woody plant mass. All living and dead non-woody biomass is clipped above the soil surface from inside each sample frame. Dry mass is determined either by drying the entire wet sample to a constant weight or by drying a subsample of the wet mass to determine a dry-to-wet mass ratio conversion factor. Because aboveground mass may be highly seasonal, the average pool must be calculated from at least two samples representing the minimum and maximum standing stocks. Alternatively, a conservative estimate of the pool may be determined from a sample taken at the time of minimum standing stock
Frequency of monitoring/recording	Every two year or before verification events
Value applied	
Monitoring equipment	
QA/QC procedures to be applied	
Purpose of data	Calculation of project emissions using the area-based quantification approach
Calculation method	Calculated as the average of sample measurements
Comments	Note that where subsampling is employed to determine a dry-to-wet mass ratio, uncertainty is calculated by treating the sample as a double sample (see parameter Up,t)

Data / parameter	$A_{burn,t}$
Data unit	Ha
Description	Area burned in the monitoring interval ending in year t
Source of data	Calculated from GIS data
Description of measurement methods and procedures to be applied	Delineation of the area burned may use a combination of remote imagery (satellite or aerial photographs) or ground survey data with GPS
Frequency of monitoring/recording	Every two year or before verification events
QA/QC procedures to be applied	Any imagery used must be geo-registered referencing corner points, clear landmarks or other intersection points.
Purpose of data	Calculations of project emissions using the area-based quantification approach
Calculation method	Calculated using GIS
Comments	

Data / parameter	$M_{wp,SF,i,t}$
Data unit	Tonnes of fertilizer
Description	Mass of N- containing synthetic fertilizer applied in the project scenario in the monitoring interval ending in year t
Source of data	Mass of synthetic fertilizer applied in the project, as recorded in land management records
Description of measurement methods and procedures to be applied	Information is monitored via direct consultation with, and substantiated with a written attestation from, the local land manager.
Frequency of monitoring/recording	Every two year or before verification events
Value applied	

Monitoring equipment	N/A
QA/QC procedures to be applied	Any quantitative information (e.g., discrete or continuous numeric variables) on management practices must be supported by one or more forms of documented evidence pertaining to the project and relevant monitoring period (e.g., management logs, receipts or invoices).
Purpose of data	Calculations of emission from nitrogen synthetic fertilizer component in project emissions
Calculation method	N/A
Comments	
Data / parameter	$M_{wp,OF,i,t}$
Data unit	Tonnes of fertilizer
Description	Mass of N- containing organic fertilizer applied in the project scenario in the monitoring interval ending in year t
Source of data	Mass of organic fertilizer applied in the project, as recorded in land management records
Description of measurement methods and procedures to be applied	Information is monitored via direct consultation with, and substantiated with a written attestation from, the local land manager.
Frequency of monitoring/recording	Every two year or before verification events
Value applied	
Monitoring equipment	N/A
QA/QC procedures to be applied	Any quantitative information (e.g., discrete or continuous numeric variables) on management practices must be supported by one or more forms of documented evidence pertaining to the project and relevant monitoring period (e.g., management logs, receipts or invoices).
Purpose of data	Calculations of emission from nitrogen synthetic fertilizer component in project emissions

Calculation method	N/A
Comments	

Data / parameter	$NC_{WP, SF, t}$
Data unit	t N/t fertilizer
Description	N content of synthetic fertilizer applied in the project in year t
Source of data	N content is determined following fertilizer manufacturer's specifications
Description of measurement methods and procedures to be applied	Not directly measured. Recorded from fertilizer manufacturer's specifications and evidenced in management records, receipts, or invoices.
Frequency of monitoring/recording	Every two year or before verification events
QA/QC procedures to be applied	Any quantitative information on management practices must be supported by one or more forms of documented evidence pertaining to the project area and relevant monitoring period (e.g., management logs, receipts or invoices).
Purpose of data	Calculations of emission from nitrogen synthetic fertilizer component in project emissions
Calculation method	Not calculated
Comments	

Data / parameter	$NC_{WP, OF, t}$
Data unit	t N/t fertilizer
Description	N content of organic fertilizer applied in the project in year t
Source of data	Published or peer-reviewed data, with preference for more recent data from the project country.
Description of measurement methods	Not directly measured.

and procedures to be applied	
Frequency of monitoring/recording	Must be monitored every 2 years or prior to any verification event
QA/QC procedures to be applied	Data referenced must be published or peer-reviewed.
Purpose of data	Calculations of emission from nitrogen organic fertilizer component in project emissions
Calculation method	Not calculated
Comments	

Data / parameter	$B_{SDW,t}$
Data unit	t d.m./ha
Description	Average biomass of standing deadwood in year t
Source of data	Field measurements
Description of measurement methods and procedures to be applied	Standing dead wood is measured via plot-based sampling. Stratification may be employed to improve precision but is not required. Sample design need not be held constant across all monitoring and verification events. Sample measurement must: 1) Be demonstrated to be unbiased and derived from representative sampling; 2) Ensure accuracy through adherence to best practices and quality assurance/quality control (QA/QC) procedures (to be determined by the project proponent and outlined in standard operating procedures governing field data collection); and 3) Apply fixed size thresholds. For each standing dead wood plant (e.g., tree, shrub), stem volume must be estimated using published allometric equations applied to one or more measured attributes. For project activities involving facilitated natural regeneration or with more than two species in a single stand, equation(s) must be

	chosen from the following as available, listed in descending order of preference: a) Equations specific to the forest type within the same ecoregion or Holdridge life-zone.⁷⁹ as the region in which the project is located or b) Global equations are specific to the forest type. Note that where global allometric equations are used, equations must have been developed from or validated with datasets including direct measurements collected via destructive sampling from within the same ecoregion of Holdridge life-zone as the region in which the project is located. Note that standing dead wood is restricted here to visible aboveground stem (bole) biomass, and must discount any missing portions of the stem (e.g., referencing visible break height in volume estimation). Attributes (e.g., DBH, total height) incorporated as independent variables in allometric equations must be directly measured in the field applying established best practices such as those found in: Kershaw Jr, J.A., Ducey, M.J., Beers, T.W., & Husch, B. (2016). <i>Forest mensuration</i>. John Wiley & Sons. Avery, T.E., & Burkhardt, H.E. (2015). <i>Forest measurements</i>. Waveland Press. Measurement protocols must be detailed in standard operating procedures. Parameter tables for all attributes (e.g., DBH, total height) incorporated as independent variables in allometric equations must be included in the project description under “Data and Parameters Monitored.” Biomass of standing dead wood must be estimated from sampled volumes using published wood densities (specific to the species, genus, family or forest type as available, in descending order of preference) and density reduction factors referencing decomposition states (e.g., procedures per Harmon et al., 2011).
Frequency of monitoring/recording	Must be monitored every 2 years or prior to any verification event
QA/QC procedures to be applied	See “Description of measurement methods and procedures to be applied”
Purpose of data	Calculations of project emissions using the area-based quantification approach

⁷⁹ [ECOSYSTEMS Holdridge present life-zone GeoTiff files](#)

Calculation method	Calculated as the average of sample measurements
Comments	
Data / parameter	$B_{LDW,t}$
Data unit	t d.m./ha
Description	Average biomass of lying deadwood in year t
Source of data	Field measurements
Description of measurement methods and procedures to be applied	Lying dead wood is sampled via line intersect sampling (Van Wagner, 1968; Warren and Olsen, 1964). Perpendicular distance sampling (Ducey et al., 2013; Williams and Gove, 2003; Williams et al., 2005) or other un-biased approaches. Stratification may be employed to improve precision but is not required. Sample design need to be held constant across all monitoring and verification events. Sample measurements must: 1) Be demonstrated to be unbiased and derived from representative sampling; 2) Ensure accuracy through adherence to best practices and quality assurance/quality control (QA/QC) procedures (to be determined by the project proponent and outlined in standard operating procedures governing field data collection); and 3) Apply fixed size thresholds. Protocols must be detailed in standard operating procedures and parameter tables under “Data and Parameters Monitored” for all lying dead wood attributes (e.g., cross-sectional diameter, length) measured and recorded. Biomass of lying dead wood must be estimated from sampled volumes using published wood densities (specific to species, genus, family or forest type, in descending order of preference) and density reduction factors referencing decomposition states (e.g., procedures per Harmon et al., 2011).
Frequency of monitoring/recording	Must be monitored every 2 years or prior to any verification event
QA/QC procedures to be applied	See “Description of measurement methods and procedures to be applied”

Purpose of data	Calculations of project emissions using the area-based quantification approach
Calculation method	Calculated as the average of sample measurements
Comments	

Data / parameter	$DM_{WP-LI,t}$
Data unit	t d.m./ha
Description	Average litter dry mass per hectare in the project scenario in year t
Source of data	Field measurements
Description of measurement methods and procedures to be applied	Litter is measured via destructive sampling. Stratification may be employed to improve precision but is not required. Sample design need not be held constant across all monitoring and verification events. Litter (dead organic surface material of less than 10 cm diameter) is collected from within fixed-area sampling frames, harvested at ground level and dried at 70°C to a constant weight to determine dry weight biomass. In cases where sample bulk is excessive, the green weight of the total sample and of a representative sub-sample are recorded in the field and the sub-sample taken for moisture content determination in the lab (i.e., oven dry weight to green weight ratio), from which the dry weight biomass of the total green weight recorded in the field is estimated. Further guidance is provided in the IPCC (2003) <i>Good Practice Guidance for Land-Use Change and Forestry (GPG-LULUCF)</i>.
Frequency of monitoring/recording	At t = 0 and every two years or before verification events
QA/QC procedures to be applied	Standard QA/QC procedures for soil inventory including field data collection and data management must be applied. Use or adaptation of QA/QC procedures available from published handbooks, such as those published by FAO and available on the FAO Soils Portal ⁸⁰ or from the IPCC (2003) GPG LULUCF is recommended

⁸⁰ [Sampling and laboratory techniques | FAO SOILS PORTAL | Food and Agriculture Organization of the United Nations](#)

Purpose of data	Calculations of project emissions using the area-based quantification approach
Calculation method	Calculated as the average of sample measurements
Comments	Note that where subsampling is employed to determine a dry-to-green weight ratio, uncertainty is calculated by treating the sample as a double sample (see parameter $U_{p,t}$).
Data / parameter	$U_{p,t}$
Data unit	Percent
Description	Percentage uncertainty (expressed as 90 percent confidence interval as a percentage of the mean) in carbon stock estimate of pool p in the project scenario in year t
Equations	See section 3.2.4.2
Source of data	Calculation from sampled field measurements
Description of measurement methods and procedures to be applied	Uncertainty in pools derived from field measurements with 90% confidence interval, calculated as the standard error of the average plot measurements in each stratum multiplied by the t value for the 90% confidence level. Where double or two-phase sampling approaches are used with a wall-to-wall remote sensing metric, parameter $U_{woody,t}$ is represented by model error in the relationship (ratio or regression) between the remote sensing metric and aboveground biomass, referencing the 90% confidence interval of the ratio or 1.645 times the root mean squared error of the regression. Where double or two-phase sampling approaches are employed for non-woody biomass and/or litter (i.e., where subsampling is employed to estimate the dry-to-green weight ratio that is then applied to a sample estimate of green weight (see parameter tables for $DM_{WP-herb,t}$ and $DM_{WP-LI,t}$), parameters $U_{p=non-woody,t}$ and $U_{p=litter,t}$ are calculated by propagating sample error of the green weight estimate and sample error of the estimate of dry-to-green weight ratio.
Frequency of monitoring/recording	Must be monitored every 2 years or prior to any verification event
Value applied	N/A

Monitoring equipment	N/A
QA/QC procedures to be applied	
Purpose of data	Calculation of project emissions using area-based quantification approach
Calculation method	Confidence interval calculated by applying unbiased estimators appropriate to sample design. For example, see Cochran, W.G. (1977). Sampling techniques. John Wiley & Sons.
Comments	
Data / parameter	$MP_{j,t}$
Data unit	Units of production
Description	Monitored production in the project area for commodity j in year t
Equations	VMD0054 (2)
Source of data	Grower records (e.g., management logs, receipts of invoices, logs or files containing machine and /or sensor data), or data derived from remote sensing methods (e.g., satellite imagery, manned aerial vehicle footage, drone imagery) where it is possible to reliably determine requisite information on production with these remote sensing methods
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	At each monitoring events
QA/QC procedures to be applied	
Purpose of data	Calculations of forgone production
Calculation method	N/A
Comments	

Data / parameter	r_j
Data unit	Percent
Description	Annual growth rate of yield for commodity j
Equations	VMD0054 (1), (3)
Source of data	Regional (where available) yield data must be obtained or calculated using published studies or official government statistics referencing the smallest territorial administrative division (e.g., district or municipality) that encompasses all areas of land included in the ARR project activity. Yield data must be disaggregated by the ownership classes for which production data are available. In the case of crop commodities, data from FAOSTAT may be used where published studies or official government statistics are not available. Alternatively, a default value of 0.025(2.5 percent) may be used.
Description of measurement methods and procedures to be applied	When using data from FAOSTAT (for crop commodities only), the following procedures must be applied to calculate the annual growth rate of yield for commodity j: 1) For year t (years elapsed since the project start date), find the yield for commodity j by selecting Group: production; Domains: crops and livestock products; Area: country where ARR project activity is located; Element: yield; and Item: commodity j. 2) Calculate annual growth rate in yield for commodity j as: $r_j = \left(\frac{yield_{j,t}}{yield_{j,t=1}} \right) - 1$
Frequency of monitoring/recording	At each monitoring events
QA/QC procedures to be applied	
Purpose of data	Calculations of baseline commodity production in the project area and the leakage mitigation area
Calculation method	
Comments	Where default value is not applied, the yield data from the year closest to year t must be used unless otherwise justified.

	Yield data for livestock commodities is expressed in terms of stocking rate and may include number of animals, animal units and forage intake units per hectare.
Data / parameter	IS
Data unit	Percent
Description	Share of leakage resulting in increased supply outside the project area
Equations	VMD0054 (6)
Source of data	
Description of measurement methods and procedures to be applied	Default value of 0.75 (75%) for agricultural commodity Where the default value is not applied, evidence justifying the use of a different value must be provided. Such evidence may include government-approved or peer-reviewed studies relevant to the agricultural commodity and region in which the project is located.
Frequency of monitoring/recording	At each monitoring events
QA/QC procedures to be applied	
Purpose of data	Estimation of area of new land brought into production
Calculation method	
Comments	See VMD0054 Appendix 2 for background on default values
Data / parameter	NL_j
Data unit	Percent
Description	Share of increased supply from new land brought into production for commodity j
Equations	VMD0054 (6)
Source of data	Default value of 0.40 (40%) for agricultural commodity

	Where the data value is not applied, evidence justifying the use of a different value must be provided. Such evidence may include government-approved or peer-reviewed studies relevant to the commodity produced and region in which the project is located
Description of measurement methods and procedures to be applied	
Frequency of monitoring/recording	At each monitoring events
QA/QC procedures to be applied	
Purpose of data	Estimation of area of new land brought into production
Calculation method	
Comments	See VMD0054 Appendix 2 for background on default values
Data / parameter	$y_{j,t}$
Data unit	Production units per hectare per year
Description	Yield on new lands brought into production for commodity j in year t
Equations	VMD0054 (6)
Source of data	Government data or published studies Productivity data must be obtained or calculated using official government statistics or published studies where available. For crop commodities, data from FAOSTAT may be used where official government statistics or published studies are not available. Where none of the above sources of data are available, or for subsistence commodities, yield in the project area during the historical reference period may be used.
Description of measurement methods	Data from the closest year to that for which leakage emissions are being calculated must be used.

and procedures to be applied	For non-subsistence commodities, the yield on new land brought into production must reference regional yields for the relevant agricultural commodities subject to foregone production. Where regional data are not available, national data must be used. In the absence of regional and national data for the commodity, the yield in the project area during the historical reference period must be used. In the case of fuelwood production, yields based on the average aboveground biomass growth rates published by the IPCC applicable to the region of the project may be used where regional or national data are not available. For subsistence commodities, the yield of new land brought into production must reference the yield in the project area during the historical reference period. When using data from FAOSTAT (for crop commodities only), the following procedure must be applied. For the year <i>t</i> (number of years elapsed since the project start date), find the yield for commodity <i>j</i> by selecting Group: Production; Domains: Crops and livestock products; Area: Country where ARR project activity is located; Element: Yield; Item: Commodity <i>j</i>
Frequency of monitoring/recording	At each monitoring events
QA/QC procedures to be applied	
Purpose of data	Calculation of area of new land brought into production
Calculation method	N/A
Comments	Where yield data are not available for year <i>t</i>, yield data from the year closest to the year <i>t</i> must be used unless otherwise justified. Yield on new lands brought into production for livestock commodities may be expressed in terms of stocking rate and may include number of animals, animal units and forage intake units per hectare per year

Data / parameter	$\Delta C_{biomass}$
Data unit	t C/ha
Description	The change in forest biomass carbon stocks equal to the regional average stock where the project is located

Equations	VMD0054 (8)
Source of data	Published studies, forest biomass inventory or FREL documents from the country in which the project is located.
Description of measurement methods and procedures to be applied	$\Delta CBiomass$ must reference the following data sources as available in the order listed, (e.g., if two data sources are available the data source nearest the top of the list must be used): • mean carbon stocks in forest biomass in the region; or • mean carbon stocks in forest biomass in the country where the project is located and must be obtained from published studies or official government statistics; or • mean carbon stocks in forest biomass derived from above-ground biomass estimates in Table 3A.1.4 of the IPCC Good Practice Guidance for Land Use, Land-Use Change and Forestry (IPCC 2003) if such information is not available. Determined based on data available at the time of the assessment and reflecting area-weighted mean stocks in above- and belowground biomass, dead wood and litter pools at the nearest date to the assessment date. If biomass is presented in tonnes of dry biomass per hectare, then carbon stocks should be determined using a carbon fraction default value of 0.47. If only aboveground biomass stocks are reported, then belowground biomass stocks should be estimated using allometric equations contained in Table 4.A.4 of IPCC (2003), unless verifiable information is provided to justify a different value. Similarly, if data presented include tree biomass only, then the carbon stocks in deadwood and litter must be estimated by multiplying the sum of aboveground and belowground biomass stocks by a factor of 1.1.
Frequency of monitoring/recording	At each monitoring events
QA/QC procedures to be applied	
Purpose of data	Calculation of leakage emissions
Calculation method	

Comments	
Data / parameter	SOC_{REF}
Data unit	t C/ha
Description	Soil organic carbon (SOC) stock corresponding to the reference condition in native ecosystems by climate region and soil type applicable to the land receiving the displaced activity
Equations	VMD0054 (9)
Source of data	Table 2.3 of the 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Description of measurement methods and procedures to be applied	Where soils vary across the region of production, a weighted average (based on relative commodity production) or the highest, most conservative value may be used
Frequency of monitoring/recording	At each monitoring events
QA/QC procedures to be applied	
Purpose of data	To calculate change in carbon stocks on new lands brought into production
Calculation method	N/A
Comments	
Data / parameter	f_{LU}, f_{MG}, f_{IN}
Data unit	Dimensionless
Description	Relative SOC stock change factors applicable to the displaced production over 20 years for land use, management practices and input respectively
Equations	VMD0054 (9)
Source of data	Table 5.5 (displaced crops, referencing tree crops in the case of fuelwood) and table 6.2 (displaced grazing) of the 2019

	Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	At each monitoring events
QA/QC procedures to be applied	
Purpose of data	Calculation of leakage emissions
Calculation method	N/A
Comments	

3.3.3 Monitoring Plan (VCS, 3.16, 3.20)

3.3.3.1 Organizational Structure, Responsibilities and Competencies

Strategic planning, project activities implementation, on-site management and monitoring are the project proponent's responsibilities. Whenever needed, the project proponent will hire qualified partners to implement and monitor each activity according to the expertise required, under its supervision.

The organizational structure of the project team is depicted in Figure 6 of Section 2.4.1 on Project Governance Structure. The responsibilities of the personnel who will be carrying out monitoring activities are described in Table 66 below.

Table 66. Team Members and Responsibilities

No	Team Member	Responsibilities
1	Board of Directors	Establishes sustainable strategies for the project development, ensuring high-quality standards and compliance with national and international regulations. They implement governance structures for transparent and accountable carbon credit generation and trading, while identifying and managing risks related to project feasibility, regulatory compliance, and carbon market volatility.
2	General Manager Carbon Project	Ensures adherence to the VCS Standard and national regulations, staying updated on changes in carbon market

No	Team Member	Responsibilities
		regulations and implementing necessary adjustments. The GM prepares periodical MRV report and manages the MRV process, ensuring accurate data collection, documentation, and the development of relevant SOPs, while serving as the key contact with Validation and Verification Bodies (VVB). Coordinating with the implementation team to align all activities with agreed TOC and standards, the GM also identifies opportunities for improving project processes and methodologies.
3	General Manager Implementation	Leads and manages the execution of all project activities in accordance with TOC and through PRA approach. The GM oversees all conservation and restoration activities, making necessary adjustments to align with project plans and standards. The GM implements community development strategies, grievance mechanism and ensures close partnership with local communities. The GM also oversees the implementation of agroforestry practices, ensuring they benefit both the environment and local communities, while liaising with local governments, regulatory bodies, and stakeholders to ensure compliance and foster strong relationships. Additionally, they manage budgets and resources allocated for site operations to ensure efficient and effective use.
4	Planning and Restoration Manager	Prepares and submits annual work plans detailing planned activities, timelines, resource, and budget requirements, along with required reports to relevant government authorities. This position will also prepare site operation weekly reports to track and communicate progress. The manager develops and implements planting strategies that align with project objectives and sustainability standards and oversee forest patrol activities to prevent illegal activities and ensure the protection of project areas. The manager also coordinates fire management efforts, including prevention, detection, and response strategies, in collaboration with the Biodiversity and Forest Conservation team.
5	Regional Manager	The Regional Manager is responsible for overseeing community development programs, ensuring effective engagement and sustainable livelihood initiatives. He/She manages biodiversity conservation activities, including habitat and species protection, while also leading fire control and prevention efforts through mitigation strategies and response plans. Additionally, He/She supervises forest patrol operations to combat illegal logging and other unauthorized activities. The role requires close

No	Team Member	Responsibilities
		coordination with local stakeholders, government agencies, and project teams to ensure compliance with conservation and sustainability goals.
6	Agroforestry Production Manager	Develops detailed plans for production activities, considering output quality and quantity, cost, available time, and labour requirements, while scheduling and coordinating production processes to optimize efficiency and meet project goals. The Manager oversees the operation of production facilities such as machineries and warehouse, ensuring adherence to quality control procedures, while recommending and implementing initiatives to improve production efficiency. Additionally, the manager support sales and stay informed in trends and demand, accordingly to the plans production in order to meet contract requirements and market demand.
7	Biodiversity & Forest Conservation Manager	Develops and implements biodiversity monitoring and forest patrol plans in coordination with the forest patrol coordinator, while creating and modifying SOPs and data collection forms. The Sr. Officer prepares and submits detailed reports on biodiversity status, forest conservation efforts, trends, and project impacts, ensuring all documentation complies with VCS standards and national regulations. The Sr. Officer engages with government agencies, NGOs, local universities, and other stakeholders. Additionally, he/she prepares educational modules for local communities and relevant stakeholders on conservation efforts and progress within the project.
8	GIS Officer and MRV officer	Collects, stores, and manages geospatial and monitoring data relevant to conservation and restoration activities, creating detailed maps that illustrate land use, vegetation cover, and biodiversity hotspots within the project areas. The officer utilizes remote sensing technologies to monitor land cover changes, deforestation, and restoration progress, integrating field data with GIS analysis to enhance accuracy. As part of Measurement, Reporting, and Verification (MRV) responsibilities, the officer support in MRV of community's aspect targets along with M&E officer. They develop interactive maps and dashboards for real-time data sharing, supporting decision-making and collaborating closely with biodiversity officers, community development, forest conservation teams, and other field staff.
9	Biodiversity Officer	The officer is in charge of collecting and analysing data on flora and fauna, including species diversity, population dynamics, and

No	Team Member	Responsibilities
		habitat conditions, while maintaining comprehensive databases to support reporting and decision-making. The officer also implements conservation plans to protect threatened species and manage invasive species. Additionally, the officer provides training to project staff and local communities on biodiversity conservation practices.
10	Forest Patrol and Fire Coordinator	The officer develops and implements comprehensive forest patrol and fire management plans to prevent illegal activities and manages slash and burn practices. He prepares detailed reports on illegal activities within the project area and submits them to relevant government bodies and law enforcement agencies. They provide training to patrol and fire response community-based teams on best practices, safety procedures, and equipment use, while ensuring teams have the necessary resources. The coordinator collects and analyses data related to patrol activities and fire events, and engages with local communities, government agencies, and stakeholders to promote collaboration and support. He conducts awareness programs on forest conservation and fire management to the local communities.
11	Agronomist	The agronomist assesses soil health and fertility, recommending soil improvement techniques to enhance productivity and developing sustainable crop management practices, including crop rotation and agroforestry systems. The Agronomist identifies and selects appropriate plant species for forest rehabilitation and agroforestry, considering local climate, soil conditions, and biodiversity goals. Additionally, the agronomist develops and tests innovative techniques to improve crop yield and pest management and implements integrated pest management (IPM) strategies to control pests and diseases in crops and trees without harming the environment.
12	Planting and Agroforestry Officers	Has the responsibility of execute planting plans for agroforestry and reforestation projects, supervising field labour and providing training to local communities on agroforestry practices and planting techniques. They address soil erosion, pest, and disease problems in coordination with the agronomist to ensure optimal growing conditions, and train field teams and local farmers on integrated pest management (IPM) techniques and

No	Team Member	Responsibilities
		safe pesticide use. Additionally, they involve community members in planting activities, fostering local ownership and support, and maintain accurate records of planting schedules, species planted, outcomes, and geo-tagging when applicable.
13	Community Development Sr. Officer	The officer engages with local communities to build trust and foster collaboration for forest conservation and rehabilitation projects, including agroforestry. They implement community development plans that align with the project's Theory of Change (TOC) and address identified needs. Develop educational materials and organize training sessions to enhance community knowledge and skills. They execute community-related activities as outlined in the TOC, ensuring active community participation and ownership. Additionally, they monitor the progress and impact of community development activities, collecting data to evaluate effectiveness and outcomes. The officer also facilitates partnerships and leverage resources to support community initiatives. Organize environmental education programs and campaigns to promote conservation ethics and practices
14	Agro production Officers	In terms of post-harvest processes, such as peeling, drying, sorting, and packaging of agricultural products, implementing quality control measures to ensure product standards are met. They supervise field workers, providing guidance and support to adhere to best practices and safety protocols. They manage the logistic including warehouse and transportation of harvested crops to processing facilities or markets. Additionally, they oversee the safety and efficiency of production machineries.
15	Agro Product Marketing Officer	The officer builds strong relationships with existing customers to ensure high levels of satisfaction and engages with potential customers to expand the client base. The officer negotiates pricing, terms, and contracts to close sales deals.
16	Forest Patrol Junior Officers	The officer conducts regular patrols to monitor and protect forest areas from illegal activities such as logging, poaching, and encroachment, while accurately documenting observations and incidents. They operate and maintain patrol equipment, ensuring it is in good working condition. Additionally, they supervise community-based forest patrol team members.

No	Team Member	Responsibilities
17	Fire Management Junior Officer	Executes fire prevention, detection, and response activities according to established protocols and guidelines, providing training and supervision to community fire brigades on safe practices and active fire response. He/she operates and maintains fire management equipment and distributes educational materials to enhance community knowledge and preparedness.
18	Nursery Staff	Collects and propagates seeds from local and suitable species, providing necessary care to seedlings to promote optimal growth, and maintain nursery facilities. The Staff supervises local field teams in nursery activities and maintain detailed records of all nursery activities, including inventory.
19	Field General Affairs and Finance	Manages the procurement of supplies and equipment for field operations and provides administrative support to field teams. Manage field petty cash. The staff oversees the maintenance and management of field offices and facilities. Additionally, she prepares and submits regular reports on field financial status and administrative activities
20	Village Facilitators	Organize various community development programs, including health, awareness raising activities on conservation and sustainable land management, agroforestry training, and other project-related activities. They monitor and document community participation and the impact of project activities on local livelihoods and environments.
21	Legal Sr. Manager	Ensures regulatory compliance and drafts, reviews, and negotiates contracts related to project activities. He identifies potential legal risks associated with project activities and develop strategies to mitigate these risks. Additionally, He prepare and support the development of internal policies and procedures to ensure legal compliance and best practices in project implementation.
22	Accounting and Finance Supervisor	Prepares and presents financial reports, manages budgets, and oversees day-to-day accounting operations. She also executes financial controls and ensure compliance with relevant accounting and financial standards. Additionally, she will coordinate with external auditor.

3.3.3.2 Avoided Planned Deforestation (APD)

Following VM0007 v1.8, there are 3 main task that will be implemented in monitoring activities, including:

- a. Monitoring Project implementation
- b. Monitoring of actual carbon stock changes and GHG emissions
- c. Ex post calculation of the net reduction of GHG emissions

As explained in section 3.2.3.1 Leakage Emission from Avoided Planned Deforestation from activity shifting and market effects leakage conservatively will be set to zero (see section 3.2.3.1 for a more detailed explanation).

Monitoring Project Implementation

- a) Technical description of monitoring tasks

All project development and execution activities for avoided planned deforestation set out in this document will be monitored in compliance with standards established in accordance with the methodology VM0007 v1.8.

All essential data adopted in the preparation and execution of the project will be documented and archived throughout the 50 years of the project, remaining available for audits and other actors involved in the process.

- b) Data to be collected and collection procedures

The monitoring project implementation shall be collecting data and information related to the geographic position of the project boundary for all areas of land, including project area, land-use/landcover, topography condition and any stratification inside the boundary. All these data may be established, recorded and archived through field survey activities or by conducting remote sensing analysis. Table below described all of the geographic data that shall be collected, along with information on its source, data type, format and procedures to generate the data needed. Please see section 3.3.3.4 for the detailed process of data collection

Table 67. Description of the data collecting procedures in the implementation of the project

Theme	Source	Format	Type	Data collection procedures
Project Area	Field Survey and GIS Spatial analysis	Vector	Primary	Produced by conducting field survey and using GIS tools to analyse the area covered by forest in PT CMI concession for at least 10 years before the project start date.
LULC	Field Survey and Remote sensing Analysis	Vector	Primary	Produced by conducting remote sensing analysis to classify the LULC condition based on

Theme	Source	Format	Type	Data collection procedures
				satellite imagery, 10 years before the project start date
Stratification	spatial analysis	Vector	Primary	Delimited through spatial analysis of the area covered by forest within the project area.
Topography	Remote sensing analysis	Raster	Secondary	Produced from secondary data (Copernicus DEM 30m)

c) Data archiving

All data and information records generated during the project implementation and monitoring will be stored and made available to all involved parties and VCS verifiers for audit purposes. Please see section 3.3.3.4 for the detail explanation of data archiving procedures.

d) Quality assurance and quality control

All procedures and records generated will strictly follow the operational protocols established by the project proponent. Please see section 3.3.3.5 for the detail explanation of quality assurance and quality control procedures for this project.

Monitoring of Actual Carbon Stock Changes and GHG Emissions

The change categories that will be subject to MRV (Monitoring, Reporting and Verification) are “forest area converted to non-forest”, “forest area in the process of reducing carbon stock” and “forest area in process of increasing carbon stock”.

a) Monitoring land-use and landcover change

Monitoring of actual changes in land-use and landcover to estimating carbon stock change in the project area will be carried out using satellite imagery, remote sensing techniques and ground truthing based on field survey, to identifying and quantifying deforestation that occurred during the monitoring period in the project area. Monitoring activities will also be conducted for burned areas (if any fire incidents occur within the project area) and other natural disturbances that would contribute to changes in carbon stocks. Satellite imageries with resolution of ≤ 30 meters (e.g., Planet Lab, Sentinel-1, Sentinel-2, Landsat, etc) with cloud cover $<10\%$ or aerial photographs obtained from field measurements using UAVs (Unmanned Aerial Vehicle) will be used. The satellite imagery to be collected consists of annual satellite images processed by a supervised classification of optical images.

b) Any deforested areas and degradation areas within the project area will be a subject of ground truthing activities by the field team. It will be conducted either by directly visiting the location and/or using UAV if the location is inaccessible. Overall classification accuracy is targeted at $\geq 90\%$ following the requirements of VMD0015 Monitoring C stock from Aboveground and Belowground biomass

The project proponent will conduct monitoring of both aboveground biomass ($C_{AB,tree,i}$) and belowground biomass ($C_{BB,tree,i}$) carbon stock during the monitoring period, which shall be implemented through field measurement and remote sensing analysis. This monitoring activity will be following the requirements of VMD0001 ($CP-AB$) and VMD0015 ($M-REDD$)

The project proponent also need to implement monitoring C stock from $C_{AB,tree,i}$ and $C_{BB,tree,i}$ to reassess the baseline emissions every ten years for ongoing planned deforestation.

DBH parameters will be used to estimate $C_{AB,tree,i}$ by using the chosen allometric equation from Manuri et al. (2017) (see section 3.2.1.1). For belowground biomass estimation, the project will use $C_{AB,tree,i}$ value multiplied by root-to-shoot ratio from 2nd FREL of Indonesia.

- c) Monitoring emissions from potential Degradation, Deforestation, Natural Disturbance and Project Emissions.

Degradation

According to VMD0015, potential degradation from tree extraction for timber and fuelwood will be monitored through PRA activities. The PRA shall consist of semi-structured interviews/questionnaires. The PRA shall evaluate whether the following activities may be occurring in the project area, including harvesting of fuelwood and timber harvesting. If $\geq 10\%$ of those interviewed/surveyed believe that degradation may be occurring within the project boundary then the limited on-the-ground degradation survey shall be triggered.

An additional output of PRA shall be a depth of penetration of degradation pressure. A maximum distance shall be recorded for penetration into the forest from access points (such as roads, rivers, already cleared areas) for the purpose of harvesting fuel wood and/or timber. It is likely that differing distances shall exist for each degradation pressure. If multiple pressures exist in the same stratum the exist in the same stratum the deepest depth of penetration shall be used to define $A_{deg\ w,i}$.

The area subject to degradation ($A_{deg\ w,i}$) shall be sampled by surveying several transects of known length and width across the access-buffer area (equal in area to at least 1% of $A_{deg\ w,i}$) to check whether new tree stumps are evident or not. if there is little to no evidence that trees are being harvested then degradation can be assumed to be zero and no monitoring is needed. As explained in section 3.2.2, the PRA activities shall be repeated every 2 years.

Where the PRA or the limited sampling indicate no degradation occurring:

$$\Delta C_{p,DegW,i,t} = 0$$

Where the PRA and the limited sampling indicate degradation is occurring:

$$\Delta C_{p,DegW,i,t} = A_{DegW,i} \times \frac{C_{DegW,i,t}}{AP_i}$$

Where:

$\Delta C_{p,DegW,i,t}$	=	Net carbon stock changes as a result of degradation in stratum i in the project area in year t; t CO ₂ -e
$A_{DegW,i}$	=	Area potentially impacted by degradation processes in stratum i; Ha
$C_{DegW,i,t}$	=	Biomass carbon of trees cut and removed through degradation process from plots measured in stratum i in year t; t CO ₂ -e
AP_i	=	Total area of degradation sample plots in stratum i; Ha
i	=	1,2,3,..., M strata
t	=	1,2,3,..., t* years elapsed since the start of the project activity

Deforestation

Net carbon stock changes in all pools as a result of deforestation estimated following the equation below:

$$\Delta C_{pools,Def,i,t} = C_{BSL,i} - C_{P,post,i}$$

Where:

- $\Delta C_{pools,Def,i,t}$ = Net carbon stock changes in all pools as a result of deforestation in the project case in land use u in stratum i in year t; t CO₂-e ha⁻¹
- $C_{BSL,i}$ = Carbon stock in all pools in the baseline case in stratum i; t CO₂-e ha⁻¹
- $C_{P,post,i}$ = Carbon stock in all pools in post-deforestation land use u in stratum i; t CO₂-e ha⁻¹
- u = 1,2,3,...,U post -deforestation land uses
- i = 1,2,3,..., M strata
- t = 1,2,3,..., t* years elapsed since the start of the project activity

The wood products are excluded in the calculation following carbon pools considered and described in section 3.1.3.2. Furthermore, the project aims to conserve the forest without conducting harvesting activity to produce the wood product, hence carbon stock from wood product may be neglected.

For each post-deforestation land use(u), carbon stock changes in the selected pools in baseline emissions must be measured and estimated following the equation below:

$$C_{P,post,u,i} = C_{ABtree,i} + C_{BB,tree,i}$$

Where:

- $C_{P,post,u,i}$ = Carbon stock in all pools in post-deforestation land use u in stratum i; t CO₂-e ha⁻¹
- $C_{ABtree,i}$ = Carbon stock in aboveground tree biomass in stratum i; t CO₂-e ha⁻¹
- $C_{BB,tree,i}$ = Carbon stock in belowground tree biomass in stratum i; t CO₂-e ha⁻¹
- u = 1,2,3,...,U post-deforestation land uses
- i = 1,2,3,..., M strata

Natural disturbance

Referring to the explanation in section 3.2.2, the project area for APD falls within the low-risk classification based on the data from InaRisk Geoportal.. The natural disturbance to consider for ex-post calculation is only forest fire and it may be omitted if they are deemed de minimis through the use of tool T-SIG or Appendix 1 of REDD-MF. When the result of T-SIG is significant, the area disturbed ($A_{disPA,q,i,t}$) shall be delineated and the area of each post-disturbance stratum must be delineated. The sum of $A_{disPA,q,i,t}$ shall be equal to the area of overlap between the delineated area of the disturbance and the summed area of planned deforestation in the project area ($D\%_{planned,i,t} * A_{planned,i}$).

$$\Delta C_{P,DistPA,i,t} = \sum_{q=1}^Q (A_{DistPA,q,i,t} \times \Delta C_{P,Dist,q,i,t})$$

Where:

- $\Delta C_{P,DistPA,i,t}$ = Net carbon stock change as a result of natural disturbance in the project case in the project area in stratum i in year t ; t CO₂-e
- $A_{DistPA,q,i,t}$ = Area impacted by natural disturbance in post-natural disturbance stratum q in stratum i , year t ; ha
- $\Delta C_{P,Dist,q,i,t}$ = Net carbon stock changes in pools as a result of natural disturbance in post natural disturbance stratum q in stratum i in year t ; t CO₂-e ha⁻¹
- q = 1,2,3, ... Q post-natural disturbance strata
- i = 1,2,3, M strata
- t = 1,2,3, t^* years elapsed since the start of the project activity

The emission per unit area is equal to the difference between the stocks before and the stocks after the natural disturbance:

$$\Delta C_{P,Dist,q,i,t} = C_{BSL,i} - C_{P,Dist,q,i}$$

Where:

- $\Delta C_{P,Dist,q,i,t}$ = Net carbon stock changes in pools as a result of natural disturbance in the project case in post-natural disturbance stratum q in stratum i in year t ; t CO₂-e ha⁻¹
- $C_{BSL,i}$ = Carbon stock in all pools in the baseline case in stratum i ; t CO₂-e ha⁻¹
- $C_{P,Dist,q,i}$ = Carbon stock in pools in post-natural disturbance strata q in stratum i ; t CO₂-e ha⁻¹
- q = 1,2,3,...., Q post-natural disturbance strata where natural disturbance included fire
- i = 1,2,3,...., M strata
- t = 1,2,3,...., t^* years elapsed since the start of the project activity

For each post-natural disturbance stratum (q) estimate the carbon stock following the natural disturbances. Carbon stocks from all pools must be measured and estimated using the equation below:

$$C_{P,Dist,q,i} = C_{ABtree,i} + C_{BBtree,i}$$

Where:

- $C_{P,Dist,q,i}$ = Carbon stock in all pools in post-natural disturbance q in baseline stratum i ; t CO₂-e ha⁻¹
- $C_{ABtree,i}$ = Carbon stock in aboveground tree biomass in stratum i ; t CO₂-e ha⁻¹
- $C_{BBtree,i}$ = Carbon stock in belowground tree biomass in stratum i ; t CO₂-e ha⁻¹
- q = 1,2,3,...., Q post-natural disturbance strata
- i = 1,2,3,...., M strata

Carbon pools excluded from the project can be accounted as zero.

Project Emissions

The project emissions are defined as emission resulted from biomass burning, fossil fuels combustion and fertilizer use. It will be calculated under the following equation:

$$GHG_{P,E,i,t} = E_{FC,i,t} + E_{BiomassBurn,i,t} + N2O_{direct-N,i,t}$$

Where:

$GHG_{P,E,i,t}$	=	Greenhouse gas emissions as a result of deforestation activities within the project area in the project case in stratum i in year t; t CO ₂ -e
$E_{FC,i,t}$	=	Emissions from fossil fuel combustion in stratum i within the project area in year t; t CO ₂ -e
$E_{BiomassBurn,i,t}$	=	Non-CO ₂ emissions due to biomass burning in stratum i in year t; t CO ₂ -e
$N2O_{direct-N,i,t}$	=	Direct N ₂ O emissions as a result of nitrogen application on the alternative land use in stratum i within the project area in year t; t CO ₂ -e
i	=	1,2,3,..., M strata
t	=	1,2,3,..., t* years elapsed since the start of the project activity

As described in section 3.2.2, , the emissions from biomass burning are set to zero in ex-ante condition. However, in the event of forest fire incident during the implementation of project activities the emissions from biomass burning will be accounted and monitored in accordance to the following steps and equations:.

$$E_{biomassburn,i,t} = \sum_{g=1}^G (((A_{burn,i,t} \times B_{i,t} \times COMF_i \times G_{g,i}) \times 10^{-3}) \times GWP_g)$$

Where:

$E_{biomassburn,i,t}$	=	Greenhouse gas emissions due to biomass burning in stratum i in year t of each GHG (CO ₂ , CH ₄ , N ₂ O) (t CO ₂ -e)
$A_{burn,i,t}$	=	Area burnt for stratum i in year t (ha)
$B_{i,t}$	=	Average aboveground biomass stock before burning stratum i, year (t.d.m. ha ⁻¹)
$COMF_i$	=	Combustion factor for stratum i (unitless)
$G_{g,i}$	=	Emission factor for stratum i for gas g (kg t ⁻¹ d.m.burnt)
GWP_g	=	Global warming potential for gas g (t CO ₂ /t gas g)
g	=	1,2,3,...G greenhouse gases including carbon dioxide ¹ , methane and nitrous oxide(unitless)
i	=	1,2,3,...M strata (unitless)
t	=	1,2,3,...t* time elapsed since the start of the project activity (years)

The average aboveground biomass stock before burning for a particular stratum is estimated as follows:

$$B_{i,t} = (C_{AB_{tree,i,t}} + C_{DW,i,t} + C_{LI,i,t}) \times \frac{12}{44} \times \frac{1}{CF}$$

Where:

$B_{i,t}$	=	Average aboveground biomass stock before burning for stratum i, year t (t.d.m. ha ⁻¹)
$C_{ABtree,i,t}$	=	Carbon stock in above ground biomass in trees in stratum i in year t (t CO ₂ -e ha ⁻¹)
$C_{DW,i,t}$	=	Carbon stock in deadwood for stratum i in year t (t CO ₂ -e ha ⁻¹)
$C_{LI,i,t}$	=	Carbon stock in litter for stratum i in year t (t CO ₂ -e ha ⁻¹)
$\frac{12}{44}$	=	Inverse ratio of molecular weight of CO ₂ to carbon (t CO ₂ -e t C ⁻¹)
CF	=	Carbon fraction of biomass (t C t ⁻¹ d.m.)
i	=	1,2,3,...M strata (unitless)
t	=	1,2,3,...t* time elapsed since the start of the project activity (years)

Emissions from fossil fuel combustion are excluded in the project's GHG emissions (Table 32) and will therefore not be considered in the project scenario (ex-post scenario).

As described in section 3.2.2, missions from nitrogen applications are considered zero. Since the project scenario aiming to conserve the forest area and does not involve any fertilization activities.

d) Frequency of Monitoring

The monitoring of the project implementation will take place at least every two years after the project registration in order to ensure the continuity of the benefits. Periodic verification and quantitative monitoring of the project will take place no longer than two years.

e) Data archiving

All of the results of the monitoring actual carbon changes and GHG emissions during project implementation will be stored and made available to all involved parties and VCS verifiers for audit purposes, following the provisions of the procedure data archiving described on section 3.3.3.4

f) Quality assurance and Quality control

Validation of the LULC used will be carried out using the confusing matrix, to estimate the general success rate period and class. There are two classes will be used: forest and non-forest. The analysis of the accuracy of measuring LULC change will be conducted using the confusion matrix and estimating overall accuracy based on the LULC classification and ground-truthing field assessment. Overall classification accuracy is targeted at ≥90% following the requirements of VMD0015.

For another monitoring results, QA/QC procedures will follow the procedures described in section 3.3.3.5.

Ex-post calculation of the net reduction of GHG emissions

The ex-post calculation of net reductions in anthropogenic GHG emissions is similar to the ex-ante calculation. The only difference being that the ex-post estimation shall be based on the results of monitoring implementation.

The data collection procedures are the same as those described in the previous steps. This step involves compiling data from previous procedures to calculate the ex-post reduction of net GHG emissions.

For this process, the procedures of data archiving and QA/QC will follow the conditions described in the section 3.3.3.4 and 3.3.3.5.

3.3.3.3 Afforestation, Reforestation and Revegetation (ARR)

Quantification Approach

As the project is using the area-based quantification approach, data will be measured using a combination of direct on-the-ground measurements and remote sensing methods and collated during each monitoring period. Direct on-the-ground measurements will be conducted for project plots and then combined with remote sensing analysis. Whereas control plots will use remote sensing analysis for monitoring. A database for those plots will be maintained and made accessible for monitoring purposes. This data will relate to:

- a. A list of project plots including unique IDs, locations, size and configuration and time series of stocking index values from $t = -10$ to $t = 0$ and $t = 0$ to time t
- b. A list of control plots including unique IDs (referencing unique ID of corresponding project plot to which they are matched), locations, size and configuration, weights, and time series of stocking index values from $t = -10$ to $t = 0$ and $t = 0$ to time t
- c. Vegetation Indices (DVI) and time stamps obtained from remote sensing analysis, to derive stocking index values. Remote sensing and field measurements will be conducted to obtain all project data and key parameters as per VM0047 requirements. All data will be aggregated, recorded and stored in the project database. At each monitoring event the data will be reported in the VCS Monitoring Report Template.

Accounting Boundary

The extent of the accounting boundary monitored is described in section 3.1.3.

Parameters to be Measured

The estimation of tree biomass is calculated using allometric equation's specific location in Kalimantan Island (Manuri et al., 2017). For the other parameters such as DBH and Height of tree (used for ex-ante estimation), used the species-specific allometric equations obtained from several peer-reviewed papers. In the measurement of year t , it is necessary to measure the required variables in all trees, such as DBH and height (H) included in the designed sampling plots.

Diameter at Breast Height (DBH) should be measured with a diameter tape, tape measure or any other technique if it provides an accurate assessment. In the case of H, the use of clinometer and laser meter, or any other technique capable of providing an accurate measurement of the variable is recommended. The details for measuring the DBH and Height of trees are described in Annex 13 Forest Inventory SOP.

Data to be Collected and Data Collection Techniques

As referenced in the VCS-ARR methodology, this project used ex-ante estimations for different tree species in year $t=0$. In this estimation, there are 2 strata that were considered to be developed, agroforestry and forest restoration. The stratification for the ARR activities for this monitoring is proposed according to the agroforestry and forest restoration planting design as described in Table 52 and Table 53.

Planting activities in the project implementation will be divided into 10 planting phases conducted annually for each stratum. Based on the planting design of the project, the total tree density is 779 trees/ha (395 trees/ha for forest restoration and 384 trees/ha for agroforestry). Monitoring the growth of trees will be conducted by applying sampling plot that have an area shall be proportionally representative of ARR eligible areas.

The number of plots is determined based on the target precision for biomass stock estimation, as outlined in VM0047 v1.0. For this project, sample plots will be selected using a random or systematic approach to ensure accurate representation of project activities. For monitoring purposes, all parameters used in calculating biomass stocks at plot level (e.g., root-to-shot ratio, wood density) are considered fixed constant. Similarly, all models used for calculating biomass stocks at plot level (e.g., allometric equations to estimate the AGB) are considered exact.

Sampling plots are not only used for measuring the biomass stock but also will be used and analyzed alongside the control plots to estimate performance benchmarks. Referring to section 3.3.2, evaluation of performance benchmarks will combine the field measurements and remote sensing analysis by using Difference Vegetation Index (DVI) as stocking indices derived from satellite imagery.

Frequency of Monitoring

Monitoring the project implementation will take place every 2 years or before verification events after the project registration to ensure continuity of the benefits. Periodic verification and quantitative monitoring of the project will take place no longer than five years.

Schedule and Procedures for Periodically Acquiring, Archiving and Processing Remote Sensing Data to Derive Stocking Index

A monitoring event is scheduled every five years during the project's crediting period. To derive the stock index and evaluate the performance benchmark during each monitoring event it is crucial to account for seasonal conditions around the project area. Monitoring should be conducted during the dry season in order to get more consistent data and avoid a large amount of cloud cover on satellite imagery.

The following steps will be undertaken for each monitoring event:

- a. Review and remove any control plots deemed invalid due to their location in areas no longer matching the project area. If a control plot is deemed invalid, it will be excluded and replaced from the donor pool selected.
- b. Search for satellite imagery (e.g., Landsat, sentinel) acquired during the dry season, conduct quality control on the imagery, derive the DVI value, and calculate the mean DVI (Stock Index) based on time series data for each project and control plot.

- c. Conduct linear regression on the Stocking Index, derive and evaluate slopes: $\Delta SI_{control}$ and $\Delta SI_{wp,t}$
- d. Test the significance of the difference between $\Delta SI_{control}$ and $\Delta SI_{wp,t}$
- e. Derive the performance benchmark.

Detailed Description of the Stocking Index and the Derivation Process

Description of the process for obtaining the Stocking Index using satellite imagery:

- a. Explore satellite imagery (e.g., Landsat, Sentinel) archive to locate suitable data. The orthorectified surface reflectance product, benefiting from atmospheric and radiometric corrections, will be employed to derive the remote sensing metric.
- b. Select the satellite imagery captured during dry season at each monitoring event to reduce the cloud cover.
- c. To ensure the inclusion of high-quality pixels, the input images will be meticulously masked to exclude clouds, shadows, or saturated pixels.
- d. For each selected image, calculate the DVI (Difference Vegetation Index) value as follows:

$$DVI = NIR - RED$$

Where:

DVI = Difference Vegetation Index
NIR = Near Infra-Red Band
RED = Red Band

- e. From the set of DVI images obtained in the previous step, two DVI composites will be generated: one for the current monitoring year and another for an intermediate year between the ongoing monitoring year and the preceding one (or $t=0$ if it is the initial monitoring period). The creation of these composites will employ the median method, which involves multiple individual DVI images into a single composite image by determining the median pixel value for each corresponding pixel location.
- f. The DVI (stocking Index) value will be computed for each control and project plot in order to derive and evaluate the slopes for the time series of the stocking index.
- g. Where SI values for plots within a matched set are not available at time t , (e.g., due to cloud cover or temporary sensor issues), the matched set of project and control plots must not be used in derivation of $\Delta SI_{control,t}$ and $\Delta SI_{wp,t}$ at each time t . To derive the $\Delta SI_{control,t}$ and $\Delta SI_{wp,t}$ shall calculated as the slope of the weighted linear regression of the accumulated time series of SI values for the respective population of plots. The weight of SI control and project plots in time series are calculated as follows (VM0047, equation A3 and A4):

$$W_{control,i,j,t} = W_{control,i,j} \times \frac{1}{\sum_{t=0}^t n_{rs_t}}$$

Where:

$W_{control,i,j,t}$ = Weight of control plot j matched to project plot i at time t (dimensionless)

$W_{control,i,j}$ = Weight of control plot j matched to project plot i (value between 0 and 1; dimensionless)
 n_{rs_t} = Number of project plots and matched control plots (i,j) with values assessed at time t

$$W_{wp,i,t} = \frac{1}{\sum_{t=0}^t n_{rs_t}}$$

Where:

$W_{wp,i,t}$ = Weight of project plot i at time t (dimensionless)
 n_{rs_t} = Number of project plots and matched control plots (i,j) with values assessed at time t

ARR leakage Monitoring Plan

To account for potential ARR leakage during project implementation (ex-post scenario), the steps outlined in VMD0054 will be applied.

Step 1: Determine Foregone Production in Project Area.

In the ex-ante scenario, the foregone production of agricultural and fuelwood commodities is assumed. However, in ex-post scenario, the ARR leakage possibility shall be considered. To assess leakage during monitoring period, the project will consider a minimum 3- year period immediately preceding the project start date. This information will be used to determine the baseline production within the project area. Baseline production is calculated according to (Equation 1 from VMD0054):

$$BP_{j,t} = \frac{\sum_{h=1}^H p_{j,h}}{H} \times (1 + r_j)^t$$

Where:

$BP_{j,t}$ = Baseline production in the project area for commodity j in year t (units of production)
 $p_{j,h}$ = Production in the project area for commodity j in year h of the historical reference period (units of productions) determined by farmers records
 H = Duration of historical reference period (3 years)
 r_j = Annual growth rate of yield for commodity j (percent) set to the default value (2.5 percent)
 t = 1,2,3,...,t years elapsed since the project start date

Foregone production must be calculated for commodities produced in the project area during the historical period. The amount of foregone production in the project area is the difference between baseline production for each commodity produced in the project area and the monitored

production of that same commodity in the project area. Foregone production is calculated according to (Equation 2 from VMD0054):

$$FP_{j,t} = BP_{j,t} - MP_{j,t}$$

Where:

- $FP_{j,t}$ = Foregone production in the project area for commodity j in year t (units of production)
- $BP_{j,t}$ = Baseline production in the project area for commodity j in year t (units of production)
- $MP_{j,t}$ = Monitored production in the project area for up to five years after the project start date for commodity j in year t (units of production)
- t = 1,2,3,...,t years elapsed since the project start date

Step 2: Determine the Impact of Leakage Mitigation Activities

Since this step is optional, the project is designed not to implementation activities outside the project area to reduce leakage. As result, the amount of foregone production that is calculated by equation 5 VMD0054 shall count only $FP_{j,t}$ since $LM_{j,t}$ is zero (equation 5 VMD0054).

$$l_{j,t} = MAX(FP_{j,t} - LM_{j,t}, 0)$$

Where:

- $l_{j,t}$ = Amount of foregone production subject to leakage for commodity j in year t (units of production); minimum value is zero
- $FP_{j,t}$ = Foregone production in the project area for commodity j in year t (units of production)
- $LM_{j,t}$ = Baseline production in the leakage mitigation area for commodity j in year t (units of production)
- t = 1,2,3,...,t years elapsed since the project start date

Step 3: Determine Amount of New Land Brought into Production

Where the amount of displaced production that may result in leakage (l) for commodity j in year t ($l_{j,t}$ as calculated above) is greater than zero, the amount of new land that is brought into production outside the project area in year t is calculated by the following equation (equation 6 VMD0054). In the case that $l_{j,t}$ is zero, this step 3 and the following step 4 and 5 along with their associated parameters to monitor are no longer applicable.

$$INL_{j,t} = \frac{l_{j,t} \times IS \times NL_j}{y_{j,t}}$$

Where:

- $INL_{j,t}$ = Area of new land brought into production in year t (ha)
- $l_{j,t}$ = Amount of foregone production subject to leakage for commodity j in year t (units of production)
- $l_{j,t}$ = Share of leakage resulting in increased supply outside the project area, or default value of 0.75 for agricultural commodities or 1.00 for fuelwood
- NL_j = Share of increased supply from new land brought into production for commodity j, or default value of 0.40 for agricultural commodities or 1.00 for fuelwood
- $y_{j,t}$ = Yield on new land brought into production for commodity j in year t (units of production/ha)
- t = 1,2,3,...,t years elapsed since the project start date

The area of new land brought into production ($INL_{j,t}$) must summed the commodity produced in the project area during the historical period to determine the total area generating leakage emissions. The area leakage emissions are calculated as follows (equation 7):

$$AL_t = \sum_{j=1}^T INL_{j,t}$$

Where:

- AL_t = Area generating leakage emissions in year t (ha)
- $INL_{j,t}$ = Area of new land brought into production for commodity j in year t (ha)
- T = Total number of commodities produced in historical reference period
- t = 1,2,3,...,t years elapsed since the project start date

Step 4: Determine Change in Carbon Stocks in New Lands Brought into Production

According to VMD0054 v1.0, the new land brought into production assumes as forested land. The change in carbon stocks in the new land that is brought into production is calculated as follows:

$$CS = \Delta C_{biomass} + \Delta SOC$$

Where:

- CS = Change in carbon stocks on new lands brought into production (t C/ha)
- $\Delta C_{biomass}$ = Change in forest biomass carbon stocks equal to the in the regional average stock in which the project is located (t C/ha)
- ΔSOC = Change in soil organic carbon (SOC) stocks in the region in which the project is located.

The change in forest biomass carbon stocks in the area generating leakage emissions assumes the complete loss of above and belowground biomass.

Step 5: Determine Leakage Emissions

The leakage emissions from new land that is brought into production and where t does not exceed 5 years beyond the project start date are calculated as follows:

$$LK_t = AL_t \times CS \times \frac{44}{12}$$

Where:

- LK_t = Cumulative leakage up to year t (t CO₂e)
- AL_t = Area generating leakage emissions in year t (ha)
- CS = Change in carbon stocks on new lands brought into production (t C/ha)
- t = 1,2,3, ..., t years elapsed since the project start date, t must not exceed five years beyond the project start date
- $44/12$ = Conversion factor from C to CO₂e

3.3.3.4 Data Collection and Management Process

The project's data collection and management process ensures the integrity of data from field measurements to the final calculation of carbon stock. This is achieved through systematic steps that transform raw data into meaningful information, with integration into a Relational Database Management System (RDBMS). The following outlines the core components of the process:

1. **Field Data Collection** Field data collection is performed by teams utilizing various tools such as GPS devices, biomass measurement tools, remote sensing datasets, and other tools.
2. **Parameter Processing** The raw data collected is processed into specific parameters. The processed data is prepared for entry into a structured system, where each piece of information is associated with an identifiable parameter.
3. **Assignment of Unique IDs** Each parameter is assigned a unique identifier (ID) that serves to classify and track data systematically. The hierarchy of IDs progresses from general to specific, starting from broad elements like a project formula parameters (APD/ARR) ID, parameter ID, grid ID, stratum ID (e.g., vegetation type), and location or time-specific IDs. This structure allows for precise tracing and referencing of each data point in the database.
4. **Data Verification** After assigning IDs, data undergoes a rigorous verification process to ensure accuracy and completeness. This step is crucial for maintaining the integrity of the information before it enters the database. If the data passes the verification, it proceeds to the next stage. However, if discrepancies or issues are found, the data is flagged, and further action is taken:
5. The team in the field is notified to confirm or correct the data.
6. Data may need to be re-collected or corrected before it is re-verified.
7. **Data Input into RDBMS** Verified data is input into a cloud-based spreadsheet system if collected by field teams in project site, or directly into the RDBMS if sourced from satellite data. The integration of field data into the RDBMS ensures that the information is securely stored

and organized. Tables hold the structured data, and relationships between them are established to enable efficient querying and analysis.

8. **Data Retrieval for Carbon Stock Calculation** The RDBMS is utilized to retrieve and query the stored data for carbon stock calculations. This includes pulling relevant information from various tables (e.g., biomass and strata) to compute carbon stocks based on established methodologies, such as those outlined by Verra Methodologies.
9. **Carbon Stock Calculation** Data from the RDBMS is processed using a set of algorithms or formulas to estimate carbon stocks for the project area. The calculation incorporates collected data/parameters to generate accurate carbon estimates.
10. **Reporting and Dashboard Integration** The final carbon stock values are compiled into reports and dashboards. The data is displayed through visual tools that allow project managers and stakeholders to monitor carbon performance, ensuring transparency and accessibility of the project's carbon outcomes.

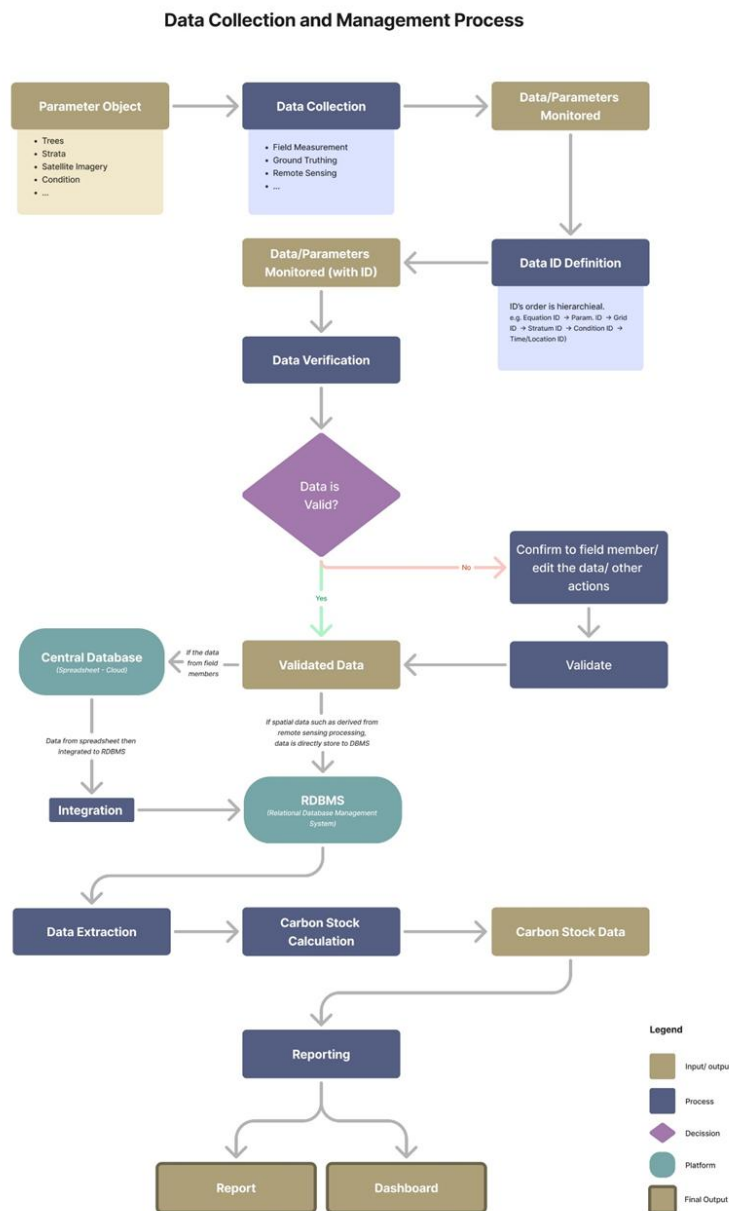


Figure 16. Data collection and management process

3.3.3.5 Data Archiving Procedures

This project is planned to be implemented over 50 years, and data archiving (maintenance and storage) will be an important part of the work. Data archiving will be performed in digital and hardcopy form, and these should be provided and maintained by the project proponent. Furthermore, all field data, data analyses and models, remote sensing data, estimates of the changes in carbon stocks, any GIS products, as well as measuring and monitoring reports should all be stored in a dedicated database in the form of a Relational Database Management System (RDBMS). Throughout data archiving process, it is necessary to comply with QA/QC procedures before data is stored in the database.

The database for this project will be developed with utilization of cloud-based storage and be facilitated through a digital application tailored for field data entry and management. The application enables efficient data capture on mobile devices, seamless synchronization with cloud-based storage, and accessible visualization via a web interface. Data retention is ensured for the entirety of the project's duration. Additionally, field data measurement also will perform manual entry data using a tally sheet, which aims to create backup data. The tally sheet of all field data should be stored in a secure location. Should any issues arise with the database, the data from the tally sheet will be used in preparing the monitoring report.

3.3.3.6 Quality Assurance and Quality Control (QA/QC) Procedures

Throughout the project period, all activities including preparation, field measurement, data processing and calculation, and report finalization will be managed through standard operating procedures and subjected to multiple quality assurance and quality control (QA/QC) assessments. All collected, recorded, stored and reported data will be reviewed and approved by the General manager Carbon Project, General Manager, Carbon Implementation, Planning and Restoration Manager, Agroforestry Production Manager, Biodiversity & Forest Conservation Senior Officer, and Carbon Officer with reference to each level of data management. In order to ensure the security and traceability of data entry and QA/QC, access to all raw, intermediate, and final stages data will be defined by pre-defined access levels for different types of users. Implementation of QA/QC flow will be provided in below.

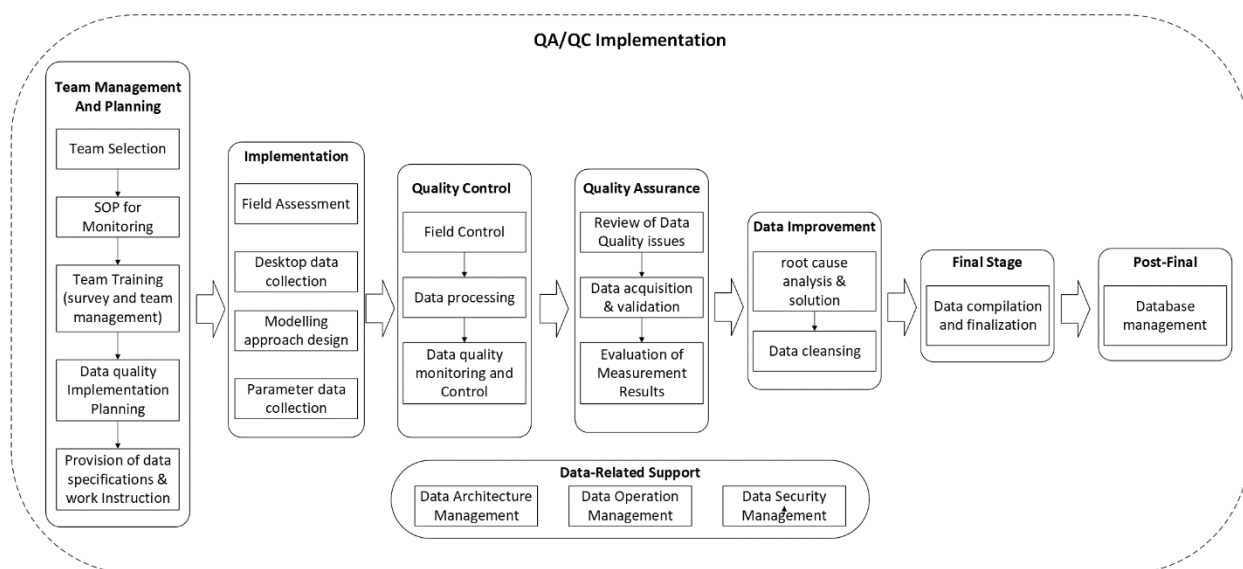


Figure 17. Data collection, QA/QC Implementation and Processing Workflow

As illustrated in Figure 17 above, there are 7 steps for data collection, QA/QC, and processing procedures that will be performed with support by data management, including data architecture, operation and security management. All data monitoring results, including the reports that have gone through the QA/QC process, will be stored in accordance with the data storage procedure until getting approval.

To ensure adherence to established standards, any non-conformances identified during monitoring activities should be addressed through corrective actions. This is crucial to minimize deviations between the actual monitoring activities and the validated monitoring plan. Addressing non-conformances is an integral part of the QA/QC process, which involves several key steps to effectively manage and rectify these issues, including:

- 1) Identification of Non-Conformance
- 2) Communicates with relevant personnel
- 3) Perform root cause analysis
- 4) Recommending corrective action
- 5) Recommending preventative measures
- 6) Document the non-conformance in a non-conformance Report

Non-conformance reports are generated, which documents all the relevant information about the issue. The report should contain details that have been or will/can be taken to mitigate the risks associated with the identified non-conformance. The report contains information such as:

- 1) Issue/non-conformance title
- 2) Description of the issue
- 3) Details of where and when the problem was identified
- 4) Associated risks
- 5) Action taken or recommended
- 6) Preventative measures recommended
- 7) Additional documents and files

The non-conformance report shall be submitted to the Validation/Verification body during the validation/verification events.

3.3.4 Dissemination of Monitoring Plan and Results (VCS, 3.18; CCB, CL4.2)

The dissemination of the monitoring plan and results, including the summary of project documents, will be shared and given access to through various channels. These include regular coordination, printed documents at the project offices and village information boards, and online resources through VERRA and the project website.

Specifically for the communities, project documents, monitoring plan and the results will be disseminated through village meetings within the project zone. These will be held prior to verification and accordingly to the community means of communication and document access.

3.4 Optional Criterion: Climate Change Adaptation Benefits

3.4.1 Regional Climate Change Scenarios (CCB, GL1.1)

Climate change scenarios and climate variability scenarios and impacts

The Indonesian government through Indonesian Agency for Meteorological, Climatological and Geophysics (BMKG)⁸¹ has observed temperature data from 1981 to 2018 at 18 meteorological observation stations spread throughout Indonesia, with two of them on Kalimantan Island. These temperature observations indicate a general increase in average surface temperatures in Indonesia of approximately 0.03 °C per year⁸² (see Figure 18).

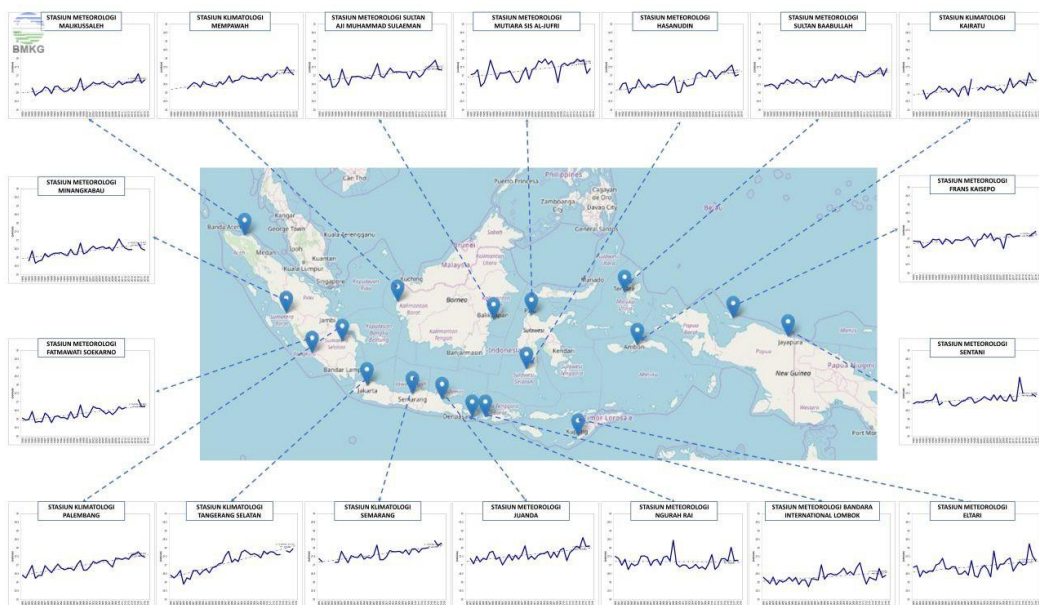


Figure 18. Average temperature trend in Indonesia

BMKG, has also projected temperature changes under scenarios of increasing greenhouse gases in the near-future period (2020-2049) relative to temperatures in 1976-2005. By using the RCP 8.5 scenarios, the projection shows that temperatures in Indonesia will increase⁸³ (see Figure 19).

⁸¹ Indonesian Agency for Meteorological, Climatological and Geophysics or BMKG is an Indonesia non-departmental government agency for meteorology, climatology, and geophysics.

⁸² [Tren Suhu | BMKG](#)

⁸³ [Proyeksi Perubahan Suhu Udara | BMKG](#)

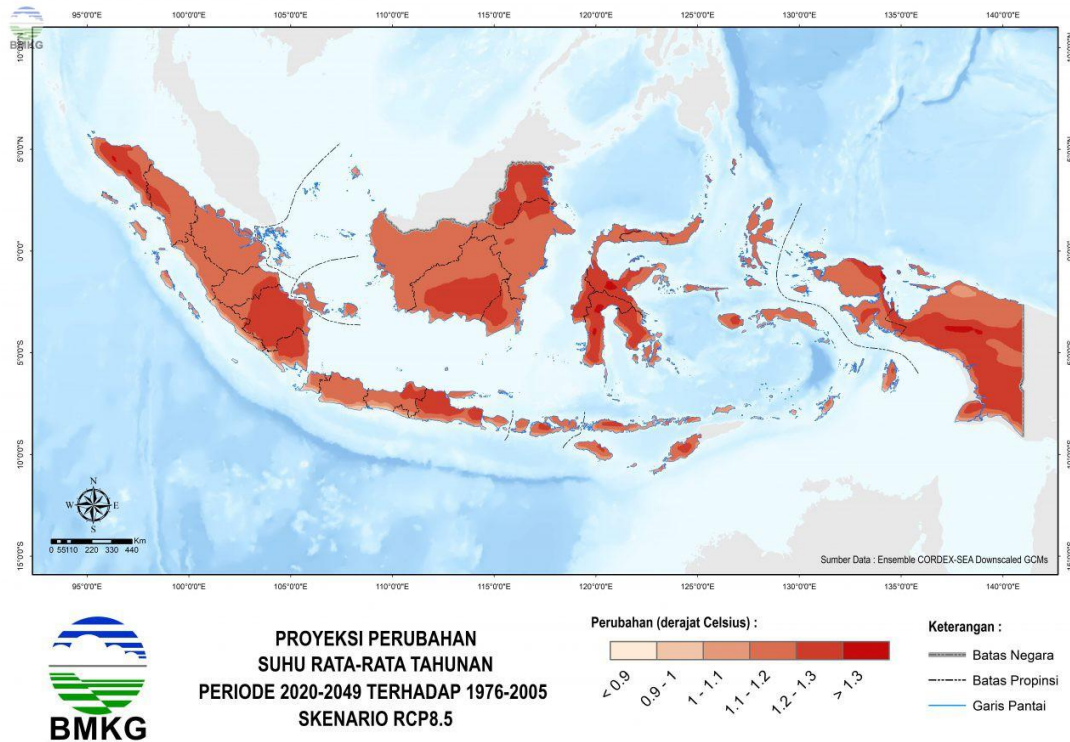
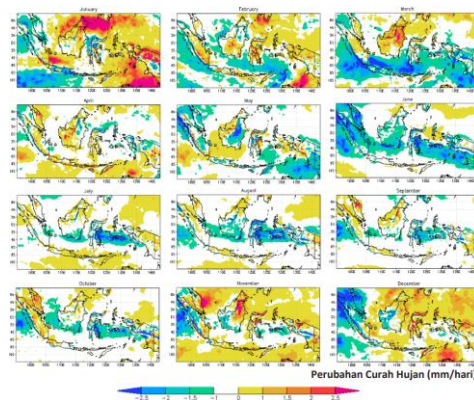
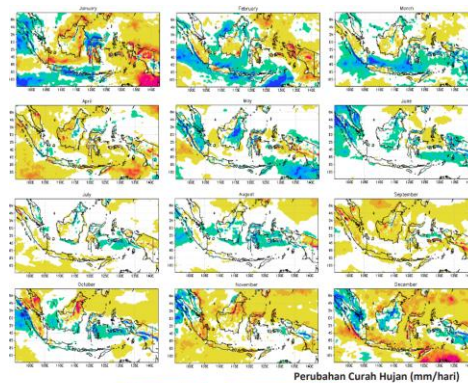


Figure 19. Projected Annual Mean Temperature Change for the Period 2020 – 2049 against 1976 – 2005 Under RCP 8.5 Scenarios

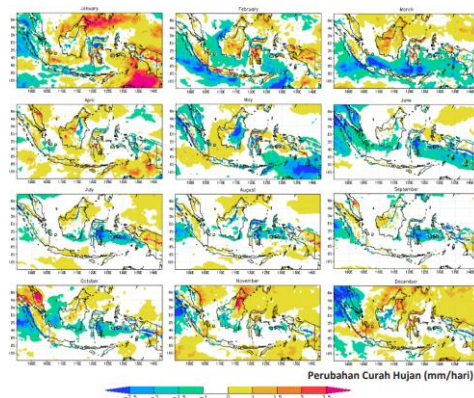
Climate change also affects rainfall patterns. Indonesia uses an ensemble model with the Intergovernmental Panel on Climate Change (IPCC) Scenario to project national rainfall patterns in response to climate change, as agreed upon in the Coordinated Regional Climate Downscaling Experiment South-East Asia (CORDEX-SEA). The IPCC scenarios used in CORDEX-SEA are the RCP4.5 and the RCP8.5 Scenario because they reflect "business as usual" conditions as the lower limit and the worst scenario as the upper limit. In addition to temperatures, these two scenarios also consider the rapidly increasing population, changes in land use and an increasing economy, which contribute to increased GHG emissions.



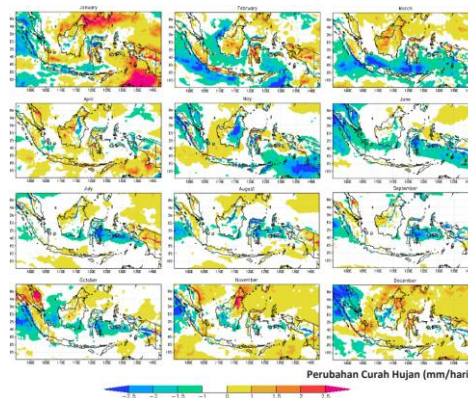
Projection of changes in daily rainfall in Indonesia for the period 2020-2035 relative to historical 1990-2005 using multi-model ensemble output data with the RCP4.5 scenario



Projection of changes in daily rainfall in Indonesia for the period 2030-2045 relative to historical 1990-2005 using multi-model ensemble output data with the RCP4.5 scenario



Projection of changes in daily rainfall in Indonesia for the period 2020-2035 relative to historical 1990-2005 using multi-model ensemble output data with the RCP8.5 scenario



Projection of changes in daily rainfall in Indonesia for the period 2030-2045 relative to historical 1990-2005 using multi-model ensemble output data with the RCP8.5 scenario

Figure 20. Projection of daily rainfall with different scenarios

The modelling above indicates a tendency toward changes in the average seasonal rainfall cycle pattern (see Figure 20). Overall, there is a predominant decrease in the percentage of rainfall across nearly all months. The most pronounced changes occur during the dry season in July and August, where there is a fairly large percentage decrease in rainfall. This indicates the potential for drought to increase during future dry seasons.

Table 68. Percentage (%) change in average monthly rainfall from the regional average throughout Indonesia for each RCP scenario in three 15-year periods calculated relative to the average rainfall data observed for the 1991-2005 period.

Wilayah	2020-2034				2031-2045				2086-2100			
	RCP-2.6	RCP-4.5	RCP-6.0	RCP-8.5	RCP-2.6	RCP-4.5	RCP-6.0	RCP-8.5	RCP-2.6	RCP-4.5	RCP-6.0	RCP-8.5
Indonesia	-4.6	-1.8	-3.8	-0.3	-1.0	-0.4	-3.6	-1.8	-3.8	-0.7	-1.6	-2.0
Sumatera	-4.2	-3.0	-1.4	-1.1	-0.7	-1.6	-3.6	-1.7	-2.0	-4.2	-1.1	-6.8
Jawa-Bali	-5.7	-3.9	-6.2	-1.1	-2.1	-5.2	-1.0	-1.5	-1.9	-1.3	-2.3	-4.1
Kalimantan	-2.3	-1.2	-1.6	0.6	1.3	-1.2	-0.3	0.4	-3.1	-1.2	0.4	-2.6
Sulawesi	-5.8	-2.2	-5.1	1.0	-1.9	-0.9	-1.8	-1.2	-4.8	-2.6	-0.9	-2.9
Papua	-6.6	-2.6	-2.3	-1.5	3.5	8.5	-2.9	-5.8	-5.4	-3.0	-2.8	7.1
Nusa Tenggara	-3.4	-1.5	-6.8	-0.6	-2.1	-3.7	-0.9	1.3	-0.4	-3.4	-3.5	-2.0
Maluku	-3.0	0.8	-2.2	-1.4	0.7	10.2	-1.0	0.2	-2.2	1.1	1.2	21.0

CH rata-rata skenario lebih rendah dari baseline observasi

CH rata-rata skenario lebih tinggi dari baseline observasi

Climate change-induced temperature increases are expected to affect the frequency of natural hazard in the future and pose more risks. Globally, drought and forest fires are natural disasters that are increasing due to the climate crisis, causing serious damage such as human and ecological, economic, and forest damages. The story is no different in Indonesia, where climate change-induced temperature increases are expected to affect the frequency of natural hazard in the future and pose more risks. In addition, many areas are experiencing legal and illegal land conversion into plantation and farmland, in which artificial fires are occurring and government efforts are being thwarted by widespread damage. The main affected areas were found to be concentrated in Sumatera and Kalimantan where oil palm plantation was concentrated. In 2019, large forest fires broke out across Sumatera and Kalimantan. From 2015 to 2019, land was damaged by 317,749 ha in central Kalimantan, 151,919 ha in western Kalimantan and 137,848 ha in southern Kalimantan.

In the event of an outbreak, massive fires can cause problems with respiratory disease in the neighbouring countries and resident through severe air quality pollution, leading to drought and contributing to serious climate crisis. Furthermore, the emission of GHG into the atmosphere resulting from forest fire contributes to acceleration of climate change, which in turn increases the likelihood of more frequent and intense drought and forest fires in the future.

Several studies have projected the climate change impact to Indonesia archipelago. A study by Heo et al. (2023)⁸⁴, specifically assess the climate change scenario in Kalimantan Island, where the project area located. This study assesses the two potential climate related hazards (drought and forest fire) in Kalimantan Island by using RCP-SSP 2-4.5 and RCP-SSP 2-8.5 (see Figure 21). This study shows the increase of percentage of area with very high and high risk for both fires and drought in the future due to climate change which align with the result of Indonesian government model in projection of rainfall.

⁸⁴ Heo, Sujung & Park, Sang-Jin & Lee, Dong Kun. (2023). Multi-hazard exposure mapping under climate crisis using random forest algorithm for the Kalimantan Islands, Indonesia. Scientific Reports. 13. 10.1038

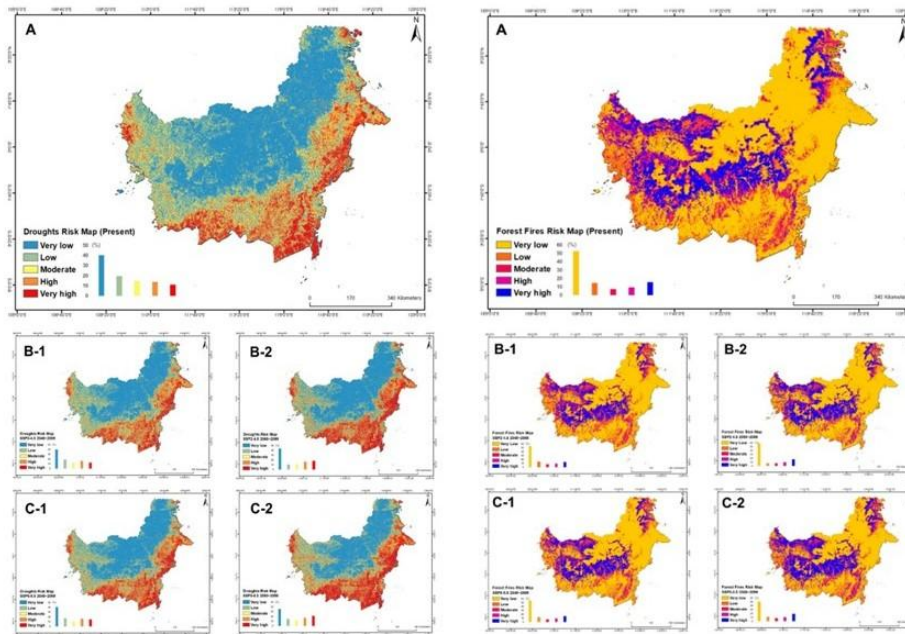


Figure 21. (Left) The risk map of drought (A Present, B-1 RCP-SSP 2-4.5 2040~2059, B-2 RCP-SSP 2-4.5 2080~2099, C-1 RCP-SSP 5-8.5 2040~2059, C-2 RCP-SSP 5-8.5 2080~2099). (Right) The risk map of forest fire (A Present, B-1 RCP-SSP 2-4.5 2040~2059, B-2 RCP-SSP 2-4.5 2080~2099, C-1 RCP-SSP 5-8.5 2040~2059, C-2 RCP-SSP 5-8.5 2080~2099).

Table 69. Risk values and ratio in forest fires and droughts

Hazard	Scenario	Risk area (%)				
		Very low	Low	Moderate	High	Very high
Droughts	Present	40.2	19.7	15.0	13.9	11.2
	2-4.5 2050	40.4	18.8	12.2	16.3	12.4
	2-4.5 2080	43.9	10.3	11.4	16.3	18.1
	5-8.5 2050	40.9	17.3	10.7	16.3	14.9
	5-8.5 2080	35.7	14.7	11.6	17.2	20.9
Forest fires	Present	52.7	14.7	7.5	9.3	15.8
	2-4.5 2050	54.2	14.6	7.9	9.8	13.6
	2-4.5 2080	58.5	7.2	7.3	9.8	17.2
	5-8.5 2050	55.8	12.5	6.7	9.7	15.3
	5-8.5 2080	51.0	11.5	8.1	10.2	19.1

This study also developed the multi hazard risk map which predict the risk probability area (see Figure 22). This map shows that the area which have high and very high risk due to drought, forest fire or both drought and forest fire would increase in the future in both RCP-SSP 2-4.5 and RCP-SSP 2-8.5 scenarios. In addition, West Kalimantan province has the highest risk of forest fires (33.02%) and South Kalimantan has the highest risk of drought (67.3%). This worsens in the worse-

case scenario, RCP-SSP5-8.5 2080, with the West Kalimantan Forest fires risk rising to 33.51% and south Kalimantan drought risk rising to 74.01%.

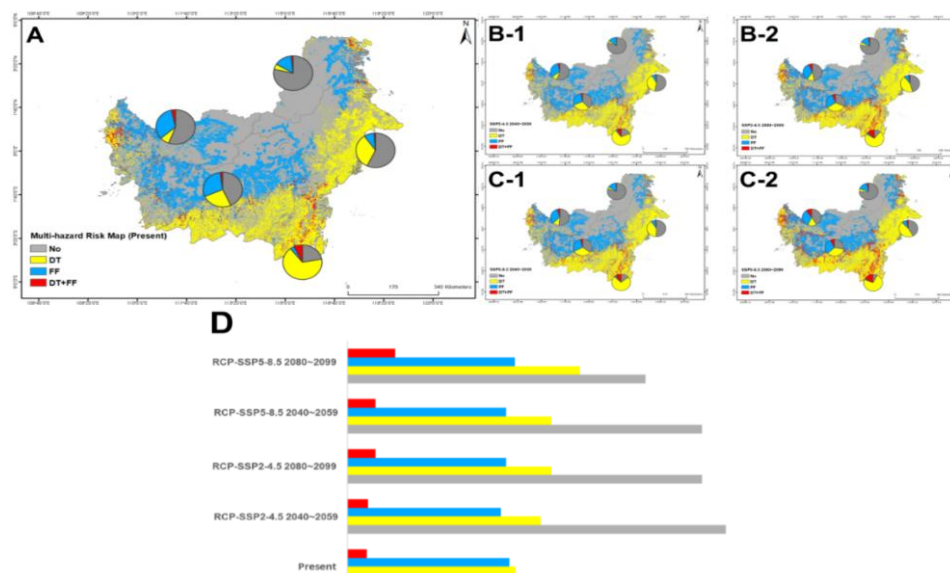


Figure 22. MH risk map (No non or low risk; DT Droughts risk area; FR Forest fires risk area; and DT+FR Droughts and Forest fires risk area) and Future climate prediction (A Present, B-1 RCP-SSP 2-4.5 2040~2059, B-2 RCP-SSP 2-4.5 2080~2099, C-1 RCP-SSP 5-8.5 2040~2059, C-2 RCP - SSP 5-8.5 2080~2099, and D Percentage by hazard type and scenarios).

Potential changes in the local land use scenario due to climate change

Climate change can affect the local land use scenario in the project area and adjacent local communities in various ways. First and foremost, climate change can affect the productivity and yield of paddy fields due to changes in temperature, rainfall, and other climatic factors⁸⁵. The impacts of climate and land use changes can also affect the livelihoods of local community, such as farmers who depends on natural resources and ecosystem services to fulfil their daily needs.

As stated by Dale, 1997, land use change is related to climate change as a major way in which the effects of climate change are expressed, For example, humans will change land use, and especially land management, to adapt to climate change and this adaptation will have ecological impacts⁸⁶. Therefore, the impacts of climate change may, in the long term, lead to changes in land use, such as adopting alternative crops or expansion of planting areas. Expanded planting areas could also contribute to increased deforestation and exacerbate the effects of climate change. However, in the short- and medium-term, it is unlikely that land use scenarios will change significantly due to the impacts of climate change.

⁸⁵ Tamrat Sinore, Fei Wang, Impact of climate change on agriculture and adaptation strategies in Ethiopia: A meta-analysis, Heliyon, Volume 10, Issue 4, 2024, e26103, ISSN 2405-8440, <https://doi.org/10.1016/j.heliyon.2024.e26103>

⁸⁶ Dale, Virginia. (1997). The Relationship Between Land-Use Change and Climate Change. Ecological Applications. 7. 753-769. 10.1890/1051-0761(1997)007[0753:TRBLUC]2.0.CO;2.

3.4.2 Climate Change Impacts (CCB, GL1.2)

Community Well being

Most of Indonesians, including people in Kalimantan, consume rice as their primary food source and is a staple for every meal, placing Indonesia as the largest consumer of rice in Asia. Rice production itself is very dependent on weather and climate conditions. Climate change has impacted on the climate pattern that causes the phenomenon of weather fluctuations, with the main climate element of climate change are rainfall and air temperature.

Climate change has numerous impacts on the agricultural sector, such as decreased yields. Research by Perdinan et al. (2008)⁸⁷ and Surmaini et al. (2010)⁸⁸ indicates that climate change has negatively affected crop yields by reducing plant productivity and the breadth of crops. Changes in climate have also impacted the growth phases and maturation of plants, leading to decreased biomass and total yields. A study conducted by Yuliawan and Handoko (2016) using the Shierary Rice Model examined the impact of climate change on rice crop yields in Indonesia. Their research found that rice yields are expected to decrease with rising temperatures, affecting both irrigated and rainfed paddy fields⁸⁹. The nationally rice for irrigated paddy field will decrease by about 11.1% while rainfed paddy field is for about 14.4% for every 1°C of temperature rise. Specifically, the paddy field in West Kalimantan decreases 7.9% for every 1°C of temperature rise (see Table 71).

⁸⁷ Perdinan, Perdinan & Boer, Rizaldi & Kartikasari, Kiki. (2008). Linking climate change adaptation options for rice production and sustainable development in Indonesia. *J. Agrometeorol.* 22. 94-107

⁸⁸ Surmaini E, Rakman, Boer R. Dampak Perubahan Iklim Terhadap Produksi Padi: Studi Kasus Pada Daerah dengan Tiga Ketinggian Berbeda. *Prosiding Seminar Nasional dan Dialog Sumberdaya Lahan Pertanian*. Bogor: Indonesian Center for Agricultural Land Resources Research and Development; 2008. In Bahasa

⁸⁹ Yuliawan, Taufiq & Handoko, I.. (2016). The Effect of Temperature Rise to Rice Crop Yield in Indonesia uses Shierary Rice Model with Geographical Information System (GIS) Feature. *Procedia Environmental Sciences*. 33. 214-220

Table 70. Decreasing rice crop yield for each province in Indonesia.

Type of paddy field	Province	Decreasing of yield (%)						Rate (%/°C)
		dt=0.5	dt=1	dt=1.5	dt=2	dt=2.5	dt=3	
IR	Bangka Belitung	-5.1	-9.4	-13.7	-17.1	-21.5	-25.7	-8.7
	Banten	-6.6	-13.0	-19.1	-23.9	-29.0	-33.8	-11.7
	Bali	-6.4	-12.2	-17.2	-22.2	-26.8	-31.0	-10.8
	Bengkulu	-5.6	-10.7	-15.6	-20.1	-24.2	-28.1	-9.7
	DIY	-8.0	-15.4	-22.2	-28.9	-34.9	-40.5	-14.0
	Jambi	-6.9	-13.3	-19.4	-24.5	-29.4	-33.9	-11.9
	Central Java	-7.1	-13.9	-20.0	-25.4	-31.3	-36.1	-12.5
	East Java	-7.8	-14.9	-21.4	-27.2	-32.7	-37.7	-13.2
	Lampung	-7.3	-13.5	-19.5	-24.5	-29.4	-34.5	-12.0
	NAD	-5.2	-9.9	-14.4	-18.7	-23.1	-27.0	-9.2
	Riau	-5.1	-10.1	-14.9	-19.4	-22.7	-26.9	-9.3
	West Sumatra	-5.8	-11.1	-16.0	-20.5	-24.8	-28.8	-10.0
	South Sumatra	-6.1	-11.9	-16.9	-22.3	-27.2	-32.0	-10.9
	North Sumatra	-5.5	-10.7	-15.4	-19.7	-23.6	-27.3	-9.5
TH	Banten	-5.6	-11.0	-16.0	-20.6	-24.9	-29.1	-10.1
	Bali	-10.0	-18.6	-25.4	-31.3	-36.6	-41.3	-14.9
	DIY	-14.5	-26.3	-36.4	-46.8	-55.2	-61.7	-22.2
	Gorontalo	-8.9	-15.6	-23.2	-30.9	-35.6	-41.3	-14.5
	Jambi	-6.1	-11.8	-17.1	-22.1	-26.8	-31.2	-10.8
	Central Java	-6.0	-11.8	-17.3	-22.6	-27.7	-32.7	-11.1
	East Java	-13.0	-24.9	-35.8	-46.4	-55.2	-62.3	-22.1
	West Kalimantan	-5.1	-8.8	-12.6	-16.2	-19.5	-22.8	-7.9
	South Kalimantan	-12.6	-24.8	-34.9	-44.2	-52.7	-60.9	-21.4
	Central Kalimantan	-12.3	-23.8	-34.2	-43.0	-52.4	-60.7	-21.1
	East Kalimantan	-7.8	-15.9	-24.7	-34.1	-43.0	-50.5	-16.9
	NTB	-19.1	-32.6	-45.0	-55.9	-66.19	-74.6	-26.8
	NTT	-7.1	-13.7	-19.9	-25.7	-31.1	-35.8	-12.5
	Papua	-13.6	-17.0	-16.0	-33.6	-31.7	-44.4	-14.4
	West Sulawesi	-8.3	-14.8	-21.4	-27.3	-31.0	-36.5	-12.9
	South Sulawesi	-11.7	-23.0	-33.2	-41.9	-49.7	-55.9	-20.0
	Central Sulawesi	-2.6	-7.5	-20.5	-28.2	-36.0	-41.6	-13.7
	Southeast Sulawesi	-11.5	-22.2	-30.8	-39.7	-47.8	-54.8	-19.2
	North Sulawesi	-8.8	-16.1	-23.0	-30.9	-38.9	-43.8	-15.2

dt = delta temperature (temperature differences, °C)

The reduction in paddy yields will impact the well-being of communities in the project area and project zone, as these communities rely on rice as their primary food source. Many residents grow rice themselves to preserve traditional practices and to meet their daily food needs. Therefore, any decline in rice production could directly affect their food security and overall quality of life.

In the absence of the project, most of the project area would have been converted into industrial timber plantation leaving communities with a limited access to the land to practice traditional agricultural activities producing rice as their staple food. The expected risk from climate change could further exacerbate these challenges. As agriculture land access decline in the project area, communities may resort to opening new forest land of area for agriculture, potentially through slash and burn practice, which could intensify ecosystem degradation, increase fire risks, and threaten both biodiversity and livelihoods.

In recent times, the impacts of climate change, particularly on precipitation and land use dynamics, has caused increased risks of hydrometeorological events in the project zone, particularly floods. Between December 2024 and January 2025, several villages within and outside the project zone of Sanggala Corridor Project experienced events of floods, such as in Balai Karangan⁹⁴ and Sepangah⁹⁵, forcing people to evacuate as their homes are inundated with water flow from the river. The increased intensity of precipitation and presence of palm oil in the ecosystem is thought to be the root cause of this extreme events. The absence of the project shall compound the floods risks in the project zone as the area would have lost its essential hydrological function. This stems from the loss of multi-layered forest canopy in a natural forest which will be replaced by single layer of canopy in an industrial timber plantation.

⁹⁵ <https://www.suarakablar.co.id/2025/01/pemkab-landak-serahkan-bansos-dan-bagikan-makanan-siap-saji-bagi-korban-banjir-di-desa-sepangah-dan-serimbu/>

Biodiversity conservation status

Changes in precipitation patterns and increasing temperature are expected to impact phenology and fruiting season in tropical forests. These changes have impact on decreased flowering and fruiting species that could reduce fruit availability, affecting many mammal species that depend on tropical forest fruits for food, including *Presbytis chrysomelas* and *Hylobates muelleri abbotti* that are endemic to the island of Borneo⁹⁶.

Moreover, the increase in temperatures, drought, and prolonged dry season leads to an increased risk of forest fires, especially in communities that still practicing slash-and-burn techniques. As mentioned previously, West Kalimantan province had the highest forest fire risk under the climate change scenario. This directly threatens the habitat of threatened species in the project area.

Furthermore, due to the decline in the productivity of paddy and other agricultural commodities such as oil palm, which are the main commodities for the community's economy in the project zone, there is a possibility of expansion of agriculture area to maintain production levels. It will pose a direct threat to the habitats of the biodiversity within the project zone and reduce chances of species survival in the future climate change scenario.

In addition to impact on animal species, climate change will significantly decrease suitable areas for biodiversity in Kalimantan. A study using maximum entropy modelling shows that climate change reduces the suitable living area for 13 bird species within the protected area, of which 7 species can be found in the project zone. These species are *Cicaria stromi*, *Aceros undulatus*, *Anthraceros malayanus*, *Buceros rhinoceros*, *Caprimulgus concretus*, *Hyrornis*, *Leptoptilos javanicus*, and *Nisaetus nanus*⁹⁷.

As previously mentioned, there have been prevalent impact of climate change on the presence of wild boar due to the spread of (ASFV). The climate change is the highest contributor on the spread of ASFV which has made wild boar hardly encountered within the project area and project zone.

The wild boar not only provide a source of protein for the community but also play an important role for the stability of biodiversity and ecosystem. The research by Erik Meijard (2000)⁹⁸ stated that the wild boars considered to be a potential ecosystem engineers, playing important roles as seed dispersers, seed predators and agents of physical disturbance. Moreover, referring to Pokorny and Jelenko (2013)⁹⁹, the wild boars have many positive roles in the environment, and the following are particularly important: creation of various habitat conditions, providing long-term increase in biodiversity, spread of plant seeds and hypogenous fungi by zoochory, reducing the abundance of some pest (invertebrate) species and providing of important food source for large carnivores. With the critical role of wild boar in the ecosystem, climate change impact might disrupt the stability of ecosystems and the sustainability of biodiversity within the project area and project zone.

⁹⁶ Numata, S., Yamaguchi, K., Shimizu, M. et al. Impacts of climate change on reproductive phenology in tropical rainforests of Southeast Asia. *Commun Biol* 5, 311 (2022). <https://doi.org/10.1038/s42003-022-03245-8>

⁹⁷ Singh M, Badcock-Scruton J, Collins CM. What Will Remain? Predicting the Representation in Protected Areas of Suitable Habitat for Endangered Tropical Avifauna in Borneo under a Combined Climate- and Land-Use Change Scenario. *Sustainability*. 2021; 13(5):2792. <https://doi.org/10.3390/su13052792>

⁹⁸ Meijaard, Erik. (2000). Bearded pig (*Sus barbatus*). Ecology, conservation status, and research methodology. Background document for the electronic conference on bearded pig research, and recommendations for a collaborative research programme.

⁹⁹ Pokorny, Bostjan & Jelenko Turinek, Ida. (2013).  Ekosistemska vloga, pomen in vplivi divjega prašiča (*Sus scrofa* L.) - Ecological importance and impacts of wild boar (*Sus scrofa* L.). Zlatogorov Zbornik. II. 2-30.

The absence of the project would aggravate this impact on biodiversity, since a complete loss of forest ecosystem is expected due to the conversion into monoculture industrial timber plantation. Wildlife will lose home range habitat and the diverse vegetation associated with natural dipterocarp forest of Borneo will cease to exist in the project area. In this scenario, it is expected that the existing wildlife will be pushed to the forest cover adjacent to the project area.

3.4.3 Measures Needed and Designed for Adaptation (CCB, GL1.3)

The project area and project zone are vulnerable to the potential impact of climate change (see sections 3.4.1 and 3.4.2), which negatively impact the ecosystem, biodiversity, and communities. To help communities and wildlife to adapt build resilience and mitigate the impacts of climate change, the project aims to conserve the remaining forest, restore non-forest and degraded forest areas, protect biodiversity, and improve the community's well-being, as described in the Theory of Change (ToC) in section 2.1.17.

To increase resilience of the ecosystem, biodiversity, and communities to climate change impacts, the project will implement the following measures. These activities have been integrated into ToC of this project and described in Table 72 below.

Table 71. Adaptation to the probable impacts of climate change

Climate Impact	Potential Impacts	Adaptation Option	Adaptation Indicator
Increased mean air temperature	Habitat loss due to reduction in natural regeneration and ecosystem resilience, which pose a threat to biodiversity	- Conserve forest habitat of threatened species - Restore degraded forest areas ¹⁰⁰ - Develop agroforestry in agricultural area ¹⁰¹ - Conduct forest patrolling - Conduct regular biodiversity monitoring	- The dynamic of forest cover is monitored - Forest conversion is avoided - Agricultural area expansion is controlled - Degraded forest areas are restored (in ha) - Agroforestry is developed in agriculture areas - The dynamic of threatened species monitored
	Increased risk of fires	- Establish fire monitoring system - Conduct community-based fire patrol	- Number of land and forest fire are controlled and reduced

¹⁰⁰ von Holle, B., Yelenik, S., & Gornish, E. S. (2020). Restoration at the landscape scale as a means of mitigation and adaptation to climate change. *Current Landscape Ecology Reports*, 5(3), 85-97.

¹⁰¹ Quandt, A., Neufeldt, H., & Gorman, K. (2023). Climate change adaptation through agroforestry: opportunities and gaps. *Current Opinion in Environmental Sustainability*, 60, 101244. <https://doi.org/10.1016/j.cosust.2022.101244>

Climate Impact	Potential Impacts	Adaptation Option	Adaptation Indicator
		- Conduct no-burn farming training to the community	
	Reduced survival rate and growth rate of seedlings in ARR activities (restoration & agroforestry)	- Plant diverse species - Plant native species - Implement best management practices to conduct restoration and develop agroforestry.¹⁰²	- Number of seedlings planted - Average survival rate of planted seedlings in restoration and agroforestry areas - Average growth rate planted seedlings in restoration and agroforestry areas
	Increased occurrence of pest and diseases in agricultural activities	Implement best practices of integrated pest management. ¹⁰³	Occurrence of pest and disease in agricultural or agroforestry areas is controlled and monitored
Agricultural and ecological drought	Increased risk of fires	Same as above	Same as above
	Reduced yields, crop failure	- Conduct agroforestry training for communities - Conduct agroforestry development in agriculture area - Improve access to weather forecast	- Number of people joined the agroforestry training - Number of hectares of agricultural area that is planted with agroforestry scheme - Number of annual yields from agroforestry
Increased in mean precipitation	Increased flood risk Increased crop failure	- Conserve forest areas to reduce flood risk.¹⁰⁴ - Install flood early warning system - Implement best management	- Number of hectares of forest deforestation and degradation - Number of flood early warning system installed in the project area

¹⁰² Riley, L. E., Steinfeld, D. E., Winn, L. A., & Lucas, S. L. (2015). Best management practices: An integrated and collaborative approach to native plant restoration on highly disturbed sites. *Natural Areas Journal*, 35(1), 45-53.

¹⁰³ Nihal, R. (2020). Global Climate change and its impact on integrated pest management. *Agro Econ. Int. J.*, 7(2), 133-137.

¹⁰⁴ Rackelmann, F., Sebesvari, Z., & Bell, R. (2023). Synergies and trade-offs in the management objectives forest health and flood risk reduction. *Frontiers in Forests and Global Change*, 6, 1208032.

Climate Impact	Potential Impacts	Adaptation Option	Adaptation Indicator
		practices for agroforestry. ¹⁰⁵	- Average survival rate of planted seedlings in agroforestry areas
Combination of all the above climate risks/hazards	Reduced GHG emission reduction potential that may affect the financial viability of the project	Conduct adaptive management concerning the project activities	Qualitative description of community meetings to review implementation of project and monitoring plan
	Increased vulnerability of local communities, and in particular women and vulnerable groups	Improve community well-being	- Number of people that have access to education under “education access is improved” output (see community monitoring plan) - Number of people that have access to healthcare under “healthcare access is improved” output (see community monitoring plan) - Number of people implementing alternative livelihood activities a result of project activities (see community monitoring plan) - Number of alternative livelihoods developed by the project (see community monitoring plan)
		Empower and develop capacity development local communities, woman and vulnerable groups	- Number of trainings per year implemented as a results of the project (see community monitoring plan)

¹⁰⁵ Rao, K. P. C., Verchot, L. V., & Laarman, J. (2007). Adaptation to climate change through sustainable management and development of agroforestry systems. *Journal of SAT agricultural research*, 4(1), 1-30.

Climate Impact	Potential Impacts	Adaptation Option	Adaptation Indicator
			- Number of communities involved in project activities - Number of women involved in /or assisted by the project activities (see community monitoring plan)

4 COMMUNITY

4.1 Without-Project Community Scenario

4.1.1 Descriptions of Communities at Project Start (CCB, CM1.1)

Administratively, the project area is located in Landak and Sanggau Regencies, West Kalimantan, Indonesia. In Landak Regency, the three villages of Sekendal, Sepangah, and Engkangin fall within the project zone. In Sanggau Regency, the project zone includes the seven villages of Sungai Illai, Mawang Muda, Thang Raya, Raut Muara, Kenaman, Engkahan, and Nekan. Map 9 depicts location of these villages and hamlets within the project zone. The project conducted a community socio-economic assessment¹⁰⁶ and an HCV assessment¹⁰⁷ to collect the relevant data. All the data in this section is based on those activities. For further details on the social parameters and socio-economic context influencing the communities within the project zone, refer to Section 2.1.15.

4.1.1.1 Well-being information

History

The communities currently residing within the project zone are believed to have originated from the mountains regions of what is now Brunei Darussalam. While there is no fixed date for their migration and settlement to the area, community elders indicate that this occurred long before Indonesia gained independence in 1945.

Economy and Income

All of the communities within these villages remain categorized as poor. Agriculture serves as the primary source of income across the villages, with an average monthly income of IDR 1,222,500. This figure is significantly below the average minimum income for the broader Landak and Sanggau Regencies in 2023, which have average monthly incomes of IDR 2,868,456 and IDR 2,789,563 respectively.

Among the villages in the project zone, Nekan village records the lowest average income at IDR 925,000 per month, while Sungai Illai has the highest, with an average of IDR 1,700,000 per month. Table 73 below details the average income for each village in the project zone.

¹⁰⁶ Community assessment report

¹⁰⁷ HCV report

Table 72. Average income of all villages in the project zone¹⁰⁶.

No	Village	Lowest Monthly Income (IDR)	Highest Monthly Income (IDR)	Average Monthly Income (IDR)
1	Engkangin	700,000	1,300,000	1,000,000
2	Sepangah	900,000	1,600,000	1,250,000
3	Sekendal	950,000	1,700,000	1,325,000
4	Mawang Muda	600,000	1,500,000	1,050,000
5	Sungai Illai	1,100,000	2,300,000	1,700,000
6	Thang Raya	900,000	1,700,000	1,300,000
7	Kenaman	850,000	1,500,000	1,175,000
8	Nekan	650,000	1,200,000	925,000
9	Engkahan	900,000	1,600,000	1,250,000
10	Raut Muara	800,000	1,700,000	1,250,000
	Average	835,000	1,610,000	1,222,500

The community is economically sustained on traditional farming methods and the collection of non-timber forest products, which generate limited cash reserves for trade. Figure 23 below illustrates the proportion of farming produce used for household subsistence and personal use, compared to the portion produced that is traded for income.

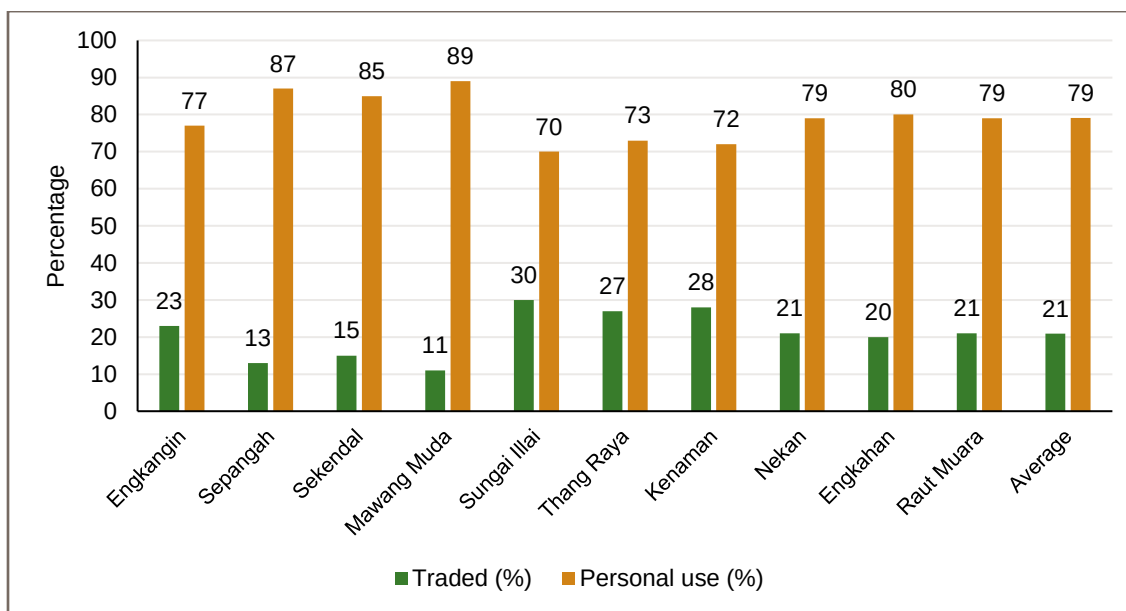


Figure 23. Percentage of farming production used for trade and personal use by the communities in the project zone

Livelihood

Agriculture is the primary livelihood for approximately 92% of the communities within the project zone. Figure 24 shows the distribution of employment across all communities in the project zone, while Figure 2 highlights the main crops cultivated, which include rice and palm oil, followed by rubber, pepper, and corn. As discussed previously and shown in Figure 23, 79% of the harvested crops are used for household consumption, with the remaining 21% sold for income or bartered for other goods.

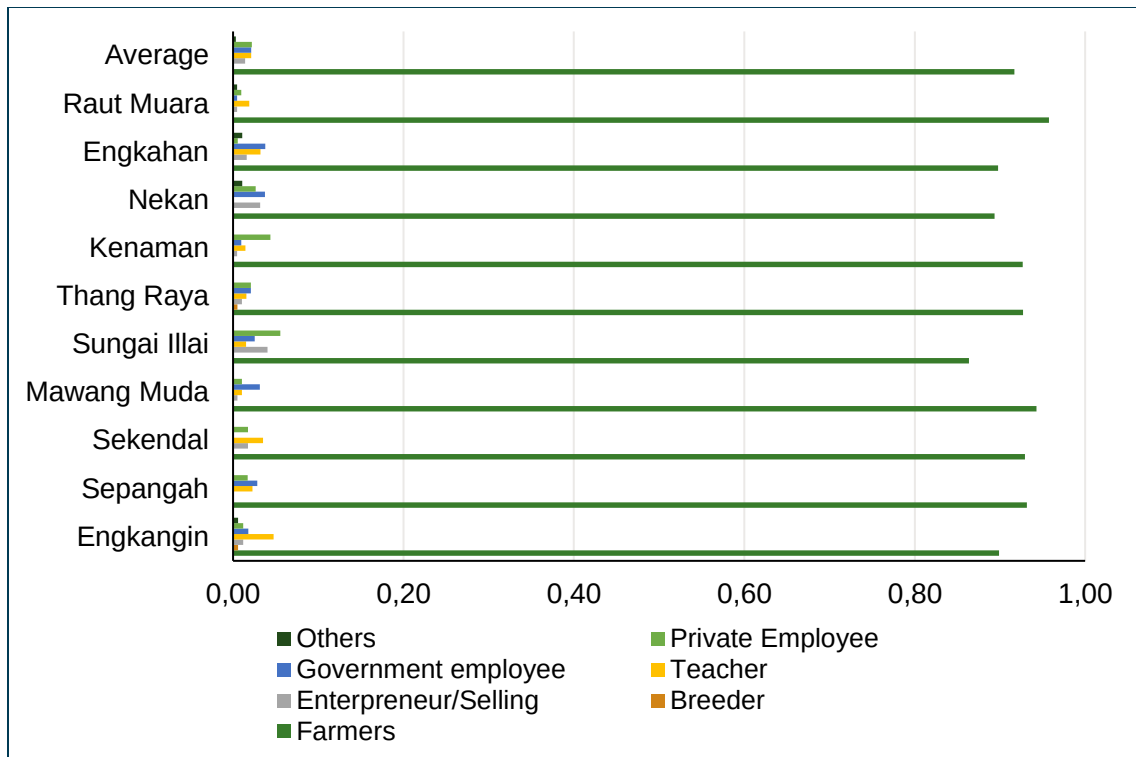


Figure 24. Percentage of jobs among the communities in the project zone

Farming knowledge and practices have been passed down through generations, deeply embedded in local culture. Like other Dayak communities, the communities in the project zone practice swidden agriculture. This traditional method involves clearing forest land, burning the vegetation, and planting crops. After a few years of cultivation, the land is left fallow to regenerate while a new forest plot is cleared for farming. This practice can be sustainable when population densities are low and fallow periods are extended (20+ years), allowing the land to recover and restore its nutrient base.



Figure 24. Paddy field and oil palm plantation in the distance

However, as the local population has grown, pressure on land resources has intensified. Fallow periods have shortened to only two or three years, and oil palm is now being planted after the cropping cycle. This shift is leading to a loss of agricultural land and increasing pressure to clear new forest areas to sustain agricultural production. Over the past decade, the area under oil palm cultivation within the PT CMI concession has expanded by 517.31 hectares.

Table 73. Oil palm plantation extent in the project area

Oil palm plantation extent in the project area		
2013	2023	Increase
825.25 ha	1342.56 ha	517.31 ha

Efforts to boost livestock production across the project zone have included the formation of livestock groups and training initiatives. Despite these efforts, a community assessment revealed that less than 1% of the population is engaged in raising livestock. Contributing factors to this low production rate include disease, theft, limited knowledge, and the absence of expert guidance.

Environment and Forest Dependence

The communities in all ten villages remain heavily reliant on the surrounding forests and natural resources to meet their daily needs, particularly for food. This includes vegetable crops like ferns and bamboo shoots, medicinal plants, and game animals such as wild boars. Over the past five years, the communities have reported a significant decline in food availability and forest resources. They attribute this decline to the reduction in forest area and ecosystem degradation, primarily due to agricultural expansion and increased hunting pressure.

Education

Elementary school facilities are located in all ten villages within the project zone. However, junior high schools are only located in Sepangah, Sekendal, Mawang Muda, Sungai Ilai, and Kenaman Villages, while high schools are limited to Sepangah and Kenaman Villages (see Section 2.1.15.4.4 for more detail).

Due to the relatively high cost of higher education and the limited availability of agricultural labor, only 44% of the community members have attained elementary education, with illiteracy remaining common throughout the project zone.

Medical

Basic health facilities are present in all ten villages within the project zone. Community health centers (*Pusat Kesehatan Masyarakat* and *Puskesmas*) serve each sub-district, while auxiliary community health centers (*Puskesmas Pembantu* and *Pustu*) are found in each village. However, there is only one hospital, located in Kenaman Village, within the project zone, (See Section 2.1.15.4.5 for more detail).

Despite having access to various medical services, several barriers prevent widespread utilization. The primary obstacle is the lack of funds, making medical treatment unaffordable for many. Transportation costs and availability also hinder access to healthcare facilities.

Government health insurance (BPJS) is available, but only in Sungai Illai, Thang Raya, Kenaman, Engkahan, and Raut Muara villages. As a result, the community often resorts to traditional medicines, sourced primarily from neighboring forest areas, to treat ailments. For more detailed information on the community's medical situation, see Section 2.

Water Access

Communities residing in project zone have relatively safe access to clean water. They predominantly source clean water from sources (e.g. water pond) and/or river flowing through village areas (See Section 4.1.3.2 below for more detail). To enhance this access, some villages (e.g. Raut Muara) constructed pipelines to facilitate water distribution to each household.

However, there are several issues regarding communities accessing clean water. First, there is unequal access of water distribution amongst communities leading to several hamlets within a village unable to access water directly in their households. In cases related to funding, there are some villages unable to construct similar pipeline facility for distribution. In addition to which, based on the landscape condition in the project zone, there is also the possibility of river flow are polluted or experiencing increased water discharge due to the hilltops (upstream area) being converted and planted with monoculture palm oil plantation. This would create the water from the river unfit for consumption and/or damage the constructed water pipeline. The presence of illegal traditional mining activities adds further the cause of water pollution in the project zone.

4.1.1.2 Community characteristics

Ethnicity and Language

All land in the project zone is customary land of the Dayak community, and six Dayak sub-tribes are currently living in the Project Zone, namely Dayak Belangin, Dayak Obi Nengeuh, Dayak Golik, Daya Keremai, Dayak Paus, and Dayak Punti (see Map 10 in Section 2.1.15.1).

Each of the Dayak sub-tribes within the Project Zone use a separate language, as detailed in Table 75. Bahasa Indonesian is used as the common language across all sub-Tribes and with other stakeholders and communities present in the project zone.

Table 74. List of language that used by each Dayak Sub-Tribe.

No	Sub-Tribe	Language
1	Dayak Belangin	Belangin Language
2	Dayak Obi Nengeuh	Bih Nengeuh Language
3	Dayak Golik	Golik Language
4	Dayak Keremai	Kerambai Language
5	Dayak Paus	Paus Language
6	Dayak Punti	Punti Language

Generally, a village is populated by one sub-tribe, but this is not always the case as shown in Table 76. Mixed communities within a village are mainly due to inter-tribal marriages.

Table 75. List of village and Dayak Sub-Tribe within it

No	Village	Sub-Tribe
1	Engkangin	Dayak Belangin
2	Sepangah	Dayak Belangin
3	Sekendal	Dayak Belangin
4	Mawang Muda	Dayak Golik and Dayak Belangin
5	Sungai Illai	Dayak Obi Nengeuh and Dayak Golik
6	Thang Raya	Dayak Golik
7	Kenaman	Dayak Paus and Dayak Bidayuh
8	Nekan	Dayak Punti and Dayak Keremei
9	Engkahan	Dayak Keramai
10	Raut Muara	Dayak Keremei

Spirituality and Mythology

As members of the broader Dayak ethnic community, the communities in the project zone hold a strong and deeply rooted belief in the spirits of nature and their ancestors. Ritual practices and worship to these spirits are believed to bring positive benefit and blessings to the community. One of the traditional and time-honored activities carried out by the Dayak sub-tribes in the project zone is *Gawai Padi*. This annual tradition marks the beginning of the new farming season and serves as an expression of gratitude to the creator for the seasons. The *Gawai Padi* ceremony involves various offerings to the rice deity, including the recitation of prayers and the offering of chickens, pigs, and fruits.

Traditional mythology also continues to play a significant role in the cultural practices of the Dayak communities. One of the myths passed down through generations is the mythology related to hornbills, which are seen as symbols of good luck and prosperity in Dayak culture. According to the legend, the hornbill bird was once a human who transformed into a bird to escape an evil spirit. Today, the hornbill bird is considered a sacred animal, and its image is widely used in traditional Dayak art and ceremonies.

For each of the communities in the project zone, these deeply held beliefs are evident in culturally significant sites or objects across their community customary lands. Table 77 below lists down some of these culturally significant sites.



Figure 25. Temegang sacred place owned by Dayak Keremei sub-tribe. A place where people convey requests so that their wishes and hopes can be granted.

Table 76. Culturally significant sites within the project zone

No.	Site Name/ Sub-tribe	Sub-Tribe	Coordinate	Sub-village/ Village/Sub-district	Description
1	Petilasan Temenggung Tua	Belangin	0.825, 110.119	Tauk/Engkangin/ Air Besar	Historical place of <i>temenggung</i> who contributed as a figure in forest clearing in Engkangin Village.
2	Penolak Penundung	Belangin	0.822, 110.119	Tauk/Engkangin/ Air Besar	A sacred area that functions as a protector of society from danger and repel enemies and protect society.
3	Pedagi	Belangin	0.824, 110.119	Engkangin/ Engkagin/ Air Besar	A place where people ask for their wishes to be granted.
4	Old Graveyard	Belangin	0.799, 110.104	Engkangin/ Engkagin/ Air Besar	The final resting place of ancestors who lived before the emergence of religion.
5	Old Graveyard (Singaraja)	Belangin	0.754, 110.105	(Kampung Lama)/Sepangah/ Air Besar	Located in Kampung Lama Sepangah (Serimbu Village). This tomb is the burial of the warlord, Singaraja.
6	Pedagi	Belangin	0.662, 110.174	Kelepuk/Sekendal/ Air Besar	A place where people ask for their wishes to be granted.
7	Old graveyard	Belangin	-	Kelepuk/Sekendal/ Air Besar	The final resting place of ancestors who lived before the emergence of religion.
8	Tembawang	Obi Nengeuh	0.657, 110.286 0.657, 110.286	Pelinduk, Emperet, Selabe, Muara Ilai/Sungai Ilai/ Beduai	Fruit plantations have existed since ancestral times and have been passed down from

No.	Site Name/ Sub-tribe	Sub-Tribe	Coordinate	Sub-village/ Village/Sub-district	Description
			0.662, 110.329 0.662, 110.379		generation to generation.
9	Old graveyard	Obi Nengeuh	0.659, 110.318	Selabe, Emperet/Sungai Ilai/Beduai	The final resting place of ancestors who lived before the emergence of religion.
10	Cave Batu Karang	Golik	0.765, 110.241	Batu Karang River/ Mawang Muda/ Beduai	Cave with large rocks in Sungai Batu Karang Sub-village.
11	Old graveyard/ Golik	Golik	0.753, 110.377	Pemodis/ Thang Raya/ Beduai	The final resting place of ancestors who lived before the emergence of religion.
12	Balai Adat	Golik	0.751, 110.377	Pemodis/ Thang Raya/ Beduai	A place for carrying out traditional activities, often containing human skulls from the Ngayau Tradition practice by their ancestors.
13	Sacred Place Temegang	Keramei	0.826, 110.303	Seka/ Raut Muara/ Sekayam	A place where people convey requests for their wishes and hopes to be granted.
14	Old graveyard Sebungah	Keramei	0.811, 110.348	Raut Muara/Raut Muara/Sekayam	The final resting place of ancestors who lived before the emergence of religion.
15	Old graveyard Kampung Selangai	Keramei	0.821, 110.277	Masa Selangin/ Raut Muara/ Sekayam	The final resting place of ancestors who lived before the emergence of religion. According to Geospatial Information Agency, the indicative village boundary is in Mawang Muda Village.
16	Old graveyard Kampung Seputah	Keramei	0.824, 110.286	Masa Selangin/ Raut Muara/ Sekayam	The final resting place of ancestors who lived before the emergence of religion. According to Geospatial Information Agency, the indicative village boundary is in Mawang Muda Village.
17	Tembawang	Keramei	0.808, 110.288	Masa Selangin/ Raut Muara/ Sekayam	Fruit plantations have existed since ancestral times and have been passed down from generation to

No.	Site Name/ Sub-tribe	Sub-Tribe	Coordinate	Sub-village/ Village/Sub-district	Description
					generation. According to Geospatial Information Agency, the indicative village boundary is in Mawang Muda Village.
18	Balai Adat Desa	Keramei	0.814, 110.344	Raut Muara/ Raut Muara/ Sekayam	A place to carry out traditional activities and usually contains human skulls resulting from the Ngayau Tradition from the time of our ancestors.
19	Balai Adat Keramei	Keramei	0.865, 110.409	Pesing/Engkahan/ Sekayam	A place to carry out traditional activities and usually contains human skulls resulting from the Ngayau Tradition from the time of our ancestors. During the survey, it was still in the process of being rebuilt.
20	Tembawang Suban	Keramei	0.853, 110.392	Engkahan/ Sekayam	Fruit plantations that have existed since ancestral times and have been passed down from generation to generation.
21	Tembawang Entubah	Keramei	0.833, 110.400	Entubah/Engkahan/ Sekayam	Fruit plantations that have existed since ancestral times and have been passed down from generation to generation.
22	Customary land Rimba Tuak Sepedak	Keramei	0.851, 110.387	Engkahan/ Sekayam	Customary land.
23	Customary land Penguduk	Keramei	0.84, 110.397	Entubah/Engkahan/ Sekayam	Customary land.
24	Sacred Stone (Sepasang)	Keramei	0.872, 110.369	Gramajaya/Nekan/ Entikong	There are two stones close to each other, symbolizing men and women. A place where people ask for their wishes to come true.
25	Tembawang Pengeret, Tembawang Tuat Kudur	Paus	0.816, 110.395 0.806, 110.392	Lomur I/ Kenaman/ Sekayam	Fruit plantations that have existed since ancestral times and have been passed down from generation to generation.

No.	Site Name/ Sub-tribe	Sub-Tribe	Coordinate	Sub-village/ Village/Sub-district	Description
26	Pedagi	Paus	0.809, 110.395	Lomur I/Kenaman/Sekayam	A place where people can make requests.
27	Old graveyard Pengireh Sibunga	Paus	0.792, 110.407	Lomur II/Kenaman/Sekayam	The final resting place of ancestors who lived before the emergence of religion.
28	Tembawang Punten, Tembawang Pedeken, Tembawang Seneyung, Tembawang Empumut	Punti	0.9, 110.346 0.894, 110.347 0.893, 110.350 0.896, 110.355	Punti Kayan/Nekan/Entikong	Fruit plantations that have existed since ancestral times and have been passed down from generation to generation. These four tembawangs circle Mount Senyang with the majority of Durian commodities.

One of the most well-known aspects of the Dayak people's history is the practice of headhunting, known as "*Ngayau*" in the Dayak language. This tradition was practiced by all six Dayak sub-tribes in the project zone. *Ngayau* was a common practice among the Dayak community, and those who could present a human head were regarded as heroes in their village. It was also a symbol of masculinity, demonstrating bravery and serving as an important offering to a woman during marriage. Furthermore, *Ngayau* was seen as a prerequisite for establishing a village, as there was a belief that a longhouse, or "*betang*," would not be sturdy if its pillars were not supported by the heads of enemies. Consequently, *Ngayau* were carried out between villages as a means of defending or expanding a community's territory.

The practice of *Ngayau* is no longer observed today, and the community now lives in peace. The cessation of the *Ngayau* was marked by a significant meeting held by the Dayak community on May 22, 1894 in Tumbang Anoi Village, Central Kalimantan to end inter-tribal conflicts. This meeting resulted in the *Tumbang Anoi Agreement*. In addition to ending both inter-tribal conflict and *Ngayau* the *Tumbang Anoi Agreement* also outlined 96 customary laws that continue to regulate the traditional practices of the Dayak people. These laws include provisions for resolving conflicts and the governance and implementation of customary practices.

Traditional Authority

The Dayak community is organized with a traditional hamlet leader, known as the *Ketua Adat*, and a village leader or *Temenggung*. Above the *Temenggung* there is the Dayak Traditional Council, or *Dewan Adat Dayak* (DAD) at the sub-district level, which is responsible for governing customary laws and resolving disputes. Similar councils exist at the district level (DAD Kabupaten/Kota) and the provincial level (DAD Provinsi).

Disputes and violation of customary laws are generally first addressed by the *Ketua Adat* before escalating to the *Temenggung*. If disputes aren't resolved at the *Temenggung* level, then resolution of the dispute will move to DAD level.

In nearly all villages, there is a strong connection with their traditional leaders, including the *Ketua Adat and Temenggung*. These traditional authorities remain powerful within the local community. Generally, the community is more subservient to the *Ketua Adat* rather than a government official such as the village head indicating these traditional institutions remain strong among the local population.

4.1.1.3 Diversity within the community

As already stated, the main factor for diversity in the community is the six Dayak sub-tribes with their distinct languages and cultural traditions.

Gender

The population in the Project Zone is 25,520, across the 10 villages, 52% of residents are male, and 48% of residents are female. Table 78 below shows the gender breakdown per village, including the gender ratio. (A gender ratio figure greater than 100 indicates a higher proportion of males to females).

Table 77. Gender breakdown for each village within the project zone

Village	Male	Female	Total	Gender Ratio
Engkangin	890	838	1,728	106
Sepangah	864	638	1,502	135
Sekendal	2,082	1,845	3,927	113
Mawang Muda	1,167	1,110	2,277	105
Sungai Ilai	1,797	1,584	3,381	113
Thang Raya	1,011	924	1,935	109
Kenaman	1,316	1,250	2,566	105
Nekan	1,279	1,218	2,497	105
Engkahan	1,358	1,293	2,651	105
Raut Muara	1,580	1,476	3,056	107
Total	13,344	12,176	25,520	110

Age

According to the Indonesia Central Bureau of Statistics (*Badan Pusat Statistik*, BPS) the productive age group in Indonesia is defined as citizens between the ages of 15 and 64. Around 70% of the community in the project zone are in their productive years. Table 79 below shows the number of people in the different age groups.

Table 78. The number of people for different age groups within the project zone

Age group	Total	Male	Female
0-4	2,174	1,137	1,037
5-9	2,133	1,115	1,018
10-14	2,097	1,097	1,001
15-19	2,199	1,150	1,049
20-24	2,287	1,196	1,091

Age group	Total	Male	Female
25-29	2,197	1,149	1,048
30-34	2,193	1,147	1,046
35-39	2,005	1,049	957
40-44	1,875	980	895
45-49	1,653	864	789
50-54	1,385	724	661
55-59	1,155	604	551
60-64	903	472	431
65-69	611	320	292
70-74	341	178	163
75+	311	163	149
Total	25,520	13,344	12,176

4.1.2 Interactions between Communities and Community Groups (VCS, 3.19; CCB, CM1.1)

The dominant grouping within the community is based on ethnicity, as seen in the six sub-Dayak groups. Although there are formal women's and youth groups located across the project zone, these groups are aligned with their ethnicities within their village.

The interaction among the six Dayak sub-tribes is harmonious, as all community members are of the broader Dayak Tribe. Traditional customary laws, governed by the *Tumbang Anoi Agreement* and customary governance structures, as seen through the *Ketua Adat*, *Temenggung* and *Dewan Adat Dayak* (DAD), effectively resolve disputes and provide work to a customary framework and laws that serve the community well.

4.1.3 High Conservation Values (CCB, CM1.2)

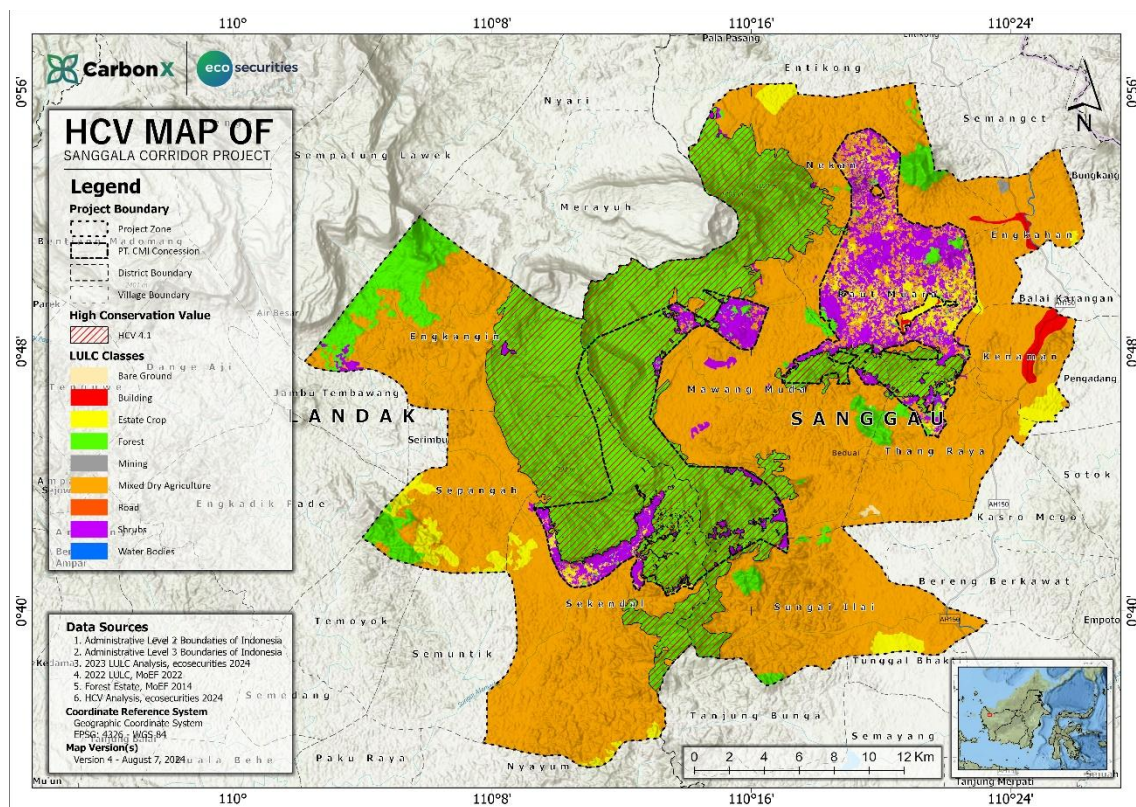
The project aims to protect and improve high conservation values (HCVs) and community well-being across the project zone. To achieve this goal, the project identified areas that provide critical ecosystem services for the community (HCV 4), areas that are fundamental for the livelihoods of communities (HCV 5), and areas that are critical for traditional cultural identity for those communities living within the project zone (HCV 6).

4.1.3.1 HCV 4: Ecosystem Services

HCV 4 are areas that provide basic ecosystem services in critical situations, including protection of water catchments and erosion control on vulnerable soils and slopes. The project team identified areas that provided:

- The provision of water for local communities and the prevention of floods for downstream communities (HCV 4.1), and
- The prevention of erosion and sedimentation (HCV 4.2).

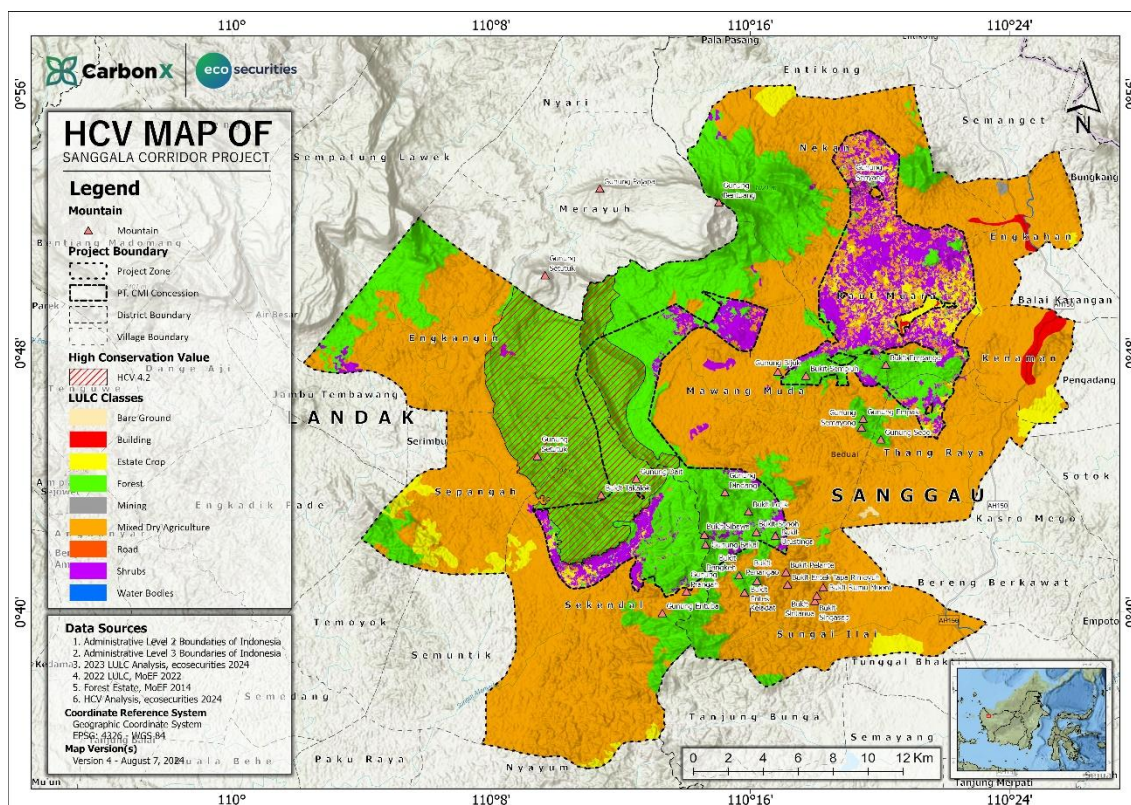
High Conservation Value	HCV 4.1 Areas Critical to Water Catchments
Qualifying Attribute	Areas identified as crucial for maintaining both the quantity and quality of the water supply to local villages through rivers and streams. These areas are also important for sustaining water supply for the Dait waterfall situated in Sekendal village, which has potential for generating income for the community through eco-tourism.
Focal Area	All the forest area in block two and forest area in southern part of block one (see Map 33).



Map 32. High Conservation Value 4.1

High Conservation Value	HCV 4.2 Areas Critical to Erosion Control
Qualifying Attribute	Areas crucial for preventing erosion and sedimentation, are forested regions located on land with high erosion potential, as identified by the Indonesian Ministry of

	Environment and Forestry ¹⁰⁸ . These areas corresponding to potential soil erosion rates exceeding 180 tons/ha/year or more if the vegetative cover is removed.
Focal Area	The forest area at Mount Dait, Takalok Hill, and Mount Setutuk have a moderate erosion risk (60-180 tons/ha/year) if the vegetative cover is removed (see Map 34).



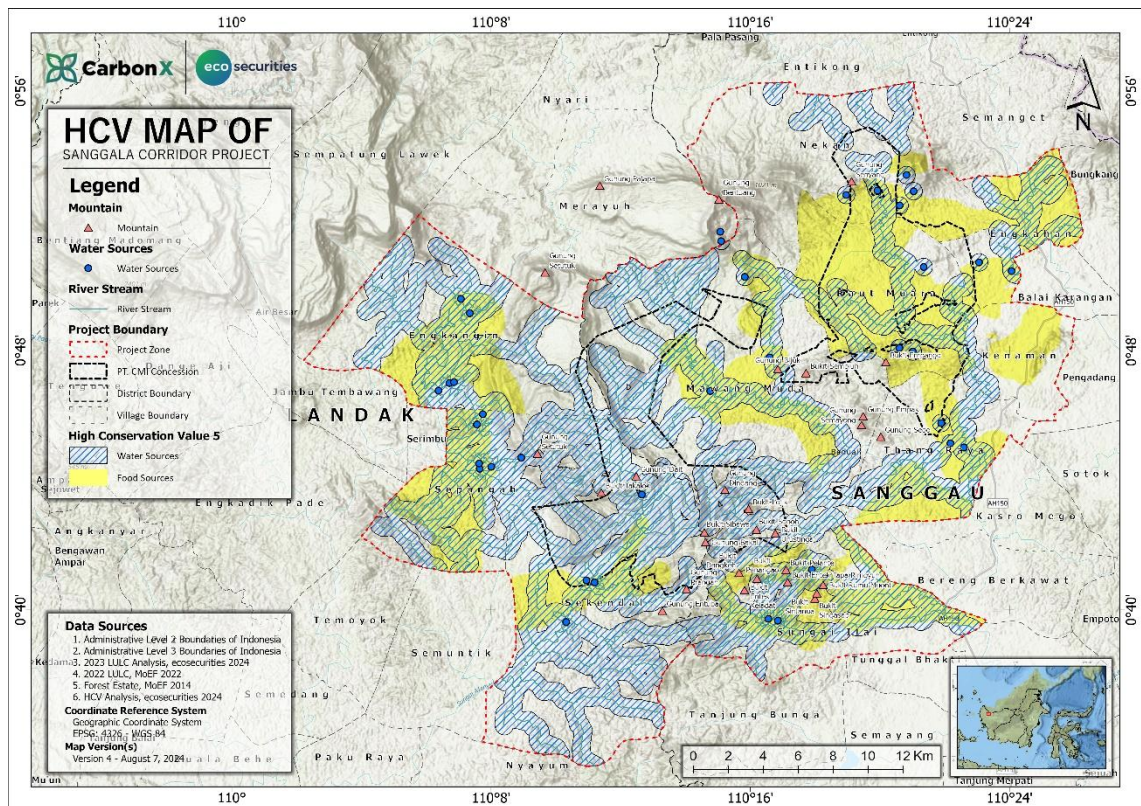
Map 33. High Conservation Value 4.2

4.1.3.2 HCV 5: Community Needs

HCV 5 refers to sites and resources that are fundamental for meeting the basic needs of local communities and indigenous peoples (i.e. health, nutrition, clean water, shelter, etc.), as identified through engagement with these communities. The project team identified areas that are crucial for providing essential food and water sources for the communities living in the Project Zone.

¹⁰⁸ [Rawan_Erosi - Overview \(menlhk.go.id\)](https://menlhk.go.id/)

High Conservation Value	HCV 5. Areas Fundamental to Meeting Basic Needs of Local Communities
Qualifying Attribute	These areas are fundamental for the local communities and indigenous peoples that rely on nature to meet their basic needs. The destruction or degradation of these resources will greatly impact the livelihoods of affected stakeholders. HCV 5 refers to areas identified within the project zone, that supply food and water to the communities to meet their basic needs.
Focal Area	See Map 35



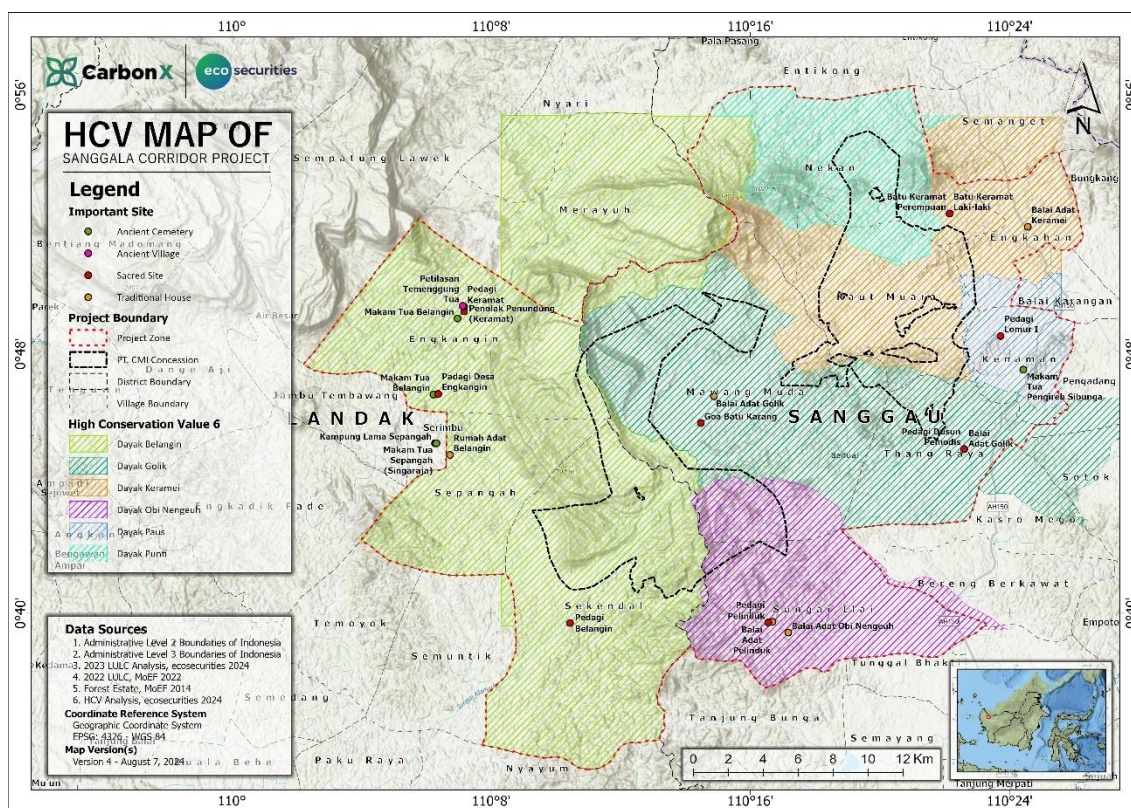
Map 34. High Conservation Value 5

4.1.3.3 HCV 6: Cultural Values

HCV 6 refers to sites, resources, habitats and landscapes of global or national cultural, archaeological or historical significance, and/or of critical cultural, ecological, economic or

religious/sacred importance for the traditional cultures of local communities or indigenous peoples. This was identified through engagement with the local communities or indigenous peoples. The project team identified the entire Project Zone as critical to the local community in maintaining local traditions and customs.

High Conservation Value	HCV 6. Areas Critical to Local Communities Traditional Cultural Identity
Qualifying Attribute	All areas within the Project Zone are located in the six traditional territories of the Belangin, Obi Nengeuh, Golik, Paus, Keramai, and Punti Dayak clans. These six traditional clans continue to uphold their customary laws and beliefs, which are based on natural resources, cultural sites and historical connection, including sacred places, art, and traditional rituals.
Focal Area	The entire Project Area (see Map 36).



Map 35. High Conservation Value 6

4.1.4 Without-Project Scenario: Community (CCB, CM1.3)

The without-project scenario involves PT CMI converting forest and non-forest areas into industrial timber plantations of *Anthocephalus cadamba* (Burflower tree) and *Hevea brasiliensis*

(Para Rubber). In this scenario, the communities in the Project Area might potentially benefit from an increase in:

- Income through the development of small rubber plantations (*Hevea brasiliensis*) with the support of PT CMI.
- Employment opportunities by working in the Industrial Timber Plantations. Most employment opportunities for the community would likely be in harvesting and plantation maintenance roles. However, these opportunities would be influenced by the educational levels of community members, suggesting that higher employment levels are less likely given the generally low levels of education within the community.

Although industrial timber plantation development may provide income and other economic opportunities for the local communities, there are several negative impacts that needs to be addressed, including:

- Community conflict. The development of rubber plantations will only occur in areas covering the Raut Muara and Nekan Village areas, as outlined in the PT CMI RKU. This could lead to an unequal distribution of economic benefits from the plantations, potentially causing resentment and jealousy between those who benefit and those who do not.
- Loss of agricultural lands. An increased area for timber plantations will reduce community access to lands for growing their own food crops, such paddy rice and other staple crops (i.e corn and vegetables). This reduction could significantly impact the community's food security status and overall livelihood opportunities if agricultural lands are converted to industrial tree crops.
- A decline in water quality and quantity. The forested areas, identified as HCV 4, currently provide the community with clean drinking water and help prevent floods, sedimentation and soil erosion. Converting these areas into industrial timber plantations may lead to significant soil disturbance and environmental degradation resulting in erosion and a decline in both the quality and quantity of water available to the community
- Low quality education services. The community will continue to face low levels of education and limited educational facilities within the communities. The PT CMI RKU does not include actions to improve community access to education services or raise the quality of education for the community.
- Low access to health care services. Limited access to health care services and their poor quality will persist. Similar with education, PT CMI has no original plan to increase community access to health care or the quality of health care under the current RKU.
- Decline in the project area's cultural values. The proposed industrial timber plantations are located within the community's customary area. This area contains significant cultural, spiritual and historical values for the community (see HCV 6) which will potentially be destroyed or degraded in the development of the Industrial Timber Plantation.

4.2 Net Positive Community Impacts

4.2.1 Expected Community Impacts (CCB, CM2.1)

While there are six Dayak clans within the Project Zone, the community demography, educational levels, cultural norms and farming practices are relatively similar across all groups. Because of

this homogeneity, only two distinct community groups have been identified: the farming community, and the community in general.

To assess the benefits, costs and risks of the proposed project for these two groups, Focus Group Discussions (FGD) were conducted by the project team during the community consultations, livelihood assessment and FPIC processes.

The proposed project is expected to have a direct and positive impact on the community groups identified. The impacts for each community group are described in Table 80 and Table 81 below.

Table 79. Expected impacts to community members that work as farmers

Community group	Community members that work as farmers
Impact(s)	Farmers in the Project Area will be impacted by the project through: • Increased income opportunities due to the agroforestry systems introduced by the project • Increased knowledge of sustainable and climate-smart farming practices, particularly on how perennial vegetation can enhance income security and diversity output. • Increased market access for agroforestry crops through CarbonX market analysis and development. • Greater security over land access and land tenure through participatory land use planning, the allocation of land for agriculture purposes, and designated forest areas of permanent protection. • Continuation of ecosystem services, such as good quality water that supports both community and farming needs. • Increased market access and profitability for a range of agricultural products analysis of value-chains to the opportunities to value add along the chain. • Restrictions on unplanned expansion of new agricultural enterprises, particularly into designated forest areas under protection. • Income instability and/or uncertainty during the transition from current traditional (subsistence) farming to farming systems based on agroforestry practices.
Type of benefit/cost/risk	Direct benefit (predicted): Training will be provided to farmers who elect to participate in the agroforestry scheme. This will enhance their income through the growth and sale of new tree crops, as well as agricultural crops benefiting from improved soil nutrients and conditions.

	Direct benefit (predicted): Employment opportunities will arise from the project, involving farmers monitoring the development of agricultural and agroforestry crops, and in marketing and selling crops from these farming systems. Indirect benefit (predicted): The permanent conservation of forest areas will provide ecosystem services, such as water, to the community. Direct cost (predicted): The community groups are not expected to incur any significant cost. They will only need to invest time in capacity building and training programs, and a possible increase in time for the management and harvesting of trees crops from the agroforestry system. Direct cost (predicted): There may be a loss of opportunities for the unsustainable or unplanned expansion of palm oil crops. Direct risk: The transition from traditional (subsistence) agriculture to tree crops in the agroforestry system presents a risk due to the shift from annual harvest cycles of subsistence crops to tree crops, which may not produce yields until the third year or later. This transition will need to be carefully managed.
Change in well-being	The well-being benefits are expected to be diverse and will encompass all farming communities across the project zone. This will include benefits from increased employment opportunities in the nursery, forest restoration work and forest patrolling, alongside the monitoring, agroforestry planting and maintenance. A total of 1,236 employment are expected to be generated through the project in the project lifetime. Increased knowledge of sustainable and climate-friendly farming practices through training and capacity-building efforts is expected to increase yields and reduce annual yield fluctuations. A total of 2,400 people are expected to join training in the project's lifetime The work of CarbonX on improving market access and increased profitability along the value chain is expected to lead to higher agricultural gross margins and increased profit. Participatory Rural Appraisal (PRA) activities will be conducted each year with the community to monitor the project's impacts.

Table 80. Expected impacts to village communities in general

Community group	Village communities in general
Impact(s)	Village communities across the project zone will be impacted by the project through: • Improved educational access and quality of education • Improved healthcare access and quality of healthcare • Improved access to clean water • Increased employment opportunities through project activities • Establishment of alternative livelihood opportunities • Increased investment in community infrastructure through development of a community development fund • Continued access to cultural places that are permanently protected within the forest area • Prohibition on the hunting of threatened species
Type of benefit/cost/risk	Direct benefit (predicted): The project will implement a broad education and capacity-building program, including school environmental programs, scholarships, and access to Package B and C educational programs to be implemented during the project's lifetime. Direct benefit (Predicted): The project will provide training to explore financially viable alternative livelihood opportunities for the community. This may include but is not limited to livestock raising, honey production, ecotourism, and the production, marketing, and sale of non-timber forest products (NTFP). Direct benefit (predicted): The project will also implement a broad health care program by supporting doctors in a community outreach program and regular (4 to 6 month) village visits for health check-ups to the community. Direct benefit (predicted): The project will create new employment opportunities for community members. While an equal opportunity approach will be adopted, preference will be given, where possible, to recruiting women, young people and other disadvantaged groups. Direct benefit (predicted): The project will establish a community development fund to be managed by community funds. This fund will receive a small proportion of revenue

	from the sale of avoidance credits, which will be allocated toward the development and maintenance of community infrastructure (i.e. community directed investment into school, health, education, religious, environmental, or other facilities) Direct cost (Predicted): Loss of hunting opportunities Indirect benefit (predicted): The project will permanently protect the forest and restore degraded forest areas, ensuring that ecosystems services are maintained, degraded landscapes are restored, and access to culturally significant places is preserved.
Change in well-being	Improved education and health outcomes for the community by introducing broad health and educational programs. The project is expected to increase healthcare access for 11,750 people and increase access to education for 5,969 people in the project's lifetime. The improvement of access to clean water is expected to create impact for 5,685 people. Increased employment opportunities, new alternative livelihood opportunities and increased income from the establishment of new community-based enterprises in such things as NTFP production, honey production, livestock management and ecotourism. A total of 2,400 people are expected to join training and a total of 1,236 employees are expected to be generated through the project in the project's lifetime. All the project activities described above are expected to increase the well-being of at least 11,750 people. PRA activities will be conducted each year with the community to monitor the Project's impacts.

4.2.2 Negative Community Impact Mitigation (VCS, 3.19; CCB, CM2.2)

The project aims to provide net positive benefits for all communities in the Project Zone. However, project actions may still result in negative well-being impacts for communities if the mitigation actions outlined below are not implemented. Therefore, a risk mitigation plan will be developed at the start of the Project and updated annually.

The potential negative impacts, as well as their mitigation actions include:

- **Loss of short-term farming income:** Most communities in the project area are farmers. Traditionally, they would increase their income by expanding their crop lands by clearing forested areas. The project will establish and monitor community agreements to protect

forest areas, prohibiting agricultural expansion into these zones. Consequently, this may prevent increases in farming income through land expansion.

To mitigate this impact, agricultural incomes will be raised by increasing productivity through the introduction of agroforestry systems, training and capacity building in alternative agricultural (i.e. livestock) and livelihood systems (i.e. honey production). The project will also explore new markets developments and opportunities along the agricultural value chain to increase returns to the farmer.

Additionally, during the transition period from growing annual traditional (subsistence) crops to a more sustainable and higher yielding (on a per hectare basis) agricultural or agroforestry systems. To mitigate any potential declines and farmer resistance during the transition period, the project will provide wages or *Hari Orang Kerja* (HOK) in Indonesian terms to support the shift from traditional farming systems to agroforestry-based systems.

- **Loss of hunting opportunities and lack of protein:** As a result of the project, the community will be prohibited from hunting threatened species in forests designated for permanent protection. This will greatly reduce the availability of animal protein for the community.

To mitigate this, the project will encourage livestock rearing and management as a commercially viable enterprise through training and capacity-building programs.

- **Uneven and unequal employment opportunities:** The project will establish credible and long-term employment opportunities that may attract people from outside the project zone or unequally benefit some community groups. This may potentially lead to conflicts and disputes within and between communities, as well as with neighboring communities outside the project zone.

To mitigate this conflict, the project will prioritize the employment of local people and ensure that job opportunities are first made available to the community within the project zone. Employment from outside the project zone will only be sought once all efforts to attract local candidates have been exhausted.

- **Oil palm plantation expansion limitations:** Under the proposed project plan, oil palm plantation expansion will be limited, which may lead to disputes between oil palm growers and those benefiting from the transition to agroforestry systems and other alternative livelihoods.

To mitigate this, all oil palm growers will be offered to join this activity to raise their income without expanding their oil palm plantation area and existing oil palm plantation areas will be excluded from the project area.

4.2.3 Net Positive Community Well-Being (VCS, 3.19; CCB, CM2.3, GL1.4)

The communities in all villages are expected to benefit from the project's activities.

Table 82 below outlines the main expected benefits of the project ('with project scenario') compared to the scenario without a project.

Table 81. Expected benefits of the project

Category	Current Condition	Without Project Scenario	With Project Scenario
Continuation of ecosystem services	In the current condition, the standing forest provides a reliable source of good quality drinking water, and abundant supplies of traditional food and medicines for the community. The threat of erosion and sedimentation is minor.	Planned deforestation leads to the conversion of forest areas into monoculture plantations, resulting in the loss or severe degradation of multiple ecosystem services. Traditional food and medicine sources are either lost or greatly diminished. Water quality and quantity decline, accompanied by an increase in erosion and sedimentation of streams and other water resources.	In the project scenario, the ecosystem services within the area are maintained or enhanced, ensuring that the community continues to benefit from accessing these protected services under a permanently protected forest regime.
Education	Communities across the project zone have a low level of education, with only 27% of the members completing nine years of compulsory schooling.	The level of education will be increased because the company is not primarily concerned about the education of villagers. No education plan is listed in the PT CMI RKU.	In the project scenario, the community living within the project zone will increase their access to education facilities. The project will implement educational programs, such as Packages B and C, environmental educational programs for high-school students, scholarship programs, and vocational studies focused on agroforestry systems and other project activities.
Employment	Communities across the project zone have limited employment opportunities, outside	The PT CMI will provide employment opportunities to a select number of villagers, mainly in jobs such as	In the project scenario, the company will provide employment opportunities that align with the educational background of

Category	Current Condition	Without Project Scenario	With Project Scenario
	subsistence agriculture.	harvesting and plantation maintenance. However, because of the limited number of positions available, not all communities will have this benefit. In fact, most will likely lose access to their agricultural lands and farming livelihoods though the establishment of industrial tree plantations.	community members applying. The project will also support the development of agroforestry systems in agricultural areas to increase crop diversity, income streams and establish more climate-resilient agricultural practices. Additionally, other livelihood activities, will be developed with the communities to increase and diversify income streams through initiatives such as livestock raising, honey production, and the sustainable collection and production of NTFPs
Health	Communities across the Project Zone face barriers to accessing healthcare due to the high costs, lack of transport and poor road conditions.	Similar to education, health services will remain unchanged as PT CMI's primary focus is not on providing primary health care for the communities. There is no plan in the PT CMI RKU to improve community access to healthcare.	In the project scenario, the community will gain increased access to healthcare services. The project will facilitate regular visits by doctors to all communities within the project zone. Additionally, training on medicinal plants will be provided to enhance local and traditional knowledge on medicinal plants. This training will be targeted at young people to deepen their cultural and environmental understanding.

Category	Current Condition	Without Project Scenario	With Project Scenario
Access to market	Low access to agricultural markets for local agricultural products.	The situation remains unchanged as there is no plan to enhance community access to markets or to introduce value-added activities for current agricultural products.	In the project scenario, the project will facilitate greater market access and create opportunities for value addition by actively seeking buyers for agricultural and agroforestry products.
Social capital and community infrastructure	There is limited investment in facilities and resources that benefit the community as a whole, such as religious and traditional ceremonies, scholarships, and road infrastructure.	The situation will remain unchanged, with no investment from PT CMI into a community management fund dedicated to development outcomes	A community fund will be established, with a portion of the funds generated through the sale of avoidance credits allocated to it. This fund will be managed by established community funds and used for community development projects and outcomes.

As outlined in Section 3.4, the project aims to achieve Gold Level for Climate Adaptation Benefit. Hence, the project will support communities in adapting to the potential impacts of climate change through forest conservation and restoration efforts, carried out in collaboration with the community. By developing community-based forest management, the project will empower local residents to manage their natural resources, enhancing local resilience. Forest conservation and restoration activities will help maintain essential ecosystem services.

Additionally, the project aims to develop agroforestry systems, which will improve soil health, enhance biodiversity, and increase water retention, thereby providing a buffer against climate extremes such as drought. Agroforestry will also offer diverse income streams through the production of fruits and cash crops as well as generating carbon credits. As the project progresses, other alternative livelihoods will be developed to further support the community's ability to adapt to future climate changes.

To enhance community resilience, the project also plans to install flood early warning system. Recent changes in precipitation patterns, coupled with the widespread of palm oil expansion have increased the risk of floods in the project zone. The December 2024 – January 2025 floods serve as recent examples of how climate change have caused adverse impact to communities in the project zone. By installing a flood early warning system in the river flowing through the project area, information regarding the risks of floods could be disseminated prior to the disaster. This will then provide the relevant stakeholders with ample time and preparation to carry out the evacuation activities.

4.2.4 High Conservation Values Protected (CCB, CM2.4)

The project will ensure that no High Conservation Values (HCV) related to the well-being of the community are negatively impacted. It will permanently protect existing secondary forest by removing the legally sanctioned threat of conversion, while employing forest patrol teams, and restoring degraded forest areas using native species. These activities will protect areas identified as HCV 4 and provide continuous ecosystem services benefits.

Additionally, conservation and forest protection activities, along with restoration, will preserve food sources traditionally obtained from the forest. The project will also monitor the water quality of rivers, to ensure that the water used by the community remains safe. These measures are part of the project's commitment to maintaining and protecting HCV 5.

The project will not relocate cultural artifacts or disrupt the ritual sites of the Dayak communities. The project acknowledges that the entire project zone is customary land. The project will respect and maintain all rights of access and will not interfere with the community conducting traditional rituals within the project zone. Thus, there will be no negative impacts on areas designated as HCV 6.

If additional values are discovered during project implementation, the project will ensure these values are not harmed.

4.3 Other Stakeholder Impacts

4.3.1 Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.1)

A list of stakeholders is outlined in Section 2.3.2. No negative impact on the well-being of these stakeholders is expected. Other stakeholders (those outside the community members) are expected to benefit through greater natural resource management planning, protection of ecosystem services and forest protection, and greater economic resilience.

4.3.2 Mitigation of Negative Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.2)

As mentioned above, negative impacts on other stakeholders through the project are not expected and are unlikely to occur. No mitigation is required.

4.3.3 Net Impacts on Other Stakeholders (VCS, 3.18, 3.19; CCB, CM3.3)

No negative impacts on other stakeholders from the implementation of the project activities have been identified.

4.4 Community Impact Monitoring

4.4.1 Community Monitoring Plan (CCB, CM4.1, CM4.2, GL1.4, GL2.2, GL2.3, GL2.5)

The project proponent will implement a community monitoring plan to identify how the project impacts communities and other stakeholders. The plan will help the project team evaluate the effectiveness of each activity in achieving the project's goals.

Table 83 details the key categories to be monitored, the indicators, the measurement frequency, and data sources. The community monitoring program will utilize both qualitative and quantitative methods, including semi-structured interviews, Participatory Rural Appraisal (PRA) techniques, and Focus Group Discussions (FGDs) with community groups. All data will be gender-disaggregated to document and clearly define impacts on women and girls.

Qualitative data collection will be carried out using Participatory Rural Appraisal (PRA) activities, while quantitative data will be gathered through surveys, with the sampling technique chosen based on the specific indicator being measured. For all community indicators, please see Appendix 2.

Table 82. Category, indicator, frequency and data source for monitoring community program activities.

Category	Indicator	Monitoring frequency	Data source
Employment	1. Number of community-based forest and fire patrolling groups established 2. Number of people employed in each forest and fire patrolling groups 3. Number of people employed in the nursery 4. Number of people employed as biodiversity team members 5. Number of community members employed in newly established alternative livelihood programs, including agroforestry programs	Annually	Internal report
Training	1. Number of people that join invasive species awareness training programs and the benefit of forest for ecosystem services 2. Number of people that join biodiversity awareness raising programs 3. Number of people that join agroforestry training programs 4. Number of people that join local culture and wisdom workshops 5. Number of community funds' staff that join development fund management training.	Annually	Internal training report
Education	1. Number of people enrolled in the Package B educational program 2. Number of people enrolled in the Package C educational program	Annually	Internal education Report

Category	Indicator	Monitoring frequency	Data source
	3. Number of people that receive a partial scholarship 4. Number of people enrolled in a vocational study program 5. Number of elementary and middle school students participating in the Environmental Education Program		
Health	1. Number of doctors visiting villages each year under the 'Doctor goes to village' program or healthcare services provision per year. 2. Number of people that visit a doctor in the 'Doctor goes to village' program or use the provided healthcare services each year 3. Number of people as beneficiaries of improved access to clean water program	Annually	Health program report
Participation of Women	1. Number of women employed by the project (calculated based on number of women employment under the Employment category) 2. Number of women who attend each training held by the project (calculated based on number of women trained under the Training category) 3. Number of women enrolled in the Package B educational program 4. Number of women enrolled in the Package C educational program 5. Number of women that receive a partial scholarship 6. Number of women enrolled in a vocational study program 7. Number of girls participating in the Environmental Education Program 8. Number of women and girls that visit a doctor in the 'Doctor goes to village' program each year	Annually	1. Internal report. 2. Internal training report. 3. Internal education report. 4. Doctor goes to village program report.
Livelihood	1. Number of alternative livelihoods created	Annual	Internal report

Category	Indicator	Monitoring frequency	Data source
	2. Number of people implementing alternative livelihood activities		
HCV 4	1. water quality monitoring through standard indicators as mandated by environmental license of the concession area 2. Percentage change in forest cover in project zone 3. Erosion related indicators	Annual	Internal report
HCV 5	1. Water quality indicators 2. Number of households with access to cultivated land to grow rice	Annual	Internal report
HCV 6	1. Number of cultural sites protected in the project zone 2. Change in condition of cultural sites 3. Number of cultural events (e.g. <i>gawai</i>) held by community in each village	Annual	Internal community report

Besides monitoring the impact of the project on the community's well-being, the project will also monitor risks to community members. The identified risks are described in sections 4.2.2 and 4.5.3. The following table shows the monitoring plans developed to monitor these risks.

Table 83. Risk monitoring indicators

Category	Indicator	Frequency	Data source
Lack of participation	Meeting Structures and attendance 1. Percentage of women, physically disadvantaged, and other marginalized groups attending meetings 2. Number of meetings structured to specifically accommodate the needs of women and marginalized community members 3. Number of meetings chaired by women or other marginalized groups	Semester	Attendance records and meeting logs Meeting records and survey or FGD

Category	Indicator	Frequency	Data source
	4. Percentage change in total number of community members attending meetings. Outreach and communication 1. Number of booklets and flyers published	Semester	Communication material logs
Poor project management	Human resources 1. Number of trainings provided on project management to project team members 2. Number of project plans that are publicly available stating project timelines, deliverables, resource allocation and budgets. 3. Number of project milestones delivered on time, within budget and reported to community. Participation in project activities: 1. Percentage of community members benefiting from project activities 2. Community feedback on the fairness of benefit distribution	Semester Annual Bi-annually	Training records and attendance logs Project planning documentation, including budget reports Survey, interviews or FGD
Elite capture	Equitable resource distribution: 1. Publication of fund allocation 2. Community feedback on fairness and transparency of resource distribution Community fund management: 1. Number of community funds operating 2. Frequency of publicly available audits of community funds Incentive payment 1. Percentage of community receiving economic benefits such as wages or HOK for transitioning to agroforestry	Bi-annually Bi-annually Bi-annually	Project financial records and publications Survey or FGD Financial audit reports Payment records or survey and training records

4.4.2 Monitoring Plan Dissemination (CCB, CM4.3)

The community monitoring plan will be disseminated to all communities and other key stakeholders, in both Indonesian and local Dayak languages. In addition, the project team will provide regular updates on project progress and the outcomes of all monitoring events. This information will be available on the project website and shared throughout village meetings and information boards.

4.5 Optional Criterion: Exceptional Community Benefits

4.5.1 Exceptional Community Criteria (CCB, GL2.1)

The project meets the Climate, Community and Biodiversity (CCB) criteria for exceptional community benefit, as over 50% of villagers in the Project Zone are below Indonesia's national poverty line of IDR 2,592,657 per month per household.¹⁰⁹ Based on the socio-economic assessment conducted by the project, the average income of the villages in the project zone ranges from IDR 1,000,000 to 1,700,000 per month per household, with an average of IDR 1,222,500 per month per household.

4.5.2 Short-term and Long-term Community Benefits (CCB, GL2.2)

In the short term, community benefits include:

1. Improved access to healthcare services through the hamlet and village health care program.
2. Improved access to education through Package B, Package C, and scholarship assistance programs.
3. Increased environmental awareness and access to education opportunities through environmental education programs delivered through the village schools.
4. Employment opportunities in the project zone are enhanced through open and transparent company recruitment policies in the company's recruitment program.
5. Increased income, and diversity of income, through newly introduced agroforestry systems and other alternative livelihood opportunities.

In the long term, the identified benefits include:

1. Protection of air and water quality and control of sedimentation, erosion, and floods through forest protection and the restoration of degraded forest areas.
2. Improved quality of life and community resilience through education, health and economic development programs.

¹⁰⁹ In 2023, the national poverty line was set at a per capita expenditure of IDR 550,458 per month per capita or 2,592,657 per household (4.71 members per household). [Indonesia Poverty Profile in March 2023 - BPS-Statistics Indonesia](#).

4.5.3 Community Participation Risks (CCB, GL2.3)

Community participation risks were identified through targeted community engagement process, community assessment studies and the FPIC process. This approach emphasizes transparency, openness, and active community involvement, providing a range of various forums for community members to openly discuss potential risks and negative impacts on their livelihoods. Regular village meetings will be served as a platform for socializing and evaluating the project with the community. The guiding principles for these meetings were:

1. Open dialog provides opportunities for everyone to be actively involved.
2. Ensuring that all community groups are represented in each meeting to guarantee equal representation and involvement opportunities.
3. Transparent and open decision-making process.

A grievance procedure will also put in place to provide a formal process for community members to raise their concerns and complaints related to the project.

Through this process, three key risks were identified:

1. **Lack of Participation:** Some groups avoid attending meetings, due to remoteness, physical or mental disadvantages, poverty or other socio-economic factors leading to marginalization. The Project will actively encourage all community members, especially the poorer and marginalized, to participate in meetings and project activities through tailored approaches. The participatory planning processes will ensure that all communities in the project-zone are involved in decision-making and community meetings. Meetings will be scheduled at various locations and times to accommodate all community members. Where appropriate, meetings will be structured and facilitated for actively engaged marginalized groups. For example, women-only meetings may be held at their homes during the evening, when they usually have spare time. Additionally, community message boards, booklets, flyers and videos, and local radio programs could also be used to reach the target audience effectively.
2. **Improper management of project activities:** Improper management of project activities due to insufficient human and financial resources, as well as ineffective monitoring and evaluation systems, may hinder full community participation. To address this, the implementation of project activities, allocation of resources, and distribution of benefits will be regularly monitored to ensure that community and marginalized households are benefiting from the project. Detailed monitoring and a stringent system of checks and balances are crucial to prevent the inappropriate allocation of resources, which could further disadvantage already marginalized communities. Training will be provided to build the capacity of local people, and community-based adaptive management planning will allow open and transparent decision-making processes, fostering democratic natural resources governance.
3. **Elite capture:** Power dynamics within a community can be challenging to assess, and often difficult to rebalance. Without carefully monitoring and control of resources, there is a risk that these resources may be unequally distributed, favoring community leaders or elites to

the detriment of poorer and marginalized households. To address this, the project will establish strong guidelines and monitoring processes to ensure that project funds are distributed fairly, equitably, and transparently

As part of benefit sharing mechanism, project shall set up a community fund as an attempt to increase project's outreach to those with limited access within the village communities. Additionally, the project will provide incentive payments directly to farmers to support their transition from traditional farming systems to agroforestry systems. Both fund allocations will be subject to careful scrutiny, so resources do not inadvertently or directly benefit community elites.

4.5.4 Marginalized and/or Vulnerable Community Groups (CCB, GL2.4)

The description for the marginalized and/or vulnerable community groups that the project is engaging with and how the communities will gain net positive benefits is provided in Table 85 and Table 86 below.

Table 84. Description of impacts and benefit access for women community group

Community Group	Women
Net positive impacts	The project benefits women by strengthening their roles and promoting gender equality within their households and in broader community. This is achieved through various measures including the provision of training, employment opportunities, improved access to health care facilities, alternative livelihood programs and initiatives to involve women in decision-making processes.
Benefit access	Cultural barriers may prevent the project from fully engaging with women, as they are often restricted to household activities and may be reluctant to actively participate in decision-making processes. This can result in their voice not being heard and their views ignored. The project will emphasize and promote the importance of gender equality in all decision-making processes throughout the duration of the project. This includes conducting targeted consultation processes with women to specifically collect their views, concerns and feedback on project design and implementation.
Negative impacts	No negative impacts on women are expected during the project design and implementation. Due to the iterative consultation and engagement process with women and all community members, it is expected that any potential negative impacts or risks will be identified early for corrective

	actions to be enacted following the project's adaptive management model.
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Table 85. Description of impacts and benefit access for smallholder farmer community group

Community Group	Smallholder farmers
Net positive impacts	Smallholder farmers will receive support through agricultural capacity building and training programs focused on sustainable and climate-resilient agroforestry systems. The project employment opportunities and financial assistance will also be provided to support smallholder farmers transition from traditional (subsistence) farming to more permanent, tree-based cropping and agroforestry systems
Benefit access	The project may face challenges in ensuring that smallholder farmers (who tend to be poorer and more vulnerable) equally benefit. However, the project is designed for all farmers, regardless of size. Project planning and monitoring will be conducted throughout the implementation stage to ensure that obstacles to participation by smallholder farmers (i.e. capital, knowledge, skills) are removed and resources are allocated fairly and equitably across the farming community. This will ensure the full involvement of the entire farming community,
Negative impacts	Farmers might experience short-term cash flow challenges and economic instability during the transition from traditional and subsistence (annual) farming to tree-cropping and agroforestry systems. To mitigate this, the project will provide wages (HOK) to support farmers during the transition to agroforestry systems yielding crops. An agreement will also be reached by PT CMI and the farming community on revenue distribution for such crops as coffee, cocoa and cash crops such as pepper and ground nuts.

4.5.5 Net Impacts on Women (CCB, GL2.5)

Based on social studies conducted at the village level, participation of women in decision-making processes and their economic opportunities are low.

One of the objectives of the project is to strengthen the role of women in the community and generate positive livelihood impacts for them. This is to be achieved by ensuring open and equal participation in all decision-making processes and their strong involvement in the project design, implementation, monitoring and reporting.

To achieve this, several activities will be established specifically for women:

1. Alternative livelihood programs that provide training and financial assistance for women's groups to establish viable business models targeted to their needs.
2. A training program for enhancing the participation of women in decision-making processes. This activity will highlight the importance of involving women in decision-making processes, where their voice is often ignored, silenced, not heard or not sought.
3. Women's involvement in all village meetings, village forums, council or representative meetings and other important processes such as the FPIC process.
4. Promoting and improving access to primary health care and healthcare services focused on women's health, while actively removing barriers such as cost, stigma, social barriers, and time availability that hinder women from accessing these services.
5. The promotion and participation of girls in all the project scholarship opportunities. The project aims to achieve at least 48% participation by girls and women in these programs.
6. The recruitment, participation and promotion for women in project-based employment. The project aims to achieve a minimum employment level of 48% women in all project-based roles.

These activities are designed to promote the role of women in village decision making processes, while increasing their access to livelihood and economic development opportunities from the project.

4.5.6 Benefit Sharing Mechanisms (CCB, GL2.6)

The benefit of the sharing mechanism was described in Section 2.5.8

4.5.7 Benefits, Costs, and Risks Communication (VCS, 3.18; CCB, GL2.7)

During the informational meeting, the project's benefits, costs, and risks were openly and transparently discussed with the community, including their perception of these issues. Short and simple brochures were produced in Indonesian, summarizing the project objectives, implementation arrangements and key underlying principles such as forest conservation, biodiversity protection, and livelihood and community development.

At the end of the informational workshop, an official report was drafted and circulated among community members to ensure they clearly understood the meeting's aims, process and outcomes before signing it. The official report of the informational meeting can be seen in Annex 4.

4.5.8 Governance and Implementation Structures (CCB, GL2.8)

The project will be partially self-governed and implemented at the hamlet (sub-village) and village level, in collaboration with customary Dayak and government structures, related institutions and the project management team. Local community involvement in the project is also embedded through participatory planning processes, community monitoring and evaluation processes, and a grievance redress mechanism.

Community members will also play a key role in project implementation through recruitment of community-based staff. This approach ensures effective communication of project planning and implementation within the community and allows for in-depth-monitoring by community members. Where possible, and aligned with their capacity and capabilities, community members will be given middle management roles, further enhancing local staff involvement in project planning and execution.

The involvement of the communities in the project activities is based on an inclusive approach that aims to ensure effective participation of all community members. This, and the other FPIC principles will ensure information disclosure and involvement of all community groups throughout all stages of the project: design, implementation, monitoring and reporting.

4.5.9 Smallholders/Community Members Capacity Development (CCB, GL2.9)

Training and capacity building is a core component of the Sanggala Corridor Project as it aims to build a sustainable and climate-resilient agroforestry system, alongside a broader green economy within the project zone. The project will conduct several training and capacity-building activities for the community and local staff. These include:

- Participatory land use planning and agroforestry training and capacity building programs for farmers.
- Alternative livelihood programs for farmers and community members, including livestock rearing and management for farmers, honey production and possibly ecotourism opportunities.
- Strengthening the capacity of established community funds' staff to manage the community development fund as well as grow and expand profitable village owned businesses and.
- Forest protection, patrolling and community fire management
- Threatened species and biodiversity protection, monitoring and recording
- Scholarship programs for community members
- Environmental education workshops for students

The training will primarily target disadvantaged community members, and smallholder farmers with limited capacity and capability to transition away from traditional, subsistence-based farming practices. By providing this training, the project team aims to build resilience and permanence, helping the community better understand the project's goals and benefits and provide incentives to transition towards a more stable, profitable and culturally appropriate 'green economy'.

The training program for local staff employed in the project will focus on developing their skills and knowledge in their respective fields, and enhancing their future employment prospects. Additionally, training will cover the project's policies and Standard Operating Procedures (SOPs), with a particular emphasis on health and safety.

All training will be evaluated to assess training delivery style, learning outcomes and training impacts. Training programs will be critically reviewed based on this feedback and continually adapted to meet the needs of those participating in the training and capacity building programs.

5 BIODIVERSITY

5.1 Without-Project Biodiversity Scenario

5.1.1 Existing Conditions (VCS, 3.19; CCB, B1.1)

Ecosystem Description

The project area is located in the tropical lowland rainforest of Borneo, where the trees from the *Dipterocarpaceae* family dominates the ecosystem in this region and the annual rainfall in the region corresponds to the tree diversity¹¹⁰. According to Koppen Geiger Climate classification, this ecosystem has equatorial rainforest climate with minimum temperature and monthly rainfall precipitation value at 18°C and 60 mm respectively¹¹¹. As demonstrated by Important Value Index in Table 83 bellow, *Shorea* spp. and *Dipterocarpus* spp. from *Dipterocarpaceae* family play critical role in the project area. In addition to the forest cover as the main feature, the project area includes non-forest area, which was mainly formed due to the intervention of activities by communities.



Figure 26. Bird eye view of the Sanggala Corridor Project Area

110 JWF, Slik, N, Raes, SI, Aiba, FQ, Brearley, CH, Cannon, E, Meijaard, H, Nagamasu, R, Nilus, G, Paoli, AD, Poulsen, D, Sheil, E, Suzuki, JLCH, van Valkenburg, CO, Webb, P, Wilkie, S, Wulffraat. (2009), Environmental correlates for tropical tree diversity and distribution patterns in Borneo. *Diversity and Distributions* 15: 523-523.

111 M, Kottek, J, Grieser, C, Beck, B, Rudolf, F, Rubel. (2006) World Map of the Koppen-Geiger climate classification updated. *Meteorologische Zeitschrift*. 15(3): 259-263. DOI: 10.1127/0941-2948/2006/0130

The non-forest area is predominantly degraded due to historical logging and the conversion of forest lands into agricultural areas. The land use in non-forest area includes agricultural land, estate plantations, bare ground, and shrubs, reflecting various stages and extent of intervention that communities have been conducting in the project area. The practice of shifting agriculture of local Dayak communities drives the dynamic of land cover, though natural regrowth of the area formerly cultivated is widely documented in the project area. On the other hand, the forest area remains in good condition and is undergoing natural succession after logging in the 1990s. The dominant plants in the forest area include *Shorea spp*, *Litsea spp*, *Syzygium spp*, and *Dipterocarpus spp*. The vegetation structure of the forest area is shown in Table 87 below.

Table 86. Vegetation composition structure of the forest area in the project area¹¹²

Vegetation level	Species type	Important Value Index ¹¹³
Big Tree	<i>Shorea spp.</i>	24.83
	<i>Litsea spp.</i>	20.95
	<i>Syzygium cerinum</i> (M.R. Hend.) I.M.Turner	16.95
	<i>Syzygium attenuatum</i> (Miq) Merr & L.M Perry	15.90
	<i>Dipterocarpus spp.</i>	13.85
Small Tree	<i>Litsea spp.</i>	41.07
	<i>Shorea spp.</i>	25.38
	<i>Syzygium cerinum</i> (M.R. Hend.) I.M.Turner	23.51
	<i>Syzygium attenuatum</i> (Miq) Merr & L.M Perry	22.99
	<i>Vatica rassak</i> (Korth.) Blume	13.70
Pole	<i>Litsea spp.</i>	37.26
	<i>Shorea spp.</i>	31.22
	<i>Syzygium attenuatum</i> (Miq) Merr & L.M Perry	28.89
	<i>Syzygium zollingerianum</i> (Miq.) Amshoff	26.16
	<i>Syzygium cerinum</i> (M.R. Hend.) I.M.Turner	24.86
Sapling	<i>Syzygium attenuatum</i> (Miq) Merr & L.M Perry	29.49
	<i>Litsea spp.</i>	28.00
	<i>Syzygium cerinum</i> (M.R. Hend.) I.M.Turner	20.91
	<i>Shorea spp.</i>	16.61
	<i>Syzygium zollingerianum</i> (Miq.) Amshoff	14.50
Seedling	<i>Litsea spp.</i>	22.62
	<i>Syzygium cerinum</i> (M.R. Hend.) I.M.Turner	17.05
	<i>Syzygium attenuatum</i> (Miq) Merr & L.M Perry	17.04
	<i>Shorea spp.</i>	14.75
	<i>Syzygium zollingerianum</i> (Miq.) Amshoff	9.84

¹¹² Forest Inventory Activity

¹¹³ Important value index is ecological measures about species dominance within an ecosystem

The Sanggala Corridor project area is adjacent to the *Gunung Nyiut Penrissen* Nature Reserve, to the north. The reserve is home of at least 44 mammal species, including the orangutan (*Pongo pygmeaus*), maroon leaf monkey (*Presbytis rubicunda*), and the southern grey gibbon (*Hylobates muelleri*), 63 bird species, including the helmeted hornbill (*Rhinoplax vigil*) and the greater green leafbird (*Chloropsis sonnerati*), 12 amphibians, including the Sambas stream toad (*Ansonia latidisca*). Hundreds of wild orchids and woody dominated plants by the Dipterocarpaceae family are also common to be found in the Nature Reserve Area ^{114, 115}.

To the western part of the project area is a protected forest that is connected to the *Gunung Nyiut Penrissen* Nature Reserve. This protected forest is particularly diverse in orchids species, hosting at least 14 orchid genera, namely *Coelogyne*, *Bulbophyllum*, *Dendrobium*, *Liparis*, *Eria*, *Thelasis*, *Mycaranthes*, *Phalaenopsis*, *Agrostophyllum*, *Cerastostylis*, *Dipodium*, *Bromheadia*, *Thrixspermum*, and *Trichotosia* ¹¹⁶. This protected forest acts as a buffer between the Nature Reserve and the production forest areas, where the Sanggala project area is located

Biotic Features of the Project Zone

Given its strategic position sharing a border with both protected forest and conservation forest, high biodiversity condition is one of key features of the Project Area, as supported by data collected on the field. From November 2023 to March 2024, a series of biodiversity assessments¹¹² were conducted by the project team through a Participatory Rural Appraisal (PRA). In conjunction with the PRA activities, direct field assessments of mammal, bird, reptile, amphibian, and vegetation biodiversity were also carried out.

Participatory Rural Appraisal (PRA) Data Collection

Collection of data regarding presence of animals within the project zone was carried out in conjunction with participatory rural appraisal aimed at determining the baseline of social condition of communities. For each villager participating in this survey, pictures of animal species were shown for them to confirm their sightings and presence in the project zone.

¹¹⁴ Rimba Nyiut (1): Rumah Flora & Fauna Mengagumkan di Titik Tertinggi KalBar | kumparan.com

¹¹⁵ Ahmad, A., Gary, D., Rodiansyah, Sinta, Srifitria, Putra, W., Sagita, N., Adirahmanta, S.N. and Miller, A.E. (2021), Leveraging local knowledge to estimate wildlife densities in bornean tropical rainforests. Wildlife Biology, 2021: 1-15 wlb.00771. <https://doi.org/10.2981/wlb.00771>

¹¹⁶ Manik, F., Suryantini, R., & Husni, H. (2017). IDENTIFIKASI FAMILI ORCHIDACEAE DIKAWASAN HUTAN LINDUNG DESA SEKENDAL KECAMATAN AIR BESAR KABUPATEN LANDAK.



Figure 27. The project team shows species list to be confirmed with the community during the PRA activity

Out of the 119 species shown to the community, only one species the presence of which is unable to be confirmed by communities in the project zone: *Nasalis larvatus*. Vast animal species presumably inhabiting the area able to be confirmed by community through sighting including *Manis javanica* (Sunda pangolin), *Buceros rhinoceros* (Rhinoceros hornbill), *Neofelis diardi* (Sunda clouded leopard), *Lanthanotus borneensis* (Borneo earless monitor), etc.

In general, the community in the project zone confirms 39 threatened species inhabiting the project zone. The list of the species confirmed by the community can be seen in Table 88.

Table 87. Threatened species list confirmed by the community living in the project zone

No	Type	Species Name	Family	IUCN Status	CITES
1	Mammal	<i>Manis javanica</i>	<i>Manidae</i>	CR	Appendix II
2	Mammal	<i>Pongo pygmaeus</i>	<i>Hominidae</i>	CR	Appendix I
3	Mammal	<i>Presbytis chrysomelas</i>	<i>Cercopithecidae</i>	CR	Not Listed
4	Bird	<i>Pseudibis davisoni</i>	<i>Threskiornithidae</i>	CR	Not Listed
5	Bird	<i>Pycnonotus zeylanicus</i>	<i>Pycnonotidae</i>	CR	Not Listed
6	Mammal	<i>Hylobates albibarbis</i>	<i>Hylobatidae</i>	EN	Appendix II
7	Bird	<i>Chloropsis sonnerati</i>	<i>Chloropseidae</i>	EN	Not Listed
8	Bird	<i>Ciconia stormi</i>	<i>Ciconiidae</i>	EN	Not Listed
9	Mammal	<i>Hylobates muelleri abotti</i>	<i>Hylobatidae</i>	EN	Appendix II
10	Mammal	<i>Macaca fascicularis</i>	<i>Cercopithecidae</i>	EN	Not Listed
11	Mammal	<i>Macaca nemestrina</i>	<i>Cercopithecidae</i>	EN	Not Listed
12	Bird	<i>Polyplectron schleiermacheri</i>	<i>Phasianidae</i>	EN	Not Listed
13	Mammal	<i>Pteropus vampyrus</i>	<i>Pteropodidae</i>	EN	Appendix II
14	Bird	<i>Rhabdotorrhinus corrugatus</i>	<i>Bucerotidae</i>	EN	Not Listed
15	Fish	<i>Scleropages formosus</i>	<i>Osteoglossidae</i>	EN	Appendix II
16	Bird	<i>Anthracosceros malayanus</i>	<i>Bucerotidae</i>	VU	Not Listed
17	Mammal	<i>Arctictis binturong</i>	<i>Viverridae</i>	VU	Appendix III
18	Bird	<i>Buceros rhinoceros</i>	<i>Bucerotidae</i>	VU	Appendix II

No	Type	Species Name	Family	IUCN Status	CITES
19	Bird	<i>Caprimulgus concretus</i>	<i>Caprimulgidae</i>	VU	Not Listed
20	Mammal	<i>Cephalopachus bancanus</i>	<i>Tarsiidae</i>	VU	Not Listed
21	Mammal	<i>Helarctos malayanus</i>	<i>Ursidae</i>	VU	Appendix III
22	Bird	<i>Hydrornis baudii</i>	<i>Pittidae</i>	VU	Not Listed
23	Herpet	<i>Lanthanotus borneensis</i>	<i>Lanthanotidae</i>	VU	Not Listed
24	Bird	<i>Leptoptilos javanicus</i>	<i>Ciconiidae</i>	VU	Not Listed
25	Bird	<i>Lophura bulweri</i>	<i>Phasianidae</i>	VU	Not Listed
26	Bird	<i>Melanoperdix niger</i>	<i>Phasianidae</i>	VU	Not Listed
27	Mammal	<i>Neofelis diardi</i>	<i>Felidae</i>	VU	Appendix I
28	Bird	<i>Nisaetus nanus</i>	<i>Accipitridae</i>	VU	Not Listed
29	Mammal	<i>Nycticebus menagensis</i>	<i>Lorisidae</i>	VU	Not Listed
30	Mammal	<i>Presbytis frontata</i>	<i>Cercopithecidae</i>	VU	Not Listed
31	Mammal	<i>Presbytis hosei</i>	<i>Cercopithecidae</i>	VU	Not Listed
32	Mammal	<i>Presbytis rubicunda</i>	<i>Cercopithecidae</i>	VU	Not Listed
33	Bird	<i>Ptilocichla leucogrammica</i>	<i>Pellorneidae</i>	VU	Not Listed
34	Mammal	<i>Rheithrosciurus macrotis</i>	<i>Sciuridae</i>	VU	Not Listed
35	Bird	<i>Rollulus rouloul</i>	<i>Phasianidae</i>	VU	Not Listed
36	Bird	<i>Setornis criniger</i>	<i>Pycnonotidae</i>	VU	Not Listed
37	Mammal	<i>Sus barbatus</i>	<i>Suidae</i>	VU	Not Listed
38	Mammal	<i>Trachypithecus cristatus</i>	<i>Cercopithecidae</i>	VU	Not Listed
39	Bird	<i>Treron capellei</i>	<i>Columbidae</i>	VU	Not Listed

The community confirmed the sightings of *Pongo pygmaeus* (Borneo orangutan) in the project zone. However, the latest sighting of the species was back in 2019 and there has not been any documentation of its sighting ever since according to the account of local communities in the area.

Wildlife Field Data Collection

The information on animal presence in the project area was further solidified through direct field assessment. The focus of the survey is mammals, birds, and herpetofauna. Standardized and simplified survey methods were used for each taxon (e.g., line transect for mammals, point count for the avifauna, and Visual Encounter Survey for the herpetofauna).

Four survey locations were selected during the assessment, three in forested areas and one in a non-forest area. The survey location in the forested areas were chosen based on primary habitat information provided by the community during the PRA activity, to increase the probability of animal encounters. For each survey location, surveyor team walked through a designated transect line of 2 km.

The field assessment conducted within the project area recorded a total of 62 avian species. The three most observed were *Megalaimidae* (10 species), *Pycnonotidae* (7 species), and *Cuculidae* (5 species). The assessment revealed 1 species classified as Endangered, 5 as Vulnerable, 14 as Near Threatened, 42 as Least Concern, and 1 as Data Deficient, according to the IUCN Red List. Species categorized as Critically Endangered, Endangered, and Vulnerable are collectively considered threatened. Six threatened species documented in the project area include *Chloropsis sonnerati*, *Aceros undulatus*, *Argusianus argus*, *Buceros rhinoceros*, *Harpactes reinwardtii*, and *Treron capellei*. In addition, the Rhinoceros hornbill (*Buceros rhinoceros*) is listed in Appendix II of CITES, which regulates international trade.

During the field assessment, the mammalian family *Sciuridae* was the most frequently encountered, with 25 species recorded in the project area, including the Endangered *Pteromys pulverulentus* and the Vulnerable *Petinomys vordermanni* and *Rheithrosciurus macrotis*. The *Tupaia* family was the second most encountered, with 5 species identified, including *Tupaia minor*, *Tupaia montana*, and *Tupaia tana*.

Other species observed in the field included *Macaca fascicularis*, *Macaca nemestrina*, *Arctictis binturong*, and *Hylobates muelleri abbotti*. Of the three Critically Endangered mammal species identified by local communities during the Participatory Rural Appraisal (PRA), only *Presbytis chrysomelas* was observed during the field assessment, although it was sighted outside the transect line. However, potential *Manis Javanica* burrow identified during the vegetation assessment. Additionally, three species identified are listed on CITES Appendix II (*Arctictis binturong*, *Diplogale derbyanus*, and *Tupaia dorsalis*), and one species is listed on CITES Appendix I (*Hylobates muelleri abbotti*).



Figure 28. Identified potential *Manis javanica* burrow found in the project area

With regards to herpetofauna, the *Dicroglossidae* family has the most species identified during the field assessment. No species of herpetofauna found in the project zone are listed as threatened species in IUCN and listed in the CITES appendix.

The field survey and PRA data collection on wildlife species have led to one conclusion: the project area harbours essential wildlife biodiversity that is inseparable with ecological function in the ecosystem.

Similar with other forest ecosystem in Borneo, the project is home for the Hornbill bird (*Buceros rhinoceros*). One of the most renowned species in Kalimantan, Hornbill bird has essential ecological function in seeds dispersal to ensure natural regeneration and survival of the forest ecosystem. It is therefore appropriate that the term of “farmers of the forest” is associated with this bird species¹¹⁷. In addition to ecological function, hornbill birds have equally important cultural values for Dayak communities – the indigenous tribe of Borneo – which regards hornbill birds as symbol of honour, bravery, and spirituality. Depictions of hornbill birds are common to be found in Dayak cultural practice, such as on tattoos, dances, and rituals¹¹⁸.

Another group of wildlife species identified to inhabit the project area and has the capacity to play critical role to facilitate sustainability of the ecosystem is the primate species. Several species identified in the area including the *Macaca fascicularis*, *Hylobates muelleri abbotti*, and *Presbytis chrysomelas*. Noting primate’s role as the herbivore in the food chain, they have equally instrumental role to act as seed dispersal with diverse potential amongst species regarding the distance to carry seeds¹¹⁹.

There is a certain wildlife species identified in the project area having an increased urgency to be protected from biodiversity conservation perspectives: Sunda Pangolin (*Manis javanica*). While other wildlife species are facing sporadic threat of illegal hunting due to embedded cultural practice, the hunting of pangolin is considered more widespread and, in some cases, could be considered as part of organized crime¹²⁰. This obviously necessitates the conservation of Sunda pangolin in the project area, since the species also plays an essential role in affecting vegetation renewal and improve habitat heterogeneity¹²¹.

Present on the top of food chain in the ecosystem of the project area is the Sunda clouded leopard (*Neofelis diardi*). The presence Sunda clouded leopard in the project zone is confirmed by local communities during PRA activities though not observed throughout field survey,

¹¹⁷ Franco, FM & Minggu, MJ (2019) When the seeds sprout, the hornbills hatch: understanding the traditional ecological knowledge of the Ibans of Brunei Darussalam on hornbills. *Journal of Ethnobiology and Ethnomedicine* 15:46. <https://doi.org/10.1186/s13002-019-0325-0>

¹¹⁸ Kurniawan, FH, Rahmansyah, R, Rahman, A, Harditanti, Kurniawan, J, Putra, ID & Hadiprakarsa, Y. (2024). Local community perceptions on the conservation of hornbill (Family Bucerotidae) in West Kalimantan, Indonesia. 25(4) : 1702-1710. DOI: 10.13057/biodiv/d250440

¹¹⁹ McConkey, KR. (2017). Seed Dispersal by Primates in Asian Habitats: From Species, to Communities, to Conservation. *International Journal of Primatology*. <https://doi.org/10.1007/s10764-017-0013-7>

¹²⁰ Pradityo, H. & Susanti, V. (2024). Kejahatan terhadap trenggiling yang melibatkan organized crime di Kalimantan Barat. *Jurnal Sosial dan Sains*. 4(1).

¹²¹ Sun, S, Wei, W, Dou, H, Chen, S, Gao, H, Yang, J, Wang, J, Zhang, Y, Zhang, Y, Gio, R, Zhang, S, Du, Y/ Kuang, Y. & Hua, Y. (2024). Identifying habitat modification by Chinese pangolin in subtropical forests of southern China.);1-15. doi: 10.1111/1749-4877.12862

underlining the critical importance of the project area as habitat range of the species. Considered as apex predator in the forest ecosystem of Borneo, Sunda clouded leopard has the potential to be considered as a species model to develop model of connectivity¹²². A species known for its secretive behaviour¹²³, Sunda clouded leopard's habitat use is highly influenced by the presence of other species such as macaques, mouse, deer, and small carnivores, which is important for this species to viably persist¹²⁴.

In summary, the biodiversity of wildlife species encountered during field assessment are reflected through the following metrics in the following table.

Table 88. Biodiversity metric along the transect line in the project area.

Metric	Transect 1	Transect 2	Transect 3	Transect 4
s (number of species discovered)	59	64	69	71
N (total number individuals)	234	239	264	377
Shanon Diversity Index	3.569	3.671	3.738	3.591
Simpson Index of Diversity (Ds)	0.9634	0.962	0.967	0.959
Species evenness index (Ed)	0.8754	0.8794	0.880	0.843
Population estimation (per ha)	5.8500	5.9750	6.600	9.425

Collectively, the consistently high diversity indices across the transects suggest that the project area supports a diverse and abundant community of species.

Flora Field Data Collection

The assessment of flora species in the project area was conducted through a rigorous field study. Sampling locations were determined using stratified purposive sampling based on the ecosystem types within the project area, which include Forest and Non-Forest Areas. A total of 94 plots were sampled, using a nested plot sampling method. Each plot measured 0.25 ha (50 m x 50 m), with sub-plot sizes varying according to the vegetation level, with 50 m x 50 m for big trees, 25 m x 25 m for small trees, 10 m x 10 m for poles, 5 x 5 for saplings, and 2 m x 2 m for seedlings¹¹².

The survey identified at least 202 flora species in the project area, 17 of which are listed as threatened by the IUCN Red List. Four species are categorized by the IUCN as Endangered, namely the tropical chestnut (*Castanopsis argentea*), Mahang (*Macaranga hypoleuca*), Meranti bunga (*Shorea uliginosa*), and White meranti (*Shorea virescens*), and 14 species categorized as Vulnerable including the Bengkirai (*Shorea laevis*), Damara (*Agathis* spp.) and the Kayu Besi

¹²² Hearn, AJ. Cushman, SA. Goossens, B. Ross, J. McDonald, EA. Hunter, LTB. & Macdonald, DW. (2019). Predicting connectivity, population size and genetic diversity of Sunda clouded leopards across Sabah, Borneo. *Landscape Ecology*. 34: 275-290. <https://doi.org/10.1007/s10980-018-0758-1>

¹²³ Hearn, J. Ross, J. Macdonald, DW. Bolongon, G. Cheyne, SM. Mohamed, A. Samejima, H. Brodie, JF. Giordano, A. Alfred, R. Boonratana, R. Bernard, H. Loken, B. Augeri, DM. Heydon, M. Hon, J. Mathai, J. Marshall, AJ. Pilgrim, JD. Hall, J. Breitenmoser_Wursten, C. Kramer-Schadt, S. Wilting, A. (2016). Predicted distribution of the Sunda clouded leopard *Neofelis diardi* (Mammalia: Carnivora: Felidae) on Borneo. *Raffles Bulletin of Zoology*. 33:149-156

¹²⁴ Tan, CKW. Rocha, DG. Clements, GR. Brenes-Mora, E. Hedged, L. Kawanishi, K. Mohamad, SW. Rayan, DM. Bolongon, G. Moore, J. Wadey, J. Campos-Arceiz, A. Macdonald, DW. (2017). Habitat use and predicted range for the mainland clouded leopard *Neofelis nebulosa* in Peninsular Malaysia. *Biological Conservation*. 206: 65-74. <https://doi.org/10.1016/j.biocon.2016.12.012>

(*Afzelia rhomboidea*). No critically endangered species were found during the assessment within the project area.

Notably, the critical importance of conservation of biodiversity in the project noting the high potential of floral biodiversity in the project area, such as on conservation status and prospects value. As previously mentioned, the project team is able to identify the presence of *kayu ulin* (*Eusideroxylon zwageri*) during forest inventory activities. An endangered tree species renowned for its quality and durability, the presence of *kayu ulin* in the natural habitat is leading to scarcity due to the high demand from the market, causing unsustainable logging practice¹²⁵. In addition to timber species, the project area also holds high potential of non-timber forest product, such as Tengkawang (*Shorea stenoptera*). The butter produced from Tengkawang has similar with cocoa butter and therefore suitable to be produced as basic materials for cosmetic products¹²⁶.

Following table are biodiversity metric that conclude the biodiversity condition in terms of flora in the project area.

Table 89. Flora biodiversity metric in the project area for each vegetation stage

Metric	Seedling	Sapling	Pole	Small Tree	Big Tree
S (number of species discovered)	88	47	61	72	140
Shanon Diversity Index	3.6646	3.2540	3.3936	3.4671	3.7156
Simpson Index of Diversity (Ds)	0.9577	0.9441	0.9475	0.9479	0.9597
Species Evenness Index (Ed)	0.8185	0.8452	0.8255	0.9479	0.7519

In conclusion, field assessment and PRA survey identified at least 274 fauna species (132 birds, 78 mammals, 64 reptiles and amphibians) and 203 flora in the project zone, including 44 fauna and 17 flora species considered threatened by the IUCN Red List, such as the Sarawak surili (*Presbytis chysomelas*) and the Sunda pangolin (*Manis javanica*). The local community reported the presence of the Bornean orangutan (*Pongo pygmaeus*) in the project zone, last seen in 2019. Table 91 shows all threatened species identified in the project zone during PRA and field assessments.

Table 90. List of threatened species in the project zone, among mammals, birds, and plants.

No	Scientific name	Family	Conservation Status		Method
			IUCN	CITES ¹²⁷	
Mammals					
1	<i>Manis javanica</i>	Manidae	CR	Appendix II. ¹²⁸	PRA

¹²⁵ Effendi, R. (2009). Kayu Ulin di Indonesia: Potensi, Manfaat, Permasalahan, dan Kebijakan yang diperlukan Untuk Kelestariannya. Jurnal Analisis Kebijakan Kehutanan. 6(3); 161-168

¹²⁶ Darmawan, MA. Curie, CA. Ramadhan, MYA. Fahriz, A. Harahap, P. Gozan, M. (2021). Preparation and Characterization of Indegenous Tengkawang (*Shorea stenoptera*) Butter for Food and Cosmetics Material. Journal of Hunan University (Natural Science). 48(7)

¹²⁷ CITES is a multilateral treaty to protect endangered plants and animals from the threats of international trade.

¹²⁸ CITES APPENDIX II lists species that aren't facing imminent extinction but need monitoring to ensure that trade doesn't become a threat.

No	Scientific name	Family	Conservation Status		Method
			IUCN	CITES ¹²⁷	
2	<i>Pongo pygmaeus</i>	Hominidae	CR	Appendix I ¹²⁹	PRA
3	<i>Presbytis chrysomelas</i>	Cercopithecidae	CR	Not Listed	PRA & Field
4	<i>Hylobates albibarbis</i>	Hylobatidae	EN	Appendix II	PRA
5	<i>Hylobates muelleri abbotti</i>	Hylobatidae	EN	Appendix II	PRA & Field
6	<i>Macaca fascicularis</i>	Cercopithecidae	EN	Not Listed	PRA & Field
7	<i>Macaca nemestrina</i>	Cercopithecidae	EN	Not Listed	PRA & Field
8	<i>Pteromyscus pulverulentus</i>	Sciuridae	EN	Not Listed	Field
9	<i>Pteropus vampyrus</i>	Pteropodidae	EN	Appendix II	PRA
10	<i>Arctictis binturong</i>	Viverridae	VU	Appendix III ¹³⁰	PRA & Field
11	<i>Cephalopachus bancanus</i>	Tarsiidae	VU	Not Listed	PRA
12	<i>Helarctos malayanus</i>	Ursidae	VU	Appendix III	PRA
13	<i>Neofelis diardi</i>	Felidae	VU	Appendix I	PRA
14	<i>Nycticebus menagensis</i>	Lorisidae	VU	Not Listed	PRA
15	<i>Petinomys vordermanni</i>	Sciuridae	VU	Not Listed	Field
16	<i>Presbytis frontata</i>	Cercopithecidae	VU	Not Listed	PRA
17	<i>Presbytis hosei</i>	Cercopithecidae	VU	Not Listed	PRA
18	<i>Presbytis rubicunda</i>	Cercopithecidae	VU	Not Listed	PRA
19	<i>Rheithrosciurus macrotis</i>	Sciuridae	VU	Not Listed	PRA & Field
20	<i>Sus barbatus</i>	Suidae	VU	Not Listed	PRA
21	<i>Trachypithecus cristatus</i>	Cercopithecidae	VU	Not Listed	PRA
Birds					
22	<i>Pseudibis davisoni</i>	Threskiornithidae	CR	Not Listed	PRA
23	<i>Pycnonotus zeylanicus</i>	Pycnonotidae	CR	Not Listed	PRA
24	<i>Chloropsis sonnerati</i>	Chloropseidae	EN	Not Listed	PRA & Field
25	<i>Ciconia stormi</i>	Ciconiidae	EN	Not Listed	PRA
26	<i>Polyplectron schleiermacheri</i>	Phasianidae	EN	Not Listed	PRA
27	<i>Rhabdotorrhinus corrugatus</i>	Bucerotidae	EN	Not Listed	PRA
28	<i>Aceros undulatus</i>	Bucerotidae	VU	Not Listed	Field
29	<i>Anthracoceros malayanus</i>	Bucerotidae	VU	Not Listed	PRA
30	<i>Argusianus argus</i>	Phasianidae	VU	Not Listed	Field
31	<i>Buceros rhinoceros</i>	Bucerotidae	VU	Appendix II	PRA & Field
32	<i>Caprimulgus concretus</i>	Caprimulgidae	VU	Not Listed	PRA
33	<i>Harpactes reinwardtii</i>	Trogonidae	VU	Not Listed	Field
34	<i>Hydrornis baudii</i>	Pittidae	VU	Not Listed	PRA

¹²⁹ CITES APPENDIX I lists species in danger of extinction because of international trade. Permits are required for import and export, and trade for commercial purposes is prohibited.

¹³⁰ CITES APPENDIX III lists species protected in at least one country when that country asks others for help regulating the trade.

No	Scientific name	Family	Conservation Status		Method
			IUCN	CITES ¹²⁷	
35	<i>Leptoptilos javanicus</i>	Ciconiidae	VU	Not Listed	PRA
36	<i>Lophura bulweri</i>	Phasianidae	VU	Not Listed	PRA
37	<i>Melanoperdix niger</i>	Phasianidae	VU	Not Listed	PRA
38	<i>Nisaetus nanus</i>	Accipitridae	VU	Not Listed	PRA
39	<i>Ptilocichla leukogrammica</i>	Pellorneidae	VU	Not Listed	PRA
40	<i>Rollulus rouloul</i>	Phasianidae	VU	Not Listed	PRA
41	<i>Setornis criniger</i>	Pycnonotidae	VU	Not Listed	PRA
42	<i>Treron capellei</i>	Columbidae	VU	Not Listed	PRA & Field
Flora					
43	<i>Castanopsis argentea</i>	Fagaceae	EN	Not Listed	Field
44	<i>Macaranga mauritiana</i>	Euphorbiaceae	EN	Not Listed	Field
45	<i>Shorea uliginosa</i>	Dipterocarpaceae	EN	Appendix II	Field
46	<i>Shorea virescens</i>	Dipterocarpaceae	EN	Not Listed	Field
47	<i>Azelia rhomboidea</i>	Fabaceae	VU	Not Listed	Field
48	<i>Artocarpus anisophyllus</i>	Moraceae	VU	Not Listed	Field
49	<i>Durio acutifolius</i>	Malvaceae	VU	Not Listed	Field
50	<i>Durio graveolens</i>	Malvaceae	VU	Not Listed	Field
51	<i>Eusideroxylon zwageri</i>	Lauraceae	VU	Appendix II	Field
52	<i>Mangifera pajang</i>	Anacardiaceae	VU	Not Listed	Field
53	<i>Shorea acuminatissima</i>	Dipterocarpaceae	VU	Appendix II	Field
54	<i>Shorea collina</i>	Dipterocarpaceae	VU	Appendix II	Field
55	<i>Shorea johorensis</i>	Dipterocarpaceae	VU	Appendix II	Field
56	<i>Shorea laevis</i>	Dipterocarpaceae	VU	Appendix II	Field
57	<i>Shorea mecistopteryx</i>	Dipterocarpaceae	VU	Appendix II	Field
58	<i>Tetramerista glabra</i>	Tetrameristaceae	VU	Not Listed	Field
59	<i>Upuna borneensis</i>	Dipterocarpaceae	VU	Not Listed	Field

Threat to Biodiversity

Three major threats to biodiversity have been identified in the project zone: habitat conversion, climate change, and poaching or hunting. This list of threats was compiled through a series of data collections and observations on the field and further supported by spatial analysis of the project zone.

Primarily regarding habitat conversion, it occurred due to communal agricultural activities, where they originally engaged in rice paddy and vegetable cultivation. Recently, there is significant changes of agricultural activities, where local community their agricultural fields into oil palm plantations due to the high economic value of palm oil, while also clearing forested areas for new paddy fields. The need to produce more palm fruit contributes to new areas being cleared by communities' agricultural activities, as shown by map below.

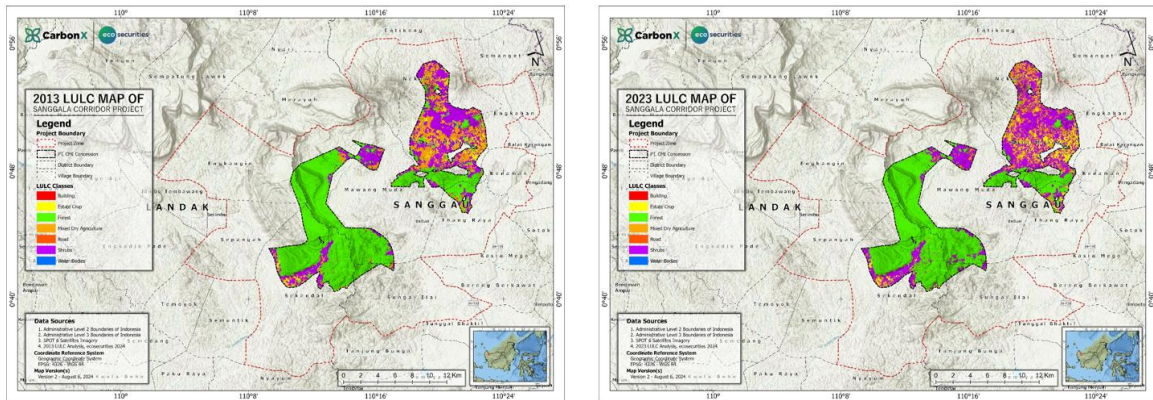


Figure 29. 2013 land use land cover map of the concession (left) and 2023 land use land cover map of the concession (right)

It is also apparent that climate change has impacted the biodiversity in the project zone, specifically on the spread of virus. As previously discussed in section 3.4, the documented temperature rise in the project zone has facilitated the spread of swine flu, subsequently driving the loss of wild boar population. Other than the loss of protein source for communities, the extent of this loss is yet to be documented. However, it is predicted that this absence of wild boar population in the project zone could further tip the balance of the ecosystem given its essential role as seed dispersal species which contribute to creation of habitat in *Dipterocarpaceae* forest¹³¹.

Hunting is the last observed threat of biodiversity that was confirmed during the PRA activity. The community confirmed during the PRA activity that hunting wildlife is still a common practice. Of the 118 species occurring in the project zone, 98 are reportedly hunted by the community. Furthermore, 33 of the 39 species listed as Threatened by the IUCN Red List are also subject to hunting. Additionally, the field team collecting data for fauna encountered the man-made hut within the forest, presumably used by those engaging in hunting activities.



¹³¹ Meijaard, Erik. (2000). Bearded pig (*Sus barbatus*). Ecology, conservation status, and research methodology. Background document for the electronic conference on bearded pig research, and recommendations for a collaborative research programme.

Figure 30. An abandoned hut, possibly a former hunter's hut (left) and remnant of a monitor lizard skin from hunting (right).

Species and habitat	The project activities focus on protecting the forested areas and restoring degraded regions to increase connectivity between all forested areas. The forest protection efforts will conserve the habitat of the threatened species and will not adversely impact the areas or habitats required by these species. In addition, the forest in block II of the concession is connected to the protected area and <i>Gunung Nyiut Penrissen</i> Nature Reserve. However, the forested area in block I of the concession is not connected to the forest in block II, the protected forest, or <i>Gunung Nyiut Penrissen</i> Nature Reserve. The project aims to bridge these areas through restoration efforts using local species. List of the species can be seen in Table 84.
Areas needed for habitat connectivity	The project will not adversely impact the protected forest area and will enhance its protection. However, it will conserve the forest production within the project zone, throughout several activities, such as forest monitoring, forest patrolling, and biodiversity monitoring. These will be conducted in order to ensure the protection of the forest area within the project and the surroundings. List of the species can be seen in Table 84.

5.1.2 High Conservation Values (CCB, B1.2)

Another approach regarding evaluation of biodiversity is through the delineation of High Conservation Value (HCV) area. This will allow the project to protect and improve high conservation values (HCVs) related to biodiversity characteristics across the project zone. To this effect, the project has identified areas with important levels of biodiversity (HCV 1), natural landscape and dynamic (HCV 2).

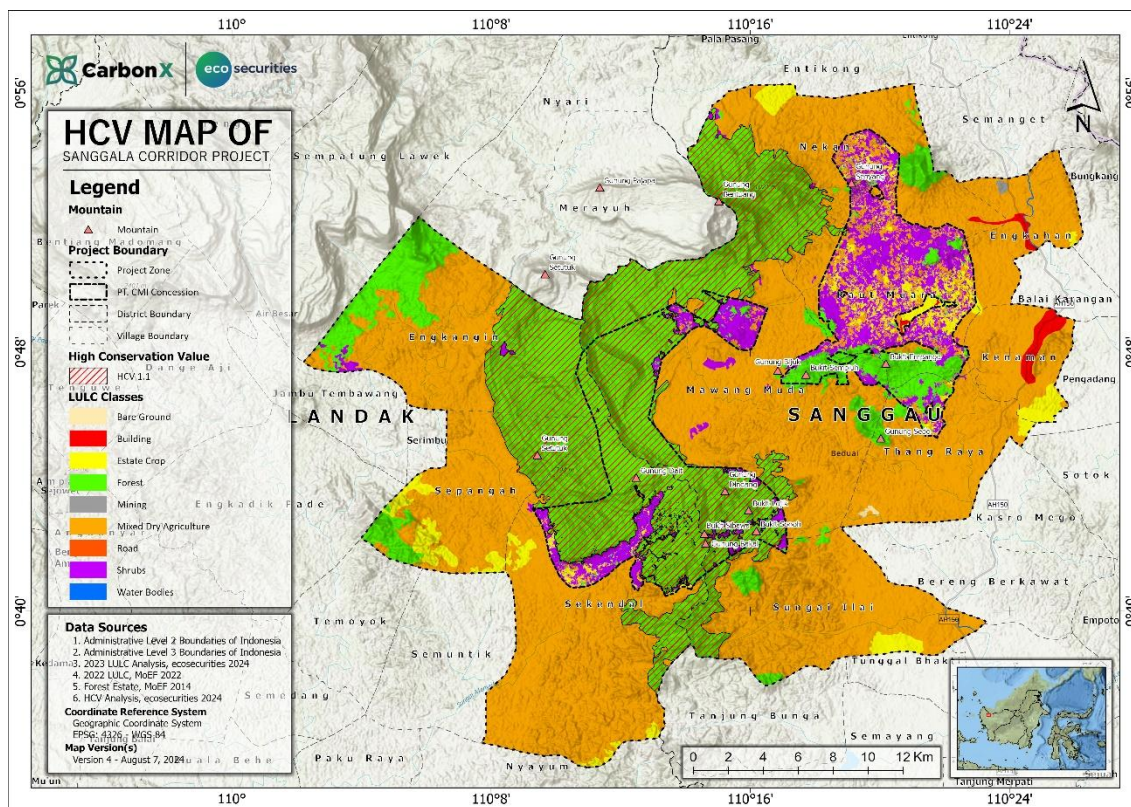
5.1.2.1 HCV 1: Areas with important levels of biodiversity (HCV 1)

HCV 1 focuses on the protection of areas that contain significant concentrations of biodiversity values, particularly those that are critical for maintaining species diversity. The project team identified areas that provided:

- Areas that contain or provides biodiversity support function to protection or conservation areas (HCV 1.1).
- Critically endangered species (HCV 1.2)

- Areas that contain habitat for viable populations of endangered, restricted range or protected species

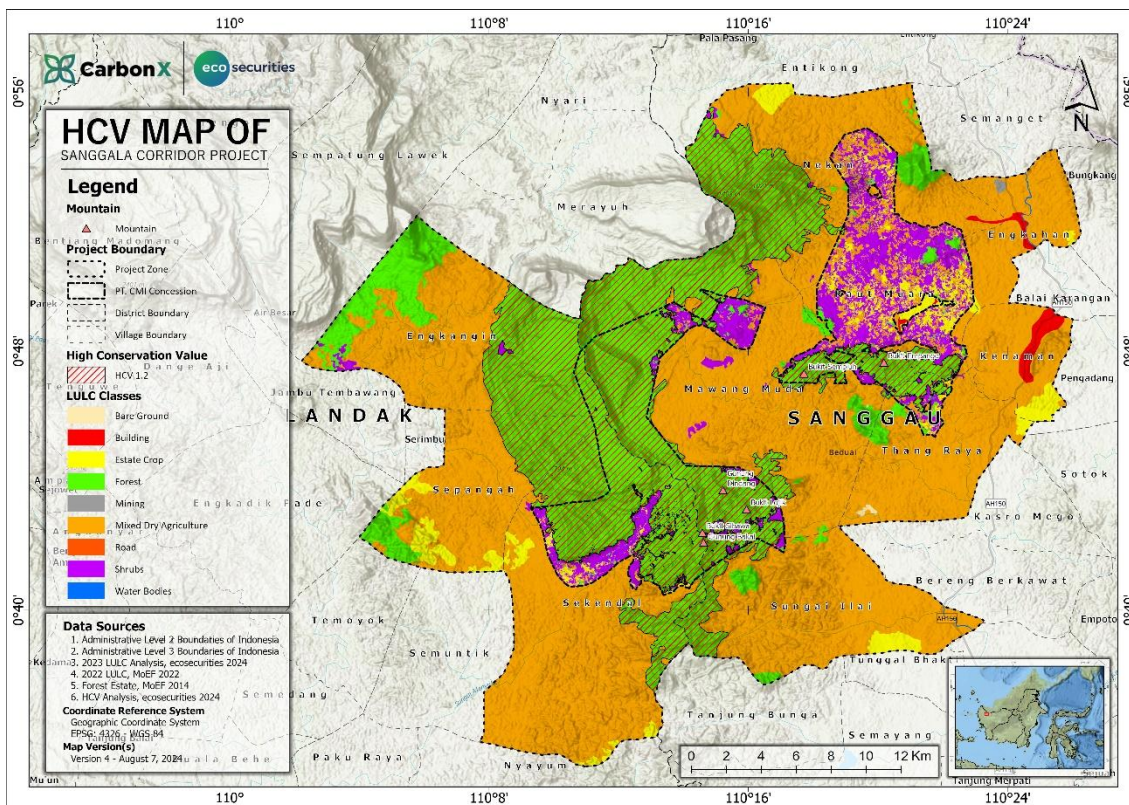
High conservation value	HCV 1.1 Area that contains or provides a biodiversity support function to protection or conservation areas.
Qualifying attribute	The western part of Block II is the forest adjacent to Gunung Nyiut Nature Reserve, a protected forest that provides biodiversity support.
Focal area	Forest area in the western part of block II in the project area, as well as the protected forest area and part of Gunung Nyiut Nature Reserve in the project zone. See Map 37



Map 36. High Conservation Value 1.1

High conservation value	HCV 1.2 Critically Endangered species
Qualifying attribute	Based on PRA activity conducted with the community within the project zone, five critically endangered species, were identified as living in the area. These species include <i>Presbytis chysomelas</i> , <i>Manis javanica</i> , <i>Pseudobis davisoni</i> , <i>Pycnonotus zeylanicus</i> , and <i>Pongo pygmaeus</i> . A rapid field survey then

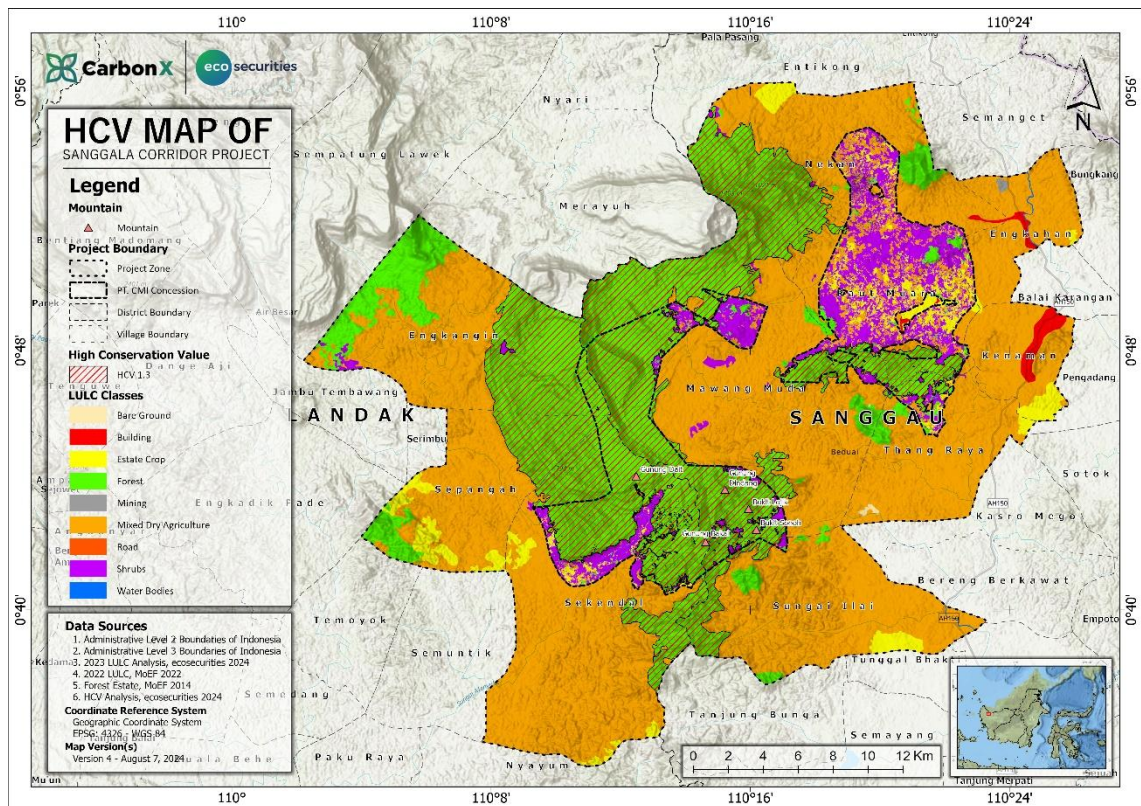
	confirmed that <i>Presbytis chrysomelas</i> is present within the project area.
Focal area	All the forest area in block I and block II, especially forested area in Mount Bakai, Mount Lojik, Mount Dindang, Mount Sempiuh, Empango Hill and Sibawa Hill. See Map 38



Map 37. High Conservation Value 1.2

High conservation value	HCV 1.3 Areas that contain habitat for viable populations of endangered, restricted range or protected species
Qualifying attribute	The project area of PT CMI is a habitat for endangered and protected species. There are 10 Endangered species, including <i>Hylobates albibarbis</i>, <i>Chloropsis sonnerati</i>, <i>Ciconia stormi</i>, <i>Hylobates muelleri abbotti</i>, <i>Macaca fascicularis</i>, <i>Macaca nemestrina</i>, <i>Polyplectron schleiermarcheri</i>, <i>Pteropus vampyrus</i>, and <i>Rhabdotorrhinus corrugatus</i>, and <i>Pteromyscus pulverulentus</i>.

	Moreover, there are 27 vulnerable species including <i>Aceros undulatus</i> , <i>Anthracosceros malayanus</i> , <i>Arctistics binturong</i> , <i>Argusianus argus</i> , <i>Buceros rhinoceros</i> , <i>Caprimulgus concretus</i> , <i>Cephalopchus bancanus</i> , <i>Harpectes reinwardtii</i> , <i>Helarctos malayanus</i> , <i>Hydrornis baudii</i> , <i>Leptoptilos javanicus</i> , <i>Lophura bulweri</i> , <i>Melanoperdix niger</i> , <i>Neofelis diardi</i> , <i>Nisaetus nanus</i> , <i>Nycticebus menagensis</i> , <i>Petinomys vordermanmi</i> , <i>Presbytis frontata</i> , <i>Presbytis hosei</i> , <i>Presbytis rubicunda</i> , <i>Ptilocichla leucogrammica</i> , <i>Rheithrosciurus macrotis</i> , <i>Rollulus rouloul</i> , <i>Setornis criniger</i> , <i>Sus Barbatus</i> , <i>Trachypithecus cristatus</i> , <i>Treron capellei</i> .
Focal area	Forest area in block I and block II especially the forest area of Mount Dait, Mount Bakai, Mount Lojik, Mount Dindang, Mount Sempih, Empango Hill and Sibawa Hill. See Map 39



Map 38. High Conservation Value 1.3

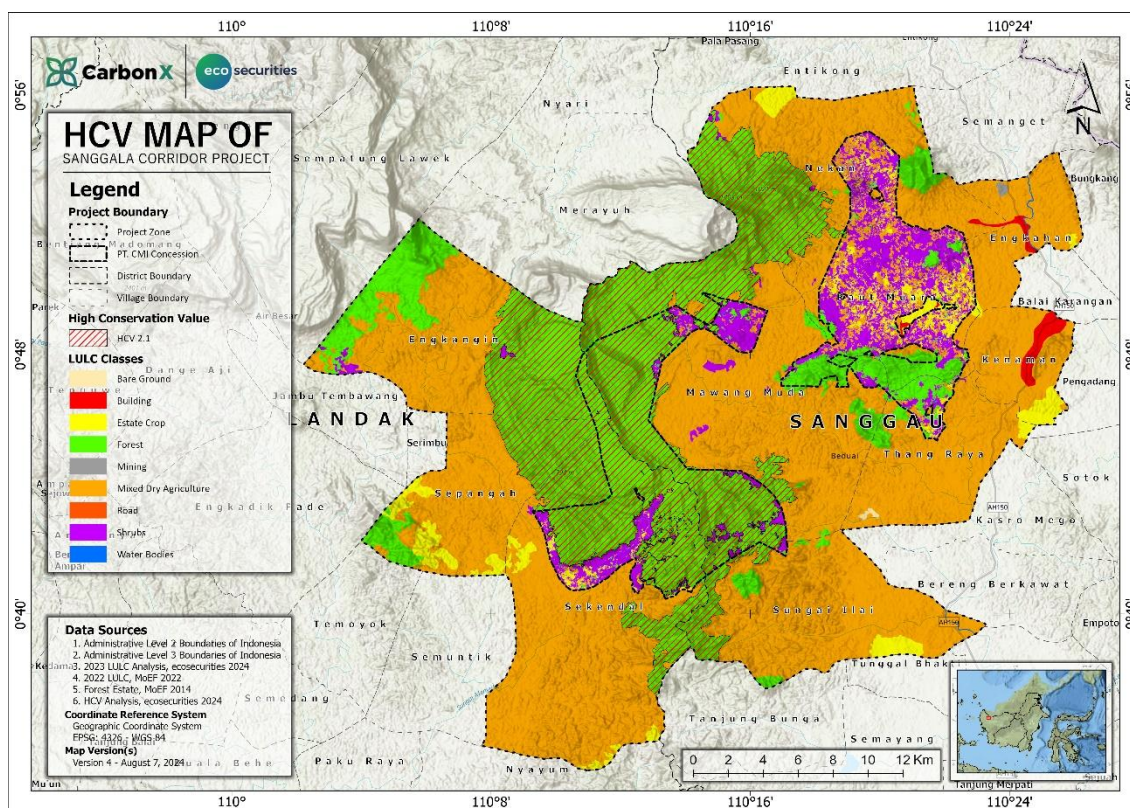
5.1.2.2 HCV 2: Natural landscape and dynamic (HCV 2)

HCV 2 pertains to large, landscape-level ecosystems and ecosystem mosaics that are significant at global, regional, or national levels. These areas are typically large and relatively undisturbed,

containing viable populations of the great majority of the naturally occurring species in natural patterns of distribution and abundance. The project team identified areas that provided:

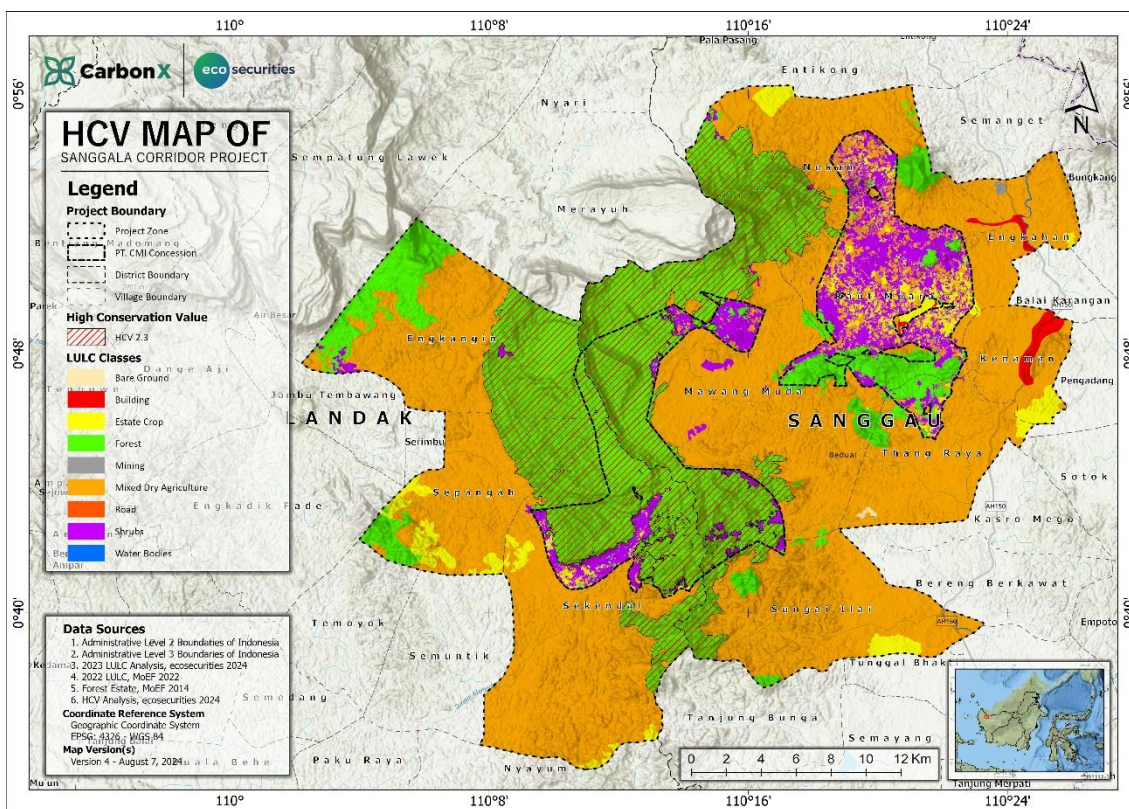
- HCV 2.1 Large Natural Landscapes with the capacity to maintain natural ecological processes and dynamics
- HCV 2.3 Areas that contain a representative population of most naturally occurring species.

High conservation value	HCV 2.1 Large Natural Landscapes with the capacity to maintain natural ecological process and dynamics
Qualifying attribute	The forest area in block II of the project site is ecologically connected to the forest in the protected area and Gunung Nyiut Nature Reserve reserve. This landscape is also categorized as Key Biodiversity Area of Gunung Nyiut. Covering approximately 22,000 hectares, this landscape is considered a large natural area.
Focal area	The forest in block II, protected forest area and Gunung Nyiut Nature Reserve. Map 40



Map 39. High Conservation Value 2.1

High conservation value	HCV 2.3 Areas that contain a representative population of most naturally occurring species.
Qualifying attribute	The project area is a habitat for fauna and flora species. There are at least 132 species of bird from 42 families, 64 species of herpetofauna from 15 families, and 78 species of mammals from 24 families. In addition, 202 species of flora from 37 families were also found. Based on these results, there seems to be a representative population of flora and fauna in the project area.
Focal area	The project area and project zone. See Map 41



Map 40. High Conservation Value 2.3

5.1.3 Without-project Scenario: Biodiversity (CCB, B1.3)

Habitat loss is a primary threat, affecting 82.1% of listed animal species and 60.5% of listed plant species in Kalimantan.¹³² In Indonesia, establishment of timber plantation is considered equal to habitat loss, as it involves a land clearing of existing standing vegetation. Timber

¹³² Budiarta, Sugeng & Meijaard, Erik. (2017). State of Kalimantan's biodiversity.

plantations have the most significant adverse impact on species richness, with an average reduction of 40% in local species richness compared to natural forests.¹³³

Without the intervention planned by the project, 9,307.62 hectares of forest within the PT CMI concession are likely to be deforested and converted into monoculture timber plantations of *A. cadamba* or *H. brasiliensis*. In other words, habitat loss would have been taking place in the project zone, creating an inhospitable environment for the diverse wildlife and plants species in the project zone to grow and develop. As land clearing would have occurred at the initial stage of forest plantation development, a complete loss of biodiversity in standing forest is to be expected, replaced by monoculture plantations.

Secondly, the loss of suitable habitat for wildlife to roam would result in increased risks of conflict with humans. As previously discussed, the community in the project zone has the tendency to hunt wildlife as a hobby or part of culture already embedded into their life. It has been already established that the forest area is home to a high variety of wildlife species. The loss of forest ecosystem to make way for industrial timber plantation would reduce the area on which wildlife species could safely roam, while simultaneously increasing the risks of encounter with human.

Third of all, it is highly probable that the risks of pests and diseases would be heightened, due to the dwindling diversity among vegetation species in the area since the establishment of industrial timber plantation in the project zone. Accounts by local communities in the project zone described the already widespread trend of cultivated pepper being attacked by pests, while the population of wild boar is disappearing due to virus. The endemic biodiversity and intact ecosystem is needed to reduce the prevalence of infectious disease¹³⁴. The establishment of monoculture plantation which by no means supporting diversity could increase the potential of other pests and or diseases emerging.

Another feature associated with biodiversity concerns the conservation of biological resources. According to Convention of Biological Diversity, biological resources include genetic resources, organisms, or parts thereof, population, or any other biotic component of ecosystem with actual or potential use or value for humanity (CBD Article 2). This means preserving any element of the ecosystem regardless of the value for humanity, while equally taking the yet to be considered value and use into consideration. The confirmation of the project zone as a reserve of genetic resources was recently established, when several individuals widely known as the corpse flower (*Amorphophallus hewittiti*) were identified following almost 2 years since project start date. Though not considered to have a decreased population trend and not protected by relevant regulations, *Amorphophallus hewittiti* is part of the genus identified to be having limited capabilities in terms of both pollinator agents and inflorescent period.¹³⁵ The limited pollinator

¹³³ Chaudhary, A., Burivalova, Z., Koh, L. et al. Impact of Forest Management on Species Richness: Global Meta-Analysis and Economic Trade-Offs. *Sci Rep* 6, 23954 (2016). <https://doi.org/10.1038/srep23954>

¹³⁴ Keesing F, Belden LK, Daszak P, Dobson A, Harvell CD, Holt RD, Hudson P, Jolles A, Jones KE, Mitchell CE, Myers SS, Bogich T, Ostfeld RS. Impacts of biodiversity on the emergence and transmission of infectious diseases. *Nature*. 2010 Dec 2;468(7324):647-52. doi: 10.1038/nature09575. PMID: 21124449; PMCID: PMC7094913.

¹³⁵ Cyrille Claudel, The many elusive pollinators in the genus *Amorphophallus*, *Anthropod-Plant Interactions* 15 (2021) 833-844, <https://doi.org/10.1007/s11829-021-09865-x>

agents and inflorescent period illustrates clearly the irreplaceability of *Amorphophallus hewittii* in the project zone. The development of industrial timber plantation in the project zone may then threaten the existence of *Amorphophallus hewittii*, as well as other unidentified organisms with potential future value in the project zone.



Figure 31. *Amorphophallus hewittii* in the project zone.

Ultimately, the absence of projects will create a fragmented habitat of the ecosystem in the project zone. The presence of industrial timber plantation in the project area will worsen the already fragmented natural landscape ecosystem, and the project zone could lose a significant portion of its carrying capacity to sustain the livelihoods in the project zone. In addition to which, the project area will have a diminished role in terms of acting as a buffer area to the nearby protected and conservation forest, due to land clearing activities would have deforested the existing standing forest in Block II of the project area allocated as production area. Whereas in Block I, there will be minimal to no connectivity between the existing forest cover within the project area and forest cover in the project zone, specifically on the northern part of Block I..

5.2 Net Positive Biodiversity Impacts

5.2.1 Expected Biodiversity Changes (VCS, 3.19; CCB, B2.1)

The Sanggala Corridor Project aims to conserve and restore forests that would otherwise be cleared for industrial monoculture timber plantations. By preventing this land conversion, the project mitigates its adverse impact on biodiversity, including the loss of natural landscape, increased human-wildlife conflict, and loss of biological resources, as outlined in Section 5.1.3. With conservation principles at the core of project activities, the project is expected to drive positive changes for biodiversity as summarized below:

Biodiversity element	Natural ecosystem
Estimated change	Increased forest cover and reduced threat
Justification of change	Forest protection activities, including several programs listed in the ToC, will be implemented as part of the project. On top of the conservation-related attempts, planting activities will be implemented, which will improve vegetation cover in the project area. These programs are expected to significantly impact the ecosystem, creating a positive domino effect for biodiversity due to improvement of habitat conditions. Primarily, the method used to ensure the expected change would be through remote sensing analysis. Subsequently, the combination of PRA survey and vegetation measurement shall be conducted as a measure to validate the findings from remote sensing analysis.

Biodiversity element	Dynamics of wildlife species
Estimated change	Reduced the hunting of threatened species, Improved condition and minimized disturbance (e.g. hunting and diseases)
Justification of change	The project activity will prevent the hunting of wildlife, especially threatened species, within the project area. Forest patrolling and monitoring will be implemented to identify illegal activities, including hunting, and collect data regarding the presence of wildlife species as well as their habitat. Additionally, the biodiversity awareness raising program will be implemented to address the root causes of community-based hunting, particularly on species identified as declining population and protected by national law and/or international treaties.. Regular PRA (Participatory Rural Appraisal) activities will be conducted to monitor the status of species and track whether they are being hunted or not. The project will conserve existing forested area and restore degraded areas as well as establish an agroforestry system using native species to mimic the natural ecosystem. As a result, the project will enhance the ecosystem, improving habitat condition by making it healthier and providing better nutrition and shelter for animals, thereby strengthening their immune systems and resilience to diseases. A diverse and stable habitat also reduces stress levels among animals, minimizing the likelihood of disease spread.

Biodiversity element	Dynamics of Floral Species
Estimated change	Gradual dispersal and increase of flora and fauna species in the project area, Improved habitat condition and minimized disturbance (e.g. spread of pests and invasive species)
Justification of change	Due to the project activities, the abundance of flora and fauna in the project area is anticipated to increase due to the restoration activities in the non-forest area which will combine several native tree species. Coupled with the already high biodiversity in the forest area, it is projected that this increased biodiversity could hamper any emergence of new pests and/or diseases as well as limit the spread of invasive species. The project's positive impact can be determined by comparing the abundance of threatened species in the project area to the number previously identified. This will be accomplished by regular patrol, and regular biodiversity monitoring of threatened species in the project zone.

5.2.2 Mitigation Measures (VCS, 3.19; CCB, B2.3)

The project activities will focus on maintaining and enhancing the natural ecosystem of the project area. Therefore, no negative impacts on biodiversity are expected to happen, and no mitigation measures are required.

As described in Section 5.2.1, the project is expected to drive positive changes in the three identified biodiversity elements, namely natural ecosystem, diversity of wildlife species, diversity of floral species and population of trigger species.

To achieve these biodiversity improvements, the project will implement targeted activities beyond forest conservation and restoration of degraded areas. These activities address the root causes of biodiversity threats in the project zone and include:

1. Biodiversity awareness and training

The biodiversity survey and analysis indicate that threats to biodiversity – such as illegal mining and the wildlife hunting – exacerbated by low community awareness and economic pressures. Many communities convert forests for agriculture land and shift traditional mixed-agriculture practices to smallholder monoculture palm oil.

The project will conduct biodiversity awareness and training that integrates scientific insights with local wisdom. This initiative aims to revive sustainable land use practices and reduce

biodiversity risks such as habitat loss, species decline, and ecosystem imbalances that can lead to increased human-wildlife conflict and the spread of pests.

2. Creation of customary laws on biodiversity

Communities in the Sanggala Corridor Project adhere strictly to Dayak customary ethics and laws. Given their strong communal structure, local enforcement of biodiversity-related customary laws can be highly effective. The Project will collaborate with tribal leaders to formalize these laws ensuring long-term community-led biodiversity conservation.

3. Regular forest patrols on HCV areas

As one of the core activities to conserve intact forest, a community-based patrol team will conduct regular monitoring specifically in HCV areas. These patrols will monitor and minimize events of forest degradation and illegal activities, collect data on ecosystem disturbances and biodiversity information (e.g. presence of wildlife species and/or protected and rare floral species), and engage with local communities encountered to discourage harmful activities such as illegal mining or slash-and-burn practices.

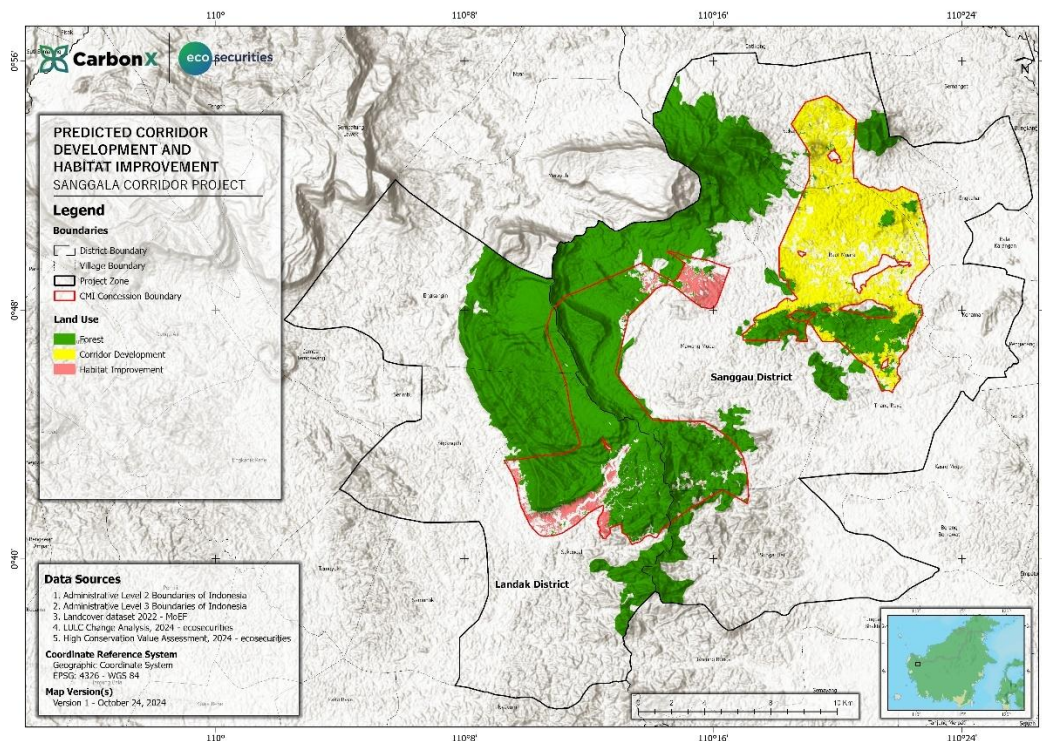
5.2.3 Net Positive Biodiversity Impacts (CCB, B2.2, GL1.4)

The Sanggala Corridor Project strives to conserve biodiversity mainly through primary project activities, namely conservation of intact forest, restoration of degraded areas, and coupled with community program. Specifically for the community program known as biodiversity awareness raising program, it is intended to incorporate knowledge about biodiversity conservation – including project's vision to protect trigger species – into community's local wisdom and knowledge. Done properly, this shall contribute to project successfully mitigating the risks already identified in section 5.1.3. On the ground, this will be reflected through a reduced conflict between wildlife and human, decreased risks of pests and diseases spreading among wildlife and vegetation in the project zone, as well as preservation of biological resources.

Secondly, Sanggala Corridor Project would have achieved a sustained function of ecosystem services in terms of conservation of biodiversity. In this case, it is expected that the diversity of wildlife and vegetation in the project zone would have at least maintained, if not increased given the attempt of ecosystem conservation and restoration. Additionally, the function of the project area as buffer/supporting area to both protected and conservation forest could be optimized. Particularly in project zone where hydrogeological adverse event such as floods are recorded, it is expected that water cycle function of the forest could be regained and thereby nullifying the frequency of similar events.

Ultimately, it is expected that the Sanggala Corridor Project could fulfill the vision of its namesake: creation of corridor for biodiversity to co-exist with human. Particularly in Block I of the concession, there exists patches of forest divided by non-forest area whereby communities conducting subsistence agriculture activities. By restoring the non-forest area through agroforestry and restoration, it is expected that connectivity between the two patches of forest

could be established, therefore creating additional habitat for wildlife species to roam between the patches of existing forest. On top of creating connection between patches of forest in the project area, the project aims to maintain and enhance the connection with adjacent conservation and protected forest areas. This creation of natural corridor would also support genetic diversity and ecosystem resilience of the area. The following map shows the area designated for restoration by the project, aimed at enhancing habitat connectivity and improving overall biodiversity.



Map 41. Designated restoration area for corridor development and habitat improvement

5.2.4 High Conservation Values Protected (CCB, B2.4)

Protecting biodiversity is a key focus in the project activities. The project contributes to preserving and improving the area and quality of forest and natural ecosystems and enhancing each HCV area. This emphasis on preserving and improving forests and natural ecosystems, the protection of them will have a significant positive impact on HCV 1-2. Additionally, project infrastructure such as basecamps, guard posts, fire towers, and nurseries will be built outside of the critical ecosystems.

5.2.5 Species Used (VCS, 3.19; CCB, B2.5, B2.6)

This table provides a non-exhaustive list of species which, in the design stage, will be planted as part of the ARR project activities. This list is subject to modification dependent on critical driving factors, such as preference of local communities, availability of seeds, marketability of commodities it produces, etc.

Table 91. List of Species for ARR Activities

Species introduced	Classification	Justification for use	Adverse effects and mitigation
<i>Coffea liberica</i>	Non-native	Increase the agroforestry economy's value to support the community's livelihood. This species is commonly used as an agroforestry commodity in Indonesia. According to the global invasive species database, it is not classified as an invasive species.	Adverse effect: Using coffee as a planted species could lead to biodiversity loss if the planting activity is conducted using a monoculture scheme. Mitigation: The planting activity will take place in non-forest areas, and the project will introduce a variety of tree and plant species into the planting system. All other species included will be native to the project area, ensuring that the planting supports local biodiversity.
<i>Theobroma cacao</i>	Non-native	Increase the agroforestry economy's value to support the community's livelihood. The species is commonly used as an agroforestry commodity in Indonesia. According to the global invasive species database, it is not classified as an invasive species.	Adverse effect: Using cacao as a planted species could lead to biodiversity loss if the planting activity is conducted using a monoculture scheme. Mitigation: The planting activity will be conducted in the non-forest area and the project will introduce a variety of tree and plant species in the planting system. All other species included will be native to the project area, ensuring that the planting supports local biodiversity.

Species introduced	Classification	Justification for use	Adverse effects and mitigation
<i>Shorea stenoptera</i>	Native	Native species that commonly found in the project area	N/A
<i>Durio zibethinus</i>	Native	Native species that commonly found in the project area	N/A
<i>Shorea pinanga</i>	Native	Native species that commonly found in the project area	N/A
<i>Artocarpus</i> spp.	Native	Native species that commonly found in the project area	N/A
<i>Pithecellobium jiringga</i>	Native	Native species that commonly found in the project area	N/A

5.2.6 Invasive Species (VCS, 3.19; CCB, B2.5)

The project reaches conclusion that there is only one species considered invasive in the project zone: *Elaeis guineensis* (Oil palm). According to Global Invasive Species Database,¹³⁶ *Elaeis guineensis* is considered an invasive species. This species is native to the west African coast from Liberia to Angola. It has been introduced to many island in the world, including Borneo.

As a mitigation measure, the project will conduct educational activities to increase community knowledge and awareness about palm oil. Additionally, project will employ member of communities for forest patrolling and monitoring, in order to identify and stop the natural growth and spread of new palm oil plants. Moreover, one of the project activities is to develop an agroforestry system in a previously degraded area with a high economic commodity, which is aimed to prevent the spread of oil palm plantation crops by providing alternative and viable livelihood opportunities to the community.

5.2.7 GMO Exclusion (CCB, B2.7)

The project will not use genetically modified organisms (GMOs) to generate GHG benefits, and planting activities to generate GHG benefits will not use seeds derived from GMOs.

¹³⁶ [GISD \(iucngisd.org\)](http://www.iucngisd.org)

5.2.8 Input Justification (VCS, 3.19; CCB, B2.8)

The project plans to implement agroforestry practices, integrating perennial tree producing fruits, cash crops, high-value agricultural commodities, and traditional agricultural products. This practice will allow a sustainable land management while providing communities with viable economic incentives to sustain their livelihoods.

As part of the initiative, the project has established a demonstration plot for agroforestry practice, followed by the development of a Feasibility Study on Agroforestry assessing technical and market viability of agroforestry in the area. Both of these initiatives have led to the conclusion that additional inputs are necessary, specifically for cash crops and high-value agricultural products, to ensure effective yield production that exceeds necessary costs. One example of required inputs is soil amendments to address land acidity, which is essential for optimizing crop growth and productivity.

The project will exhaust options to use sustainable inputs and minimizes the use of chemical fertilizer. However, due to technical challenges, the complete elimination of chemical inputs may not always be feasible. A detail justification and potential adverse impacts associated with the use of additional input are provided in the table below. As field conditions evolve, the project may adjust this list to reflect changing needs. Accordingly, relevant SOPs will be developed to mitigate any adverse impacts and to ensure that sustainability remains a core consideration in the project's agricultural practices.

Name	Controlled used of chemical input for soil amendment
Justification of use	• Poor soil condition of the soil in the field require intervention to enhance fertility. • Ensure productivity of cash crops and high-value agricultural commodities that could exceed necessary costs. • Facilitate communities in maximizing economic incentives from their agroforestry efforts.
Potential adverse effect	The identified potential adverse effects from the use of chemical input include: • Increased project emissions due to synthetic input application. • Imbalance of soil chemical compositions. • Increased river and/or water source pollution due to runoff.

5.2.9 Waste Products (VCS, 3.19; CCB, B2.9)

The project will identify and categorize waste products into two main groups: organic (fruit residues, paddy straw, pruning residues) and plastic (nursery containers, irrigation components, mulch film). These will be further classified into recyclable and non-recyclable categories. Understanding the waste composition will allow the project to tailor strategies and implement adaptive management for effective waste management.

The project adopts the principles of Reduce, Reuse and Recycle. Organic waste will be separated and either composted or disposed of through burial. Inorganic waste will be separated into recyclable components, which will be processed by local partners, while residual inorganic waste will be removed from the site and disposed of at the nearest government-run waste disposal facilities.

5.3 Offsite Biodiversity Impacts

5.3.1 Negative Offsite Biodiversity Impacts (CCB, B3.1) and Mitigation Measures (CCB, B3.2)

Negative offsite impact	Mitigation measure(s)
The project activities are designed to deliver positive biodiversity effects, and they are not anticipated to lead to negative impacts inside or outside the project zone.	Due to no negative impact to biodiversity outside the project zone is expected, no mitigation measures needed to be performed.

5.3.2 Net Offsite Biodiversity Benefits (VCS, 3.19; CCB, B3.3)

The net biodiversity benefit is expected to be positive due to the connectivity with the surrounding landscape, particularly with the Gunung Nyiut Nature Reserve. The project main focus is protecting and enhancing biodiversity condition.

5.4 Biodiversity Impact Monitoring

5.4.1 Biodiversity Monitoring Plan (CCB, B4.1, B4.2, GL1.4, GL3.4)

A biodiversity monitoring plan is essential for ensuring the long-term sustainable and ecological integrity of the Sanggala Corridor Project. The project aims to conserve the project area and provide a continuous, suitable habitat for threatened species that live within and surrounding the project area. All biodiversity monitoring data will be compiled and analyzed to produce reports that will track the efficacy of project intervention in safeguarding biodiversity and maintaining both HCV 1 and HCV 2 values that are presented in the project zone.

Following section 5.2.1 regarding expected biodiversity changes, the framework for monitoring of biodiversity will conform to the already identified biodiversity elements, namely natural ecosystem, dynamics of fauna species, and dynamics of floral species. These three biodiversity elements are indicator for both climate adaptation and mitigation action. For each of the identified biodiversity elements, the indicators to be monitored are the following:

Biodiversity Elements	Natural Ecosystem
Indicators to be monitored	• Forest cover • Events of fire in the project zone
Monitoring area	Project zone, HCV 1 and HCV 2 areas.
Monitoring Activities	• LULC analysis • Hotspot monitoring • Forest patrols • Camera trap analysis • PRA survey

Biodiversity Elements	Dynamics of wildlife species
Indicators to be monitored	• Biodiversity index for wildlife species • Records of diseases in wildlife species
Monitoring area	Project zone, HCV 1 and HCV 2 areas.
Monitoring Activities	• Camera trap • Field surveys methods: ○ Line transect method to survey mammals ○ VCP or point count to survey bird ○ Visual encounter survey to survey herpetofauna • Forest patrols • PRA surveys

Biodiversity Elements	Dynamics of floral species
Indicators to be monitored	• Biodiversity index for floral species • Records of pests in vegetation
Monitoring area	Project zone, HCV 1 and HCV 2 areas.
Monitoring Activities	• Field surveys, specifically vegetation plot inventory • Forest patrols • PRA surveys

The following assumptions should be taken into consideration as well:

- For each indicator, standardized methodology and/or technique (e.g. nested sampling for vegetation, point count for bird monitoring, remote sensing analysis for forest cover) shall be employed to collect data reflecting the impact of the project.
- Biodiversity Monitoring Report will be produced every 2 years, as data will be collected in the times prior.
- Determination of biodiversity and/or survey monitoring focus area shall consider project's dual needs in creating time series data and collecting new information regarding the vast biodiversity potential in the project zone
- Relevant SOPs referred to during the baseline data collection will be regarded as main guideline for data collection.

Monitoring of Trigger Species

On top of the monitoring of main biodiversity elements, Sanggala Corridor Project has identified three trigger species, the monitoring of which shall have specific measures. The three identified trigger species are:

- *Manis javanica* (Sunda pangolin);
- *Presbytis chrysomelas* (Bornean banded langur); and
- *Hlyobates muelleri abboti* (Abbot's grey gibbon).

For all of these trigger species, the primary focus shall be emphasized on monitoring of the threat of the species. It is identified that the population of animal species in the project zone is threatened by habitat loss and hunting activities. Accordingly, primary monitoring activities for trigger species shall be focused on the habitat and the identified threats.

Like any other wildlife species identified during baseline assessment, the main habitat of trigger species is the existing remaining forest cover in the project zone. Currently, this natural forest cover is threatened by human-induced activities such as forest degradation and expansion of palm oil, which drives the loss of natural habitat for wildlife species. By conserving the intact

natural forest area, it is assumed that the trigger species shall face no additional threat of habitat loss and therefore their population in the project zone is at least maintained if not enhanced. In this habitat monitoring, the primary activity undertaken by project is spatial analysis to monitor the forest cover and expansion of palm oil estate in the project zone, supported by other activities such as forest patrol and documentation from camera traps.

Second of all regarding monitoring hunting activities as one of the identified main threats, the tool is PRA survey. In surveys directed to communities, information regarding trigger species' dynamics will be collected, specifically through the following two guidelines:

- Whether there are any trigger species being hunted and/or kept as pets;
- Whether measures from project to limit hunting of targeted trigger species in the project zone already mainstreamed and acted upon accordingly.

The PRA activity shall also monitor and evaluate the behavior of the communities following the biodiversity awareness raising program regarding the importance of conservation and danger of hunting and the enactment of customary law related to biodiversity conservation.

Completing the habitat monitoring activity, Sanggala Corridor Project would conduct monitoring of each of the trigger species population at the appropriate time when project would have conducted proper measures to address the threat faced by identified trigger species. Respectively each of the trigger species, the survey methods are the following:

- Survey of *Manis javanica* shall employ burrow count method
- Survey of *Presbytis chrysomelas* shall employ permanent transect method and
- Survey of *Hylobates muelleri abboti* shall employ triangulation method.

Further, information from camera traps will be analyzed to produce further data regarding population of the trigger species.

5.4.2 Biodiversity Monitoring Plan Dissemination (CCB, B4.3)

The result of biodiversity monitoring will be distributed in two ways: first, by distributing annual reports to relevant government stakeholders (e.g., Forest Management Unit (KPH), BKSDA, Provincial Forestry Agency); second, by consistently sharing information via social media and the project's website. To reach the village community, the annual report will be distributed to the Village Head's office. In addition, the project will regularly publish information sheets and display them on the village information boards.

5.5 Optional Criterion: Exceptional Biodiversity Benefits

The project aims to achieve exceptional biodiversity benefits and be validated to the Gold Level based on the presence of ecologically important IUCN Red List species in the project area.

5.5.1 High Biodiversity Conservation Priority Status (CCB, GL3.1)

There are 59 species categorized as threatened by the IUCN Red List in the project zone. Overall, there are 5 species with the status of Critically Endangered (CR), 14 as Endangered (EN), and 40

with Vulnerable status. Besides, there are 16 species that are also included in the CITES appendix.

5.5.2 Trigger Species Population Trends (CCB, GL3.2, GL3.3)

As defined in the CCB Standards, trigger species must meet the criteria of “vulnerability” or “irreplaceability”. Vulnerability is defined as regular occurrence of IUCN Red List of threatened species (species listed as CR, EN, or VU) in the project area, while irreplaceability is defined as presence of 1-5% of species global population (at any stage of species lifecycle) in the project area.

The project’s initial biodiversity assessment that deploys Participatory Rural Appraisal and field assessment has identified within the project zone 59 species categorized as threatened species by the IUCN Red List species, which are also considered as vulnerable and trigger species. The project selected three trigger species to be monitored more intensively, namely *Presbytis chrysomelas*, *Manis javanica*, and *Hylobates muelleri abbotti*. The remaining species will also be monitored through regular standardized protocol (e.g., point-transect for birds, line transect for mammals, permanent vegetation plot for plants, etc.) to determine species’ occurrence and abundance in the project area.

Presbytis chrysomelas

The Bornean banded langur is classified as Critically Endangered on the IUCN Red List species. This primate is endemic to the island of Borneo, particularly in Brunei, Indonesia (Kalimantan), and Malaysia (Sabah and Sarawak). It has one of the smallest distribution ranges of any langur on the island, and can be found in the north and northeast of Kapuas River, as well as in northwestern Kalimantan and through Sarawak and Brunei.¹³⁷

Manis javanica

The Sunda pangolin, or *Manis javanica*, is classified as Critically Endangered on the IUCN Red List Species. This species is native to Indonesia, with home range which could cover 607 hectares.¹³⁸

Hylobates muelleri abbotti

The Abbott’s grey gibbon also known as the western grey gibbon is a primate in the gibbon family that is endemic to the western part of Borneo, where it is found both in Kalimantan and Sarawak. This species is categorized as Endangered species on the IUCN Red List Species.

¹³⁷ Nijman, V., Cheyne, S., Traeholt, C. & Setiawan, A. 2020. *Presbytis chrysomelas*. The IUCN Red List of Threatened Species 2020: e.T39803A17955321. <https://dx.doi.org/10.2305/IUCN.UK.2020-2.RLTS.T39803A17955321.en>

¹³⁸ R.J. Gray, D. Van Le, H.T.T. Nguyen, L.N. Cao, T. Van Nguyen, T. Van Pham, D. Willcox, T. Chen, T. Van Nguyen, Home ranges and activity patterns of sunda pangolins *Manis javanica* (pholidota: manidae) in Vietnam, *J. Asia-Pacific Biodivers* 16 (2023) 421–431, <https://doi.org/10.1016/j.japb.2023.05.005>.

Trigger species	<i>Presbytis chrysomelas</i>
Population trend at start of project	The species is endemic to Borneo and classified as a Critically Endangered species by the IUCN Red List, with current population trends decreasing. Only one individual was met during the field assessment. According to the IUCN, this species is now considered as one of the rarest primates in the world. Which occurs in less than 5% of its historic range. Habitat conversion has historically been the main threat to this species, resulting in its disappearance from most of its home range. It has been affected by expanding plantations.
Without-project scenario	Without the project scenario, this species' habitat will continue to decrease as the project area would have been converted to an Industrial Timber Plantation.
With-project scenario	To maintain and/or enhance the population status of <i>Presbytis chrysomelas</i> , the project will conserve and protect the remaining forest in the project area. Restoration efforts will be conducted to increase the forest connectivity and enlarge the habitat connection between each forest patch. Additionally, forest patrolling will be undertaken to detect illegal activity, such as hunting, that will impact this species. Customary law also will be developed with the local community to protect this species from hunting activity.

Trigger species	<i>Manis javanica</i>
Population trend at start of project	The species is endemic to Borneo and classified as a Critically Endangered species by the IUCN Red List, with current population trends decreasing. In this case, none of this individual species was seen during the field survey, but the community the presence of burrow usually associated to be the home nest of this species, and the IUCN home range map collectively point

	to the fact of the occurrence of this species within the project zone.¹³⁹ The species population trend is decreasing due to targeted hunting, being in Indonesia, including Kalimantan, the biggest hunting regions. In the last two decades thousand pangolins have been hunted¹⁴⁰. The secondary threat to this species is habitat loss, driven by commercial operations such as forest conversion to agriculture and plantation³⁹.
Without-project scenario	Without the project scenario, this species' habitat will continue to decrease as the project area would have been converted to an Industrial Timber Plantation.
With-project scenario	To maintain or enhance the population status of <i>Manis javanica</i> . The project will conserve and protect the remaining forest in the project area. Restoration efforts will be conducted to increase the forest connectivity and enlarge the habitat connection between each forest patch. Forest patrolling will be undertaken to detect illegal activity, such as hunting, that can impact this species. Customary law also will be developed with the local community to protect this species from hunting activity.
Trigger species	<i>Hylobates muelleri abbotti</i>
Population trend at start of project	<i>Hylobates muelleri abbotti</i> is considered as an Endangered species by IUCN Red list. The population trend of this species is reducing. In the period between 2004 to 2018, the population decreased around 50%.¹⁴¹ Between 1973 and 2017, Borneo lost an average of 1% of its forest coverage annually, if this situation continues 75% of suitable habitat for this

¹³⁹ Challender, D., Willcox, D.H.A., Panjang, E., Lim, N., Nash, H., Heinrich, S. & Chong, J. 2019. *Manis javanica*. The IUCN Red List of Threatened Species 2019: e.T12763A123584856. <http://dx.doi.org/10.2305/IUCN.UK.2019-3.RLTS.T12763A123584856.en>

¹⁴⁰ Pantel S, Chin SY (2009) Proceedings of a Workshop on Trade and Conservation of Pangolins Native to South and Southeast Asia. TRAFFIC Southeast Asia, Petaling Jaya, Malaysia.

¹⁴¹ Cheyne, S.M. & Nijman, V. 2020. *Hylobates abbotti*. The IUCN Red List of Threatened Species 2020: e.T39889A17990882. <https://dx.doi.org/10.2305/IUCN.UK.2020-2.RLTS.T39889A17990882.en>

	species can be lost ¹⁴² . Another threatening factor for this species is the continued practice of hunting activities, which is common by the community within the project zone.
Without-project scenario	Without the project scenario, this species' habitat will continue decreasing as the project area would have been converted to an Industrial timber Plantation.
With-project scenario	To maintain or enhance the population status of <i>Hylobates muelleri abbotti</i> . The project will conserve and protect the remaining forest in the project area. Restoration efforts will be conducted to increase the forest connectivity and enlarge the habitat connection between each forest patch. Additionally, forest patrolling will be undertaken to detect illegal activity, such as hunting, that will impact this species. Customary law also will be developed with the local community to protect this species from hunting activity.

¹⁴² Gaveau, D.L.A., Locatelli, B., Salim, M.A., Yaen, H., Pacheco, P. and Sheil, D. 2018. Rise and fall of forest loss and industrial plantations in Borneo (2000-2017). Conservation Letters 12

ANNEX 1: PROOF OF CARBONX OWNERSHIP

ANNEX 2: PROJECT ACTIVITIES AND THEORY OF CHANGE

Outcome	Output	Key Activity	Monitoring Indicator	Monitoring Frequency	Data Collection Method
Conserve d forest areas	The dynamic of forest cover is monitored	1. Develop LULC map 2. Conduct LULC analysis 3. Conduct community-based forest monitoring program using participatory rural appraisal tools and techniques to track forest degradation	1. Yearly LULC map 2. Yearly LULC analysis report 3. Bi-annually PRA for forest degradation report	Annually for LULC analysis and bi-annually for PRA activities	1. Remote sensing and ground truthing 2. Remote sensing and ground truthing 3. Internal PRA report
	Forest conversion is avoided	1. Revise the concession management plan for conservation and protection activities 2. Establish community-based forest and fire patrol team 3. Conduct periodic patrols for the prevention of illegal activity 4. Establish forest patrol guard post 5. Establish basecamp	1. A new management plan to conserve and protect the forest area 2. Number of community-based forest and fire patrol groups established 3. Number of people employed as forest and fire patrol team 4. Number of hectareage forest patrolled 5. Number of guard post constructed 6. Number of basecamps constructed	Annually except for No. 1	1. Company management plan 2. Village head decree on group establishment 3. Internal employee report 4. Internal forest patrol report 5. Construction report 6. Construction report
	Agricultural area expansion is controlled	1. Document the location and area of palm oil and other agricultural area 2. Conduct training to increase the community awareness about invasive species and the benefit of forest for ecosystem services	1. Annual map of oil palm and agricultural areas 2. Annual new area of oil palm or agriculture in the project area 3. Number of trainings 4. Number of people that join the training	Annually	1. Remote sensing and ground truthing 2. Remote sensing and ground truthing 3. Internal training report 4. Internal training report

Outcome	Output	Key Activity	Monitoring Indicator	Monitoring Frequency	Data Collection Method
Increased trees cover in non-forested areas	Tree nurseries are established	1. Construct nursery ground 2. Hire community members to work on the nursery 3. Cultivate the seedling	1. Number of nursery ground constructed 2. Number of people employed in nursery 3. Number of seedlings produced annually	Annually	1. Nursery internal report 2. Internal employee report 3. Nursery internal report
	Degraded forest areas are restored	1. Conduct restoration in the degraded forest area	1. Shrub areas are planted or undergoing restoration (in ha). 2. Number of seedlings planted for shrub restoration 3. Growth indicator of planted seedlings in degraded forest area monitored 4. Survival rate of planted seedlings in degraded forest areas	Bi-annually	1. Reforestation internal report and drone photos/satellite images of reforestation area 2. Restoration internal report 3. Restoration internal report 4. Restoration internal report
	Agroforestry is developed in agricultural areas	1. Develop agroforestry in agricultural area	1. Agricultural areas are planted with agroforestry scheme (in ha) 2. Number of seedlings planted for agroforestry 3. Growth indicator of planted seedlings in agroforestry area monitored 4. Survival rate of planted seedlings in agroforestry areas 5. Occurrence of pest and disease in agricultural or agroforestry area 6. Number of yields from agroforestry	Annually	1. Reforestation internal report and drone photos/satellite images of agroforestry area 2. Agroforestry internal report 3. Agroforestry internal report 4. Agroforestry internal report 5. Agroforestry internal report 6. Agroforestry internal report

Outcome	Output	Key Activity	Monitoring Indicator	Monitoring Frequency	Data Collection Method
Reduced fires and water pollution	Land and forest fires are controlled and reduced	1. Establish community-based forest and fire patrol team 2. Provision of firefighting equipment 3. Establish fire monitoring system 4. Conduct "pertanian tanpa bakar" (no slash and burn for agriculture) training to the community	1. Number of community-based forest and fire patrol groups established 2. Number of people employed as forest and fire patrol team 3. Availability and condition of equipment 4. Fire monitoring system established 5. Number of people join the training 6. Number of fire hotspot	Annually	1. Head of village decree regarding establishment of community-based forest and fire patrol group 2. Internal employee report 3. Internal report 4. Internal report 5. Internal training report 6. Remote sensing and ground truthing
	Water quality is controlled and monitored	1. Conduct periodic patrols for the prevention of illegal activity 2. Conduct water quality monitoring 3. Conduct waste, pesticide, and fertilizer monitoring 4. Develop waste collection point	1. Number of identified mining areas in river during patrol activities 2. Availability of water quality data 3. Availability of waste, pesticide, and fertilizer data 4. Waste collection point developed in each village	Every 3 months	1. Internal patrol report 2. Water quality internal report 3. Waste, pesticides, and fertilizer internal report 4. Waste, pesticides, and fertilizer internal report

Outcome	Output	Key Activity	Monitoring Indicator	Monitoring Frequency	Data Collection Method
Enhanced population of threatened species and maintained habitat	The dynamic of biodiversity is monitored	1. Establish a biodiversity monitoring team 2. Provision of biodiversity monitoring equipment 3. Conduct regular biodiversity monitoring 4. Establish MoU with universities/research agencies 5. Conduct joint research on the biodiversity in the project area	1. Number of people employed as biodiversity monitoring team 2. Availability and condition of biodiversity monitoring equipment, including permanent biodiversity monitoring transect/plots 3. Number of biodiversity monitoring activities conducted 4. Number of MoU with universities/research agencies 5. Number of research report disseminated to partner institution	Bi-annually	1. Internal employee report 2. Internal biodiversity report 3. Taxonomy based survey for mammals, bird, herpetofauna, and vegetation 4. MoU document 5. Research document
	Hunting of threatened species is controlled	1. Conduct training to increase the community awareness of biodiversity 2. Facilitate the creation of customary law related to conservation of biodiversity 3. Conduct periodically PRA survey to determine the species that is still being hunted	1. Number of people join the training 2. Number of created of customary law related to conservation of biodiversity 3. Number of threatened species that is still being hunted	Bi-annually	1. Internal training report 2. Customary law document 3. Internal PRA report
Improved community livelihood	Alternative livelihood options for the community are developed through green economy activities	Develop alternative livelihoods for community as substitution of oil palm plantation and development of new agricultural areas	1. Number of alternative livelihoods established 2. Number of people implementing alternative livelihood activities	Annually	1. PRA with the community with questionnaire, FGD and interview 2. Internal training report

Outcome	Output	Key Activity	Monitoring Indicator	Monitoring Frequency	Data Collection Method
	Communities' involvement in agroforestry and restoration is maintained	1. Document the land user for agroforestry and restoration area 2. Develop community agricultural groups 3. Involve the community in agroforestry and restoration development	1. The land user in agroforestry area 2. Number of groups established 3. Number of people involved in agroforestry and restoration activities	Annually	1. Agroforestry and restoration internal report 2. Agroforestry and restoration internal report 3. Agroforestry and restoration internal report 4. Company internal report
	Capacity to manage natural resources is improved	1. Conduct green economic activities training 2. Conduct development fund management training for established community funds	1. Number of people join the green activities training 2. Number of community funds received a training 3. Number of community funds staff join the training	Annually	1. Internal training report 2. Internal training report 3. Internal training report
Improved community wellbeing	Education access is improved	1. Provide package B program to the community 2. Provide package C program to the community 3. Provide a partial scholarship for community members who are accepted to state universities 4. Establish the vocational study 5. Conduct environmental educational program	1. Number of people provided by package B program 2. Number of people provided by package C program 3. Number of people that got partial scholarship 4. Number of people that got vocational study 5. Number of elementary and middle school student participate in Environmental Education Program	Annually	1. Community internal report 2. Community internal report 3. Community internal report 4. Community internal report 5. Community internal report

Outcome	Output	Key Activity	Monitoring Indicator	Monitoring Frequency	Data Collection Method
	Healthcare access is improved	1. Conduct doctor goes to village or other healthcare services program 2. Conduct traditional medication from forest resource workshop	1. Number of people that visit doctor on doctor goes to village program or use the provided healthcare services 2. Number of doctor goes to village program or healthcare services provision per year 3. Number of community member that join the workshop	Annually	1. Doctor goes to village internal report 2. Workshop internal report 3. Workshop internal report 4. Community Internal Report
	Local culture and wisdom are preserved	1. Document all the important places for culture and local wisdom 2. Conserving all the important places for culture and local wisdom 3. Provision cultural activities and/or equipment's that promote local culture and wisdom 4. Facilitate the cultural event in the project zone	1. Availability of map of location of key place for culture and local wisdom 2. Condition of the important place for culture and local wisdom 3.1. Number of people that join the event 3.2. Number of women that join the event 3.3 Number of cultural equipment sets provided to the community 4. Number of cultural events conducted in a year	Annually	1. Map of key place for culture and local wisdom 2. Community internal report 3. Community internal report 4. Community internal report
	Water access is improved	1. Improve communities' access to clean water	1. Number of people benefiting from enhanced access to clean water	Annually	1. Community Internal Report
	Flood early warning system is ready for operationalization	1. Install early warning system for flood	1. Number of early warning system ready for operationalization	Annually	1. Internal report

ANNEX 3: PT CMI NEW MANAGEMENT PLAN UNDER CARBONX

ANNEX 4: OFFICIAL REPORT OF INFORMATIONAL MEETING

ANNEX 5: INFORMATIONAL MEETING NOTES

ANNEX 6: GRIEVANCE PROCEDURE

ANNEX 7: HEALTH AND SAFETY SOP

ANNEX 8: PERFORMANCE BENCHMARK CALCULATION

The methodology implemented in this appendix follows the requirements specified in Appendix 1: Performance Method of the VM0047 v1.0. This section aims to explain the methodology implemented to compute the changes in the stocking index and determine the performance benchmark.

Select Stocking Index (SI)

For this project, it was decided to use the vegetation indices the Difference Vegetation Index (DVI). DVI is a vegetation index measuring the difference between near-infrared band and red band from satellite imagery reflected from plants. It is considered a simple vegetation index that is sensitive to the amount of vegetation and can distinguish between soil and vegetation.¹⁴³ The Study from Purnamasari et al, 2021 used the DVI, NDVI and EVI to estimate the correlation between the stocking index and above-ground carbon stock, and it was generated the results of DVI has the highest correlation and accuracy in estimating above-ground carbon stock based on the stocking index. Accordingly, the project used DVI as a stocking index that is being used for performance benchmark estimation and as the parameter used to estimate the correlation between the stocking index and aboveground biomass (AGB).

DVI was generated through remote sensing analysis by utilizing the Google Earth engine platform. Landsat imagery is used to produce DVI and it is acquired from 2013 – 2023, to get historical stock index conditions.

Select Project Plots

To select the project plots, the project area was gridded into 0.36023 Ha plots and 30 plots were selected based on a systematic random approach. The grid 0.36023 Ha was created and filtered based on spatial analysis and the information of LULC within the project area. From The 30 plots selected, 15 plots were in the shrub area and the rest were in the agriculture area. Detailed location of selected project plots explained in ER Model (Annex 16).

Define Donor Pool Area of Control Plots

To determine the donor pool area, firstly the 100km buffer was created based on the centroid point of the project plots. Secondly, was gridded into 0.288384 Ha plots and thirdly the plots were filtered based on different raster and vector geospatial layers. The donor pool area plots were filtered based on the factors detailed in table A1 of VM0047 v1.0 and based on LULC conditions.

Table 92. Description of the criteria implemented to determine the Donor Pool Area

Factor	Layer(s)	Criteria	Justification
Distance from Project Plots	100 km buffer shape. Source: spatial analysis	Areas beyond 100 km buffer of the centroid points were excluded	Methodology Requirement

¹⁴³ Tucker, C. "Red and Photographic Infrared Linear Combinations for Monitoring Vegetation. *Remote Sensing of Environment* 8 (1979): 127–150.

Factor	Layer(s)	Criteria	Justification
Jurisdictional Boundary	Administration Boundary of Indonesia (Region and Provincial Levels), shapefile layers Source: Geospatial Information Agency of Indonesia (Badan Informasi Geospasial – BIG) ¹⁴⁴	Plots outside West Kalimantan Province were excluded	The project area is located in West Kalimantan Province
Ecoregion	Ecoregion dataset from RESOLVE, 2017, shapefile layers Source: RESOLVE, 2017 ¹⁴⁵	Plots outside tropical & sub-tropical Moist Broadleaf Forest were excluded from Donor Pool Area	The project area is located in Tropical & subtropical moist broadleaf forest ecoregion. Following VM0047 v1.0, any area not within the same ecoregion shall be excluded
Policy Environment	-	No plots were filtered based on this factor	No plots were filtered based on this factor as the project area is a concession area for industrial timber plantations. Thus, donor pool area selection must have the same conditions as the project area
Outside any Registered AFOLU Projects	-	No plots were filtered based on this factor	There are no Registered AFOLU projects in West Kalimantan within industrial timber plantation areas
Land Tenure	Concession areas of Industrial Timber Plantation of Indonesia, shapefile layers Source: Concession Areas of Industrial Timber Plantation ¹⁴⁶	Plots outside industrial timber plantations were excluded from donor pool areas	VM0047 v1.0 stated that the donor pool area must have the same condition as the project area (the project area is a concession area for industrial timber plantation).

¹⁴⁴ <https://atlas.big.go.id/#/atlas/Wilayah%20NKRI~Territory%20of%20NKRI>

¹⁴⁵ Dinerstein, Eric & Olson, David & Joshi, Anup & Vynne, Carly & Burgess, Neil & Wikramanayake, Eric & Hahn, Nathan & Palminteri, Suzanne & Hedao, Prashant & Noss, Reed & Hansen, Matt & Locke, Harvey & Ellis, Erle & Jones, Benjamin & Barber, Charles & Hayes, Randy & Kormos, Cyril & Martin, Vance & Crist, Eileen & Saleem, Muhammad. (2017). An Ecoregion-Based Approach to Protecting Half the Terrestrial Realm. *BioScience*. 1. 1-12. 10.1093/biosci/bix014.

¹⁴⁶ Ministry of Environmental and Forestry Indonesia

Factor	Layer(s)	Criteria	Justification
LULC	LULC raster dataset (2013,2016,2020,2023). Source: Remote sensing analysis and LULC from MapBiomass. ¹⁴⁷	Plots outside non-forest and other agriculture were excluded from the donor pool area.	the selected project plots are in the shrub (non-forest) and agriculture area (other agriculture). Hence the plots outside these areas shall be excluded from the donor pool area

After filtering the donor pool area based on all these criteria, the donor pool area of control plots with similar biophysical and political characteristics to the project plots was obtained

Stocking Index Temporal Series in Plots

Once the project plots and the filtered donor pool area are defined, the annual time series of the stocking index in the target period is obtained for each plot for the period t-10 to t0, where t0 is the year of the project start date. A minimum of three time points must be used to select the control plots and evaluate the control plots matched with project plots through the regression of Stocking Index (SI). the three time points include: t-10 and t-8, between t-8 and t-1 and in t0. This project used four times points, including t-10, t-7, t-3 and t0.

Selecting Control Plots, Evaluate Match Quality and Finalize Matching

Control plots from the donor pool area were selected and matched with project plots through a K-Nearest Neighbour matching approach without replacement. The match was done with a spatial analysis tool, using the Euclidean distance as a distance metric and five as K (number of nearest neighbors per project plot). The nearest neighbors were determined by the Euclidean distance for each project plot, and five control plots with the lowest Euclidean distance values were matched. The SI values used as covariates were the same as indicated in step 4: t-10, t-7, t-3 and t0. Based on this process, 30 control plots were selected (detailed 30 control plots selected with the geolocation can be found in Annex 16. ER Model GHG Calculation).

After selecting control plots, it followed up with assessing the quality of the match by calculating standardized difference of means (SDM). The SDM for each covariate was calculated as shown in equation A2, VM0047 v1.0 (please see section 3.1.4.2 for a detailed explanation of SDM). The result of the SDM calculation shows a mean value of 0.08 or less than 0.25. Based on this, the 30 control plots matched to the 30 project plots are deemed valid.

Ex-ante Estimations

The calculation of ex-ante estimates involves:

- Estimating performance benchmark from historical period (t-10 – t0)
- Generating a regression of SI on above-ground biomass
- Estimating the SI increase 50 years ahead, for project ($\Delta SI_{lp,t}$) and control plots ($\Delta SI_{control,t}$).
- Calculating ex-ante estimation of Performance Benchmark

¹⁴⁷ <https://platform-map.nusantara.earth/>

a) Estimating performance benchmark from historical period (t-10 – t0)

Performance benchmark from the historical period was calculated based on $\Delta SI_{wp,t}$ and $\Delta SI_{control,t}$. Both parameters were estimated across the sample populations of project and control plots. The rate of increase in stocking index in $\Delta SI_{wp,t}$ and $\Delta SI_{control,t}$ is calculated as the slope of the weighted linear regression of the accumulated time series of SI values for the respective population of plots. Based on this process, generated $\Delta SI_{wp,t}$ of 0.00757 and $\Delta SI_{control,t}$ of 0.00529.

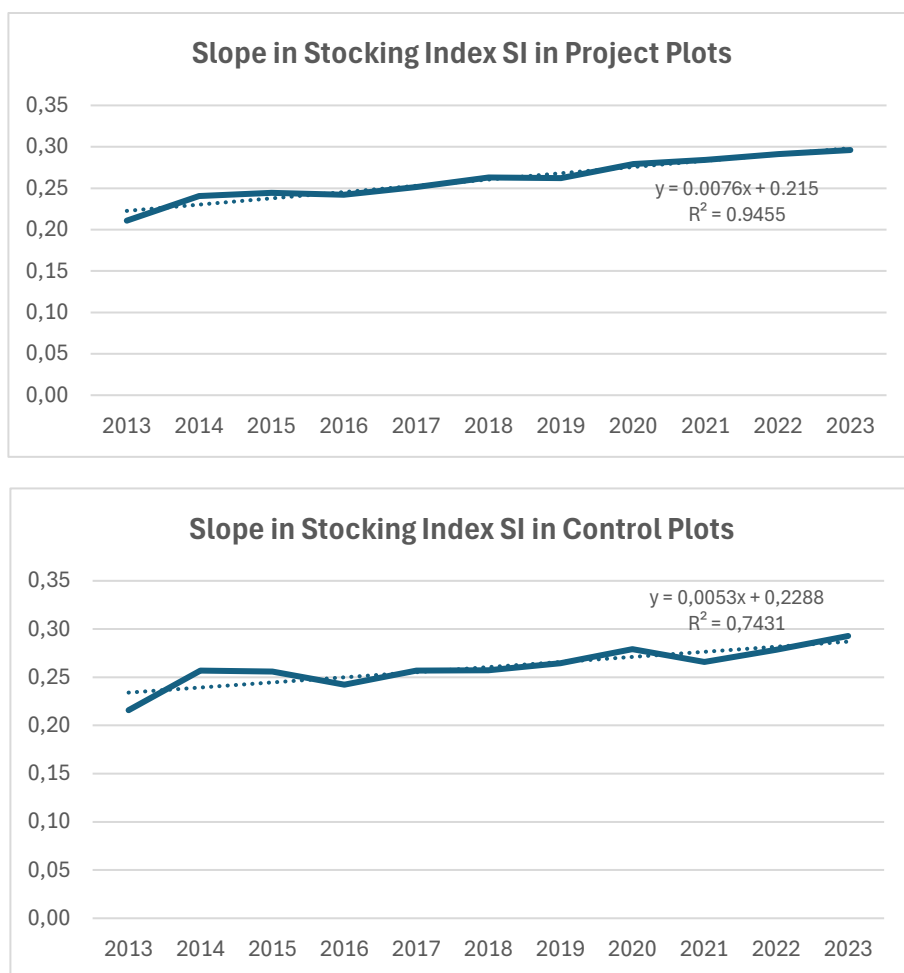


Figure32. The result of Linear Regression for Estimating $\Delta SI_{wp,t}$ and $\Delta SI_{control,t}$

After estimating the results for $\Delta SI_{wp,t}$ and $\Delta SI_{control,t}$, significant differences were assessed using the z-test calculated from the following equation A5 in VM0047 v1.0 (see section 3.1.4.2 for the explanation of the equation). The z-test required the standard error from the average annual increase (slope) in the stocking index in control plots and project plots. The standard error was estimated based on statistical analysis, and the results show a value of 0.00069 for project plots and 0.00074 for control plots. By those parameters, the z-test calculation results show a value of 2.27, which means the $\Delta SI_{wp,t}$ and $\Delta SI_{control,t}$ are deemed significantly different (Z value < 1.96 the parameters are not significantly different).

Performance benchmark from historical period estimated based on $\Delta SI_{wp,t}$ and $\Delta SI_{control,t}$ with following the equation A6 in VM0047 v1.0. Based on the estimation performed, the performance benchmark of the historical period is obtained as 0.70.

b) Generating a regression of SI on aboveground biomass

Following VM0047 v1.0, to generate $\Delta SI_{wp,t}$ for ex-ante estimation shall perform the regression model between aboveground biomass and stocking index. The linear regression between SI and aboveground biomass was calculated from the AGB dataset published by the NASA Carbon Monitoring System (2014) within the project area¹⁴⁸ and the stocking index data in the same year. The linear regression analysis was conducted only on shrubs and agriculture areas following the eligible areas for the ARR project. The linear regression model obtained from this analysis has an R2 of 0.812 with the following equation:

$$Y = 0.005740 + 0.001377x$$

Where y is the stocking index and x is the aboveground biomass value in tonnes of dry matter per hectare.

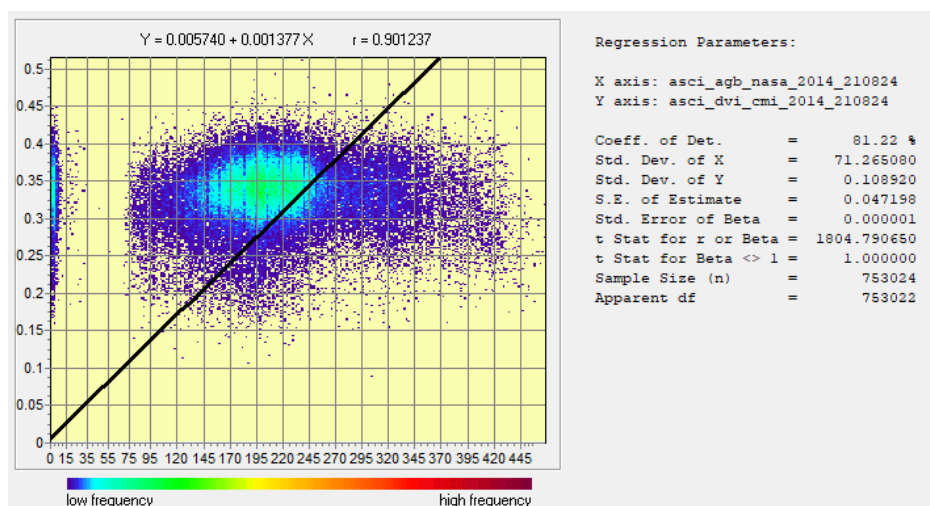


Figure 33. Scatter plot between the Stocking Index and the Aboveground Biomass

c) Estimating the SI increase 50 years ahead, for project ($\Delta SI_{wp,t}$) and control plots ($\Delta SI_{control,t}$). From the regression of SI as a function of biomass, SI values over time was estimated for the project scenario and ΔSI_{wp} was calculated as the slope of SI from $t_0 - t_{10}$, $t_{10} - t_{20}$, $t_{30} - t_{40}$ and $t_{40} - t_{50}$. To do this, x was replaced by the biomass per hectare values obtained from the biomass growth curves in the project scenario. As explained in section 3.2.2.2, the calculation of biomass is done not only based on planting but also on assisted natural regeneration (only for shrub areas). Based on actual conditions, the shrubland area has already started to grow, hence the calculations need to begin at year $t-9$, and natural growth should be estimated as part of the project scenario starting at year t_0 or during the growth in year 10 to align the growth conditions with the inventory results.

Based on the process above, the results of ex-ante estimation for ΔSI_{wp} are provided in the table below:

¹⁴⁸ Ferraz, A., S.S. Saatchi, L. Xu, S. Hagen, J. Chave, Y. Yu, V. Meyer, M. Garcia, C. Silva, O. Roswintarti, A. Samboko, P. Sist, S.M. Walker, T. Pearson, A. Wijaya, F.B. Sullivan, E. Rutishauser, D. Hoekman, and S. Ganguly. 2019. Aboveground Biomass, Landcover, and Degradation, Kalimantan Forests, Indonesia, 2014. ORNL DAAC, Oak Ridge, Tennessee, USA. <https://doi.org/10.3334/ORNLDAC/1645>

Table 93. Estimation of $\Delta SI_{wp,t}$ From Aboveground Biomass by Linear Regression

Period	Slope ($\Delta SI_{wp,t}$)
Year 1-10	0.0172
Year 11-20	0.0194
Year 21-30	0.0167
Year 31-40	0.0131
Year 41-50	0.0099

The ex-ante value of $\Delta SI_{control,t}$ for the period t_0 to t_{50} was estimated from the slope of linear regression of the SI over the last 10 years on the control plots. For example, if the slope of the regression is 0.00529 over the last 10 years, then the control plots are expected to maintain the same growth rate over the next 50 years.

d) Calculating ex-ante estimation of Performance Benchmark

Performance benchmark values were estimated based on the ex-ante values of ΔSI_{wp} and $\Delta SI_{control}$ for 50 years. The performance benchmark is estimated following equation A6 in VM0047 v1.0 (See section 3.1.4.2 for the explanation of the equation). The results of the performance benchmark estimation are provided in the table below.

Table 94. Ex-ante Estimates values of ΔSI_{wp} , $\Delta SI_{control}$ and Performance Benchmark

Years	T	AGB (tdm/ha)- planting + Natural Regeneration	SI _{wp}	ΔSI_{wp}	SI _{control}	$\Delta SI_{control}$	PB
2023	0	210.85	0.30	0.01	0.29	0.00529	0.70
2024	1	218.90	0.31	0.02	0.30	0.00529	0.31
2025	2	228.74	0.32	0.02	0.30	0.00529	0.31
2026	3	239.77	0.34	0.02	0.31	0.00529	0.31
2027	4	251.71	0.35	0.02	0.31	0.00529	0.31
2028	5	264.34	0.37	0.02	0.32	0.00529	0.31
2029	6	277.51	0.39	0.02	0.32	0.00529	0.31
2030	7	291.10	0.41	0.02	0.33	0.00529	0.31
2031	8	305.00	0.43	0.02	0.34	0.00529	0.31
2032	9	319.12	0.45	0.02	0.34	0.00529	0.31
2033	10	333.40	0.46	0.02	0.35	0.00529	0.31
2034	11	347.76	0.48	0.02	0.35	0.00529	0.27
2035	12	362.16	0.50	0.02	0.36	0.00529	0.27
2036	13	376.55	0.52	0.02	0.36	0.00529	0.27
2037	14	390.88	0.54	0.02	0.37	0.00529	0.27
2038	15	405.13	0.56	0.02	0.37	0.00529	0.27
2039	16	419.26	0.58	0.02	0.38	0.00529	0.27
2040	17	433.25	0.60	0.02	0.38	0.00529	0.27
2041	18	447.07	0.62	0.02	0.39	0.00529	0.27

Years	T	AGB (tdm/ha)- planting + Natural Regeneration	Siwp	ΔSiwp	Sicontrol	ΔSicontrol	PB
2042	19	460.70	0.64	0.02	0.39	0.00529	0.27
2043	20	474.14	0.66	0.02	0.40	0.00529	0.27
2044	21	487.36	0.68	0.02	0.40	0.00529	0.32
2045	22	500.35	0.69	0.02	0.41	0.00529	0.32
2046	23	513.11	0.71	0.02	0.41	0.00529	0.32
2047	24	525.62	0.73	0.02	0.42	0.00529	0.32
2048	25	537.89	0.75	0.02	0.42	0.00529	0.32
2049	26	549.89	0.76	0.02	0.43	0.00529	0.32
2050	27	561.64	0.78	0.02	0.44	0.00529	0.32
2051	28	573.13	0.79	0.02	0.44	0.00529	0.32
2052	29	584.36	0.81	0.02	0.45	0.00529	0.32
2053	30	595.32	0.83	0.02	0.45	0.00529	0.32
2054	31	606.02	0.84	0.01	0.46	0.00529	0.40
2055	32	616.46	0.85	0.01	0.46	0.00529	0.40
2056	33	626.63	0.87	0.01	0.47	0.00529	0.40
2057	34	636.55	0.88	0.01	0.47	0.00529	0.40
2058	35	646.21	0.90	0.01	0.48	0.00529	0.40
2059	36	655.62	0.91	0.01	0.48	0.00529	0.40
2060	37	664.77	0.92	0.01	0.49	0.00529	0.40
2061	38	673.68	0.93	0.01	0.49	0.00529	0.40
2062	39	682.34	0.95	0.01	0.50	0.00529	0.40
2063	40	690.77	0.96	0.01	0.50	0.00529	0.40
2064	41	698.96	0.97	0.01	0.51	0.00529	0.53
2065	42	706.91	0.98	0.01	0.51	0.00529	0.53
2066	43	714.64	0.99	0.01	0.52	0.00529	0.53
2067	44	722.15	1.00	0.01	0.53	0.00529	0.53
2068	45	729.44	1.01	0.01	0.53	0.00529	0.53
2069	46	736.52	1.02	0.01	0.54	0.00529	0.53
2070	47	743.39	1.03	0.01	0.54	0.00529	0.53
2071	48	750.05	1.04	0.01	0.55	0.00529	0.53
2072	49	756.52	1.05	0.01	0.55	0.00529	0.53
2073	50	762.79	1.06	0.01	0.56	0.00529	0.53

ANNEX 9: FINANCIAL ANALYSIS

ANNEX 10: PT CMI MANAGEMENT PLAN

ANNEX 11: RISK OF ABANDONMENT CALCULATIONS

ANNEX 12: PT CMI AGREEMENT WITH PT INDONESIA WOOD INVESTMENT

ANNEX 13: FOREST INVENTORY SOP

ANNEX 14: BIODIVERSITY ASSESSMENT SOP

ANNEX 15: AFOLU NON PERMANENCE RISK TOOL

ANNEX 16: ER MODEL GHG CALCULATION

ANNEX 17: COMMERCIAL SENSITIVE INFORMATION

Section	Information	Justification
Section 2.1.22, 3.1.5.2.2	Project implementation plans and financial plans	Competitive positioning
Section 3, Section 4, Section 5	Original data, computer code, and GIS information	The intellectual property of the project will be available upon request from VVB
Section 2.1.8, 2.5.7, 3.3.2.1	Agreement and contract between project proponent and other entities	Confidentiality clauses
Section 2.5.8, 4.5.6	Benefit sharing mechanism	Competitive positioning

ANNEX 18: SIGNED FIELD WORK CONSENT LETTER

ANNEX 19: SIGNED PROJECT CONSENT

ANNEX 20: FOREST INVENTORY REPORT